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OECD ECONOMIC SURVEYS

2003-2004

Switzerland

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BASIC STATISTICS OF SWITZERLAND

THE LAND

Area (1 000 sq. km)	41.3
Cultivated land, grassland and pastures (1 000 sq. km)	15.3
Forests (1 000 sq. km)	12.7

Major cities (1 000 inhabitants, 31.12.2001):	
Zurich	340.9
Basel	164.9
Geneva	176.0
Bern	122.5

THE PEOPLE

Population (thousands, 31.12.2002)	7 138
Number of inhabitants per sq. km (2002)	177
Net natural increase (thousands, 2002)	10.6
Number of foreign workers (thousands, 2002)	1 058

Civilian employment (thousands, 2002)	4 172
Agriculture (%)	4.1
Industry and construction (%)	25.1
Services (%)	70.8

PRODUCTION

Gross domestic product, current prices (2002)	
CHF billion	417.3
Per head (USD)	36 630

Gross fixed investment, current prices (2002)	
% of GDP	18.9
Per head (USD)	6 912

THE GOVERNMENT

Public consumption (% of GDP, 2002)	15.2
General government (% of GDP, 2002)	
Expenditure	40.3
Revenues	40.1
Debt	55.4

Composition of Parliament (No. of seats)	National Council	State Council
Socialists	52	9
Central Democratic Union	55	8
Radical Democrats	36	13
Christian Democrats	28	14
Other	29	2

Last elections: October 2003
Next elections: October 2007

FOREIGN TRADE

Exports of goods and services (% of GDP, 2002)	37.3
Commodity exports (billion CHF, 2002)	130.4
Distribution by area (% of total, 2002)	
To OECD countries	82.5
To EU countries	60.0
To OPEC countries	3.0
Distribution by categories (% of total, 2002)	
Raw materials and semi-finished goods	25.1
Capital goods	32.4
Consumer goods	42.2
Energy	0.3

Imports of goods and services (% of GDP, 2002)	33.1
Commodity imports (billion CHF, 2002)	123.1
Distribution by area (% of total, 2002)	
From OECD countries	91.3
From EU countries	80.4
From OPEC countries	1.5
Distribution by categories (% of total, 2002)	
Raw materials and semi-finished goods	26.2
Capital goods	26.6
Consumer goods	42.9
Energy	4.3

THE CURRENCY

Monetary unit: Swiss franc

Currency unit per USD, average of daily figures	
Year 2002	1.5568
October 2003	1.3235

This Survey is published on the responsibility of the Economic and Development Review Committee of the OECD, which is charged with the examination of the economic situation of member countries.

•

The economic situation and policies of Switzerland were reviewed by the Committee on 5 November 2003. The draft report was then revised in the light of the discussions and given final approval as the agreed report of the whole Committee on 12 December 2003.

•

The Secretariat's draft report was prepared for the Committee by Claude Giorno and Miguel Jimenez under the supervision of Peter Hoeller.

•

The previous Survey of Switzerland was issued in May 2002.

Assessment and recommendations

Boosting growth is the main policy challenge

Although trend growth of GDP has been meagre for two decades (1¼ per cent against 2¾ per cent for the OECD average), per capita income in Switzerland remains among the highest in the OECD. In recent years, output growth was less resilient than in most other OECD countries, with the economy being in recession during the first semester of 2003. This is partly due to the global slump in the financial and capital goods sectors which are very important for the Swiss economy, but the unsatisfactory growth performance is largely attributable to the timid pace of structural reforms over many years. Against this background, there are three major challenges for economic policy:

- In the short term, the most urgent task is to support the projected upturn and to minimise deflation risks, inflation being currently very low. The recovery should not be constrained by a fiscal tightening, while the National Bank should maintain easy monetary conditions, even though it has no margin left for lowering interest rates.
- In the medium term, the main challenge is to increase the economy's speed limits, mainly by raising productivity growth. Substantial efforts are needed to deal with the serious shortcomings in the functioning of the product markets, which are reflected, *inter alia*, in the very high level of prices compared with other countries. Moreover, the strict implementation of the new legal framework for competition will be vital.
- In the longer term, the challenge is to keep the public finances healthy. Underpinning this objective is the new fiscal rule that requires that the Confederation's accounts remain balanced in structural terms. If trend

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growth does not pick up, a difficult trade-off presents itself between curtailing spending, especially social spending, or raising tax pressure, which could well impair growth potential further. In this light, a further reform of the pension system to ensure its financial viability beyond 2015 appears essential.

The economy is in recession and inflation is very low

The downturn in activity has been severe, with almost no growth in 2002 and probably a small decline in 2003. Apart from the slump in the financial and capital goods sectors, exports were hit by the sluggishness of German demand and the franc's appreciation until spring 2003. In addition, final domestic demand has been depressed by the fall in investment since early 2001, following an investment boom in the late 1990s. Private consumption has also lost steam with the saving ratio drifting up because of mounting uncertainty. The unemployment rate has risen, reaching 3.8 per cent in autumn 2003, the highest in five years, but it is still low in international comparison. Because of weakness in the labour market and the negative output gap, inflation, which was already very low at the start of the recession, fell to ½ per cent (year-on-year) in November 2003.

The monetary stance should remain easy and the authorities should stand ready to intervene, should deflation become a serious risk

Responding to sagging activity, policy has been eased. Monetary policy started to be relaxed at the beginning of 2001 and the 3-months LIBOR rate was cut successively by 3¼ percentage points to 0.25 per cent. As measured by a monetary conditions index, the easing more than offset the real effective appreciation of the Swiss franc, which took place till the spring of 2003, partly in response to international uncertainties. While monetary conditions have become even easier more recently as the franc has weakened, the economic prospects for the rest of the year remain bleak. Activity, which has started picking up in the third quarter of 2003, pulled by exports, will however firm with the international recovery, and growth could reach 1¼ per cent next year, close to potential output growth. Reflecting a large negative output gap, inflation could fall further and even be negative for a few months in the near future. Deflation risks should not be exaggerated but need close surveillance. They could intensify if the negative risks surrounding the

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outlook materialise, in particular if the exchange rate were to strengthen substantially. In that event, and given that there is no room for further rate cuts, the Swiss National Bank (SNB) should stand ready to use unorthodox instruments to avoid deflation from becoming entrenched in expectations. In this respect, interventions in foreign exchange markets would be most effective. In any case, even in the absence of deflation, the monetary stance should remain easy until the recovery is firmly established. Overall, the monetary policy framework adopted since 2000, with its focus on inflation prospects over the next three years, has allowed the National Bank to respond flexibly and effectively to economic developments.

The slight fiscal easing in 2004 is appropriate

Following the consolidation of the 1990s, public finances have deteriorated uninterruptedly in the past few years, even though the deficit remains below the euro area average and that of the United States. The budget balance moved from a surplus of 2½ per cent of GDP in 2000 to a deficit of almost 2 per cent of GDP in 2003. While partly explained by sluggish activity, more than half of this swing is the result of a substantial decline in the cyclically-adjusted balance as conventionally measured. While discretionary measures only played a limited role, it is mostly due to the marked downturn in financial asset-related federal revenues, which have distorted the trend in and calculation of the structural balance. A structural deficit of ½ to 1 per cent of GDP is likely to be recorded in 2003 for government as a whole and the Confederation, instead of the previously projected balanced budget. Fiscal policy in 2004, which is based on a growth assumption of 1 per cent, is in principle bound by the debt containment rule. Its strict application would require the elimination of the federal structural deficit. As this could further weaken an already depressed economic situation, and in view of the uncertain outlook, the government has proposed to relax this constraint to avoid an immediate fiscal tightening, with the structural deficit then being gradually eliminated between 2005 and 2007. Under this scenario, the fiscal stance as a whole would remain slightly expansionary next year, largely because of the planned cut in unemployment insurance contributions. All told, the general government deficit could increase to

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2½ per cent of GDP in 2004. The federal authorities' strategy of taking any restrictive steps only when the recovery is underway seems appropriate, as does the reduction in social insurance contributions as this does not jeopardize the long-term financial viability of the unemployment insurance system. On the other hand, a further structural deterioration in the public accounts would be neither desirable nor a particularly effective way of stimulating activity in a small and outward-oriented economy.

Control over public spending needs to be improved

The proposed temporary deviation from the debt containment rule has been set in the context of a multi-annual consolidation plan, which is designed to avoid harming the credibility of the Confederation's fiscal strategy. There is consensus on a strategy to balance the federal accounts, but two constraints exist. First, the increase in tax pressure, which has been greater than in other countries over the past decade, needs to be halted. The consolidation plan, which amounts to ¾ per cent of GDP up to 2006, rightly attaches priority to controlling spending. Second, as the Confederation's budget accounts for only about 10 per cent of GDP, there are constraints on the expenditure side as well, which prevent a too rapid reduction in federal spending. These constraints restrict the scope of the proposed measures which, moreover, will not allow the debt containment rule to be applied fully in 2005 and 2006. This militates in favour of reforms that do more to tackle the causes of the spending increases than does the consolidation plan, which was established in a short time and did not focus on far reaching reforms. The objective should be to alter outlay trends on a lasting basis to bring them in line with the medium-term revenue trajectory. The severity of such a spending adjustment will depend on what happens to trend growth in the future.

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The implementation of the debt containment rule has been difficult, but fiscal discipline needs to be maintained in the medium term

The initial difficulties in implementing the debt containment rule must not undermine the Confederation's fiscal discipline which the rule is designed to guarantee in the medium term. The difficulties reflect problems in estimating the structural balance, which can be affected by temporary fluctuations. The authorities have therefore made technical adjustments to limit the impact of transitory revenue fluctuations linked to financial asset cycles. The amended fiscal rule should now be allowed to operate for some time so that more information on its functioning can be gathered before making further adjustments. However, lessons have been learnt from the initial implementation of the rule. For example, it might be desirable to identify more clearly the non-cyclical irregular component of the budget balance, instead of making *ad hoc* adjustments. The debt containment rule should also be embedded in a medium-term fiscal framework, to avoid annual fine-tuning and to reconcile the objectives of public spending control and efficient allocation. The principles underlying the debt containment rule should also be respected in the four-year financial plan.

The long-term financial viability of the basic and invalidity pension systems has to be guaranteed, without penalising growth

The structure of the pension system, which combines both funded and pay-as-you-go schemes, limits the fiscal impact of population ageing better than in most other countries. However, the funding of first pillar public pensions is not guaranteed beyond 2015. The recent opening of discussions to limit the borrowing requirement – put at 2¼ per cent of GDP between 2015 and 2040 – is timely. The time required to implement a reform should not be underestimated, the more so as reforms have to be phased in gradually so as to provide people with time to adapt. These could include the raising of the retiring age from 65 to 67 and a reform of the pension indexation system, even though this latter measure might be difficult to sustain in the long run. For reasons of equity between generations, it would seem right to focus on both revenues, for instance an increase in VAT, and benefits. Choosing measures which do not hamper growth and which extend working life would be rational and compatible with the lengthening of life expectancy, while the implementation of an effective growth policy would limit the scale of the

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needed adjustment. Apart from the reform of basic pensions, it is also important to guarantee the financial viability of the invalidity pension system, which has worsened in recent years with the sharp rise in the number of pensions paid. A reform, implemented in 2004, will improve its financial situation by raising revenues, and will also reinforce the Confederation's surveillance of the granting of invalidity pensions in the cantons, but further measures may be needed to check the rise in expenditure and fund the system until 2020. It is important to avoid invalidity leading to long-term benefit dependency for those who might subsequently be able to re-enter the labour market, and such re-entry should be facilitated.

Too rapid an adjustment in pension fund balance sheets could hamper the upswing

A reform of Switzerland's funded occupational schemes is under way. It provides in particular for a change in the method of calculating pensions to take account of the increase in life expectancy. Such an adjustment is clearly appropriate. Measures have also been adopted, or are under discussion, to cushion the serious impact of the financial market meltdown on the solvability of pension funds, whose funding levels have fallen by between 20 and 30 percentage points since the late 1990s and is now on average less than 100 per cent of liabilities. It is therefore important to take corrective measures, such as on the one hand lowering the minimum return on assets of compulsory occupational pension schemes while also expanding the instruments enabling pension funds to adjust their balance sheets gradually. It may not be realistic to rely on a sharp financial market recovery to maintain the system's solidity. But it is important to avoid too rapid an adjustment in pension fund balance sheets as this could weigh on the present fragile economic situation. Better information would improve the surveillance of these schemes and would enable the authorities to monitor changes more closely and so prevent a faster-than-necessary adjustment in those pension funds that are in difficulty. In the longer term, it is vital to improve the transparency of pension fund accounts and surveillance to maintain the public's confidence in the system. Better supervision of the funds would help to prevent the "excess return" over the minimum rate benefiting only pension fund owners and those who reach retirement

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age during a high-yield period, and would ensure that it is better shared between the insured. This should be accompanied by flexible management of the minimum rate of return, both upwards and downwards.

Enhancing competition is essential for boosting growth

The main medium-term challenge identified by the government is to raise potential growth. This is an appropriate focus even if international comparisons of GDP growth may overstate the Swiss "growth gap" insofar as national income has been rising somewhat faster than GDP. According to indicators taking into account the terms-of-trade gain that Switzerland has enjoyed over the last decades and the revenues earned on foreign investments, real income has risen by more than ½ percentage point a year faster than growth as usually measured by real GDP. Yet, even on the basis of these indicators, performance has been weak in international comparison, while prices are on average 40 per cent above the EU level. Improving the functioning of product markets and reinforcing competition, which is the subject of the special chapter of this *Survey*, would lead to stronger productivity gains, which are vital to stimulate potential growth. Vigorous competition spurs companies to continuously invest in producing new and better products, reduces rents accruing to dominant incumbents and raises pressures on businesses to allocate and utilise resources in the best way. Competition would also be stimulated by further economic integration with the European countries and worldwide.

Strict implementation of the new legal framework for competition is vital

The authorities have undertaken a number of reforms to strengthen competition policy. The most important is the recent revision of the cartel law, which should come into force in early 2004 and provides for penalties against anti-competitive behaviour, without waiting for the firm to re-offend, which will increase the dissuasive effect of the legislation. This is an important step and the new legal provisions now need to be strictly applied. The Competition Commission (COMCO) has since 2002 given clear signals that the legislation will be strictly adhered to, which is a departure from the timid approach adopted previously – for example with regard to vertical restrictions. It is important to persevere with this course of action. To succeed, however,

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COMCO will need funding and staff at least similar to those in comparable countries, which will mean increased resources. With the application of direct penalties and the leniency programme, the independence of COMCO's members ought also to be better guaranteed and part-time membership abolished in favour of full-time membership at least for the presidency.

Network industries need faster and carefully designed reforms

Liberalisation of *network industries* has been only partial and varied across industries, leaving room for further productivity gains and price reductions, prices being generally well above the OECD average. In some cases liberalisation has been slow, only responding with lags to EU directives and often pursued with less ambition. The long lags for designing and approving legislation inherent in the political system have contributed to slowing the reform process.

- In the electricity market, the previous reform, which provided for a wholesale electricity market and for the separation of transmission from other market segments, was rejected in a referendum in 2002. A new project that adheres to this approach, together with the setting-up of a strong regulatory body, is necessary. Security of supply and environmental issues, while legitimate concerns, ought not be considered as barriers to reform.
- Liberalising the natural gas market at the same time would avoid more delays and lead to synergies between the sectors; current initiatives by some private companies to set up a gas market are insufficient as they are only partial and do not provide for independent regulation and supervision of the separation of transport, storage and distribution.
- In telecommunications, the most urgent task is the unbundling of the local loop, which would raise the market penetration of new participants.
- The calendar for liberalising postal services should be accelerated to match the EU schedule and reforms in railway transport should avoid protecting incumbent operators from competition by creating an independent regulator. Moreover, competition in

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tendered lines of regional railway transport should occur on a regular basis without favouring public firms.

The service sector's dynamism would be strengthened by the opening of cantonal markets

Competition also needs to be stepped up in the *service sector* which is suffering from both the excessive segmentation of cantonal markets, notably for professional services, and insufficient openness to the pressures exerted by foreign competitors. The Domestic Market Act (DMA) needs to be amended to create a genuinely unified domestic market and to allow freedom of establishment in all cantons. COMCO's role also needs to be strengthened so that it can appeal before the courts in the event of restrictive practices and systematically monitor and enforce the consistency of cantonal legislation with the principles of the DMA. The opening up of services to foreign competition in the context of a bilateral agreement with the European Union must also remain a priority, while the pace of the negotiations now in progress needs to be stepped up. The obstacles to effective foreign competition for patent-protected products, including drugs, should be reduced. The possibility for the COMCO to act within the framework established in the new cartel law will only affect specific companies and products on a case-by-case basis. Negotiations should therefore be initiated with the European Union with a view to adopting the EU principle of regional exhaustion with respect to patent law. Coupled with the reduction of obstacles to the building of hypermarkets and effective measures to counter vertical agreements, these reforms would no doubt deliver a sharp fall in prices, leading to increased output within the sector concerned and more markedly so for the economy as a whole.

More competition in the health sector and improved provider payment methods are likely to reduce the excessive cost of health care

The most serious competition problems are concentrated in the sectors least exposed to foreign competition, and particularly in *health care*. Here, government intervention plays an important role, as it does in most other OECD countries. A reform of this sector should include the abolition of the obligation for insurers to contract in the ambulatory sector and the decartelisation of health care price setting. This is likely to lower the very high level of medical costs and strengthen control over spending growth, to the extent that insurers are able to negotiate lower

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average prices. Doing away with this obligation, which Parliament is considering, must not however perpetuate, *de facto*, an administrative control over supply that would not improve efficiency. If it is to be effective, the reform must treat all health care providers in the same way, both those recently qualified and those already in practice, who would then also run the risk of no longer being able to work for the compulsory health insurance system. Competition in the drugs market would be stimulated, if doctors were to prescribe active substances rather than branded products, and by a less restrictive policy with regard to putting pharmaceutical products on the market. In addition, reforms to hospital and physician payment methods should be pursued that provide incentives to improve efficiency.

The fragmentation of the public procurement market needs to be reduced further

Competition is being enhanced in *public procurement* contracts, which constitute a large share of public spending. Switzerland has signed the WTO agreement that obliges the international tendering of large projects. For smaller projects, however, the market is still fragmented, with different bidding and appealing procedures across cantons, which introduce transaction costs and reduce competition. Bid-rigging seems also to be widespread due to the lack of effective deterrence from the current cartel law. To harmonise legislation and enhance transparency, an inter-cantonal Agreement on Public Procurement has already been signed, which should be quickly implemented by all cantons and communes. The thresholds above which lower levels of administration are obliged to tender public contracts should be lowered to raise competition, and provisions should be made to avoid the splitting of large public works into smaller contracts to escape from this obligation. In addition, harmonisation with the federal level should also be pursued.

Support to agriculture should be cut

Agricultural policy is another area where public intervention impedes the adequate functioning of a market. The total level of support to agriculture has barely changed in recent years and is the highest in the OECD since the mid-1990s, resulting in food prices far above international prices. Although aid to agriculture has been more linked to environmental goals, these goals, together with other objectives (like social cohesion or rural development) could

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be met in a much cheaper way through other instruments, while public support should concentrate on encouraging alternative activities in rural areas where Switzerland has a comparative advantage.

The financial sector is adapting quickly to the stock market losses but supervision must be enhanced

The *financial sector* has suffered a heavy negative shock since 2000 due to stock market losses but has been reacting rapidly with cost cuts, limiting the impact on profitability, while maintaining solvency ratios. In the medium term, it is not likely to grow at the rates of the 1990s if only because global growth of this sector will probably remain subdued for some time. In addition, the sector will have to adapt to increasing international competition and additional administrative costs. These include the implementation of the new Basle Accord which will require enhanced risk evaluation methods, and the application of the draft agreement with the European Union on taxation of savings income, which includes the introduction of a withholding tax on interest income of EU individuals. Switzerland may however find itself under continued pressure to provide more complete access to information to foreign tax authorities, as recently illustrated by a statement of the G7 Finance Ministers and by the OECD Council's consideration regarding a draft recommandation on improving access to bank information for tax purposes. However, the Swiss Government has shown its determination to resist such pressures and to defend alternative approaches. Regarding supervision, the merger of the Federal Banking Commission and the supervisory body of private insurance as recommended by the Zimmerli Commission is a positive step, but should be complemented by the incorporation of other types of financial intermediaries and pension funds within this supervisory framework, and by additional personnel for the new authority. Members of the new authority should also be independent and have no other interests in the sector.

Incentives for labour force participation of old and female workers should be sharpened

Despite the current cyclical peak of unemployment, the rate of structural unemployment is one of the lowest in the OECD thanks to the flexibility of the *labour market*, which will be further enhanced by the reform of unemployment benefits adopted this year. Participation rates, which are high in international comparison, have been falling in recent years

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for older workers, while female labour supply is lower when taking into account the very high share of part-time work. To raise participation, the higher social security contribution rates paid by firms for older workers should be reduced, while tax incentives or higher pension accrual rates could be introduced for prolonged activity. This should be combined with greater emphasis on measures improving the employability of older workers. This would increase both labour supply and demand for such workers, while reducing pension outlays. Part-time work of women, when involuntary, could be reduced by measures that make it easier to reconcile work and family life and additional support for early child education.

Reforms in education should be implemented without raising the cost per student

An increase in enrolment in *education* at early ages, which is currently low, would have a positive impact both on female labour market participation and on education outcomes later in life. The OECD PISA report has detected insufficient reading abilities for 15 year-old students, especially for those of foreign origin, a wide dispersion of results and a strong correlation with parents' socio-economic background, which suggests that the capacity of the education system to integrate children should be improved. Reform measures already outlined by Swiss experts to cope with these problems, including bringing forward the age of scholarisation, more testing of students, enhancing the autonomy of schools and focusing attention on the worst students, especially the language skills of immigrants, go in the right direction and should be implemented rapidly. Such reforms should aim at using the available resources better rather than increasing spending per student, which is already by far the highest in the OECD. University education is very expensive in budgetary terms and spending is programmed to rise further, only in part because of the rising number of students. Current and future reforms should therefore aim at raising the efficiency and quality of education at the university level. Raising somewhat the contribution of students to the cost of their education would appear equitable, since private rates of return to education are usually high and exceed real interest rates. Students with liquidity constraints could be helped by grants or loans.

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Climate change policy should harmonise abatement costs

Business-as-usual projections suggest that Switzerland will achieve greenhouse gas abatement of 4 per cent against a Kyoto Protocol target of 8 per cent for the period 2008-12. To date, the costs of policy measures undertaken to reduce emissions have been high and variable across sectors, suggesting that there are considerable efficiency gains to be made. The authorities have always envisaged the possibility of introducing a carbon tax and coupling it with an emission trading programme, in which participation would be voluntary but which would lead to exemption from the carbon tax. The existing law provides that a decision on the introduction of such a system be taken in 2004. There now seems little option but to adopt such policies if the Kyoto target is to be met. In designing the system, the authorities should bear in mind that markets work best with no price discrimination between sectors. Hence, carbon taxes should be equal across sectors and linked to the price of permits. The trading system has been well designed with possible international linkages but has the drawback that unused permits cannot be carried forward, so increasing compliance costs. With such a programme in place, the budget for voluntary initiatives could be safely reduced, as proposed. Finally, support for renewable energy should be tied tightly to either the cost of emission permits or the carbon tax.

Waste management and water pollution control could benefit from greater use of economic instruments

In the areas of waste management and water pollution, considerable progress in achieving policy targets has been made over the past decade. While the authorities have examined the cost-effectiveness of measures to achieve targets in these areas, greater use of cost-benefit analysis could help. Water quality has improved quite markedly in a number of water bodies and concentrations of pollutants are low in comparison with many OECD countries. However, diffuse pollution from agriculture remains a problem in many rural areas. Reducing agricultural support and introducing a tax on farms' nutrient balances would avoid such a problem if the new policy of direct payments to farmers does not have the envisaged effects. Waste management policy has led to a large share of municipal waste being recycled. In promoting recycling, the authorities should determine an upper limit on recycling cost premia, based on the cost of incineration and the avoided

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production externalities, so that recycling occurs to the point that it is profitable taking into account external costs and benefits. Finally, the authorities should examine the feasibility of introducing competition in the market for incineration.

Summary

Although Switzerland enjoys a very high living standard, relatively low unemployment and low inflation, the protracted period of meagre productivity growth performance is a cause for concern. The severity of the downturn since 2001, even if partly linked to specific factors, highlights the cost of low trend output growth, and the corresponding urgency of a more aggressive approach to structural reforms. The expansionary stance of macroeconomic policy in recent years has been appropriate because of the cyclical downturn. It is important to keep the monetary stance easy until the recovery is firmly established. Moreover, the National Bank should stand ready to intervene in the foreign exchange market should deflation become a real danger. The slightly accommodative fiscal stance in 2004 is also appropriate, even if it means a temporary deviation from the strict application of the fiscal rule. However, the longer-term sustainability of the public finances needs to be ensured. In this context, the generosity of Switzerland's welfare programmes, and especially pensions, will have to be reviewed, given the ageing of the population. Increasing potential growth, which is the main challenge for economic policy, would help to limit the scale of sacrifices. Raising the dynamism of the economy implies that the scope and pace of structural reforms will have to be stepped up significantly. The key to increasing productivity and thus potential output will be the authorities' ability to strengthen competition and improve the regulatory framework – above all in the sheltered sectors. Since the 1980s, openness to international trade has increased although less rapidly than in the small EU countries, which might be partly the consequence of the negative 1992 referendum on membership in the European Economic Area. Competitive pressures seem particularly low in the network industries, health, agriculture, business services, public procurement and distribution. Illustrative OECD Secretariat estimates suggest that ambitious reforms in these sectors could increase GDP by some 8 per cent over a 10-year period. A clear shift towards a pro-competitive

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environment should be accompanied by measures to increase labour force participation of older workers and women, while a reform of the education system should aim at achieving better results for the large amount of money spent. Slow growth is not an inevitable fate. Determined action combining ambitious structural reforms and the pursuit of a stable macroeconomic policy is needed for Switzerland to maintain a standard of living that is among the highest in the OECD.

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I. Macroeconomic developments and main policy challenges

Switzerland was in recession during the first semester of 2003, having been hit harder than the majority of the other OECD countries by the sluggishness of international activity since 2000-01, despite a rebound of activity in the third quarter of 2003. This is partly due to the nature of the downturn, which has strongly affected sectors that are particularly important to the Swiss economy. However, these poor results also reflect the negative trend growth differential that has existed for some time *vis-à-vis* other countries. Against this background, the most immediate challenge for the authorities is to create conditions that will support the recovery as foreign trade picks up. This chapter reviews recent developments in the Swiss economy and also, more generally, highlights the reasons for the poor medium-term growth performance. It goes on to assess the role of fiscal and monetary policies in the present phase of the cycle, and also their room for *manœuvre* and effectiveness should deflationary pressures fail to abate. Finally, the chapter reviews the main challenges that low trend growth poses for the economy.

Recent trends and short-term prospects

The downturn in activity has been more pronounced than in other countries since 2000

Following the brief fall in activity in the second half of 2001, Switzerland suffered its second recession in two years during the first semester of 2003 (Table 1).[1] Despite a rebound of growth in the third quarter, the decline in output in the first three quarters of 2003 (–¾ per cent over the corresponding period of the previous year), is the biggest in ten years. To a large extent, cyclical developments reflect the state of the world economy, but the downturn was more pronounced than in the United States or the euro area. It was especially marked in the capital goods and financial sectors, in which Switzerland is specialised. The drop in manufacturing exceeded 5 per cent in 2002 and was much sharper than in other OECD countries. The financial market slump also seriously affected the banking and insurance sector, which accounts for around 12 per cent of GDP and had been one of the engines of growth in the second half of the 1990s (Chapter IV). With the

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Table 1. **Demand and output**

Percentage changes from previous period, s.a.a.r., volume (1990 prices)[1]

	Average 1991-2000	1999	2000	2001	2002	2003 Q1	2003 Q2	2003 Q3
Private consumption	1.1	2.2	2.0	2.1	0.7	0.9	1.6	1.5
Government consumption	1.3	1.2	2.1	2.4	1.9	1.2	2.3	3.1
Gross fixed investment	0.8	2.7	5.8	–3.3	–4.1	0.3	–3.2	2.9
Construction	–1.0	–3.3	2.5	–3.2	2.1	1.9	1.9	1.3
Machinery and equipment	2.9	8.7	8.8	–3.4	–9.3	–1.2	–7.9	4.5
Final domestic demand	1.0	2.2	3.0	0.7	–0.3	0.8	0.5	2.1
Total domestic demand	0.9	2.5	2.5	0.7	–1.2	–0.9	0.3	0.8
Exports of goods and services	4.0	5.1	10.1	–0.0	–0.4	–3.7	0.3	5.2
Exports of goods	4.0	3.6	9.3	2.1	1.1	–5.5	–1.3	5.3
Exports of services	4.2	11.3	12.7	–7.5	–6.0	4.0	7.2	4.6
Imports of goods and services	4.2	7.4	8.4	–0.3	–3.5	–1.3	2.6	4.9
Gross domestic product	0.9	1.5	3.2	0.9	0.2	–2.0	–0.7	1.0
GDP price deflator	1.6	0.7	1.2	1.1	0.6	5.7	–2.6	1.9
Private consumption deflator	1.9	0.3	1.0	0.7	0.8	2.3	–0.7	1.0
Memorandum items:								
Index of industrial production[2]	2.4	3.5	8.4	–0.7	–5.1	0.9	–2.8	
Government consumption (% of GDP)	15.2	14.7	14.7	14.9	15.2	15.4	15.6	15.6

1. Adjusted for seasonal and random variations by the Secrétariat d'État à l'économie (SECO) except for the price deflators. All data in volume are provided already deflated by the Swiss authorities.
2. Excluding construction.

Source: OFS; SECO; OECD, *Main Economic Indicators*.

fall in asset prices, the volume of bank business and revenue dropped sharply. The downturn in activity in these sectors then spread to others in the second half of 2002 and early 2003, which led the entire economy into a recession in the first semester. More recently, however, activity picked up somewhat in the financial sector and in industry.

The depressed state of the economy is not only due to the deterioration in the external environment. Domestic demand has also weakened appreciably because of the decline in investment and inventories. Corporate investment, which has been down for 10 consecutive quarters up to the second quarter of 2003, has fallen by more than 15 per cent since end-2000, despite the increase in the third quarter of 2003. As in many OECD countries, the steep increase in information technology-related machinery and equipment over the second half of the 1990s led to over-investment which resulted in a particularly sharp adjustment as of 2001.[2] In addition, the franc's real effective appreciation until early 2003 hit corporate profitability, which has fallen since 2000, especially in industry. Firms

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have responded with cost-saving measures by reducing their inventories and also by laying off staff.

The fall in domestic demand has been limited, however, by the growth of public consumption and, to a lesser extent, of construction expenditure which has been stimulated to a certain degree by the increase in public investment and the fall in interest rates.[3] In contrast with previous cycles, household consumption has risen uninterruptedly, although purchases of goods of a more cyclical nature are showing signs of flagging.[4] In fact, private consumption has lost momentum over the last two years and the savings ratio rose by 3 percentage points since 2000 to some 11 per cent in 2002, its highest level for ten years. Household confidence remained at a low level in autumn 2003 (Figure 1). The confidence problem stems primarily from labour market trends, in particular the fear of job losses. But it is also due to the likely scaling back of future pension benefits, notably those belonging to the second pillar of the system. The system is managed by funds that have announced substantial changes in contributions and benefits to adjust to the financial market losses (Chapter II).

The downturn in exports in 2002 and in early 2003 resulted in market share losses. Apart from capital goods, which account for one-third of exports, revenue from tourism, international transport and financial services has fallen. This is due mainly to the slowdown in foreign, and particularly German demand, but also to the real effective appreciation of the exchange rate, which reached 10 per cent between the first quarters of 2000 and 2003 (Figure 2). Since spring, the franc did however depreciate by about 4½ per cent in real effective terms. In view of the decline in imports prompted by the downturn in total demand as well as the terms-of-trade gains stemming from the franc's appreciation, the current account surplus rose somewhat in 2002, reaching more than 9 per cent of GDP, and was still at a very high level in early 2003.

The labour market has deteriorated

The deterioration of the labour market has continued until the third quarter of 2003 (Table 2), with the fall in employment affecting strongly the manufacturing and construction sectors. In services, the decline in employment was also particularly pronounced in financial intermediation. The bulk of the job losses concerned mainly prime-age male workers, working full time, but the worsening labour market situation was also reflected in a fall in the participation rate among men aged over 55, many of them taking early retirement (Chapter IV). On the other hand, part-time working, mainly by women, continued to increase despite the cyclical slowdown.[5] According to standardised data, the unemployment rate reached 4.2 per cent in the third quarter of 2003, *i.e.* 1.6 percentage points above the structural rate estimated by the OECD Secretariat. The share of the jobless with high skill levels has risen appreciably, reflecting the special nature of

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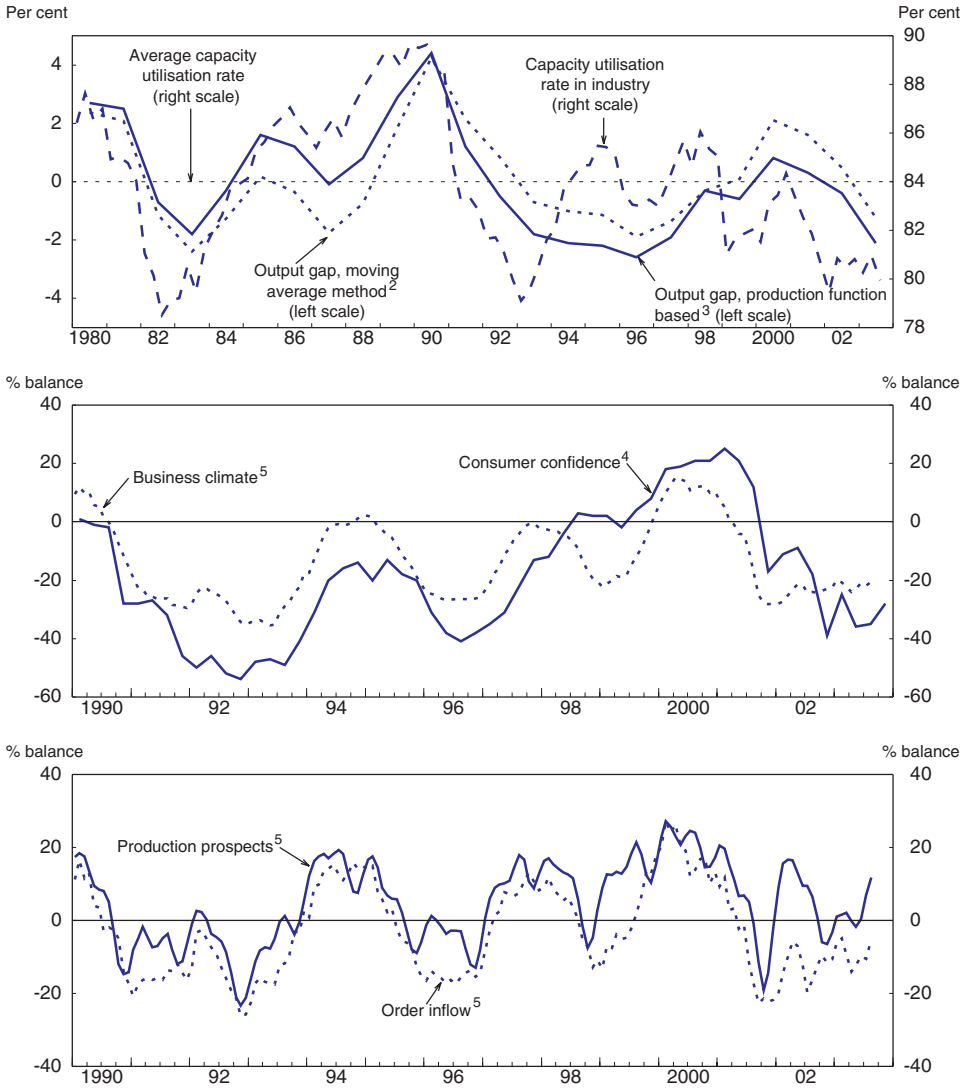
Figure 1. **Output gap[1] and economic activity indicators**



1. Deviation of GDP from potential. Data for 2003 are OECD projections.
2. Method used by the national authorities to compute their cyclically-adjusted fiscal balance. See Amman, Y. (1995), "Le budget du plein-emploi – un réexamen", Office fédéral des questions conjoncturelles, Study No. 20.
3. Estimate based on OECD method.
4. Balance of positive and negative responses.
5. Three-month moving average.

Source: KOF/ETH, Konjunktur; OECD, *Main Economic Indicators*.

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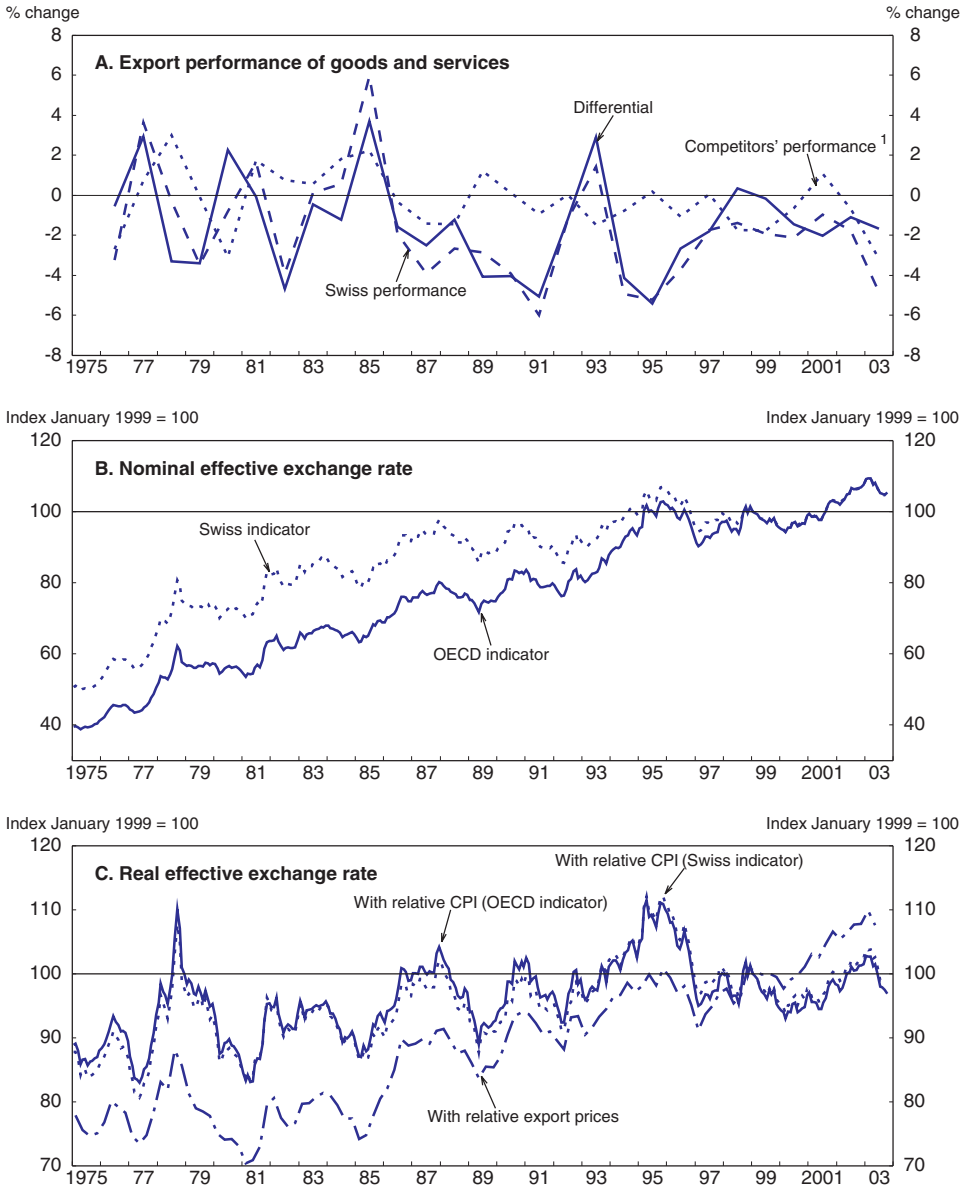
Figure 2. **Competitiveness indicators and export performance**



1. Represented by France, Germany, Italy, the United Kingdom and the United States.
Source: SNB and OECD.

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Table 2. **Labour market trends**

Percentage changes from the same period of the previous year, unless otherwise indicated

	1999	2000	2001	2002	2002 Q4	2003 Q1	2003 Q2	2003 Q3
Employment, total	**0.8**	**1.0**	**1.7**	**0.6**	**0.3**	**0.1**	**0.0**	**–0.6**
By nationality								
Swiss	1.4	0.4	0.2	0.7	0.5	0.3	0.5	–0.5
Foreign	–1.0	3.0	6.1	0.0	–0.4	–1.1	–1.2	–0.8
By sex								
Males	0.7	1.0	1.2	–0.5	–0.3	–0.0	–0.0	–1.0
Females	0.9	1.2	2.3	2.0	1.1	0.0	0.1	0.0
By sector								
Agriculture	2.5	–4.3	–5.4	–1.3	1.6	1.7	–1.8	–1.4
Industry	–0.5	2.3	1.2	–2.1	–3.4	–4.5	–4.7	–4.6
Manufacturing	–1.1	2.6	1.6	–2.6	–4.7	–5.4	–5.5	–5.3
Construction	0.7	1.7	0.7	–0.9	–0.4	–2.7	–3.2	–3.4
Services	1.2	1.0	2.3	1.7	1.6	1.5	1.8	0.9
Labour force, total	**0.3**	**0.7**	**1.6**	**1.2**	**1.2**	**1.1**	**1.2**	**0.4**
By nationality								
Swiss	1.1	0.2	0.3	1.1	1.1	1.0	1.1	
Foreign	–2.1	2.0	5.4	1.3	1.4	1.2	1.3	
By sex								
Males	0.2	0.6	0.9	0.5	0.8	1.0	0.9	
Females	0.4	0.8	2.5	2.0	1.6	1.0	1.5	
Unemployment rate, total[1]	**2.9**	**2.5**	**2.5**	**3.1**	**3.5**	**3.8**	**4.1**	**4.2**
By nationality								
Swiss	2.0	1.9	1.9	2.3	2.6	2.8	2.8	
Foreign	5.6	4.7	4.1	5.2	6.6	7.1	7.0	
By sex								
Males	2.5	2.1	1.9	2.9	3.5	3.7	3.6	
Females	3.4	3.1	3.3	3.3	3.8	4.1	4.3	
Memorandum items:								
Registered vacancies	8.8	–6.3	–8.1	–25.3	–30.2	–14.7	–14.4	
Labour productivity[2]	0.7	2.2	–0.8	–0.3	0.6	–0.7	–1.0	–0.1
FTE employment[3]	0.9	1.9	1.9	–0.2	–0.9	–1.2	–1.3	–1.7

1. Level.
2. Real GDP per person in employment.
3. Full-time equivalent employment in the industry and services sectors.

Source: OFS, *Statistique de population active occupée* and *Statistique d'emploi*; OECD, *Main Economic Indicators*.

the recession which has hit activities with skilled personnel, such as banking, insurance and information and telecommunications technologies (Table 3).

Inflation, already low, has slowed still further

The slowdown in nominal wage increases caused by the deteriorating employment situation seems to be relatively moderate in 2003. According to wage negotiations, pay should increase on average by 1.4 per cent, which is slightly down

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Table 3. **Structure of registered unemployment**

	1999	2000	2001	2002	2003[1]
	Rate[2]				
Total unemployment rate	**2.5**	**1.8**	**1.7**	**2.5**	**3.7**
By citizenship					
Swiss	1.7	1.3	1.2	1.8	2.7
Foreign	5.2	3.7	3.4	5.0	6.8
	Share in per cent				
By duration					
0-6 months	52.7	59.0	65.0	65.5	58.8
7-12 months	21.5	20.9	19.4	22.0	25.5
More than one year	25.8	20.1	15.7	12.5	15.7
By function					
Self-employed/homeworkers	1.4	1.4	1.3	1.1	1.0
Executives	5.5	6.0	6.4	6.8	7.0
Experts	49.4	51.1	51.0	51.8	54.0
Other workers	38.3	36.7	36.9	36.0	33.0
Apprentices/students	5.4	4.8	4.4	4.3	5.0

1. Average of the first eleven months of the year.
2. Official unemployment statistics include those who are partially unemployed. Official unemployment rates are expressed as a percentage of the workforce in the 2000 population census.

Source: SECO, Labour market statistics and *La situation sur le marché du travail en novembre* 2003, press document, December 2003.

on 2002 (1.8 per cent) and 2001 (2.5 per cent). As in previous years, however, per capita wage income could rise faster, as the upward trend in average wages prompted by higher qualifications is not taken into account by these indicators. While remuneration in some sectors, such as financial services, has fallen sharply, the rise in real wages could again outstrip productivity and increase unit labour costs, as happened in 2001 and 2002. These wage increases are relatively brisk, given the economic situation, and no doubt reflect a catch-up following the long period of restraint in the second part of the 1990s. According to Fehr and Goette (2003), nominal wage rigidity for those remaining in their jobs is quite high in Switzerland, including during periods of low growth, even though wages, which are negotiated on a decentralised basis, are more flexible than in the majority of OECD countries and tend to move in line with productivity in real terms.[6] Despite this recent increase in labour costs, inflation has fallen even further and is lower than in Switzerland's main partner countries (Figure 3). The rise in consumer prices, which was around 1 per cent in the fourth quarter of 2002 (year-on-year), was down to no more than ½ per cent in the fourth quarter of 2003. This is due partly to the fall in prices of imported goods, but entrepreneurs are also being forced to cut margins, especially in the industrial sector, because of the persisting sluggishness of sales in both the domestic and export markets (Table 4).

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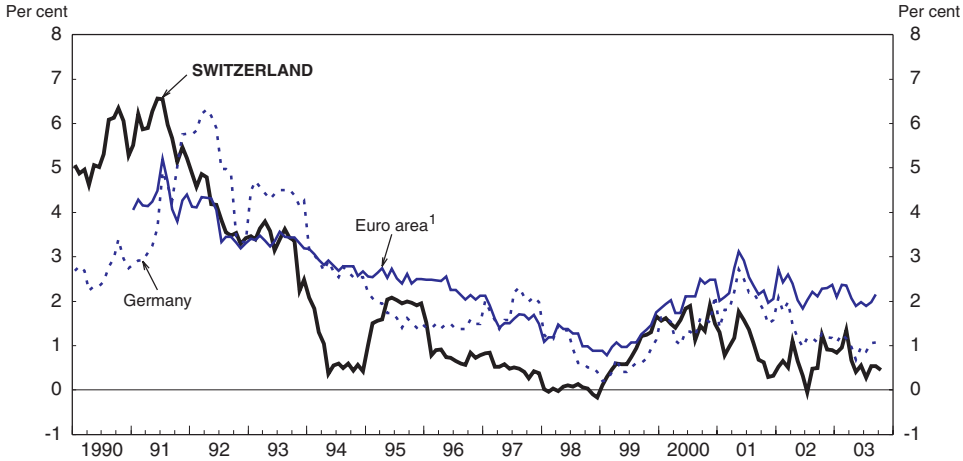
Figure 3. **Consumer prices: an international comparison**

Year-on-year percentage changes



1. Harmonised CPI.
Source: OECD, *Main Economic Indicators*.

The strength of the recovery will depend on the external environment

The strength of the projected upturn will depend on the rapidity of the turnaround in the international situation, whose first positive effects have been felt in exports in the third quarter of 2003. This is consistent with the information provided by the leading indicators (Figure 4), which suggest that Switzerland has bottomed out. According to the latest KOF institute surveys, inflows of orders have also picked up, although order books remain bare. With output slow to pick up, activity could decline by ½ per cent on average in 2003, and the output gap could reach around 2 per cent, its highest level since 1996 (Table 5).[7] This low rate of factor utilisation, which is confirmed by business surveys,[8] is hardly likely to encourage firms to increase their capital expenditure strongly, particularly since there may still be a capital overhang from the earlier investment boom. Nor do the available surveys point to any rapid improvement in the labour market situation, and this will continue to weigh on households' consumer behaviour. All in all, the increase in GDP, which will probably not exceed 1¼ per cent in 2004, *i.e.* a rate close to potential growth, will depend mainly on exports, and is predicated on the assumption that the exchange rate will remain stable at its early autumn 2003 level. In 2005, growth could accelerate to reach about 1¾ per cent. This outlook does not presage a rapid fall in unemployment, which would not fall below 3½ per

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Table 4. **Price trends**

Percentage changes from the same period of the previous year

	Weight	1999	2000	2001	2002	2003 October	2003 Nov.
	May 2003						
Total supply index[1]	100.0	–1.4	2.7	–0.1	–1.2	–0.3	
Producer price index	65.8	–1.0	0.9	0.5	–0.5	0.2	
Import price index	34.2	–2.2	6.7	–1.6	–2.8	–1.6	
	May 2000						
Consumer price index	100.0	0.8	1.6	1.0	0.6	0.5	0.5
Goods	42.9	0.7	3.1	–0.1	–0.5	0.1	0.2
Non-durables	26.6	1.0	5.0	0.9	0.0	0.9	1.1
Food, beverages, tobacco	13.6	0.3	1.6	2.0	2.2	1.9	1.9
Energy	5.9	2.1	11.4	0.4	–3.2	0.0	0.8
Services	57.1	0.9	0.4	1.8	1.5	0.7	0.7
Rents	19.7	0.7	1.5	2.8	1.0	0.3	0.3
Home produced goods and services	74.5	0.7	0.7	1.7	1.4	0.8	0.8
Imported goods and services	25.5	1.0	4.1	–1.2	–1.7	–0.5	–0.4
Core inflation	78.4	0.7	0.6	1.0	0.8	0.4	0.4
Export prices[2]	. .	1.3	2.7	1.4	–1.9		
Import prices[2]	. .	–1.3	5.9	1.0	–2.6		

1. The total supply price index is the weighted average of producer and import price indices.
2. Change in the price deflator of goods and services, national account basis, 1990=100.

Source: SNB, *Bulletin mensuel de statistiques économiques* and OFS, Communiqué de presse, series 5, December 2003.

cent in 2005. With the output gap still negative, inflation could even get close to zero per cent for some time. In this context, deflation cannot be excluded if the conjuncture evolves substantially less favourably than foreseen. Deflation risks and policy responses are discussed in the monetary policy section.

Medium and long-term performance

Switzerland suffers from insufficient trend growth…

The poor growth performance of the last few years, by comparison with the OECD average, has characterised the Swiss economy for two decades. Between 1980 and 2002, the annual growth differential *vis-à-vis* the euro area averaged ¾ of a percentage point, while *vis-à-vis* the United States it was over 1½ percentage points. Moreover, these differentials have widened since the early 1990s, reaching 1 percentage point on average with the euro area and over 2 percentage points with the United States. An international comparison reveals that, with the exception of general government consumption spending, the increase in all components of demand was lower in Switzerland than in its main trading partner countries

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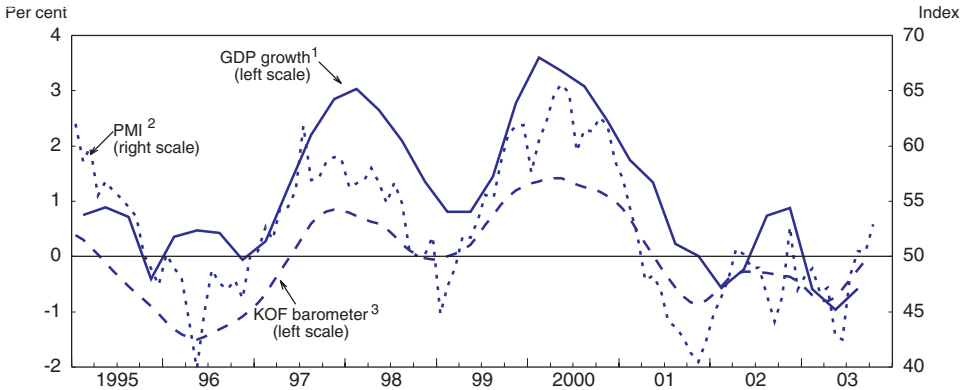
Figure 4. **Leading indicators**



1. Real terms. Year-on-year percentage changes.
2. The Purchasing Managers' Index (PMI) is an index based on the response of 200 purchasing managers at Swiss industrial companies about their performance in the current month compared with the prior month. An index below 50 indicates that production is falling.
3. The KOF barometer is a leading indicator of future GDP growth, with an average lead of 6 to 9 months.

Source: KOF and OECD, *Quarterly National Accounts*.

(Figure 5). The growth differential is, moreover, particularly marked as regards exports of goods and services because Switzerland has suffered bigger export market share losses than have other countries. Since 1980, the performance of Swiss enterprises on foreign markets has been on average 1¼ percentage points per year below that of their competitors (Figure 2). The trend rise in the real effective exchange rate, which has been ½ to 1 per cent per year since 1980, is related to this relatively limited increase in exports. The latter has been accompanied, however, by terms-of-trade gains so that income has risen faster than output.

… although income growth has been stronger than real output

Between 1980 and 2002, Switzerland's terms of trade improved by some 37 per cent, the biggest increase among OECD countries. However, the positive impact of this trend on real income is not properly accounted for by real GDP developments, which measure increases in output. For a number of years now, various studies have been carried out to account more accurately for the positive economic effects of these terms-of-trade improvements by basing estimates of real value added on a Törnqvist rather than Laspeyres index, which is usually used in national accounting (Kohli, 2003). Estimates based on such indices, which are approximated by the *command* GDP indicators (Box 1), suggest that traditional real GDP series have been underestimating real income growth by ½ percentage point per year on average since 1980. According to these indicators, growth of real

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Table 5. **Short-term projections**
Percentage changes, s.a. at annual rates, 1990 prices

	2000		2001	2002	2003	2004	2005
	Current prices, CHF billion	% of GDP					
Demand and output							
Private consumption	242.0	59.6	2.1	0.7	0.4	1.2	1.8
Government consumption	59.7	14.7	2.4	1.9	0.9	0.4	0.4
Gross fixed investment	84.1	20.7	–3.3	–4.1	–2.1	0.6	3.0
Construction	40.6	10.0	–3.2	2.1	0.8	0.6	1.9
Machinery and equipment	43.5	10.7	–3.4	–9.3	–4.8	0.5	4.2
Final domestic demand	385.7	95.1	0.7	–0.3	–0.1	0.9	1.9
Stockbuilding[1]	–1.1	–0.3	0.0	–0.9	–1.2	0.4	0.0
Total domestic demand	384.6	94.8	0.7	–1.2	–1.4	1.3	2.0
Exports of goods and services	178.2	43.9	–0.0	–0.4	–0.5	3.8	5.9
Imports of goods and services	157.1	38.7	–0.3	–3.5	–2.4	4.4	6.5
Foreign balance[1]	21.2	5.2	0.1	1.4	0.8	–0.2	–0.1
GDP at constant prices			0.9	0.2	–0.5	1.2	1.8
GDP price deflator			1.1	0.6	–0.1	–0.1	0.3
GDP at current prices	405.8	100.0	2.0	0.8	–0.6	1.0	2.1
Memorandum items:							
Consumer price index			1.0	0.6	0.6	0.3	0.2
Unemployment rate (level)			2.5	3.1	3.9	3.9	3.6
Potential output, whole economy			1.4	1.2	1.3	1.0	1.2
Output gap, whole economy			0.6	–0.5	–2.2	–2.0	–1.4

1. Contribution to GDP growth.
Source: OFS and OECD.

income in Switzerland has in fact been quite close to that of Germany, France and Italy, even if it remains one of the lowest among OECD countries, especially in the 1990s[9] (Figure 6). This observation remains valid even when the income that Switzerland earns on its foreign investment is taken into account. Income growth measured on the basis of GNP, which includes investment income, has on average exceeded that of GDP by only 0.1 of a percentage point since 1980, although the differential widened to 0.2 of a percentage point per year on average between 1990 and 2002. Overall, the rise in average living standards, measured either by GDP, command GDP or GNP per capita, has been less rapid than in European countries and in the United States, especially since the beginning of the 1990s (Figure 7).

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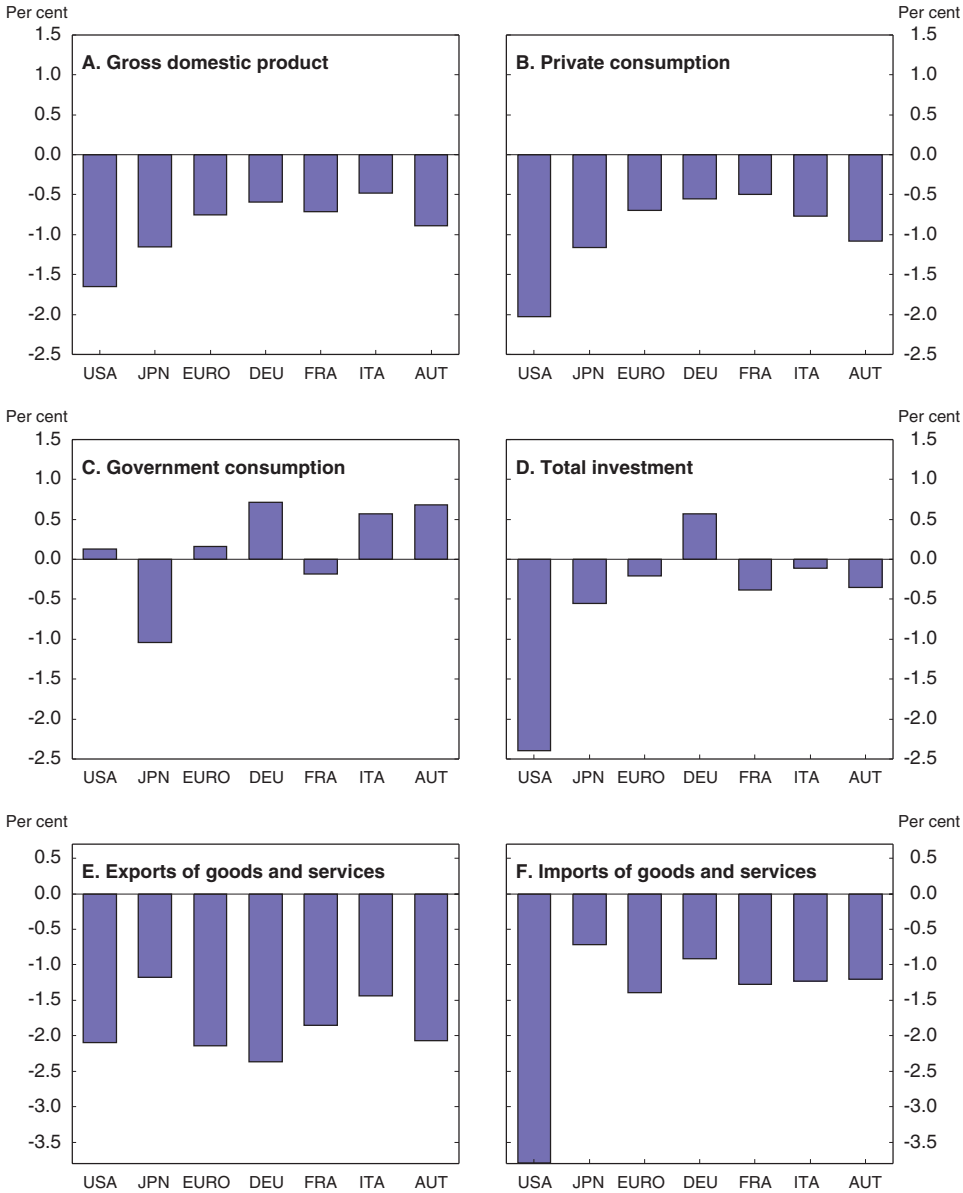
Figure 5. **Average growth differential between Switzerland and its main trading partners**

1980-2002



Source: OECD.

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Box 1. *Command* GDP: a real income indicator

Real GDP, as calculated traditionally by the national accounts, does not account adequately for the positive effect of an improvement in the terms of trade on real income. Terms-of-trade gains deriving from a fall in import prices result in an increase in import volumes leading, other things being equal, to a decline in real GDP. However, volume GDP, which measures production, underestimates the increase in real income that is amplified in this case by the improvement in purchasing power gained from the fall in import prices.

To correct this bias, which is substantial in the case of Switzerland in view of the trend improvement in the terms of trade observed since 1980, Kohli (2003) proposes an alternative estimate of GDP based on a translog production function, which allows Törnqvist indices to be used to deflate the nominal figures. Moreover, an approximation of the estimates obtained by using this method can be derived by means of a simple indicator (*command* GDP) calculated on the basis of the components of GDP and deflating nominal exports by the import price deflator.* Thus:

Command GDPV = TDDV + XGSV*(PXGS/PMGS) – MGSV

where TDDV is real domestic demand, XGSV and MGSV are, respectively, volume exports and imports, and PXGS and PMGS are the export and import deflators.

The idea behind this concept is that exports are only useful for the imports that they allow. So, if the terms of trade improve, more can be imported with the same volume of exports. If, moreover, domestic demand, foreign trade and therefore standard GDP remained unchanged in volume terms in the event of improved terms of trade, *command* GDP would increase and better account for the growth of real income.

The differential between the growth of the latter indicator and that of standard GDP between 1980 and 2002 is larger in Switzerland than in other OECD countries. Switzerland is in fact the country with the biggest improvement in its terms of trade and whose share of exports in production is larger than in other countries that have experienced similar trends in foreign trade prices (Italy and Japan, see Figure 6). According to the *command* GDP indicator, the real income growth rate is probably ½ percentage point higher than is indicated by the growth of standard GDP. That said, even on that count, Switzerland is still one of the OECD countries with the poorest long-term growth performance.

* This indicator is similar to the *command-basis* GNP published by the Bureau of Economic Analysis since 1981. See Denison (1981). See also SECO (2002a) for a discussion of the statistical questions regarding Switzerland's poor growth performance.

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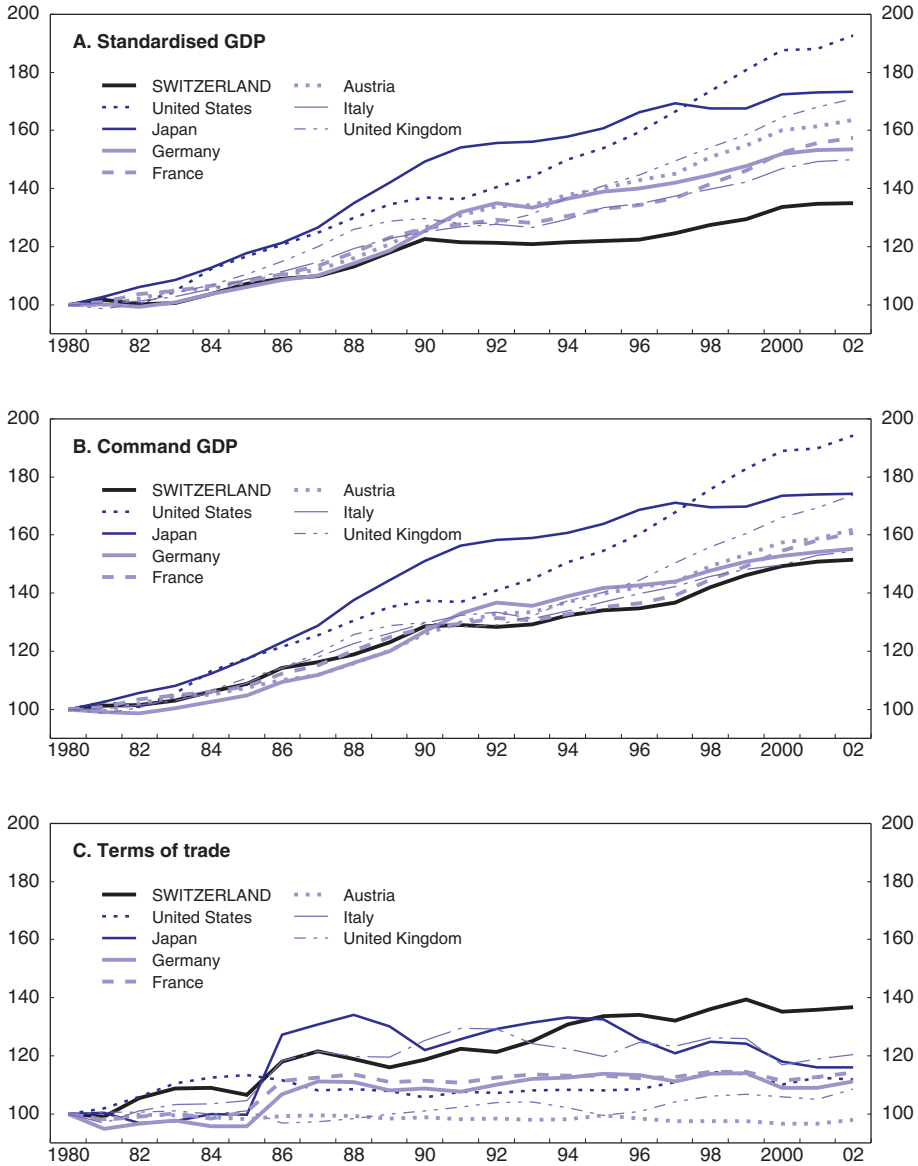
Figure 6. **Various concepts of GDP**
In real terms, index 1980 = 100



Source: OECD.

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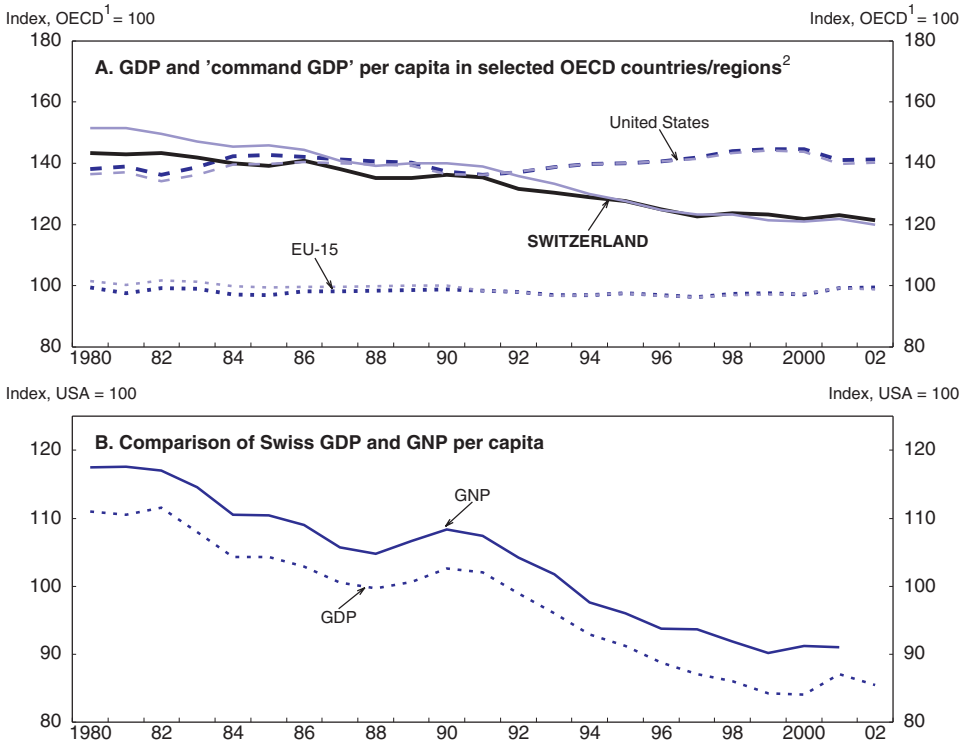
Figure 7. **Indicators of income and production per capita**

At constant 1995 prices and PPPs



1. Excluding Czech Republic, Hungary, Poland and Slovak Republic.
2. The "command GDP" curves are in dark bold, with command GDP defined as:
   Command GDP = GDP + (Terms of Trade – 1) * Exports
   Since the terms of trade are equal to 1 in the 1995 base year, GDP and command GDP are the same for that year. Prior to 1995, the lower level of the command GDP reflects that the terms of trade (expressed in 1995 prices) are below unity during that period. The trend decline in per capita command GDP has been smaller than that of per capita GDP.

Source: OECD, *National Accounts*.

This poor growth performance is essentially due to sluggish productivity growth

Analysed in the traditional framework of growth determinants, slow potential output growth reflects an insufficient increase in factor efficiency (Table 6). The rate of capital accumulation, which is encouraged by low interest rates, has in fact been similar to that of other countries over the last few decades. Also, employment has risen more than in Switzerland's main neighbouring countries, although it has been lower than in the United States, especially if account is taken of the trend in the number of hours worked. Average hours worked has fallen in Switzerland, no

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Table 6. **Decomposition of potential output growth**
Annual averages, per cent

	Switzerland	United States	Three major European countries[1]	Austria
1980-03				
Potential output growth	**1.5**	**3.0**	**2.1**	**2.3**
Potential output growth of the business sector	1.5	3.1	2.2	2.6
of which contribution of:				
Capital stock	0.8	0.9	1.0	0.9
Trend labour efficiency	0.2	1.2	1.2	1.4
Trend hours	–0.2	0.0	–0.3	0.0
Potential employment	0.7	1.0	0.3	0.3
of which:				
Working-age population	0.5	0.8	0.3	0.4
Trend participation rate	0.3	0.2	0.1	0.0
Structural unemployment	–0.1	0.0	–0.1	–0.1
1991-03				
Potential output growth	**1.2**	**3.1**	**1.8**	**2.3**
Potential output growth of the business sector	1.3	3.2	1.9	2.6
of which contribution of:				
Capital stock	0.8	0.9	0.9	0.9
Trend labour efficiency	0.2	1.4	1.0	1.4
Trend hours	–0.1	0.0	–0.3	0.0
Potential employment	0.4	0.9	0.3	0.5
of which:				
Working-age population	0.4	0.8	0.1	0.4
Trend participation rate	0.1	0.0	0.2	0.1
Structural unemployment	0.0	0.0	0.0	0.0

1. For Germany, there is a break in series in 1991. Therefore data of that specific year are not included.
Source: OECD estimates.

doubt partly as a result of the increase in part-time work. Under these conditions, strengthening the medium-term growth performance, which is the main challenge for economic policy means increasing first and foremost factor productivity. This need to boost growth has been clearly recognised in a recent report by the Ministry for Economic Affairs (SECO, 2002b).

Monetary policy

While structural reforms appear necessary to improve the medium-term growth performance, in the short term, the main task of the authorities is to promote the cyclical recovery with appropriate macroeconomic policies, both monetary and fiscal.

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Monetary policy has been appropriately accommodative over the last two years

In 1999, the Swiss National Bank (SNB) adopted a new monetary policy framework. The major goal is to ensure price stability (defined by the SNB itself as price inflation of less than 2 per cent and avoiding deflation). An inflation forecast with a three-year horizon is used as main indicator to set interest rates at the right level to meet the price stability objective in the medium term. The published forecast, which is conditional on fixed interest rates, is based on a wide set of indicators and models. The new framework differs from the previous approach which had relied mainly on controlling monetary aggregates. The operating target is the 3-month LIBOR rate, which can fluctuate within a range of usually 100 basis points. As argued in the previous S*urvey*, this framework has improved the conduct of monetary policy (OECD, 2002).

In the past two years, the monetary stance has been accommodative, with the interest rate being cut successively and aggressively in response to the severe downturn and the appreciation of the franc linked to safe haven effects. The central value of the target band had been 3½ per cent at the peak in mid-2000, but had been lowered to 1¾ per cent by the end of 2001 due to the sharp deterioration in the international environment, and to the subsequent appreciation of the franc. However, as the recovery did not materialise, and to counter the further appreciation of the franc, interest rates were cut again by a total of 100 basis points by summer 2002, and were left unchanged until March 2003. In March 2003, due to global economic and political uncertainties linked to the outbreak of the Iraq conflict, rates were cut again pre-emptively by 50 basis points while the target range was narrowed to 0-¾ per cent. After this move, the franc depreciated by more than 5 per cent against the euro. In June and September 2003 rates were not changed, but they now fluctuate in the low end of the target range, around ¼ per cent. In its most recent report, the SNB considered that inflationary pressures would only revive at the end of the projection period. While the current low interest rate was thought not to be sustainable in the medium term, there would be sufficient time to raise it once the recovery is firmly established. Overall, the interest rate differential with the euro area has widened over the last two years, as the SNB has reduced interest rates faster than the ECB (Figure 8).

Other indicators confirm the accommodative stance of monetary policy. A monetary conditions index, which takes into account interest rate and exchange rate developments,[10] suggests that monetary conditions are easy (Figure 9). Money aggregates have shot up, in part because of the easy monetary stance but also reflecting the greater demand for less risky assets after the slump in stock markets. Indeed, the increase in money holdings has not been accompanied by strong credit growth, which was negative in the second quarter of 2002 before recovering to a moderate 1½ per cent by mid-2003. Yield curves, which can be used to gauge market expectations of future inflation, remain relatively flat at the two-year horizon, while ten-year bond yields are below 2¾ per cent (Table 7).

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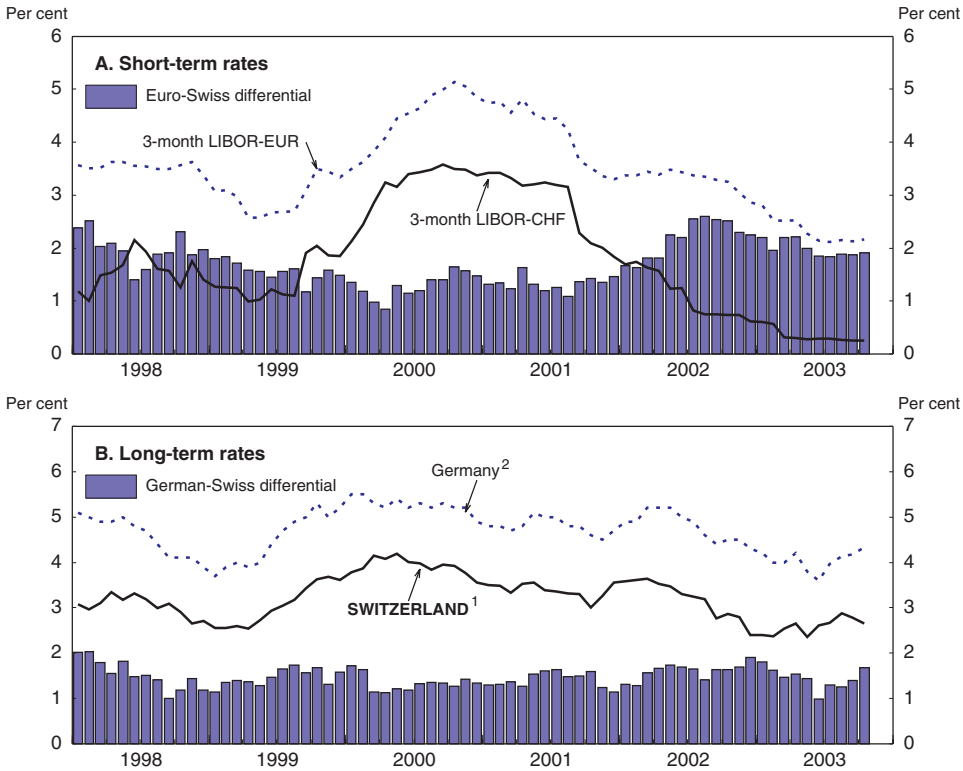
Figure 8. **Interest rates**



1. Confederation bonds of 10 years.
2. Listed federal securities with residual maturities of over 9-10 years.
Source: SNB and OECD.

Deflation risks should not be exaggerated…

Past periods of stagnation or recession were associated with higher rates of inflation than during this most recent episode. In the recession of 1983, for instance, inflation was near 3 per cent, while at the trough of the 1995 recession inflation was hovering around 2 per cent. The current business cycle conditions, with output remaining weak and a negative output gap estimated at 2 per cent, are likely to pull inflation further down from the already very low level. If negative risks materialise, delaying the recovery, inflation could enter into negative territory. Temporary deflation is not necessarily worrying as long as deflation does not become entrenched in expectations; indeed, in the recent past there have been sporadic episodes of negative inflation. Moreover, the risk of prolonged deflation should not be exaggerated, despite the current low rate of inflation, as the decline

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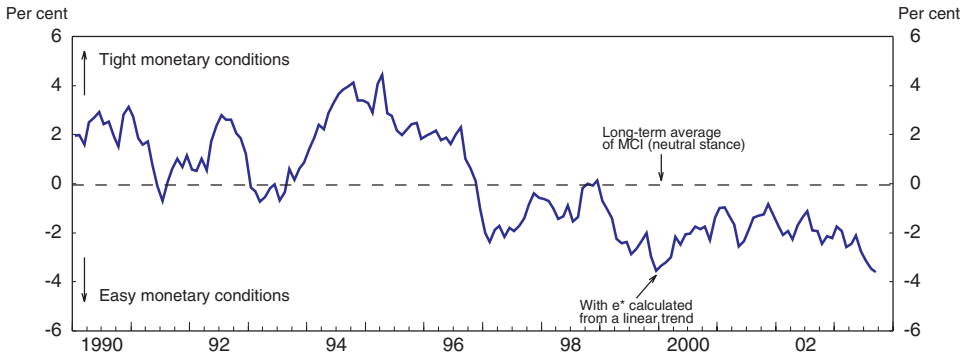
Figure 9. **Monetary conditions index (MCI)**[1]



1. The MCI is defined as MCI = (r – r*) + (e/e* – 1) * w, where r is the real (CPI deflated) three-month LIBOR rate, r* the long-term average of r, e the real effective exchange rate (CPI based) and w is the weight of the exchange rate deviation from its average. It is set at ⅓. Since the real exchange rate in Switzerland is characterized by a significant upward trend, the MCI uses the deviation of the real exchange rate from its (linear) trend e* rather than from its long-term average. A negative MCI implies easy monetary conditions (relative to the long-term average).

Source: OECD.

Table 7. **Yield curve**

Libor rates and bond yields, per cent[1]

	Libor 1 month	Libor 3 months	Libor 12 months	Bond 2 years	Bond 10 years
1998	1.25	1.41	1.71	1.81	3.07
1999	1.59	1.85	2.21	2.06	3.01
2000	3.37	3.37	3.37	3.52	3.88
2001	1.79	1.84	2.00	2.85	3.36
2002	0.61	0.62	0.69	1.84	3.22
2003[2]	0.20	0.25	0.43	0.73	2.78

1. End-of-year values.
2. End September figures.

Source: SNB.

in Swiss prices would boost competitiveness and exports. A large negative shock to activity would be necessary to lead to a sizeable fall and prolonged decline in prices. In this context, the relatively high degree of wage rigidity limits deflation risks. Specifically, simulation exercises suggest that a shock to external demand induced by a delayed recovery in the European Union, or a temporary shock to household consumption, leading to lower GDP growth of ¾ percentage point below the baseline, would not affect inflation in 2004 and would only have a small impact of about ½-¾ percentage point from 2005 onwards (Box 2). Only a substantial and sustained appreciation of the franc would result in a rapid and significant fall of

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Box 2. Deflation risks

In order to evaluate deflation risks, several medium-term scenarios have been elaborated using the OECD's INTERLINK model. The baseline scenario, which is used as a benchmark for comparison, is an extension of the current short-term projections until 2007. Around this, two simulations aim at illustrating the effects of a slowdown of external or internal demand on activity and prices with respect to the baseline.

The baseline scenario assumes that annual GDP growth would be 1¾ per cent between 2004 and 2008, significantly above potential growth to close the output gap by 2007. Inflation would decelerate to ¼ per cent in 2004-05 before increasing again with the acceleration of activity, while the unemployment rate would remain at around 4 per cent in 2004 and would fall afterwards to the estimated structural rate of 2½ per cent.

The scenario of lower external demand than in the baseline assumes a slower than projected recovery in the euro area, with growth 1 per cent lower in 2004, offset by an acceleration of ½ per cent above the baseline in 2005 and 2006. In this case, growth in Switzerland would be ¾ percentage point lower in 2004 than in the baseline, before progressively increasing as the external environment improves (Figure 10). Inflation would be scarcely affected in 2004, but could decelerate by ½ percentage point with respect to the baseline as from 2005 and remain close to zero.

The scenario of weaker than projected internal demand assumes that household consumption decelerates more than projected due to a deterioration of the labour market. Employment is assumed to grow *ex ante* ¾ percentage point less than in the baseline scenario in 2004, while the household savings rate would increase by 1¼ and ½ percentage point in 2004 and 2005 above the assumptions of the main scenario. In that case, and in the absence of a reaction of monetary policy, GDP growth would be ¾ percentage point lower than projected in 2004 to reach only ½ per cent, before increasing rapidly to 2½ per cent in 2005. Inflation could fall by ¼ percentage point in 2004 and ½ percentage point in 2005 with respect to the baseline, falling temporarily below zero. Overall, the impact on inflation of a delayed recovery due to internal or external reasons would be limited and unlikely to generate deflation.

An appreciation of the exchange rate, however, would have a stronger and more rapid effect on inflation. According to standard simulations, the reaction to a 10 per cent rise in the effective exchange rate would induce a reduction of the output level by around 1½-2 per cent below the baseline during two years and would decrease inflation by 2 percentage points. Such a shock, in the absence of a reaction from monetary policy, would put Switzerland in deflationary territory.

inflation, due to the large degree of openness of the Swiss economy. Specifically, an effective appreciation of 10 per cent would induce a fall of inflation of about 2 percentage points in the absence of a monetary policy reaction.

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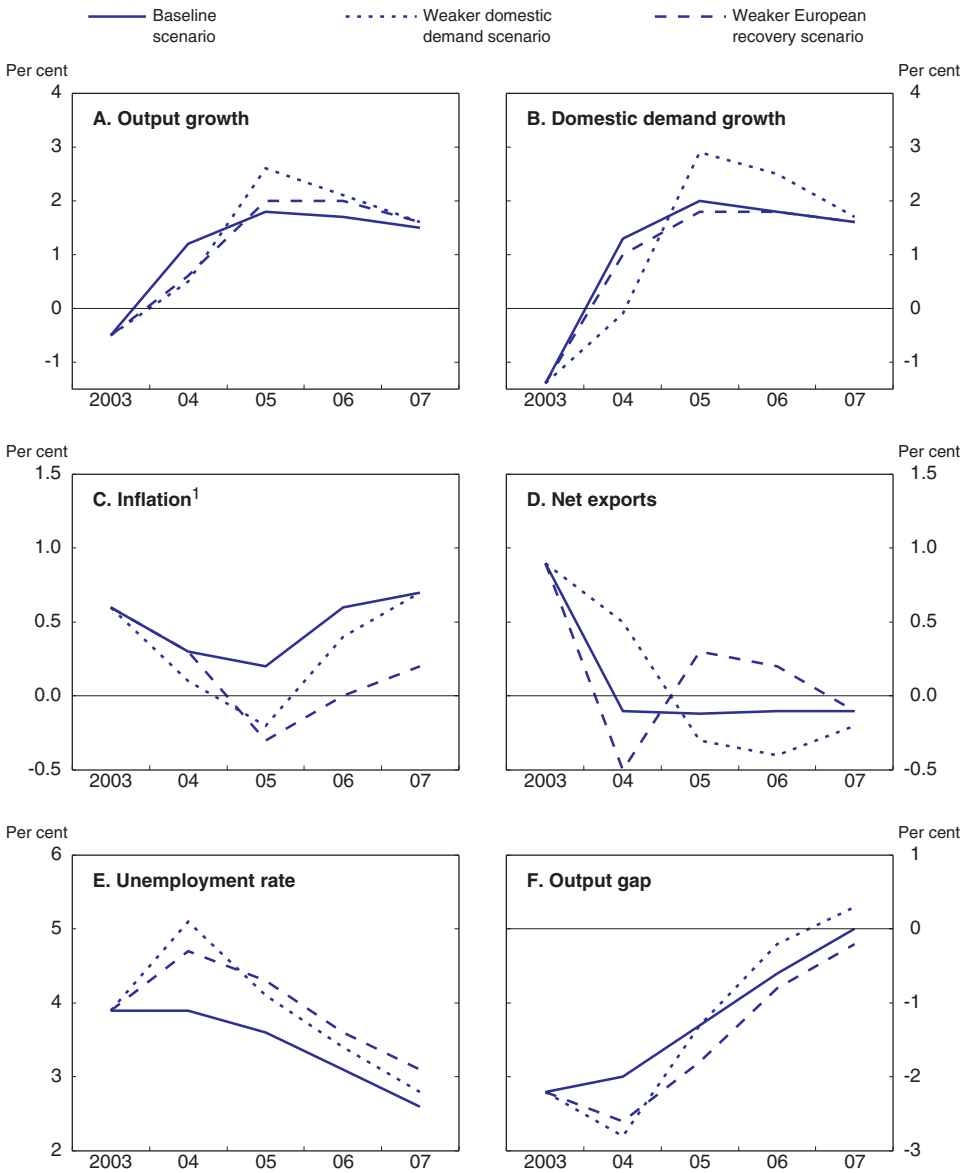
Figure 10. **Downside risks around the projections**



1. Change of the CPI over the previous year.
Source: OECD.

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In the event that a protracted decline in prices were incorporated in expectations, demand could be affected further, with investment and consumption of durables being delayed in order to profit from lower prices in the future. At the same time, if falling prices were to affect real or financial assets, negative wealth effects could further damage demand and undermine the financial position of indebted households and businesses, which could destabilise the banking system. This situation could generate a mutually reinforcing process of deflation and recession; the experience of Japan in the past few years provides an example of such a situation.

However, the characteristics of the Swiss economy are quite different from those of Japan and provide a cushion against a prolonged period of deflation. F*irst*, Switzerland is a small open economy, whose cyclical dynamics depend to a large extent on those of the neighbouring countries, which makes an isolated deflationary process unlikely. If deflation and depression hit the European Union (its main trading partner) as a whole, policy action there could be expected. S*econd*, both households and companies have sound financial accounts, with small debt burdens, implying that temporary deflation would not feed quickly into a large number of defaults. T*hird*, the Swiss banking system is robust, which implies that the risk that deflation would generate a credit crunch is small. Finally, broad money growth has accelerated markedly and credit growth picked up recently, confirming that in Switzerland the monetary transmission mechanism is working as expected (Table 8).

Table 8. **Growth of money supply and domestic credit**
Year-on-year percentage change

	2001	2002	2003		
			Q1	Q2	Q3
M1	1.0	8.7	15.0	25.3	23.5
M2	–2.3	8.1	13.4	19.7	18.6
M3	3.1	3.8	6.8	9.2	8.8
Domestic credit	1.5	–0.5	–0.3	0.3	0.7

Source: SNB, *Bulletin mensuel de statistiques économiques*, October 2003.

… but require continued surveillance

While the effectiveness of monetary policy against high inflation has been repeatedly confirmed, fighting deflation appears to be more difficult. Once a deflationary process starts, traditional monetary policy instruments lose their power to pull the economy out of deflation, since interest rates hit the zero bound

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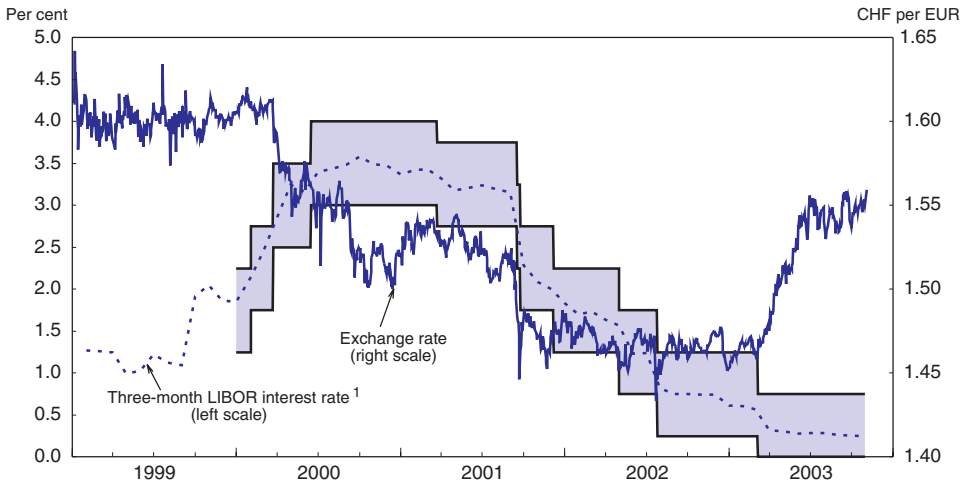
Figure 11. **Three-month LIBOR and the CHF/EUR exchange rate**



1. The shaded area corresponds to the National Bank's intervention corridor.
Source: SNB, *Bulletin mensuel de statistiques économiques*, October 2003 and OECD.

and increases in the monetary base may not translate into higher demand. In this context, pre-emptive action is preferable. However, at the extreme, unorthodox instruments of monetary policy to raise inflation expectations may have to be used. The SNB has indeed reacted rapidly and pre-emptively to the appreciation of the franc with interest-rate cuts to cushion a negative external shock in a period of weak demand and inflation. In particular, the cut of the LIBOR by 50 basis points in March 2003 was beyond market expectations, sending a strong signal that it would counter appreciation pressures on the franc in a period of stagnation. Indeed, the franc depreciated against the euro after this move (Figure 11). Communication policy has also helped, as the SNB has made clear that it is ready to use unsterilised interventions in the foreign exchange markets if additional pressures on the franc materialise. The SNB rightly considers that intervention in foreign exchange markets would be a more effective instrument to generate inflation expectations than the purchase of other assets, like long-term bonds, this market being relatively illiquid in Switzerland.

The exchange rate is an important variable in the management of monetary policy

Over the last two decades the Swiss franc has shown a trend appreciation both in nominal and real effective terms (Figure 2). The main forces behind this trend are not clear-cut (Box 3). It is often argued that the relatively high productivity

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Box 3. The Swiss franc: ever stronger?

The Swiss situation is unusual. *First*, while the nominal effective exchange rate has clearly risen on both the official and OECD measure, the real effective exchange based on relative CPIs has shown a more modest rise, even though the rate based on relative export prices has shown a significant upward trend since the mid-1980s. *Second*, while Switzerland has a very large current account surplus, only about half of it is due to the external balance on goods and services. Other countries with a large current account surplus (Finland, Ireland, Korea and the Netherlands) showed a strong export performance, but Switzerland did not. *Third*, Switzerland benefited most of all OECD countries from terms-of-trade gains.

There are two prominent approaches to explain real exchange rate developments. These are: the purchasing power parity (PPP) model and structural models that determine medium to long run equilibrium exchange rates that satisfy stock/flow equilibrium.

The purchasing power parity hypothesis states that national price levels should be equal when expressed in a common currency. There are various reasons why PPP may not hold in the short run, but a large body of empirical work has examined the validity of PPP in the long run by either testing whether nominal exchange rates and relative prices move together in the long run or by testing whether the real exchange rate has a tendency to revert to a stable equilibrium level over time. Whether PPP holds in the long run remains contentious, but recent work modelling the real exchange rate in a non-linear fashion have found more evidence for mean reversion. These models suggest that the exchange rate becomes increasingly mean reverting with the size of the deviation from the equilibrium level. For Switzerland, Guerra (2003) and Kilian and Taylor (2001) find evidence for mean reversion, while Baum *et al.* (2001) do not. Mean reversion of the Swiss real exchange rate, when calculated with export prices, is however unlikely to hold.

PPP, the equalisation of aggregate prices, is not a necessary implication of open international markets. Theory would suggest only that at the commodity level, the price of traded goods should be equalised across borders, after allowing for transport costs and trade barriers. Also, differences of relative prices of traded goods may be due to quality effects. In addition, many goods are not traded, and their prices could differ systematically across countries. Trend differences in relative prices of non-traded goods across countries may account for trends in real exchange rates based on aggregate price indices. The Balassa-Samuelson hypothesis, for instance, ascribes trends in real exchange rates to the differences in relative price trends of non-traded goods and services, which are driven by differences in productivity growth. Aebersold and Brunetti (1998) found that this is important for Switzerland. In the Swiss Franc – Deutsche Mark case, there is a large divergence of the development of tradable and non-tradable prices, with the difference in productivity between the tradable and non-tradable sector being larger in Switzerland than in Germany. The Swiss central bank has established a similar empirical regularity in the case of the euro exchange rate.

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Box 3. **The Swiss franc: ever stronger?** (*cont.*)

In the NATREX (NATural Real Exchange) approach, the real exchange rate is explained by changes in thrift and productivity, and for small countries by changes in the terms of trade and world interest rates. Thrift changes savings and productivity affects investment. Under the assumption of relatively high long-term capital mobility these factors drive the real exchange rate to produce the necessary change in the current account (Stein *et al.*, 1995). The fundamental equilibrium exchange rate (FEER) model is similar in spirit: it determines the real exchange rate that would bring the current account in line with some measure of desirable capital flows, the latter being investment and saving that are not distorted by public policy. No research applying the NATREX or FEER model to the Swiss case could be found. But clearly the stylised facts also accord with this model. When growth slowed down appreciably this implied a slower growth in the capital stock and a sizeable downward adjustment in the investment ratio. Indeed, the Swiss investment ratio has shown the steepest trend decline in the OECD, after Japan, while national saving has stayed very high. Thus, a large current account surplus emerged, and this was pushed higher by the large factor income flow from abroad. For reasons that are not well understood, the rise in foreign assets has not led to stronger consumption over the medium term, so that a significant rise in the real exchange rate was needed to hold exports back and encourage imports in order to keep the rise in the current account balance in check. According to the model, a further appreciation will be needed for the external position to stabilise, but the dynamics will depend strongly on whether it is possible to revive the Swiss economy. Stronger growth would imply higher imports, a smaller current account surplus and thus probably less of a medium-term appreciation that in itself tends to stifle growth.

growth in the exporting sector is the main explaining factor. Nevertheless, the loss of export market shares, common to many developed countries due to the catching-up of less developed ones, has been stronger in Switzerland than in other OECD countries. The high propensity to save of Swiss households and firms, reflected in a very large trend current account surplus of around 10 per cent, may also contribute to the appreciation trend.

Irrespective of the factors that explain the real trend appreciation, it is clear that the real exchange rate is an endogenous variable that reacts to shifts in underlying forces and it cannot be influenced by monetary policy. The nominal exchange rate is not a target of Swiss monetary policy, although nominal rates are an important information variable for the SNB, not least because of the high elasticity of inflation and activity to the exchange rate. The role of the franc as a safe haven currency in periods of international economic or geopolitical uncertainty complicates the implementation of monetary policy, as these exchange rate movements may lead to

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monetary conditions that are not appropriate for the underlying economic conditions in Switzerland. In this sense, lower fluctuations of the exchange rate around a long-term trend may be desirable, although the negative impact of exchange rate volatility on output is difficult to pin down in empirical studies.

Over the recent past a more stable exchange rate *vis-à-vis* the European countries (which account for 60 per cent of Swiss exports) has been apparent. This greater stability reflects to some extent the reaction of monetary policy to pressures on the franc that were considered to be linked to safe haven considerations. It is also the result of the introduction of the euro, which implies that there is a reduced risk of currency misalignments among European currencies. The anti-inflationary reputation acquired by the European Central Bank since its creation may also have had a positive stabilising effect. The latter two factors may contribute in the future to reduce pressures on the franc as a temporary safe haven, which should facilitate the management of monetary policy and be beneficial for the Swiss economy as a whole.

Fiscal policy

Following the fiscal consolidation of the 1990s, public finances have deteriorated uninterruptedly over the last few years, though the deficit remains smaller than in the euro area on average and in the United States (Figure 12). In international comparison, fiscal policy remains relatively sound, the more so as public debt is also low by international standards and contingent liabilities related to population ageing are smaller than in other countries. Nevertheless, recent budget trends are a cause for concern. The budget balance moved from a surplus of 2½ per cent of GDP in 2000 to a deficit of almost 1 per cent in 2002 (excluding exceptional receipts). It affected all levels of government, particularly the Confederation, and continued in 2003, with the overall deficit expected to reach some 2 per cent of GDP (Figure 13). The deterioration is partly due to the negative impact of the economic situation. But it also reflects a marked widening of the cyclically-adjusted budget deficit which could reach approximately ¾ per cent of GDP in 2003. In particular, federal revenue declined markedly following a brisk increase until 2000, partly due to fluctuations in the financial asset cycle which distorted both developments in and estimates of structural balances. According to the authorities, the Confederation's balance (financial accounts definition) could show a deficit of ¾ per cent of GDP in 2003, instead of the balanced budget that had been expected.[11] Overall, the fiscal stance has eased substantially since 2000. While this probably has supported economic activity, it emphasises the importance of ensuring that medium-term fiscal soundness is not put at risk.

Fiscal policy is bound by the debt containment rule which, starting in 2003, requires the federal accounts to be in balance in structural terms by capping expenditure.[12] Strict implementation of this rule in 2004 would require a

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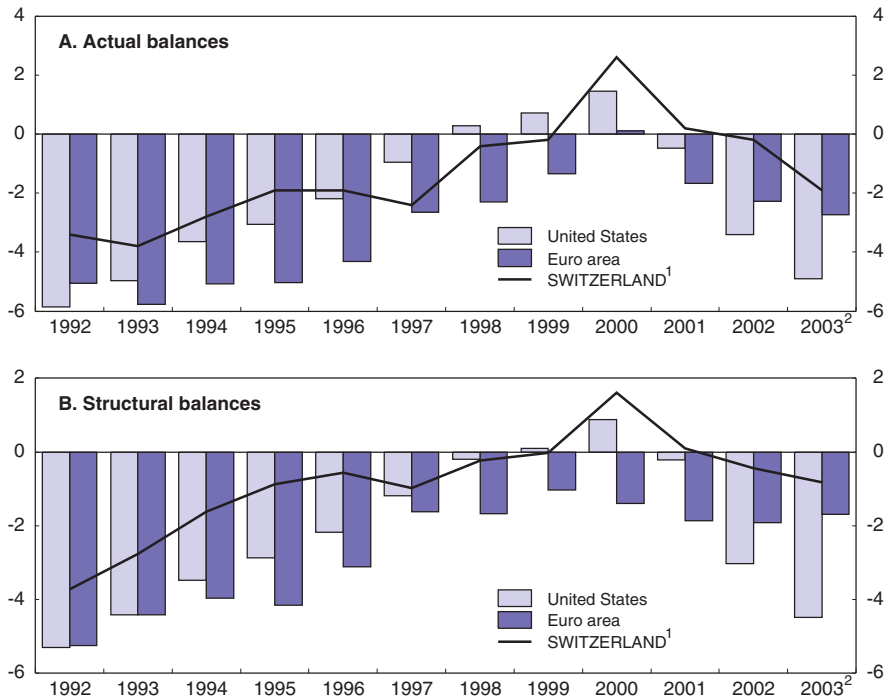
Figure 12. **Fiscal balances of general government**
As a percentage of GDP



1. "*Statistique financière révisée*" basis. The Swiss structural balance excludes exceptional factors such as the sale of shares in Swisscom in 2002.
2. Estimates.

Source: Administration fédérale des finances and OECD.

fiscal tightening to eliminate the 2003 structural deficit. Given the risk that such a policy could further weaken an already depressed economic situation, and with the uncertain outlook, the Federal Council has suggested easing the short-term constraint imposed by this rule. It has proposed to Parliament to raise the expenditure ceiling on a temporary basis. Although consistent with the spirit of the constitutional rule on debt containment, this measure must not damage its credibility or call into question the authorities' commitment to restore public finance equilibrium on a sustainable basis. This is why these measures are part of a medium-term fiscal consolidation strategy (Chapter II). In the short term, however, the fiscal stance could remain slightly accommodating.

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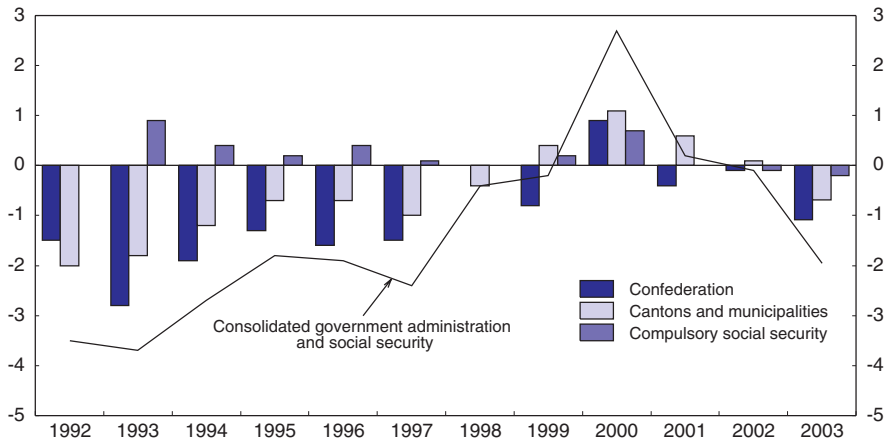
Figure 13. **Budget balances by level of government**[1]
As a percentage of GDP



1. "*Statistique financière révisée*" basis. Data for 2003 are estimates.
Source: Administration fédérale des finances.

The public accounts worsened in 2002 *and the federal budget is in structural deficit in* 2003

Whereas a balanced outturn for general government was projected for 2002, a deficit of some CHF 650 million (0.2 per cent of GDP) was recorded. However, this includes exceptional revenues of CHF 3.7 billion (0.9 per cent of GDP) from the sale of Swisscom shares. Excluding this, the deviation from the forecast is in excess of 1 per cent of GDP. The slippage is attributable to the marked downturn in federal tax revenue and the optimistic forecast of social security revenues, whereas expenditure by these levels of government was contained (Table 9). According to preliminary information on the 2003 budget outturn, the forecasts for general government, which anticipated a deficit of ¾ per cent of GDP, will not be met again because of further slippages in the federal and social security accounts. The deficit could reach nearly 2 per cent of GDP. Federal tax revenue has remained sluggish since the second half of 2002 and total resources of the Confederation could be more than 7 per cent down on the budget forecast (*i.e.* CHF 3.7 billion). Tax revenues from financial and stock market activities, which are important for the central government, have remained very depressed,[13] while the unforeseen stagnation of economic activity in 2003 has also weakened other revenues, particularly corporate income tax.

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Table 9. **Government accounts**

	Outcomes		Budgets		Estimates		Financial plan[1]		
	2000	2001	2002	2003	2002	2003	2004	2005	2006
Confederation[2] (CHF millions)									
Expenditure	48 208	51 136	52 530	52 605	51 970	52 075	53 193	55 308	56 992
Revenue	51 994	49 436	51 226	51 147	51 430	47 517	48 804	52 188	54 341
Balance	3 786	–1 700	–1 304	–1 458	–540	–4 558	–4 389	–3 120	–2 651
Expenditure excl. special proceeds[3]	48 208	50 056			51 281	52 075			
Revenue excl. special proceeds[3]	50 362	47 531			46 577	47 517			
Balance excl. special proceeds[3]	2 154	–2 525			–4 704	–4 558			
Cantons (CHF millions)									
Expenditure	60 151	64 001	64 400	67 000	67 000	67 000	67 093	67 870	68 660
Revenue	62 818	65 257	63 800	65 000	67 000	65 000	64 888	66 378	67 900
Balance	2 667	1 256	–600	–2 000	0	–2 000	–2 205	–1 492	–760
Municipalities (CHF millions)									
Expenditure	40 599	41 709	42 800	44 400	43 500	44 400	44 008	44 224	44 440
Revenue	42 068	43 033	42 400	43 700	44 000	43 700	42 525	43 203	43 890
Balance	1 469	1 324	–400	–700	500	–700	–1 483	–1 021	–550
Compulsory social security (CHF millions)									
Expenditure	44 388	46 179	48 949	51 501	48 625	52 671	54 117	54 047	54 621
Revenue	47 105	46 309	51 383	52 196	48 009	51 842	51 815	54 893	57 142
Balance	2 717	130	2 434	695	–616	–829	–2 302	846	2 521
Expenditure excl. special proceeds[4]	43 338	44 479			47 475	52 671			
Revenue excl. special proceeds[4]	47 105	46 309			48 009	51 842			
Balance excl. special proceeds[4]	3 767	1 830			534	–829			
Consolidated account of general government[5] (CHF millions)									
Expenditure	152 524	159 977	165 802	168 396	168 014		170 291	172 370	178 937
Revenue	163 163	160 986	165 932	164 933	167 358		165 760	172 752	179 188
Balance	**10 639**	**1 010**	**130**	**–3 463**	**–656**	**–8 087**	**–10 379**	**–4 787**	**–1 440**
Balance excl. special proceeds[3, 4]	**10 057**	**1 885**			**–3 670**	**–8 087**	**–10 379**	**–4 787**	**–1 440**
Balance (% of GDP)									
Confederation	0.9	–0.4	–0.3	–0.3	–0.1	–1.1	–1.0	–0.7	–0.6
Cantons	0.7	0.3	–0.1	–0.5	0.0	–0.5	–0.5	–0.3	–0.2
Municipalities	0.4	0.3	–0.1	–0.2	0.1	–0.2	–0.3	–0.2	–0.1
Compulsory social security	0.7	0.0	0.6	0.2	–0.1	–0.2	–0.5	0.2	0.5
Consolidated general government	**2.6**	**0.2**	**0.0**	**–0.8**	**–0.2**	**–1.9**	**–2.4**	**–1.1**	**–0.3**
Balance excl. special proceeds[3, 4] (% of GDP)									
Confederation	0.5	–0.6			–1.1	–1.1			
Cantons	0.7	0.3			0.0	–0.5			
Municipalities	0.4	0.3			0.1	–0.2			
Compulsory social security	0.9	0.4			0.1	–0.2			
Consolidated general government	**2.5**	**0.5**			**–0.9**	**–1.9**			
Debt (% of GDP)	51.2	51.4			55.4	55.7			

1. Partially estimated. Includes the expected outturns of the 2003 fiscal reduction plan proposed by the Federal Council.
2. Includes railway infrastructure financing fund and Swiss Federal Institutes of Technology.
3. Excluding proceeds from the sale of shares in Swisscom (CHF 3.7 billions) in 2002. The Confederation account was also adjusted for the reimbursement of loans granted to the unemployment insurance which was included in the social security account.
4. Excluding transfers related to grants and their subsequent repayment from the Confederation to the unemployment insurance system.
5. Double counting not included in the total.

Source: Administration fédérale des finances.

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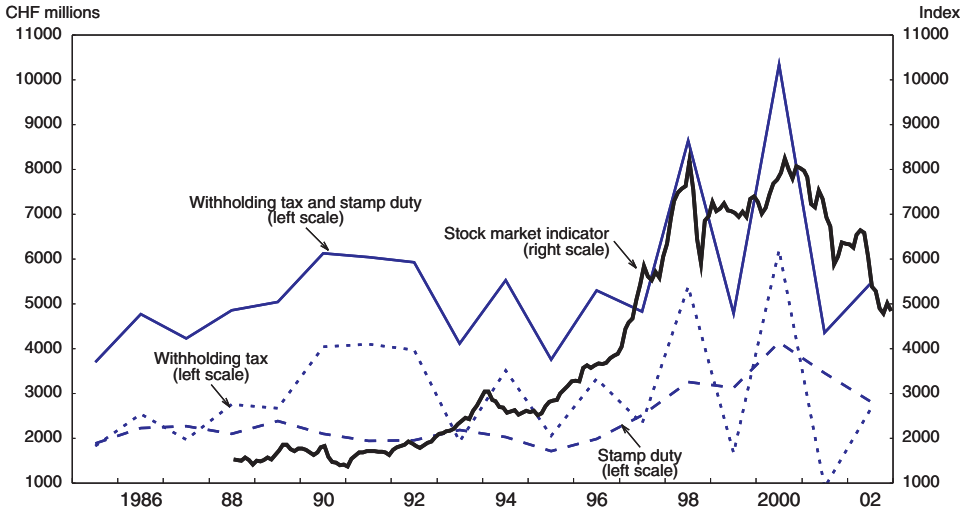
Figure 14. **Federal tax revenues and the stock market**



Source: Datastream and OECD.

According to official estimates, over half (2½ percentage points) of the deterioration in public finances between 2000 and 2003 (4½ per cent of GDP, excluding exceptional factors) is attributable to a reduction in the cyclically-adjusted budget balance, which is largely (1¼ per cent of GDP) due to the federal level. The structural balance of the cantons and communes also worsened by almost 1 per cent of GDP as a result of the easing of fiscal policy following the exceptional results achieved in 2000, while the balance of the social security account remained more or less stable.[14] The easing of federal fiscal policy between 2000 and 2003, as measured by the cyclically-adjusted balance, does not reflect the adoption of discretionary measures by the authorities but the end of the temporary factors that had swelled revenues. In Switzerland as in other countries, the financial asset cycle and other specific factors temporarily increased non-cyclical tax revenue in the late 1990s and early 2000s. The high levels of revenue during this period, especially from stamp duty and withholding tax, was partly due to the exceptional financial market situation and to corporate profits, which have dipped sharply since (Figure 14). Corporate profits were also boosted until the early 2000s by a change in accounting standards which resulted in the elimination of hidden reserves. This temporarily increased the cyclically-adjusted budget balance, but that phenomenon came to an end during the recent period, notably as a result of the bursting of the financial bubble.[15] This development led to the re-assessment of the Confederation's structural budget balance, with the

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authorities now forecasting a deficit of between ½ and 1 per cent of GDP in 2003 instead of a balanced budget as previously projected.

The official figures may slightly underestimate (by about ½ per cent of GDP) the scale of the structural deterioration in the consolidated government accounts in recent years.[16] The OECD Secretariat's estimate suggests that it could have amounted to almost 3 per cent of GDP between 2000 and 2003 because the structural weakening of the social security accounts may have been underestimated.[17] Official estimates indicate no deterioration in the social security's cyclically-adjusted balance despite the continuous worsening of the disability insurance budget and the cut in the rates of contribution to unemployment insurance in 2003, which was only partially offset by the reduction in the maximum duration of compensation and the increase in the minimum contribution period. Also, there could be a slight distortion in the official estimates of the structural balances if, as the OECD Secretariat's estimates suggest, the output gap is larger in 2003 (–2¼ per cent) than the authorities' estimate indicates (–1¼ per cent). This difference suggests, that the cyclically-adjusted deficit of the Confederation for 2003 could be towards the bottom end of the range of the official estimate (–½ per cent of GDP). This slight overestimation of the structural deficit, which also affects the cantons and communes, is however offset by the social security balance being overestimated in 2003.

Fiscal policy will be slightly expansionary in the short term

The federal authorities want to avoid pursuing a pro-cyclical fiscal policy

The difficulty to take account of the transitory nature of certain factors, which led to the cyclically-adjusted federal budget deficit being underestimated for 2003, make the management of the Confederation's finances more complex. The debt containment rule requires permanent structural balance and strict application of the rule would mean eliminating the deficit as of 2004. Such an adjustment would mean adopting a restrictive policy, and could weaken an already depressed economic situation or to check an incipient upturn. The government has recognised the need to temporarily amend the fiscal rule and has submitted a proposal to Parliament which would allow the non-cyclical federal deficit to be reduced gradually. According to the Federal Council this strategy, which does not invoke the exceptionality clause provided by the debt containment rule,[18] would be consistent with the spirit of the constitutional mandate to take the economic situation into account in applying the fiscal rule. Under the proposal, to be discussed this autumn, the ceiling on expenditure permitted by the debt containment rule would be raised, allowing an overshoot of CHF 3 billion in 2004 (¾ per cent of GDP), CHF 2 billion in 2005 (½ per cent of GDP) and CHF 1 billion in 2006 (¼ per cent of GDP). These measures are

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accompanied by a medium-term consolidation plan, which will also be discussed by Parliament this autumn (Chapter II).

Main features of the 2004 budget

The 2004 budget will be based on GDP growth of 1 per cent, which is below potential growth. With the fiscal policy options under discussion until the autumn, it is difficult to forecast the consolidated general government budget. The provisional estimates suggest that it could worsen slightly on 2003 and show a deficit of 2½ per cent of GDP. For the Confederation, the objective for 2004 is to stabilise spending in real terms, which is to be achieved at least partly through the early implementation of some of the measures included in the medium-term consolidation programme. The result should be savings of CHF 800 million, achieved by a number of measures, in most areas of federal spending. In view of the expected revenue trend, the federal deficit is likely to be reduced only marginally, so that fiscal policy will probably be neutral (Table 9). The social security accounts, on the other hand, could be slightly expansionary. As in 2003, unemployment insurance contributions are scheduled to fall, in accordance with the reform adopted in autumn 2002 (Chapter IV). These adjustments are compatible with continuing equilibrium of the unemployment insurance accounts over an entire cycle. However, further slippage of the disability pension system is probable, corrective measures being unlikely to come into effect before 2005. Overall, the social security deficit could rise by ¼ per cent of GDP compared to 2003, despite the projected upturn. Greater uncertainty surrounds the policy of the cantons and communes. According to the autumn 2002 financial plan it was designed to maintain the fiscal positions achieved in 2003 and will probably be slightly expansionary. That said, increased fiscal stringency as regards expenditure in particular cannot be ruled out, especially at cantonal level, following the deterioration of the last few years.

Assessment

The Federal Council's decision not to tighten the Confederation's budget by proposing to lift the debt containment constraint temporarily is judicious, bearing in mind the persisting sluggishness of the economic situation and the uncertainty about the strength of the economic upturn in 2004. The macroeconomic assumptions underlying the budget seem slightly optimistic, as they rely to a large extent on a swift euro area recovery which may be more muted. In this context, it is better not to risk weakening activity any further through budgetary restrictions at federal level, particularly if the latter were to be complemented by the cantons and communes. This strategy allows the authorities to remain in line with the spirit of the debt containment rule by allowing the automatic stabilisers to play their role, and to wait for the upturn before adopting a more restrictive stance. The slight

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reduction in the cyclically-adjusted social security balance also seems justified inasmuch as it reflects a decline in unemployment insurance contributions that does not endanger the equilibrium of the accounts in the long term. While the recovery must not be hindered in the short term by a restrictive fiscal policy, it is however important to avoid any further major structural deterioration in the government accounts, which would be neither desirable nor an effective way of stimulating activity. Recourse to fiscal activism in the past, as in 1997, was not as effective as expected, so that focusing on budgetary sustainability over the medium term is no doubt the best option. It is a strategy that ought to be applied at all levels of government.

The main medium and long-term policy challenges

Overall, the macroeconomic policy followed by the authorities appears well adapted to the current cyclical position as it supports the recovery, while minimising deflation risks. However, the difficulties ahead in ensuring the long-term viability of public finances should not be underestimated, especially if trend growth remains weak. In particular, efforts are required to better control government spending. Medium and long-term issues concerning the sustainability of the public finances are taken up in Chapter II. The main policy challenge is however to strengthen trend growth. To that end, stepping up productivity gains will no doubt require important reforms to the functioning of the product markets and a strict implementation of the new competition policy framework. This should be a priority on the reform agenda and is indispensable for raising potential output growth. The persisting problems in this domain are analysed in Chapter III, which also provides a set of recommendations to correct the main shortcomings. Chapter IV of the Survey focuses on the other structural reforms required to reinforce employment performance, including raising education results. Measures stimulating the participation of females and older workers in the labour market would also be desirable to raise trend growth, even though the employment ratio is already high in international comparison.

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Notes

1. Recession is usually defined as two or more consecutive quarters of negative GDP growth. However, the Swiss authorities do not apply the same definition for a recession.
2. Measured in constant francs, the ratio of investment to GDP rose by 4½ percentage points between 1993 and 2000, *i.e.* substantially more than in other European countries, reaching its highest level since 1970. Despite having fallen by 2½ percentage points since 2000, the investment ratio is still quite high in a historical perspective. Measured in current francs, the rise in the investment ratio in the late 1990s was much the same as in France or Germany, while the recent fall has been more pronounced than in these countries.
3. However, the effect of interest rates on construction investment is limited as a quarter of this expenditure is made by government.
4. In the third quarter of 2003, new car registrations had fallen by 13 per cent since end-2000.
5. During both the expansion and the contraction phases of the cycle, part-time work has increased. Since the start of 1998, it is up by 175 000 persons (+20 per cent), whereas full-time employment has fallen by 35 000 people (–1.4 per cent).
6. Nominal downward wage flexibility being greater for people changing jobs (and those working part time), the present phase of redundancies ought to be followed by a period of greater wage restraint during the initial improvement in the labour market.
7. These projections are consistent with those presented in the OECD *Economic Outlook*, No. 74, December 2003.
8. The capacity utilisation rate reached 81 per cent in industry in the first half, but remains well below the multi-year average (84 per cent).
9. However, according to standard national accounting data, this increase in income does not seem to have helped consumption to grow faster than output, compared to other countries (Figure 5).
10. The index of monetary conditions uses the fluctuations of real exchange rates around its long-run trend, which is ½ per cent per annum.
11. Swiss government accounts are not available on a national accounts basis, as in other countries, nor with the same degree of detail. The consolidated accounts are based on the revised financial statistics. Moreover, where the Confederation is concerned, budget discussions are based on the financial accounts data. The latter does not incorporate special funds such as that for major railway projects. The debt containment rule applies to the financial account of the Confederation.
12. The debt containment rule requires federal accounts to be in balance once cyclical effects on the budget have been taken into account. To achieve this, a ceiling on

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expenditure has been set on the basis of estimated cyclically-adjusted revenue so that, while revenue can fluctuate cyclically, the rule prevents these fluctuations from having an impact on expenditure (OECD, 2002).

13. In 2003, revenue from withholding tax and stamp duty, which makes up some 15 per cent of federal tax receipts, could be over 40 and nearly 25 per cent down, respectively, on budget projections.
14. Official estimates suggest that there is a gap between the sum of the structural balances of the different levels of government and that of government as a whole. This statistical problem accounts for approximately ¼ per cent of GDP of the fall in the consolidated general government balance.
15. The increase and then the fall in revenue due to financial market developments result from an "automatic stabilisation" effect of the financial asset cycle stemming from the impact of the tax system on their valuation and on financial transactions. This effect is not, however, taken into account by the traditional methods of estimating structural budget balances.
16. The change in the output gap by some 2¾ percentage points between 2000 and 2003 is likely to have increased the budget deficit by 1¼ to 1½ per cent of GDP. Standard estimates indicate that a 1 percentage point fall in the output gap reduces the budget balance by ½ per cent of GDP in the majority of OECD countries. In this case, more than 3 percentage points of the deterioration in the general government balance of 4½ per cent of GDP between 2000 and 2003 would be attributable to non-cyclical factors.
17. The method used by the authorities to estimate the structural balance of the social security system is not consistent with that used for the other levels of government, being based on an evaluation of the cycle different from that measured by the output gap. Whereas, for example, the authorities estimated the output gap in 2002 to be zero, the cyclical budget balance of the social security was not zero, in contrast with that of the other levels of government.
18. The budgets voted by Parliament may exceed the expenditure ceiling as defined by the debt containment rule for reasons stipulated by law. These include exceptional events that cannot be controlled by the Confederation (such as a serious recession, a natural disaster or some other specific event), but also payment peaks due to accounting practices as well as adaptations of the accounting system. In this case, a qualified majority vote by Parliament (*i.e.* an absolute majority of the members in the two Chambers) is necessary.

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II. Medium and long-term fiscal policy challenges

Looking beyond the short term, one of the important tasks for economic policy is to keep the public finances on a sound footing, which could prove challenging if growth should remain weak over the medium term. Fiscal management faces three major challenges. The first relates to the implementation of federal fiscal policy. Because of the difficulties encountered in implementing the new debt containment rule in 2003, the government has had to adjust its financial plan to restore its accounts to structural equilibrium in the medium term. A number of lessons should be drawn from the initial application of the debt containment rule in order to improve this new budgetary framework. The second challenge concerns the reform of the pension system. Adjustments are still needed to ensure the long-term financial viability of the basic old-age and invalidity pension systems, while the financial market slump also necessitates reforms of the (second pillar) occupational benefit plans which are funded schemes. Third, the public sector must pursue its efforts to improve efficiency. These will pay off not just for general government, whose financial resources will probably not grow fast in the medium term, but also for the private sector which should benefit from improved services.

Medium-term management of federal finances

The new fiscal policy framework has experienced difficulties stemming mainly from problems in estimating the structural budget balance which is now a key federal policy indicator. Consolidation of the Confederation's finances was less far-reaching than had previously been thought and the debt containment rule requires the federal accounts to be restored to structural equilibrium (Chapter I). To this end, the authorities have prepared a medium-term consolidation plan which is being discussed by Parliament. The programme has prompted a debate on the strategy for improving public finances, questions being raised in particular concerning the scale of the proposed programme which ought to be implemented in conjunction with a tax cut, which some think, should be more far-reaching. However, tax measures to offset the recently recorded revenue shortfalls have also

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been suggested, thus raising the question as to what would be the most appropriate measures to adopt. Moreover, suggestions have been made to improve the mechanisms of the new fiscal rule.

A *medium-term fiscal consolidation programme is necessary*

An adjustment programme is necessary…

The Federal Council reacted promptly to the abrupt deterioration in the Confederation's finances in the second half of 2002, presenting a medium-term consolidation strategy in three parts in early 2003. The first part consisted of a CHF 2 billion fiscal consolidation programme (½ per cent of GDP) until 2006, mainly involving reductions in expenditure. The second component consisted of postponing the tax reform which was under discussion at the time, while the third part involved tax increases. The combination of these measures was designed to offset the revenue losses recorded which, measured against the autumn 2002 financial plan, was estimated at some CHF 5 billion up to 2006, *i.e.* 10 per cent of total federal revenues (1 per cent of GDP). Parliament's refusal to postpone the tax reform (Annex I) and its reluctance to increase taxation caused the authorities to raise the first part of the consolidation plan. All told, the fiscal consolidation programme now involves a CHF 3.3 billion adjustment by 2006 (¾ per cent of GDP). It relies mainly on reining in expenditure (CHF 2.8 billion) and, backing these up, an increase in revenue (CHF 0.5 billion).

The plan presented to Parliament consists of a wide-ranging set of 70 measures affecting the Confederation's main spending areas. The effects of the measures will be felt above all in 2005-06, when the economy should have recovered. However, according to estimates by BAK Basel Economics, the impact on activity will be limited to a cumulative fall of approximately ¼ of a percentage point by 2006. These measures will lower the nominal rise in spending from 3.8 to 2.1 per cent per year on average between 2002 and 2006, *i.e.* lower than the expected rate for nominal GDP growth (Table 10). They include a reduction in payroll costs, with job cuts and reduced indexation of old-age pensions. The savings will also affect priority sectors such as training and research, though expenditure here will rise more than the average (4 per cent per year until 2006). Revenues will be increased due to higher profits being redistributed by the Swiss central bank and to higher taxes on tobacco and alcohol. The plan also takes into account the 0.8 percentage point increase in VAT to finance invalidity insurance (see below), which is scheduled in 2005, with the Confederation keeping its share in the revenue which amounts to CHF 0.7 billion, as confirmed recently by the Parliament.[1]

… but its characteristics are prompting debate

While discussions about the programme have shown that there is a consensus on the need to balance the federal accounts in the medium term,

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Table 10. **Effects of the 2003 budget reduction programme**
CHF billions

	Outcome	Budget	Financial plan			Average annual growth
	2002	2003	2004	2005	2006	2002-06
Revenue	47.4	47.2	48.2	51.2	53.3	3.0
Expenditure	50.7	50.6	52.8	56.2	58.8	3.8
Expenditure cuts			–0.8	–1.8	–2.8	
Cuts in federal grants or maintenance of the Confederation shares in the increase of the VAT revenue			–0.4	–0.7	–0.8	
Cuts in the remaining expenditure in order to stabilise the outlays in real terms in 2004			–0.2			
New expenditure	50.7	50.6	51.4	53.7	55.2	2.1
Revised balance	–3.3	–3.4	–3.2	–2.5	–1.9	
Structural balance according to the debt containment rule	–3.3	–2.8	–2.9	–2.6	–2.3	
Balance entitled by the debt containment rule		–0.6	–0.3	0.1	0.4	
Ceiling excess allowed by the Federal Council proposition			3.0	2.0	1.0	
Other necessary adjustments				–0.6	–1.3	
Memorandum items:						
Output gap (official estimates)	0.0	–1.2	–0.5	0.2	0.3	
Output gap (OECD estimates)	–0.5	–2.2	–2.0	–1.4		

Source: Département fédéral des finances.

opinions differ as to the scale of the savings to be made and the relative importance of measures affecting expenditure and revenue. One of the main criticisms is that the plan is not ambitious enough, particularly the spending measures. The programme will indeed not allow the debt containment rule to be respected in 2005 and 2006, even taking into account the temporary raising of the expenditure ceiling (Chapter I). The official estimates suggest that an additional adjustment of CHF 1.3 billion (¼ per cent of GDP) would be needed in 2006 (Table 10). The scale of the adjustment would also need to increase in subsequent years because of the obligation to achieve structural balance as of 2007 and because of the further fall in revenue expected as of 2008 when the reform of the real estate tax will be implemented (Annex I). The calls for a more ambitious plan also reflect demands for the recently voted tax cuts to be stepped up, the increase in tax pressure having been greater than in most OECD countries during the 1990s (Figure 15).

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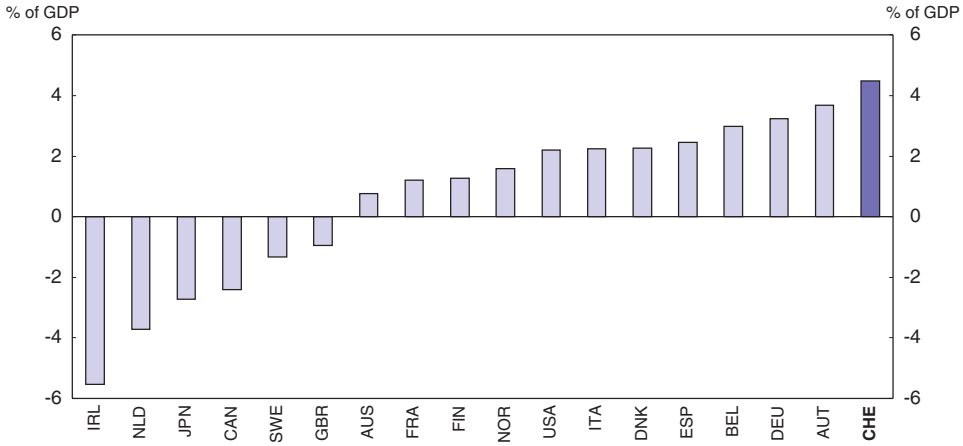
Figure 15. **Tax revenue development in OECD countries**

As a percentage of GDP, 1990-2002[1]



1. Difference between the tax revenue as a percentage of GDP in 2002 and 1990 for all countries, except for Australia, Japan and the United States (1990-2001).

Source: OECD, *Revenue Statistics*.

Moreover, the reductions in tax adopted for the period 2004-06, especially with respect to household income, will be more than offset by the scheduled increase in VAT.

The way the authorities' plan is conceived shows that they recognise the need to give priority to controlling expenditure in order to achieve fiscal consolidation. Measured in relation to GDP, the increase in public expenditure for government as a whole, and for the Confederation in particular, was higher over the past decade than in other countries, despite more effective control in recent years (OECD, 2002). That trend, which pushed up tax pressure more rapidly than in other countries, was due especially to the surge in social spending (Table 11). If expenditure is reduced too much or too quickly, however, there could be problems as regards the quality of the services provided by the Confederation, whose budget represents only 10 per cent of GDP. In fact this problem has already been highlighted by some of the reactions to the fiscal consolidation plan, and this could lead to expenditure cuts deemed excessive in areas such as education and transport being called in question when they are discussed in Parliament. This is why the Federal Council is advocating offsetting measures, mainly on the revenue side, if the scope of the consolidation plan were to be reduced.

There is also a consensus that Switzerland should remain a low tax country so as to guarantee, and if possible enhance, the country's economic attractiveness.

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Table 11. **Changes in public expenditure over the 1990s**

	Level		Change	Average annual growth
	2001		1990-2001	1990-2001
	CHF billion	% of GDP	% of GDP	Per cent
Total[1]	**160.0**	**38.6**	**5.4**	**3.9**
By function				
National defence	5.3	1.3	–0.8	–2.0
Culture and entertainment	4.2	1.0	0.0	2.7
Environment	4.9	1.2	0.0	2.7
Public economy	7.2	1.7	0.1	3.0
Education	24.1	5.8	0.6	3.4
Administration	8.8	2.1	0.2	3.5
Justice and police	7.2	1.7	0.2	3.5
Transport	14.1	3.4	0.5	4.0
Health[2]	17.7	4.6	0.7	4.2
Foreign affairs	2.7	0.6	0.2	5.0
Social insurance[2]	55.2	14.2	4.0	6.1
Treasury and tax	10.5	2.5	0.8	6.1
By level of government				
Confederation	51.1	12.3	2.4	4.5
Cantons	63.9	15.4	2.5	4.1
Municipalities	41.7	10.1	0.5	3.0
Social security	46.2	11.1	2.8	5.2
Memorandum item:				
Nominal GDP	414.9			2.5

1. Consolidated data for the total; unconsolidated data for all detailed items.
2. Last year of available data is 1999.
Source: Administration fédérale des finances.

For this to happen, it is important to stop, or even reverse, the increase in tax pressure seen in the last few years. Though it reached 35.7 per cent of GDP in 2000, that pressure was nevertheless a little below the OECD average (37.4 per cent) and short of the EU average (41.6 per cent), despite the trend increase in taxation over the past decade. Although caution is needed with these comparisons because some compulsory spending – such as sickness insurance and second pillar occupational benefit schemes – is mostly privately financed, the present level of taxation does not seem to be a serious problem for Swiss competitiveness. In addition, a tax cut has been recently voted by the Parliament, as indicated above. It includes three parts:

– A *reduction and change in income tax*. At present, income tax discourages married women from working, takes insufficient account of costs pertaining

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to dependent children and is more favourable towards common-law spouses than towards married couples.

– A *cut in taxation on financial transactions*. Stamp duty undermines the competitiveness of the Swiss financial market. Buyers and sellers are able to avoid it by carrying out their transactions abroad.

– A *reform of housing taxation*. The rental values on which this tax is based are controversial because they vary from one canton to another. It is also thought that encouragement should be given to home ownership, which is less widespread than in other countries.

This tax package, the total amount of which is estimated at between CHF 3 and 4 billion (¾ to 1 per cent of GDP), should alleviate the tax burden, even though it will be challenged by a referendum called by a number of cantons, which consider its budgetary cost excessive (Annex I). Moreover, an endogenous reduction in effective tax pressure cannot be ruled out as the recent structural fall in revenue shows. This could well persist in the coming years because of the possibilities for carrying over corporate losses and the probable increase in (deductible) contributions to the occupational benefit scheme which will serve to offset, in part, the effect of the financial crisis on the second pillar (see below).

A *new tax reform ought to improve the efficiency of the tax system and simplify it*

A new tax reform designed to both improve the efficiency of the tax system and simplify it could have a beneficial impact on the economy. A bill has therefore been tabled, that aims to provide the Confederation with a new financial regime. The changes proposed are modest, however, and would not substantially alter the tax burden. The main objective is to guarantee the Confederation's main sources of income (VAT and direct federal tax) when the present regime comes to an end in 2006. It is proposed to do away with the time limit imposed by the Constitution, but maximum rates would continue to be governed by it.[2] When Parliament assesses this project in the second half of 2003, the opportunity arises to discuss other modifications of the tax system. For example, technical measures could be envisaged with a view to lessening the chronic instability of the withholding tax on dividends (without modifying its rate or tax base), which poses problems for fiscal management.[3] Also worth discussing is whether to shift the balance of the tax structure between direct and indirect tax. In international comparison, direct taxes (27.4 per cent of GDP in Switzerland and 25.8 for the OECD average) predominate, compared to indirect taxes (7.1 per cent of GDP in Switzerland and 11.6 for the OECD average). An increase in VAT offset by a reduction in the tax burden on enterprises might help to boost investment and growth and would not be a problem as regards inflation, even though, as in other countries, VAT is not entirely neutral.[4]

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The debt containment rule should be better incorporated in medium-term management

Recent problems in estimating the structural budget balance…

The implementation of the debt containment rule has been marked by errors in predicting the structural budget balance which is required for implementing the rule. The inaccuracy of the estimates of structural balances is raising doubts about using this type of indicator as a guide for federal policy.

Measuring the structural budget balance also poses problems in other OECD countries such as Finland and the United States (Girouard, forthcoming). The standard methods used to estimate these balances calculate discretionary fluctuations in a residual manner. All the temporary factors not taken into account in estimating the cyclical fiscal component are considered to be structural. These methods do not take account of the impact of fluctuations in asset prices on revenues, and financial cycles are not necessarily correlated with the economic cycle. This seems to have been a major cause for the error in estimating the Confederation's structural balance in 2003, but there are numerous other distorting factors.[5] Under these conditions, the cyclically-adjusted balance can be affected by temporary movements, and this undermines the stabilising effect of a fiscal policy with a medium-term perspective, which is what the debt containment rule is supposed to achieve.

… have required technical adjustments

The Swiss authorities are aware of these difficulties, which need to be taken into account in order to protect federal policy against the risk of erratic movements dictated by temporary fluctuations in the non-cyclical balance. *Ad hoc* technical measures have hence been adopted to avoid strong fluctuations of some taxes, notably those related to financial activities, undermining fiscal policy objectives. In future, projections of those taxes will depend on their average level estimated over a longer period. In addition, it would be useful to ensure that the increase in expenditure does not exceed the trend growth of the economy in cases where, initially, the budget was structurally in equilibrium and no tax changes were scheduled.[6] More fundamentally however, overcoming the problem of the instability of the non-cyclical balance could require breaking the budget balance down into three components (cyclical, structural and irregular non-cyclical), the breakdown being based on estimates of structural revenue ratios as proposed by recent research (Bodmer and Geier, 2003). Despite being based partly on a subjective expert evaluation, which limits its transparency, this type of approach is an interesting development which might usefully be followed up in the future.[7]

As part of its fiscal consolidation plan, the Federal Council has also adopted some slight changes to the method of calculating the output gap used in

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the debt containment rule. They concern the Hodrick-Prescott (HP) filter used to estimate trend output and are intended to make the output gap more sensitive to revisions in GDP growth. The HP filter's traditional problems with end-of-period estimates mean that a 1 percentage point downward revision of effective GDP growth results in only a 0.4 percentage point fall in the output gap. The proposed adjustments would double the degree of sensitivity. This reduces the risks of fiscal policy being pro-cyclical.[8] Allowing bigger cyclical fluctuations will also, however, require bigger surpluses when activity is buoyant. However, despite these improvements, there are doubts regarding the reliability of output gap estimates provided by a simple HP filter (Chapter I). In the future, consideration could be given to having these estimates carried out by a group of experts who would be independent of the Finance Ministry and would incorporate a wider variety of economic indicators. Such a reform would no doubt make the estimates more reliable than the results produced by a mechanical and imperfect method. This group of experts could also be responsible for evaluating if economic developments do or do not justify proposing to Parliament, recourse to the exceptionality procedure included in the debt containment rule. The notion of "serious recession", which underlies the exceptionality clause is ambiguous and would need an economic assessment.[9]

The debt containment rule needs to be better incorporated in medium-term fiscal management

The authorities have observed that the initial difficulties in implementing the debt containment rule must not be allowed to call into question the use of the new rule or the Confederation's fiscal discipline. Following the adjustments adopted, one should let the new budgetary rule function for some time and accumulate more information concerning its implications before making further modifications. Some lessons drawn from its initial application would be worth thinking about and could provide useful information for future reforms. They concern the risks and limitations of year-by-year fiscal management insofar as the containment rule seems to require better projections of expenditures and revenues that are technically possible. With that in mind, it could be envisaged to subject the consolidation plan to a fiscal rule, which would prevent decisions on expenditure being regarded as separate from those on taxation. However, this would imply the need to modify the budget process to better integrate the rule in a multi-annual framework. Taking medium-term trends into account when applying the containment rule would help to reconcile the objectives of public expenditure control and efficient allocation.

The constraint that the debt containment rule imposes on spending on the basis of trends in revenue also makes it inappropriate to cut taxes before curbing expenditure. In the short term, this sort of policy can have harmful crowding-out effects on certain outlays, bearing in mind the trend rise in social spending

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linked to demographic developments. In the long term, a reduction in taxes that was neither tenable nor credible would not have a positive effect either. This militates in favour of reforms that do more to tackle the causes of the trend rise in expenditure than does the consolidation plan. For this trend to be changed on a sustainable basis and made compatible with trend revenues, structural reforms have to be taken further, in particular those relating to social expenditure or education, which are the largest items. Renewed efforts to step up efficiency and ensure better management of procurement contracts would also improve control over expenditure. More fundamentally, the fiscal problem highlights the consequences of economic growth that is too slow and needs to be boosted. Persistent low potential growth would imply the need to revise welfare programmes downwards compared with other countries. On the other hand, the dynamics of rising public expenditure linked to population ageing would mean tax pressure continuing to increase, stifling growth even more.

The financial viability of the pension system must be ensured

The main social security systems that protect people against the risks of old age and disability are facing serious problems. These result in part from the ageing of the population, the effects of which will start to be felt in 2005 and will culminate in 2035 or thereafter.[10] While the architecture of the Swiss pension system, which combines pay-as-you-go and fully funded features, is better designed than that of many other countries to limit the fiscal impact of the demographic shock, expenditure on first-pillar public pensions (AVS) could grow by 3.5 per cent of GDP (representing 5.3 points of VAT) by 2035-40.[11] Reforms are therefore necessary to ensure the financial viability of this system. Adjustments will also have to be made to the second pillar, the funded occupational benefit plans, to reflect growing life expectancy and the fallout from the financial market slump. Finally, steps will have to be taken to stabilise the financial situation of disability pension plans, which have worsened in recent years with the sharp growth in the number of beneficiaries. Reforms already adopted or under discussion in Parliament for addressing these problems will not be enough to ensure the long-term financial viability of the old age, survivors' and disability insurance systems, and new measures will have to be taken.

Financing of the AVS *must be assured beyond* 2015

The first pillar of the pension system, which is financed on a pay-as-you-go basis and guarantees a minimum livelihood to all elderly or widowed persons, is directly affected by ageing. The 11th revision of this system, which has been voted by Parliament in October 2003, is intended to ensure the financing of pensions to 2015, through a 1 percentage point increase in the VAT, planned for 2010, and expenditure reductions. The latter include raising the retirement age

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for women from 64 to 65 years (to make it the same as for men) as of 2009, reducing survivors' pensions, and shifting the indexing of pensions from a two-year to a three-year cycle.[12] This reform, which is not expected to come into force before the beginning of 2005, might be challenged by a referendum. But even if it is implemented, it will not guarantee the financing of basic pensions beyond 2015. While the measures currently envisaged are designed to fill a funding gap representing about 1¾ percentage points of VAT, the deficit between 2015 and 2025 is estimated at 2 points of VAT, with an additional 1.5 points between 2025 and 2040. This official forecast is based on an assumption of real wage (or productivity) growth of 1 per cent per year, which would seem realistic in light of past performance.

Technical studies have been conducted on measures that might resolve this financial problem. The results of those studies will be used to prepare the 12th revision of the AVS, which is to be released for consultation in the second half of 2004, with a view to tabling legislation in the course of 2005. These studies, which focus on reforms that will penalise growth as little as possible, have pointed to three types of supplementary measures:

- Increasing the standard retirement age: an increase of two years, to 67 years, would reduce financing needs by 1.3 points of VAT in 2025 and 1 point in 2040.
- Reforming the pension indexing system: a move to indexing that is dependent exclusively on prices and that would apply only to persons already retired would reduce spending by a ¾ point of VAT in 2025, and by nearly 1 point in 2040.[13]
- Raising revenues: an increase in the VAT is favoured, because this will not penalise labour demand, and will enlist the entire population, including retirees, in financing the system.

However, the scope of the necessary adjustments could be limited through an effective growth policy. According to Schluep (2003), a ½ percentage point addition to annual productivity growth would reduce the AVS deficit by the equivalent of a ½ point of VAT by 2025, and by nearly 1 point by 2040.

The debate on the financing of the AVS beyond 2015 is a welcome development. It is important, indeed, not to underestimate the time needed to implement pension reforms, given the difficult measures that may be required, and to which insured people will need time to adapt. Given the importance of the deficits to be covered, considerations of intergenerational equity would seem to justify distributing the effort by making changes on both the revenue and the expenditure side. Choosing measures that will not penalise growth, such as an extension of working life, is also reasonable and compatible with individuals' growing life expectancy and improved health status. A longer working life could be achieved in a flexible manner that will take account of the increased risk of dis-

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ability with age, or the strenuousness of certain occupations, given their consequences in terms of life expectancy. An increase in the employment rate among persons over 60 years of age, which is already high by international comparison, might however be difficult to achieve simply by raising the retirement age, unless two problems are overcome. First, the demand of firms for older persons must be sufficient and better adapted to the supply, which would seem in large measure to be under exploited (Widmer *et al.*, 2002). Second, early retirement must be made less attractive, and here the second pillar plays an important role: early retirement is in fact more frequent among well-off households that have accumulated significant assets in the second pillar.[14] Under these conditions, it would be desirable to reduce the progressive nature of contributions to the second pillar, which now rise as a function of age, making older workers less attractive to firms. Specific incentives to employ older workers who have no access to the second pillar could also be considered, although care must be taken not to distort the value of pensions compared to their actuarial value (Chapter IV). It would also be useful to improve the possibility of accumulating partial pension rights through part-time work, as proposed in the occupational pension reform. Such measures would encourage both labour supply and demand for older workers and would be desirable regardless of any decision on the legal retirement age.

The occupational benefit system must be reformed

Reforms are also needed to the occupational benefit scheme, which operates on a funded basis, in order to deal with the consequences of ageing and the financial market slump, despite the recent improvement. The growth in life expectancy since the mid-1980s makes it necessary to reduce the conversion rate used for calculating pensions on the basis of accumulated retirement assets (Annex II). This rate, which has remained unchanged since 1985, requires pension funds to pay out more in benefits than they can finance with their available capital (Crédit Suisse, 2003).[15] The pension institutions have also been affected, some in a sharper way than others, by the combined impact of the stock market collapse in recent years, the decline in bond rates, and the appreciation of the Swiss franc, which has reduced investment income in foreign currency (Figure 16).[16] This trend has weakened the balance sheets of the pension plans, as well as their investment income, which between 1995 and 2000 represented a more important source of financing than did employers' or employees' contributions. Although recent statistics are lacking, the yield on pension fund investments since 2000 does not seem to have achieved the minimum rate of 4 per cent imposed until 2002, and their performance, which takes account of capital gains or losses, was undoubtedly negative in 2001 and 2002. According to recent surveys, at the end of 2002, 45 per cent of the pension funds were unable to cover their future commitments fully from their own assets, as the law requires, and the average funding ratio was estimated between 92 and 98 per cent depending on the surveys, whereas it

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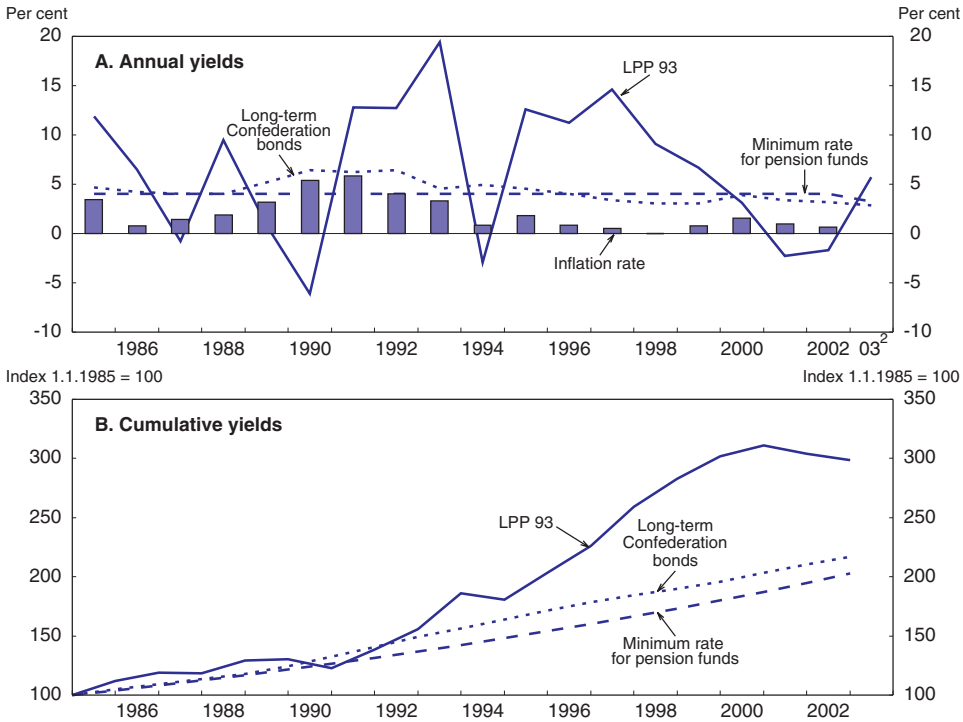
Figure 16. **LPP index and interest rates**[1]



1. The LPP index is an index calculated by the Pictet bank and which is used as a reference to measure the performance of pension funds. The rate of return on ten-year Confederation bonds and the minimum legal rate for pension funds are also shown. In panel B, the yields of all series are cumulated since 1985.
2. 2003 figures are estimates based on the annualised rates of the series at end-October 2003.

Source: SNB and banque Pictet.

exceeded 120 per cent between 1997 and 1999. This decline of some 20 to 30 percentage points in the funding ratio is similar to that recorded in the Netherlands during the same period (van Ewick and van de Ven, 2003).

In general, this funding problem has not had any consequences in terms of near-term solvency, since at the system's current state of maturity, most institutions' revenues are higher than their expenditure.[17] Some pension funds are more exposed than others, however: this is the case with institutions that cover a high proportion of retirees compared to active workers, or those that were recently created and have not had time to build up fluctuation reserves or have been unable to earn the minimum interest rate through low-risk investments. This situation applies to certain public sector funds, such as that for the federal railways (CFF), where the number of workers is slightly lower than that of retirees. The funding

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ratio for this plan stood at 80.5 per cent at the end of 2002, representing a deficit of CHF 2.7 billion (0.6 per cent of GDP). The funding deficit for the post office pension plan or for the federal employees' plan, which became independent in June 2003 and which must now be funded, constitute further financial problems for the government. In total, the demand for financing presented to the Confederation by these public plans amounts to CHF 18 billion (4.3 per cent of GDP).[18]

A number of measures have been taken to deal with this situation. The revision to the occupational benefit law (LPP), which has been before Parliament since March 2000, has been adopted in September 2003, and includes a progressive reduction in the pension conversion rate from 7.2 to 6.8 per cent over a ten-year period. In order to prevent pensions from dropping, an increase in contributions will no doubt also be adopted as of 2004, with a broadening of the contribution base and a reduction in the minimum salary for access to the second pillar, which will also allow an additional 180 000 individuals (in reality 100 000 individuals, 80 000 being already members of the extra compulsory part of the second-pillar system), primarily part-time female workers, to benefit from the system. Moreover, this reform will provide greater flexibility in choosing between retirement and part-time work, and will bring greater transparency to pension fund management accounts and their supervision. In response to the financial difficulties, the minimum guaranteed interest rate applicable to pension plans was reduced from 4 to 3.25 per cent in 2003. This rate will decline further to 2¼ per cent in 2004 and will be revised every year by the government in light of federal bond rates, yields on other investments, and the financial situation of the pension plans. Other measures are under discussion for resolving the most serious problems of under-funding in social security institutions (Federal Department of the Interior, DFI, 2003).[19] These measures, which could be implemented in mid-2004, are intended to broaden the instruments available to the pension funds for eliminating their funding gaps while giving them sufficient time to meet this obligation. The measures include the possibility of increasing employer and employee contributions,[20] remunerating insured accounts at a rate less than the minimum rate on a temporary basis, and requiring contributions from pensioners under certain circumstances for funds with a high proportion of pensioners to active workers. The authorities are also examining similar measures for the public pension funds. Details of the plan, which could include higher contributions and elimination of inflation indexing for pensions, will be decided in the second half of 2003. As to the funding of the new independent federal employees' fund "PUBLICA", the plan will also include an injection of CHF 7 billion, which the Confederation is committed to pay as employer over a period of eight years, and CHF 5 billion to cover the loss in capital assets under the previous plan. With these transfers, PUBLICA should be able to start with retroactive effects from June 2003 with a funding ratio of 100 per cent.

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The impact of the financial slump on the occupational benefits system has been severe, although it seems to be more limited than in the United Kingdom or the United States. Even if uncertain, the scope of the adjustment that will be needed to bring the funding ratio of the pension plans back to 100 per cent could be in the order of 8 to 10 per cent of GDP, according to some estimates. In this context, it was important to take measures such as making the minimum yield of the plans more flexible or reducing the conversion rate to prevent the situation from becoming worse. However, this downward revision of the conversion rate could prove to be insufficient if it does not take into account the likely increase of the future life expectancy in an appropriate manner.[21] It would also be desirable to plan corrective measures to restore the financial health of social security institutions gradually. It would be unwise to rely on a swift recovery of financial markets despite their recent improvement, if the robustness of the pension system is to be preserved.[22] Yet it is essential to avoid too abrupt an adjustment to pension fund balance sheets, which could destabilise the economy in light of its current weakness and the risk of deflation. Pension plan adjustment could significantly reduce the incomes of businesses, employees and pensioners, and this would also affect public revenues. Moreover, the increased contributions on low-paid jobs that would result from extending the coverage of the system could reduce businesses' demand for labour. The scope of measures, which should be spread out over time, is nonetheless difficult to calibrate, given the uncertainty surrounding financial market developments. Recent reactions in the area of "supplementary" or "beyond-requirements" benefits, which are not subject to the legal obligations of the compulsory LPP, have sparked fears that insurance companies active in this sector might adjust abruptly. This could also affect the savings behaviour of households and, in the short term, increase incentives to retire early, even if pension cutbacks are likely to diminish these incentives in the long run.[23] Better statistical information on the system would serve to strengthen surveillance of occupational benefit plans and would allow the authorities to monitor the situation more closely and so prevent an overly hasty adjustment in pension plans that face problems. Reports on the financial situation of both private and public pension funds should be presented this autumn.

Over the longer term, greater transparency will be required in pension plan accounts and their surveillance, which now relies on ex-post controls, in the light of questions that have been raised about the management of insurance companies in this sector. More than 20 per cent of the CHF 500 billion pension plan assets are managed by private companies, and they are thought to have favoured their shareholders to the detriment of their insurees during the stock market's heyday, while leaving the policyholders to bear the brunt of the current financial crisis. Greater accounting transparency would strengthen public confidence in the system. While it is true that the "surplus yield" over the guaranteed minimum rate may have benefited insurees in the form of lower contributions or

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higher benefits, these advantages have accrued sometimes in a disproportionate way to those who reached retirement age during the high-yield phase. This situation points to the short-sightedness of pension funds and their difficulty in smoothing out the impact of financial market fluctuations on pension systems (Butler, 2002). Smoothing the effect of big swings in results on pension paid out would help to prevent the hard-to-predict situation where the size of a person's supplementary benefits, beyond the legal minimum, depends on the timing of retirement. Such a development must be accompanied by greater flexibility when it comes to either raising or lowering the minimum yield rate. If this rate is not properly set it could create negative incentives, pushing the pension plans to take excessive risks if it is too high, or to adopt an overly conservative management stance if it is too low. Adjusting the minimum interest rate upwards more slowly than downwards would also reduce the incentives of insurers to constitute enough reserves. Finally, the wider possibilities of adjustment of pension plans, which could also affect current pensioners, lead to the more fundamental issue of the dividing line between the second and the third pillar.

The growth in the number of disability pensions must be curbed

Disability insurance (DI) has faced recurrent financial problems since the beginning of the 1990s, because of the persistent and rapid growth in the number of beneficiaries. DI expenditure stood at nearly CHF 10 billion (2.4 per cent of GDP) in 2002 and the system's indebtedness, which reached CHF 4.5 billion, has been growing recently at a rate of CHF 1 billion per year. While it is relatively low by international standards, the proportion of DI beneficiaries in the under-64 population, which rose from 3.1 to 4.8 per cent between 1990 and 2002, has been higher than in many other countries (OECD, 2003a). This increase is due only in part to population ageing, and the fact that people face a higher probability of becoming disabled between the ages of 50 and 64 (Prinz and Nydegger, 2003). The risk of disability has in fact risen for all age groups since 1990, particularly among women and people of intermediate age, which means that new beneficiaries are relatively younger, and this increases the length of time during which benefits must be paid. The proportion of mentally disabled persons has also risen sharply, and such illnesses indeed have a severely debilitating effect.[24] The causes underlying these trends are complex and are not yet clearly understood. While higher unemployment rates no doubt bear a relation to the growing number of disability beneficiaries, the use of DI as a substitute for unemployment insurance does not seem to be either automatic or institutionalised (OECD, 2000). Available studies covering the situation up to the mid-1990s show that only about 2 per cent of unemployed people received disability benefits once their unemployment benefits ran out. Moreover, the studies have found no link between the disability insurance system and early retirement, nor do they point to any abuse of the system

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(Donini and Eschmann, 1998). One explanation that is sometimes offered for the increasing number of mental disability claimants is the changing notion of illness. This argument, however, cannot fully explain the important discrepancies in the mental disability rates among cantons, which varied from 1 to 5.7 in 2001. On the other hand, there appears to be some correlation between these regional disparities and the availability of psychiatric services, which varies widely and has risen rapidly during the past decade.[25]

In the face of these difficulties, the fourth revision of the DI was adopted in March 2003, and will come into effect in January 2004. This reform is intended primarily to stabilise the system's financial situation with supplementary revenues. Disability insurance will receive a transfer of CHF 1.5 billion to reduce its debt, and a planned 0.8 per cent VAT hike will increase its revenues beginning in 2005. Second, this reform strengthens federal surveillance over the conditions under which disability pensions are granted, in order to standardise application of the law and to curb spending by the more profligate cantons. This will be done through the creation of regional medical offices under the direct supervision of the federal services, which will be responsible for applying the same medical criteria to all insurees. Third, this reform broadens the supply of benefits somewhat so as to enhance the autonomy of disabled persons, and it expands the role of DI in helping to find employment opportunities for such persons. Finally, funding will be provided for studies to delve more deeply into the causes behind the rising number of claimants, so that supplementary measures can be taken if necessary to restrain their growth on a permanent basis. In fact, the authorities themselves suspect that cost control measures and the planned revenue increase may not be enough to cover the accumulated debt and to finance spending until 2020 (DFI, 2003). Given the current probability of becoming disabled, expenditure could rise by 2.3 per cent a year until 2020, which would exceed the economy's growth potential and would generate a deficit representing 0.1-0.2 VAT points. Under these circumstances, it would seem necessary to explore other reform avenues. These might include introducing a system of providing benefits by stages, initially for a limited period, as suggested by the Federal Department of the Interior (DFI). It would be useful to examine the conditions of these pensions in order to prevent patients from becoming permanently dependent on them: the longer the time on pension, the more difficult it is to reintegrate the individual into the labour market. It would also be useful to involve the social partners, particularly employers, more thoroughly in the process of reintegrating disabled persons, perhaps through an active labour market integration programme more in keeping with the philosophy of programmes for the unemployed (OECD, 2003a).

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Further efforts are needed to make the public sector more efficient

While all levels of government will no doubt continue to face a tight financial constraint over the medium term, efforts have been launched to make the public sector more efficient. Among the measures to improve the management of public spending, which were reviewed in the previous OECD S*urvey* (OECD, 2002), the reform of the federal system itself is the most ambitious, and could produce significant benefits. More recently, the authorities have also introduced a programme to alleviate the administrative burden on business. Finally, further reforms are still required in the health sector, where government intervention plays an important role.

T*he reform of the federal system is making headway*

The reform of the federal system of public finances and its financial equalisation regime should lead to a more effective use of public sector funds. This ambitious reform, which is now being examined by Parliament, will reorganise 40 per cent of the responsibilities that are now shared by the federal and regional administrations. It will untangle a whole range of activities that will now become the exclusive preserve either of the Confederation or of the cantons. The portion of resources that is freely available to the cantons for performing their new tasks should rise from 25 to 40 per cent of total transfers from the Confederation, which will increase flexibility and allow them to improve the quality and the targeting of their services. The tasks to be assumed jointly by the Confederation and the cantons will be better organised. Earmarked subsidies will be replaced by global grants, the amount of which will be linked to the results to be achieved, and no longer to the costs of the activity, which will reduce wastage. Moreover, co-operation among the cantons in certain areas will be institutionalised, and cantons that provide services to their neighbours will be guaranteed fair compensation.

Another objective of the reform is to reduce fiscal disparities among the cantons, by revising the financial equalisation formula. The total volume of equalisation funds, which stood at CHF 2 billion at the end of the 1990s, will rise to CHF 3.4 billion: this will allow the poorer cantons to shoulder their new duties and will reduce existing regional funding disparities by 25 per cent. These transfers will be financed in part by the richer cantons and the Confederation under the equalisation formula, in part by federal compensation payments for costs related to topographical or socio-demographic factors, and by a Temporary Cohesion Fund (Table 12). The creation of this CHF 0.4 billion fund to ease the transition to the new system implies that the overall project will not be limited to a financially neutral transfer of activities and costs, but will in fact increase federal expenditure slightly. Parliamentary debate has resulted in only minor amendments to the government's initial proposal, and it has retained its overall coherence.[26] Certain aspects of this reform could still be improved, however, in particular to take greater account of the effects of a decline in certain social services, such as

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Table 12. **The new financial equalisation system: charges, payments and equalisation effects**

Charges (+) and payments (–) in 1 000 CHF

	Resource equalisation		Cost compensation					
	Charges to cantons with above average potential of resources (horizontally)	Payments to cantons with below average potential of resources (horizontally and vertically)	Geographic and topographic cost compensation	Socio-demographic cost compensation	Total	Resources indicator before transfers (Switzerland = 100)	Resources indicator after transfers (Switzerland = 100)	Cohension fund: charges
Zurich	567 587	–	–	–59 500	508 087	146.7	140.4	24 090
Berne	–	–759 414	–14 691	–9 900	–784 005	71.1	83.4	18 917
Lucerne	–	–211 031	–5 712	–5 628	–222 370	74.7	84.3	6 835
Uri	–	–30 849	–8 952	–	–39 802	69.5	86.7	688
Schwyz	24 504	–	–3 419	–	21 085	119.2	116.7	2 523
Obwald	–	–22 456	–3 814	–	–26 270	72.9	85.2	636
Nidwald	13 577	–	–810	–	12 767	136.9	131.7	728
Glaris	–	–107	–3 953	–	–4 060	98.3	99.9	763
Zoug	134 752	–	–67	–	134 685	238.0	217.3	1 933
Fribourg	–	–185 141	–11 706	–781	–197 628	71.3	83.9	4 683
Soleure	–	–150 375	–	–	–150 375	74.5	83.8	4 815
Bâle-Ville	85 262	–	–	–35 238	50 024	144.1	140.2	3 831
Bâle-Campagne	26 787	–	–	–	26 787	110.4	108.9	5 088
Schaffhouse	–	–7 997	–	–1 879	–9 876	89.3	91.3	1 456
Appenzell Rhodes-Extérieures	–	–5 308	–14 639	–	–19 946	89.8	95.4	1 063
Appenzell Rhodes-Intérieures	–	–5 756	–5 982	–	–11 738	79.7	91.7	289
St-Gall	–	–131 507	–1 911	–5 847	–139 264	82.5	87.1	8 879
Grisons	–	–2 664	–103 315	–	–105 979	96.2	104.6	3 736
Argovie	–	–45 717	–	–	–45 717	90.6	91.8	10 690
Thurgovie	–	–172 437	–3 352	–	–175 789	71.8	83.4	4 513
Tessin	–	–118 074	–10 720	–25 125	–153 919	79.9	87.4	6 051
Vaud	–	–24 533	–	–53 735	–78 268	93.6	95.5	12 426
Valais	–	–431 527	–61 075	–	–492 602	57.8	84.9	5 406
Neuchâtel	–	–49 152	–17 868	–8 516	–75 535	82.4	89.2	3 308
Genève	147 530	–	–	–68 851	78 679	136.3	133.4	8 048
Jura	–	–77 374	–3 014	–	–80 388	65.2	83.0	1 340
Total cantons	**1 000 000**	**–2 431 419**	**–275 000**	**–275 000**	**–1 981 419**			142 738
Total Confederation		**1 431 419**	**275 000**	**275 000**	**1 981 419**			285 475

Source: Administration fédérale des finances.

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programmes to overcome social exclusion or childcare services, which are financed primarily by the municipalities. The project still has a long institutional process ahead, and it is not expected to be implemented before 2007.[27]

Efforts to reduce the administrative burdens for businesses are welcome

To enhance the efficiency of the federal administration, the authorities recently tabled a reform proposal that would reduce the resources that businesses must devote to bureaucratic tasks (Federal Council, 2003). The significance of these administrative burdens, which are proportionately heavier for SMEs than for large businesses, should not be underestimated. According to the authorities, their cost could amount to some 2 per cent of GDP.[28] These costs, which would appear relatively high in comparison with the OECD average at the end of the 1990s (Nicoletti *et al.*, 2000), fall into three principal categories: withholdings for basic old age insurance, tax declarations (in particular the VAT), and the enforcement of commercial law provisions (Balastèr, 2003). The simplifications proposed in the draft reform are intended, first of all, to strengthen coordination among the various administrative entities and in particular those responsible for taxes and fees. It is proposed that VAT returns be calculated and paid via an annual rather than a quarterly declaration, so that this can be processed in combination with the social insurance forms. Creation of a consultation system accessible by Internet is also planned, for purposes of defining the salary level at which AVS contributions kick in, while procedures for reimbursement of VAT on services related to cross-border trade will be simplified. In general, the use of on-line information systems will be expanded to complement the introduction of electronic administrative forms and the development of "one-stop windows". As part of the effort to alleviate the administrative burden, government intervention should be examined in order to permit *ex post* controls instead of compulsory prior authorisation. Finally, federal administrative procedures are to be better integrated with those of the cantons.

These measures certainly go in the right direction, and they could in some cases be accelerated. The introduction of annual rather than quarterly VAT filings, for example, is planned only for 2006. A more ambitious reform of the system for delivering administrative authorisations could also be considered, with adoption of the principle used in several OECD countries whereby "silence means consent". This approach would help to shift the burden from businesses to the government, which would thereby find an incentive to raise efficiency (OECD, 2003b). Such a development would be compatible with the introduction of public services that are increasingly geared to results. The fear that such a system might lead to uneven or discriminatory application of the law should not be exaggerated,[29] provided that adequate procedures for recourse are in place. Moreover, any such risk must be viewed against the benefits to be gained from swifter administrative

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procedures. On the other hand, these measures, which are intended to reduce the costs of applying regulations, should not be allowed to divert attention from the need for a true and in-depth regulatory reform (Chapter III).

Control over the provision of health care services remains inadequate

Since the 1996 reform, the Swiss health system has guaranteed access to quality care for the whole population, while solidarity between the insured has been strengthened (OFAS, 2001). It is, however, the second most expensive in the OECD, behind the US system. Total medical spending reached 10¾ per cent of GDP in 2000. These expenditures, most of which are covered by compulsory social insurance, are less well controlled than in other countries (Table 13). The rapid rise in spending in recent years has resulted in a steep rise in households' compulsory sickness insurance premiums – by 6.3 per cent per year in real terms between 1999 and 2003. Financing the system by means of per capita premiums paid to private insurance funds does reduce the impact of the rise in expenditure on public accounts and prevents an increase in companies' social insurance contributions,[30] but does push up the cost of living. The high price level of health care by comparison with other countries also indicates regulatory problems. The main features of the health care system are highlighted in Annex III.

This section reviews the various measures already adopted or under discussion, which aim at a better control over the rise in health spending. The rise, which cannot be attributed to the impact of population ageing, is due in large part to inadequate control of health care supply which creates its own demand and a lack of price control.[31] Chapter III assesses in greater detail both the need to boost competition in this sector so as to find a lasting solution to the problem, and also possible ways of doing so.

The authorities have taken measures to achieve better control over health care supply and to sharpen incentives to contain expenditure. *First*, pharmacists' remuneration has since July 2001 been partially disconnected from drug prices and is based on fixed rates negotiated at national level between insurers' and pharmacists' associations. The reform is thought to have saved some CHF 100 million by removing the incentive to sell expensive drugs, and should encourage the use of generics which are still uncommon compared with other countries.[32] *Second*, the number of ambulatory care providers allowed to work for the mandatory health insurance system was frozen for a 3-year period in July 2002. This moratorium reflects the authorities' fear of a further increase in outlays when the agreement on the free flow of people with the European Union came into force on 1 June 2002. Under this agreement, the 3 500 European doctors employed in hospitals (20 per cent of all practitioners) are allowed to open practices in Switzerland.[33] There is indeed a close correlation between ambulatory costs and medical density, and the opening of a practice on average generates additional

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Table 13. **Expenditure, resources and utilisation in health care**

	Expenditure, in % of GDP			Health employment			Acute-care hospitals	
	Level	Change over period		Practising physicians per 1 000 population	Total hospital employment per 1 000 population (persons)	Total hospital employment per 1 000 population (full-time equivalent)	Beds per 1 000 population	Average length of stay
	2000	1990-95	1995-00	2000	2000	2000	2000	2000
International comparison								
Switzerland	**10.7**	**1.5**	**0.7**	**3.5**	**20.6**	**16.0**	**4.1**	**9.3**
Australia	8.9	0.4	0.7	2.4	21.4		3.8	6.1
Austria	8.0	1.5	–0.6	3.1	15.6		6.2	6.3
Canada	9.2	0.2	0.0	2.1		12.2	3.2	7.2
Denmark	8.3	–0.3	0.1	3.4		16.1	3.3	3.8
France	9.3	0.9	–0.2	3.3	18.7		6.7	8.5
Germany	10.6	2.1	0.0	3.3	14.9	11.3	6.4	9.6
Italy	8.2	–0.6	0.8	4.1		11.2	4.3	7.0
Japan	7.6	0.9	0.8	1.9		12.9		
Netherlands	8.6	0.4	0.2	3.2	17.1	11.9	3.5	9.0
Norway	7.7	0.2	–0.2	2.9		17.3	3.1	6.0
United Kingdom	7.3	1.0	0.3	2.0	22.3		3.9	6.9
United States	13.1	1.4	–0.2	2.7		15.8	2.9	5.8
EU average[1]	8.1	0.8	0.0	3.2			4.3	7.0
OECD average[1]	7.9	0.7	0.1	2.9			4.2	7.0
Swiss health system								
Expenditure by category								
In-patient care	5.0	0.7	0.2					
Out-patient care	3.2	0.5	0.3					
Non-durable medical goods	1.3	0.1	0.2					
Other	1.1	0.1	0.0					
Expenditure by financing source								
Compulsory	5.9	. .	0.6					
State	1.6	. .	0.0					
Social insurance	4.3	. .	0.6					
LAMal	3.5	. .	0.5					
Private	4.7	. .	0.1					
Memorandum items:								
Share by type								
Compulsory	55.6	. .	1.8					
Private	44.4	. .	–1.8					

1. Unweigthed average.

Source: OFS, "Coûts de la santé en Suisse : évolution de 1960 à 2000", Actualités OFS, série 14, June 2003 and OECD, OECD Health Data 2003.

health insurance costs of some CHF 500 000 per year.[34] The moratorium has been accompanied by an increased access to hospital careers, where a larger number of staff would reduce the excessive hours worked by doctors. *Third*, patients' direct contributions to their health care consumption under the mandatory health

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insurance system will be raised in 2004 in an effort to check the growth of consumption.[35] It is also planned to reduce the maximum cuts in premiums of the insured who agree to cover a larger proportion of their expenditure directly. The reductions in premiums awarded at present actually exceed the amount of the extra risk assumed. The effect of these measures should reduce the growth of compulsory insurance premiums by 3.2 per cent in 2004, with an offsetting rise in the patients' direct contributions to their medical spending.

The review of the reform of the health system that has been in progress for more than two years has also been continuing in Parliament. Given the state of progress of the discussions, however, the reform is not expected to come into effect before early 2005. However, some reform proposals are likely to be implemented. F*irst*, hospital funding should in future be based on a payment-per-case system rather than the refunding of costs linked to the number of days of hospitalisation. The cantons will have to meet 50 per cent of these outlays, including for patients covered by complementary insurance. S*econd*, a consensus has been reached on raising mandatory health insurance premium subsidies to low-income households. The premiums will be capped at between 2 and 12 per cent of income, depending on the level thereof and whether or not there is a dependent child. An agreement also seems close on the *third* and more controversial point, on easing the obligation for health insurers to contract with all providers, which implies restricting the freedom of choice of health care provider. Abolishing the obligation would allow insurers to choose the providers authorised to practice for the health insurance scheme, and this is meeting with opposition from doctors who are disputing the legitimacy of a selection system based solely on economic criteria. The model under consideration would let the cantons decide on the minimum number of practitioners with whom insurers would have to contract in each region for a period of at least four years. Many safeguards and exceptions would however be introduced to limit the breaking of contracts, which will need to be duly motivated and subject to the right of appeal up to the Federal Tribunal. Under such an arrangement, incentives would be introduced to encourage newly trained doctors to join managed care systems with several specialists, which would have a budgetary allocation and would act as an interface between health funds and patients. Such systems are still underdeveloped (8.1 per cent of the insured bought HMO-insurance policies with restricted choice of provider in 2001) and the reform envisages obliging insurers to provide these forms of insurance throughout the country.[36]

Several aspects of these measures ought to improve the degree of control over total health spending and the incentives for health care providers to be efficient. For example, the switch in hospitals to a payment-per-case system, *i.e.* based on outputs rather than inputs, should prompt them to minimise their costs per case. According to the objective of the legislator, which includes an identical financing system of health services provided by public or private institutions, this should promote competition between public and private hospitals, while it could

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also cut the excessive length of hospital stays (Table 12). To be effective, however, such incentives require that the price-per-case structure is established in a manner that prevents cost overruns and does not generate rents.[37] The planned adoption by hospitals of cost accounting is a welcome change in this respect. However, the joint involvement of the cantons and insurers in the payment of hospital services is liable to undermine the quest for proper pricing. This is because the continued use of cantonal subsidies to cover hospitals' deficits prompts the regions, which control hospital prices, to use them to limit their contributions.[38] Moreover, this direct subsidising of hospitals, which reduces the prices borne by insurers, encourages excessive use of hospital care that is costly compared to ambulatory care. Being aware of these difficulties, the authorities are looking at the consequences of changing the system of subsidies, including awarding the subsidies directly to insurers (monistic system) so as to eliminate these distortions.

The scheduled increase in patients' participation in their medical expenditure should promote more responsible consumer behaviour, while equity problems should not arise since the level will remain moderate.[39] The effectiveness of this measure would doubtless be enhanced if it targeted the most elastic health care demand, *i.e.* ambulatory and pharmaceutical spending, where the health element of the services provided is often less important than in the case of hospital care. Consideration could then be given to introducing, as in other countries, a drug refund system based on reference prices encouraging the purchase of products that are the least expensive in their category. These incentives could be combined with a financial incentive for doctors to prescribe generics, which would encourage their more widespread use. Lower co-payment rates could also be accorded to individuals buying policies with restricted choice of provider to encourage participation in this type of arrangements, which could lead to cost savings. As a general rule, it is also important not to increase the range of refunded services indiscriminately, as it is broad in some sectors by comparison with other countries.[40]

The impact on expenditure of the moratorium on the setting up of medical practices could be slight because a large number of applications to open up practices were filed just before the measure was introduced.[41] There is the danger that applications could surge when the moratorium is lifted. When it ends, this check on the number of doctors could, however, be replaced by an easing of the obligation to contract. The more the new system currently discussed in the Parliament will protect practitioners already established, the more likely it will discriminate against newcomers. This sort of system would be tantamount to giving doctors already in practice a rent, which would not improve efficiency. It would be preferable that the lifting of the obligation to contract be accompanied by increased price flexibility, and the introduction of real competition between suppliers, while the moratorium on the opening of new practices should not be perpetuated and turned into a mechanism for exercising direct control over supply (see Chapter III).

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Notes

1. Federal subsidies finance 17 per cent of basic old-age insurance expenditure (AVS) and 37.5 per cent of invalidity pensions (AI). In order to cover the additional AVS spending caused by demographic factors, the rate of VAT was raised from 6.5 to 7.5 per cent in 1999. The Confederation managed to secure a 17 per cent share of the additional revenue to finance its legal contribution.
2. According to the Constitution, the Confederation's prerogative to levy direct federal tax and VAT, which account for 60 per cent of its income, will cease at end-2006. The Constitution also stipulates the maximum rates of VAT (7.6 per cent) and of direct federal income tax and corporate tax (11.5 and 9.8 per cent, respectively). In the case of tax on corporate profits, the reform will lower the ceiling to the level of the legal rate which has already been reduced to 8.5 per cent.
3. In order to smooth the instability of the withholding tax, the tax on dividends collected in a given year could depend for instance on the dividend earned during the past two or three years rather than the current year.
4. According to Bodmer (2002), VAT is distorted, because both part of construction investment and of the financial sector are excluded. It is not a neutral tax, weighing on intermediate inputs and hence on costs in both of these sectors.
5. See Girouard (forthcoming) for examples among OECD countries. Where Switzerland is concerned, Bodmer and Geier (2003) mention the case of the recent change in accounting standards which resulted in the disappearance of hidden reserves and a temporary increase in companies' reported profits and hence their taxation.
6. In principle, on the basis of a budget balance which is structurally in equilibrium, the debt containment rule requires that growth of expenditure remain equal to that of structural revenue. However, revenue growth is the same as that of trend GDP if it is assumed, as is often the case, that the average long-term GDP elasticity of tax revenues is one.
7. In that case, the rule ought to be for the structural balance plus the average of the non-cyclical irregular component to be in equilibrium.
8. The adjustments to the HP filter involve reducing the weighting of the present values in the filter, using the logarithm of GDP instead of its absolute values in the estimation and not taking account of the financial plan projections. Like that of the output gap, the estimation of revenue also plays an important role in the debt containment rule, but the tendency to overestimate revenue at times of recession and to underestimate it when activity is buoyant encourages a counter-cyclical fiscal policy.
9. Several years of stagnation leading to a large output gap could in theory justify a recourse to this exceptional procedure in the same way as a strong recession for a given year.

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10. The ratio of persons older than 64 years to those aged between 20 and 64 should increase from 26 per cent to 44 per cent between 2005 and 2040, *i.e.* by a proportion comparable to other OECD countries.
11. On the basis of similar assumptions concerning productivity gains, public pension spending should grow by 1¾ per cent of GDP less in Switzerland than the OECD average by 2040.
12. The AVS reform also includes a reduction in the reserves of the system from 100 to 70 per cent of annual basic pension spending. The resources provided by this measure amount to ½ VAT point.
13. Currently, pensions are subject to a hybrid index calculated as an average of wage and price trends.
14. Most of the second-pillar pension plans provide "à la carte" benefits between 62 and 65 years for men and between 57 and 62 years for women, and two-thirds of them pay transitional pensions to their members until they reach the legal AVS retirement age (IDA ForAlt, 2003).
15. In order to pay the minimum 4 per cent on insurees' accounts, as required until 2002, these plans had to earn a return of 4.8 per cent (OECD, 2000).
16. The proportion of equities in the pension plan portfolios rose from 7 per cent to around 30 per cent between 1987 and 2000, before declining to 12 per cent in mid-2002. At the end of 2000, this proportion was lower than in the United States (60 per cent), the United Kingdom (70 per cent) or the Netherlands (40 per cent).
17. Insurance institutions' total revenues amounted to more than 150 per cent of their outlays in 2000. The number of pensioners was less than one-quarter of those insured.
18. These shortfalls result in part from non-payment by the Confederation of its employer contribution, which is no longer possible for the independent plans. From 1999 to 2002, the Confederation provided CHF 20.3 billion to the plans of federal enterprises that had become independent, and this increased the public debt.
19. A pension plan is considered to be in serious trouble if its funding ratio is less than 90 per cent. The cantons have applied a standard definition of this funding ratio since July 2003. At the end of summer 2003, 6 per cent of the 9 000 pension funds would be in such a difficult situation (*Le Temps*, 2003).
20. Higher contributions are already possible, but there would be a guarantee that these supplementary resources would remain in the pension fund if the employee changes companies.
21. The main insurance companies have revised their conversion rates to 5.8 per cent as of 2004 for the pension plans belonging to the part of "beyond-requirement" benefits, which are not submitted to the same legal obligation as the compulsory LPP. This more important reduction than the one decided for the compulsory part has been validated by the authorities. It is based on a projected increased of life expectancy and on a technical discounting rate of ½ to 1 percentage point lower than the one used in the context of the revision of the LPP.
22. *De facto*, the stock market increase between the spring and the autumn 2003 does not appear to have improved the situation of the pension funds managed by insurance companies, which have substantially reduced the weight of equities in their portfolios, whereas it has benefited the autonomous pension funds.
23. The five main insurers in the occupational pension market, covering almost half of the employees, have announced their intention to raise their premia and reduce the rents

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of the second pillar as from 1 January 2004. The Winterthur insurance company, for example, is going to reduce the interest it pays as of 2004, as well as the conversion rate for the optional benefit portion, so that these pensions will decline by 20 to 25 per cent and total rents by 7 to 10 per cent. However, this pension reduction will be avoided by an insured who prefers to get back the retirement capital rather than receive a rent at the end of the contribution period. These changes could lead to a rise in early retirement in the very short term to avoid this pension reduction.

24. In 2001, more than 45 per cent of disability claimants suffered from mental illness, compared to an average of 35 per cent in OECD countries at the end of the 1990s. These illnesses are the most likely to lead to 100 per cent disability, and measures to readapt victims to vocational life are less effective than those for other non-congenital diseases.

25. The number of urban psychiatrists rose more strongly than that of other physicians from 1991 to 2000 (*Le Temps*, 2002). The correlation between the density of psychiatrists in each canton and the rate of mental disability is 0.75. This may, however, merely reflect the high urban concentration of psychiatric therapists and patients. Studies have been launched by the authorities to explain the regional disparities of disability rates.

26. The Upper Chamber has proposed eliminating the Cohesion Fund in accordance with a strict schedule, and placing a ceiling on contributions by the richer cantons at 80 per cent (instead of 100 per cent) of transfers paid by the Confederation under the equalisation formula. These proposals will be negotiated in fall 2003.

27. Final parliamentary approval of the constitutional amendments is still needed, which will be followed by a referendum. Then further changes to specific federal laws will be necessary, which might also be submitted to another referendum.

28. According to a 1998 study (Muller, 1998), an SME with 10 to 20 employees devoted 55 hours a month to administrative tasks, whereas the figure in 1985 was only 30 hours. The implied financial cost, equivalent to CHF 4 000-5 000 per employee for an SME, falls to CHF 1 000 for businesses with more than 50 employees.

29. According to Balastèr (2003), applying the principle that "silence means consent" could induce a government agency that, for opportunistic reasons, preferred not to apply the law to a nonconforming project, to let it pass by simply "overlooking" the deadline.

30. There is, however, pressure on public spending because hospitals are funded in part by the cantons which also subsidise the premiums of low-income households, in conjunction with the Confederation.

31. According to a 2002 KOF study, the increase in insurance premiums linked to ageing is likely to be in the region of 0.4 per cent per year until 2035. The impact of technical progress, which contributes to the rise in expenditure, is not in principle any more marked than in other countries.

32. Generics accounted for only 3.9 per cent of the drugs market in 2002, compared with 20-25 per cent in Germany and the United States. When a generic product is substituted for the original drug, the pharmacist receives 40 per cent of the saving made.

33. The cantons can authorise departures from this moratorium, but any exceptions have to be justified.

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34. According to Domenighetti and Crivelli (2001), the big difference in medical density between regions has a clear influence on health care costs per insured, but has no impact on the degree of satisfaction of the population or on health outcomes.
35. As of 2004, the insured will have to pay the first CHF 300 of their annual expenditure on health (the deductible used to be CHF 230 until now). In addition, the 10 per cent mandatory co-insurance payment with respect to expenses in excess of that amount will be payable up to CHF 700 per year (instead of 600).
36. The obligation to contract would also be maintained for the practitioners working in a managed care system and willing to be remunerated with a system of standard payments depending on the number and type of patients.
37. Prices that are too low or are set at too aggregated a level can, for example, encourage hospitals to select risks in such a way as to take only the least expensive patients. For example, France's experience in this respect saw private clinics cream off the surgical care market and leave public hospitals with the task of handling the most serious and costly cases (OECD, 2003c).
38. In their capacity as planners, the cantons also decide which hospitals provide services covered by sickness insurance. Vis-à-vis their constituents, they often refuse to close hospitals even when there is excess capacity.
39. To limit equity problems when raising cost-sharing, care should also be taken to ensure that poor and very sick populations are not disproportionately affected by cost-sharing.
40. Basic insurance covers, for example, Chinese forms of health care and occupational therapy and refunds balneotherapy prescribed by a doctor, including a contribution towards non-medical daily expenses. With cover being so extensive, only 20 per cent of the insured have complementary insurance.
41. The number of applications to open practices filed in 2002 was more than 50 per cent up on the average for the last few years.

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III. Product market competition and economic performance

Introduction

As highlighted in Chapter I, the relatively poor performance of the Swiss economy over the past 20 years has mainly structural roots, with cyclical factors and negative shocks playing a lesser role. Human resources are well used and highly qualified, thanks to a flexible labour market and a good education system, which explains to a large extent the high living standard. However, several markets have not functioned well and the competition framework has not provided conditions conducive to greater dynamism. This has resulted in sluggish productivity growth which largely explains the lacklustre performance of the Swiss economy. While there is a long-standing tradition of maintaining a business-friendly climate, this often translates into a mild attitude towards anti-competitive behaviour. Since the early 1990s, reforms have aimed to improve the competition framework, although the range of problems addressed is sometimes narrow, the pace of reform is slow and initiatives are often taken in reaction to developments in neighbouring countries. Moreover, reform attempts have sometimes been blocked by popular vote, perhaps because the large potential benefits of liberalisation and their compatibility with legitimate security concerns are not well understood and need to be better explained. This chapter will highlight the potential gains from reforms, pointing to key areas where efforts would be most beneficial.

The main links between stronger competition and macroeconomic performance are reviewed in the first section of this chapter, while Section 2 lays out the competition policy framework and the recent revision of the cartel law. In Section 3, competitive conditions and recent reform initiatives are analysed for a wide range of sectors. Competition in several public markets, including the key areas of public procurement and health services, are discussed in Section 4, as well as the need for deeper reform of the agricultural sector. The last section provides a summary of the recommendations and a quantification of the likely impact of reforms on GDP growth.

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Table 14. **Output, employment and productivity**

	Switzerland		Austria	France	Germany	Italy	Japan	United States
A. Growth decomposition, 1990-2001[1]								
Average GDP growth		1.1	2.4	2.0	1.5	1.6	1.7	2.8
Productivity		0.4	1.9	1.2	1.4	1.4	1.3	1.5
Employment		0.7	0.5	0.7	0.1	0.2	0.4	1.3
Unemployment[2]		–0.2	–0.1	0.1	–0.2	0.1	–0.2	0.0
Labour force		0.8	0.6	0.7	0.3	0.2	0.6	1.3
B. Labour productivity growth, 1990-2001[3]	(90-96)	(97-01)						
Agriculture	–1.2	3.8	5.3	3.7	7.2	5.3	–0.1	2.2
Mining and manufacturing	4.4	1.7	3.8	3.5	2.4	2.0	2.4	3.4
Electricity, gas and water	7.3	1.6	3.0	2.6	5.4	3.7	2.2	1.1
Construction	–1.4	0.0	1.9	–0.1	0.0	–0.1	–3.1	0.2
Financial services	1.8	–0.1	3.3	–1.1	3.2	2.0	4.2	3.7
Non-financial services	–0.2	1.1	0.4	0.4	1.0	0.6	0.8	1.4
Memorandum items:								
GDP per capita[4]		85.1	80.3	76.6	74.9	75.2	75.6	100.0
GDP per hour worked[4]		87.8	98.1	105.4	99.1	108.0	72.3	100.0

1. 1992-2001 for Germany.
2. A positive sign indicates that unemployment has declined and contributed to boost output growth.
3. 1992-2001 for Germany and as indicated in the columns for Switzerland. For Switzerland, the sectoral productivity growth estimated in panel B is not comparable to the overall productivity growth of panel A because the production series by sector is affected by a break and the employment series used are different. The employment series used for the overall productivity is based on the number of jobs (SPAO: Employed labour force statistics) while the sectoral employment series are expressed in full-time equivalent terms and are derived from the employment data of the Entreprises' Survey.
4. 2001 levels, PPP-based. United States = 100.

Source: OFS and OECD.

Macroeconomic performance and competition in goods and services markets

Over the last two decades, Swiss output growth was among the weakest in the OECD due to slower productivity growth (Table 14). This largely reflects problems with the functioning of markets and a lack of competition, but the increase in the fiscal burden and in the cost of services offered by public provision or financed by compulsory payments is also to be mentioned. Other potential causes can to a large extent be ruled out: workers have a high level of training, infrastructure is well developed and information and communication technologies are used widely. Moreover, despite the high living standard the level of productivity, calculated in 2001 PPP terms on an hourly basis, is lower in Switzerland than in the United States or comparable European countries, with the room for catch-up to the best performing countries still to be exploited. Another indication of a lack of

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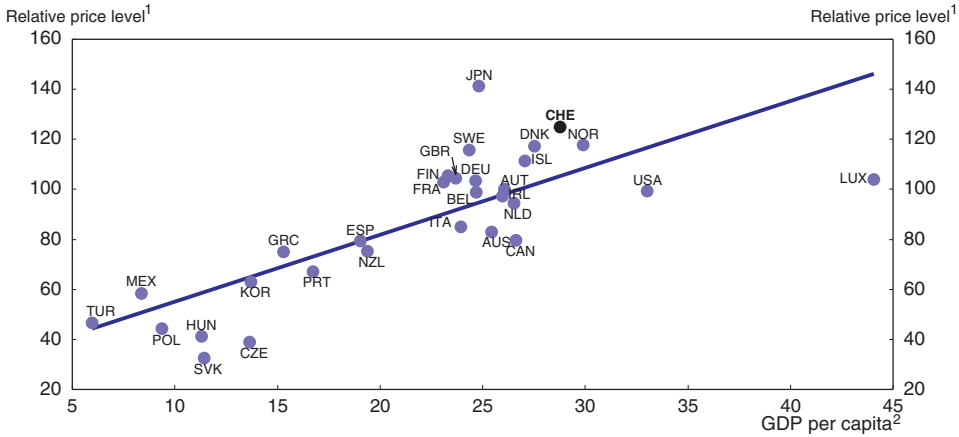
Figure 17. **Relative price levels and GDP per capita**
1999



1. Purchasing power parities divided by the exchange rate, OECD = 100.
2. In thousand USD, converted with the PPPs.
Source: OECD.

competition can be gleaned from international price comparisons, though these indicators need to be interpreted with caution.[1] The Swiss overall price level is among the highest in the world, and the difference in the price level compared with the EU average has barely fallen since the early 1990s and stood at 40 per cent at the start of this decade. One of the reasons might be the high real exchange rate.[2] Part of the differential also appears to stem from a higher average profit ratio than in the majority of OECD countries (Salgado, 2002), whereas less than half of this price differential can be explained by the higher standard-of-living in Switzerland than in the other countries (Figure 17).[3]

Problems relating to competition and the functioning of product markets do not affect every sector to the same degree. Although identifying the most affected industries is complicated by the lack of detailed statistical information,[4] the most acute problems seem to be concentrated in the sectors least exposed to foreign competition, where government intervention plays an important role. For example, the largest price differentials concern rents, health services and food (Figure 18). Also, as the sectoral differences in productivity growth show, efficiency gains were especially small in non-financial services and construction (Table 14). This latter sector has however experienced a fall in prices in the 1990s, which was at least partly explained by the adjustment following the boom in housing market at the end of the 1980s.

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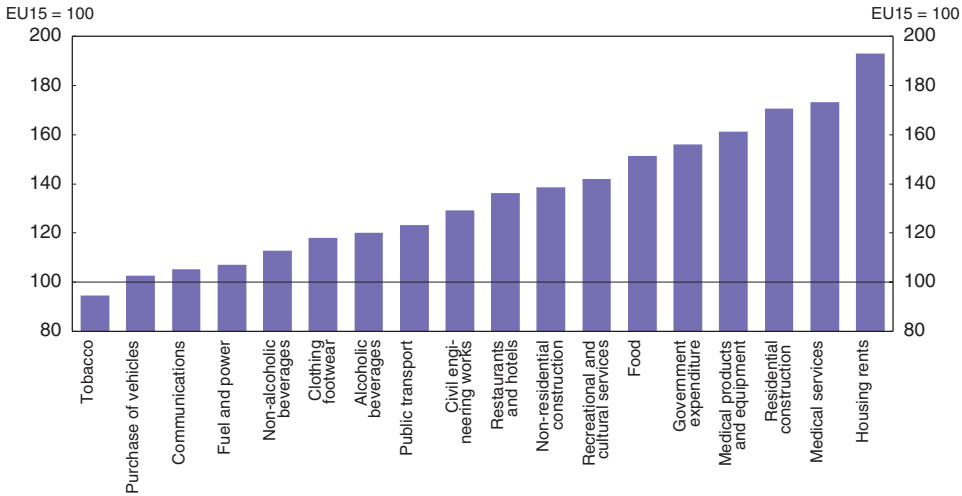
Figure 18. **Price levels in Switzerland relative to the EU**
2001



Source: Eurostat.

The strength of competitive pressures depends to a large extent on how exposed industries are to international competition. Situated at the crossroads of Western Europe, Switzerland has traditionally had a liberal foreign trade system. Tariff barriers are below the OECD average, while non-tariff barriers have remained limited as a result of efforts to lower technical hindrances to trade.[5] External competitive pressure is further increased by the underlying real appreciation of the Swiss franc which, during the 1990s, served as a strong incentive for the industries most exposed to international trade to increase their efficiency (Gagales, 2002). However, although the average penetration rate of imports in manufacturing is similar to that of comparable economies, the degree of openness for goods and services as a whole is not particularly high when the size of the economy, the standard of living and transport costs are taken into consideration (Figure 19). Moreover, if the high real exchange rate were thought to account for the high price level, the low level of import penetration would appear even more surprising.[6] Since the 1980s, openness to international trade has increased less rapidly than in the small EU countries, which might be partly the consequence of the negative 1992 referendum on membership in the European Economic Area.[7] On the other hand, the degree of sectoral openness to foreign trade is no more heterogeneous than in other countries, although some industries are notable for having a particularly low rate of import penetration (Table 15). These include certain capital goods industries in which Switzerland is specialised,

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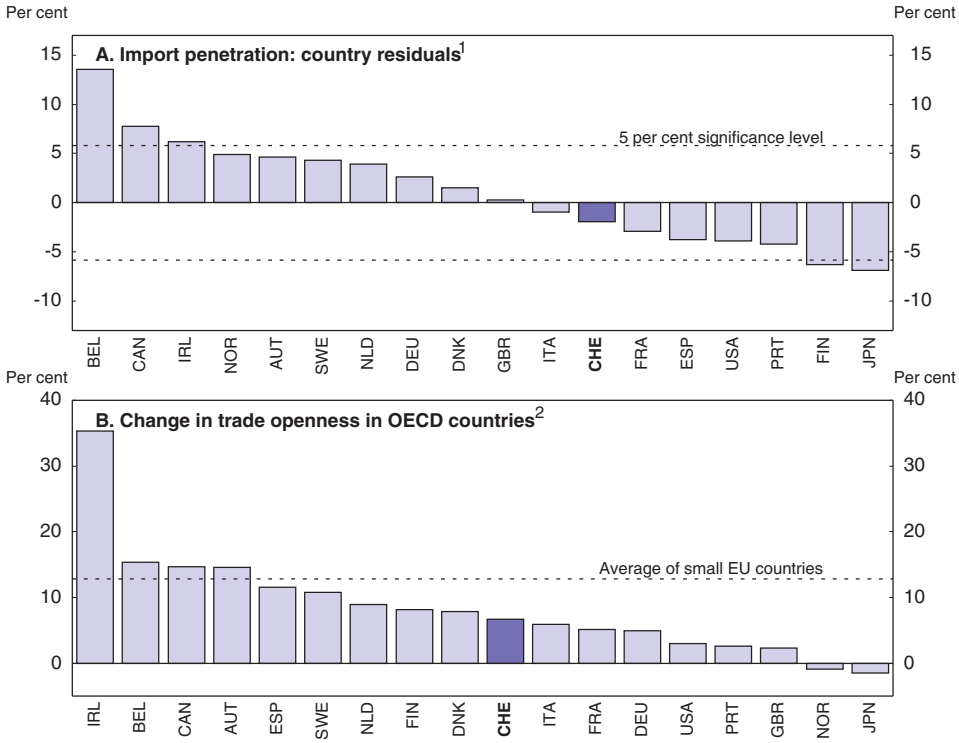
Figure 19. **Indicators of trade openness**



1. Residuals after control for effects of country size, GDP per capita and transportation costs.
2. The change in trade openness for each country is defined as the difference between the average share of exports and imports to GDP in the early 2000s and the 1980s.

Source: OECD.

but also the textiles, food products, the beverage and tobacco industries, which suggests that there could be barriers to competition.

Imports are not always capable of enhancing competition in every sector: agricultural support programmes result in strong tariff[8] and non-tariff protection, so that domestic food products are insulated from international price competition; conditions of access to certain services such as life insurance are more exacting than in the EU countries; and patented products (such as pharmaceuticals) are protected against parallel imports.[9] Moreover, numerous products are still imported via exclusive networks that have long been protected by vertical agreements, which restrict direct imports and perpetuate price differentials with other countries.[10] All in all, the degree of integration in the European market is limited by the existence of a significant border effect (de Serres *et al.*, 2001),

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Table 15. **Import penetration by manufacturing industry**[1]
2000[2]

	Switzerland	Small European countries[3]	France	Germany	Italy	Japan	United States
Total manufacturing	65.2	64.5	37.0	39.3	31.4	9.2	22.6
	Percentage points difference with total manufacturing average						
Segmented, high R&D							
Chemicals and pharmaceuticals	64.6	–95.9	3.3	15.8	19.0	7.3	–29.7
Electrical and office machines	13.7	141.7	91.3	37.8	50.5	61.9	143.8
Communication equipment	6.2	33.3	76.6	145.8	66.5	18.4	72.3
Motor vehicles	57.9	69.7	1.2	–18.7	85.0	–70.0	52.0
Other transport equipment	21.6	25.0	19.5	142.0	74.5	70.0	4.8
Fragmented, high R&D							
Medical and optical instruments, watches and clocks	9.1	–29.1	16.7	48.8	89.8	226.4	5.5
Machinery and equipment	4.3	26.0	47.0	–7.1	17.3	–43.3	12.8
Other manufacturing	47.5	–1.6	–9.7	–2.0	–41.5	–30.5	63.5
Segmented, low R&D							
Basic metals	138.5	14.4	26.3	20.5	38.0	–27.0	–4.0
Rubber and plastic	–14.8	5.1	–17.3	–25.9	–30.6	33.7	–46.7
Food, beverages and tobacco	–70.3	–53.2	–48.8	–51.1	–40.0	15.8	–71.2
Fragmented, low R&D							
Textile, wearing apparel and leather	37.4	69.9	53.7	111.5	–21.1	226.1	84.6
Wood products	–70.4	–45.0	–38.9	–47.7	–45.1	154.0	–45.6
Paper products and publishing	–43.1	–56.2	–43.6	–49.3	–45.5	–68.5	–74.1
Non-metallic products	–38.8	–53.0	–49.1	–53.6	–71.9	–56.9	–42.4
Fabricated metal products	–59.9	–50.8	–61.8	–62.0	–76.3	–66.9	–61.6

1. The import penetration rate is defined as the share of imports in total demand (estimated as production minus exports plus imports).
2. Or latest available year.
3. Non-weighted average of Austria, Belgium, Denmark, Finland, Netherlands, Norway and Sweden.

Source: OECD, Bilateral Trade database.

resulting in prices that are appreciably higher than in other European countries – this border effect also applies to tradable goods that are not much protected by trade barriers.

Coupled with the relative isolation of some markets is the segmentation of the domestic market in a large number of sectors, which is partly due to Switzerland's federal organisation and to its linguistic disparities. While competition policy is the Confederation's domain, the cantons do have extensive powers to intervene in markets, and often control the supply and the pricing of electricity, water, gas and regional transport, resulting in large disparities between

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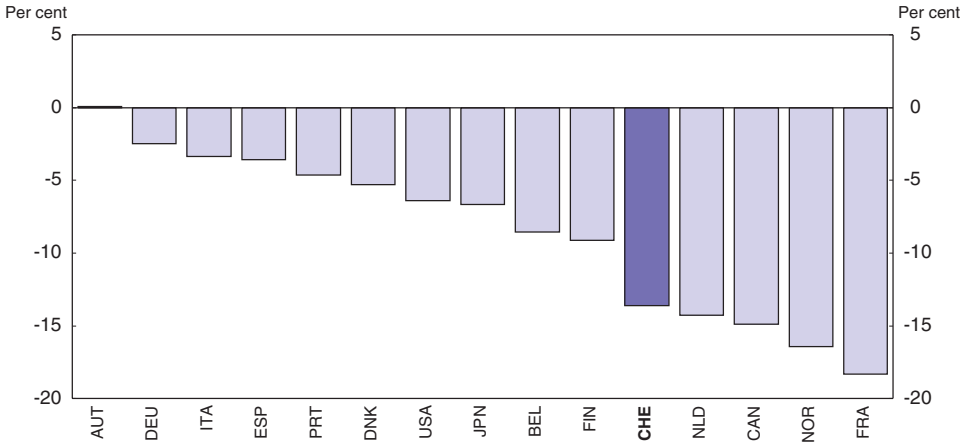
Figure 20. **Productivity in the construction and private non-financial service sector**[1]

Relative to total economy, 2001[2]



1. Construction and non-financial services excluding public administration and defence.
2. Or latest available year.

Source: OFS and OECD, STAN database.

public utility prices. Cantons also have a marked influence on industries such as construction and professional services with very diverse regulations that *de facto* constitute entry barriers. The effect of a bias in public procurement in favour of local firms is enhanced by the large share of outlays decided by the cantons and municipalities. The lack of a level regulatory playing field in the domestic market deprives firms of economies of scale and reduces productivity in private non-financial services and construction, thus accentuating the dichotomy between the sheltered and the exposed sectors (Figure 20). Given this segmentation of domestic markets, the average concentration of firms, measured on the basis of employment data, does not seem particularly high compared with other economies of similar size, such as Austria (Figure 21). Switzerland's economic fabric is characterised by a relatively large number of big multinational firms, but also comprises many SMEs (Table 16).

International comparisons indicate the restrictive nature of Switzerland's product market regulations in the late 1990s and the limited progress of reforms in this area – compared with other countries – over the last two decades (Figure 22). These rigidities limit the benefits that Switzerland can derive from the globalisation process. The degree of regulation and segmentation of the domestic product market, combined with the relative isolation of some markets from foreign competition for some product categories, has the effect of reducing the expected returns on investment, which in turn curbs direct investment inflows which are not

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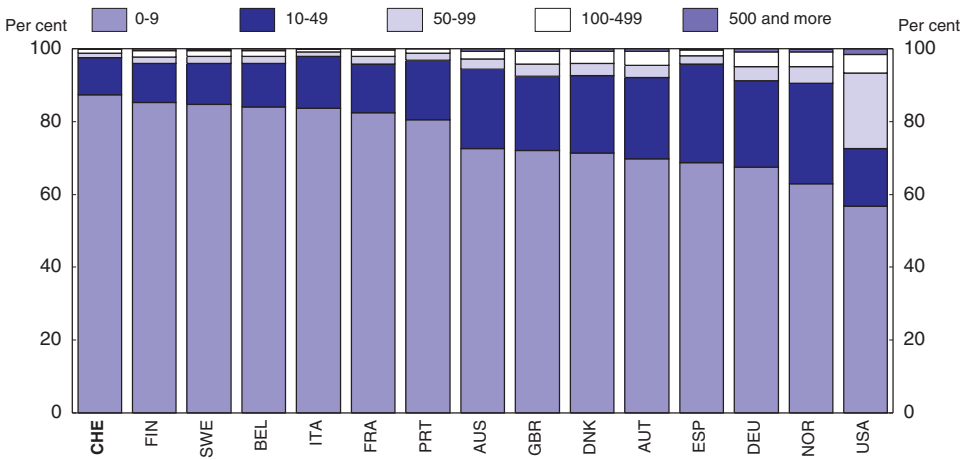
Figure 21. **Distribution of enterprises by class size**
Total economy, 1999 or nearest year



Source: OFS and OECD, Statistics on Enterprises by Class Size (SEC) database.

low, but concentrated in the financial sector.[11] That trend is reinforced by the high price level, which is keeping wages higher than in other countries even though productivity calculated over all sectors appears barely greater than elsewhere.[12]

With high labour costs, a highly skilled labour force and the trend appreciation of the currency, the Swiss economy tends to specialise in niche activities with high value added and moderate or low price elasticity, such as pharmaceuticals, precision instruments and the financial sector (Table 17). In manufacturing, this specialisation is based on dynamic research activity financed mainly by business. It is a strategy which allows new markets to be won thanks to quality improvements that can be protected by patents and other barriers to entry, limiting price competition. The number of patents per head is by far the highest in the OECD, while R&D expenditure as a proportion of GDP is also high, especially in the industrial sectors exposed to international competition (Table 17). However, firms often relocate their production facilities abroad to take advantage of economies of scale and gain better access to larger markets and lower production costs.[13] Between 1995 and 2000, the average ratio of corporate investment to GDP was 1½ percentage points below the OECD average, despite interest rates being lower than in other countries. Capital outflows, which accelerated in the 1990s, were doubtless also stimulated by the attraction of the EU area for companies following the rejection of EEA membership in the 1992 referendum. All told, outward direct investment is relatively high and the difference between direct

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Table 16. **Hirschmann-Herfindahl indices of industry concentration**
Based on employment

	Switzerland		Austria	France	Italy
	1991	2001	1999	1999	1999
Manufacturing industry					
***Segmented, high* R&D**					
Chemical products	347.0	179.4	297.1	33.5	33.6
Office and computing machinery	306.5	2 341.5	–	1 006.2	700.2
Electrical machinery	137.0	207.2	161.3	72.2	16.9
Radio, TV and communication equipment	485.5	396.2	737.6	172.2	120.6
Motor vehicles	421.7	380.2	554.8	198.3	168.5
Other transport equipment	1 000.6	1 098.1	882.9	207.4	247.4
***Fragmented, high* R&D**					
Medical appliances, optical instruments, watches and clocks	65.1	71.8	118.9	28.7	15.1
Machinery and equipment	68.4	45.4	41.8	17.4	7.1
Furniture and other manufacturing	34.1	32.5	51.4	31.6	2.1
***Segmented, low* R&D**					
Coke and petroleum products	3 392.7	2 737.9	–	532.2	678.9
Basic metals	375.0	262.8	419.0	117.9	59.9
Plastic and rubber products	85.1	86.7	88.7	43.6	17.0
Food and beverages	77.3	101.5	23.0	6.4	5.2
Tobacco products	1 920.1	3 088.1	–	–	2 211.7
***Fragmented, low* R&D**					
Textiles	58.3	92.8	136.5	22.5	5.3
Wearing apparel	129.1	285.6	224.0	14.6	3.2
Leather and footwear	758.8	259.6	454.0	45.5	3.4
Wood products	9.1	14.6	49.6	10.4	1.4
Paper and pulp products	191.2	165.9	252.8	48.7	41.8
Publishing and printing	43.5	44.1	57.4	10.3	15.3
Non-metallic products	58.1	82.1	158.8	43.9	10.2
Fabricated metal products	13.1	10.3	29.8	4.7	0.8
Non manufacturing industry					
Electricity and gas	220.7	221.4	359.6	1 377.4	398.7
Water	3 033.1	1 064.5	572.0	711.5	255.9
Construction	3.3	3.9	9.1	1.1	0.3
Land transport	283.6	243.5	140.5	103.9	122.7
Water transport	393.4	470.5	645.8	428.7	–
Air transport	3 431.0	3 029.3	–	1 323.3	–
Post and telecommunications	4 782.1	1 741.4	–	1 116.2	796.1
Sale and repair of motor vehicles	11.0	17.4	26.0	4.3	–
Wholesale trade	10.7	10.6	16.6	2.3	–
Retail trade	24.3	49.0	31.3	8.2	–
Hotels and restaurants	7.7	7.4	2.5	5.5	157.1

Source: OFS and OECD, Statistics on enterprises by class size (SEC database).

investment in Switzerland and abroad, which on average represented one-third of annual corporate investment in Switzerland between 1990 and 2000, is more unbalanced than in other countries (Figure 23). These outflows, which may in some

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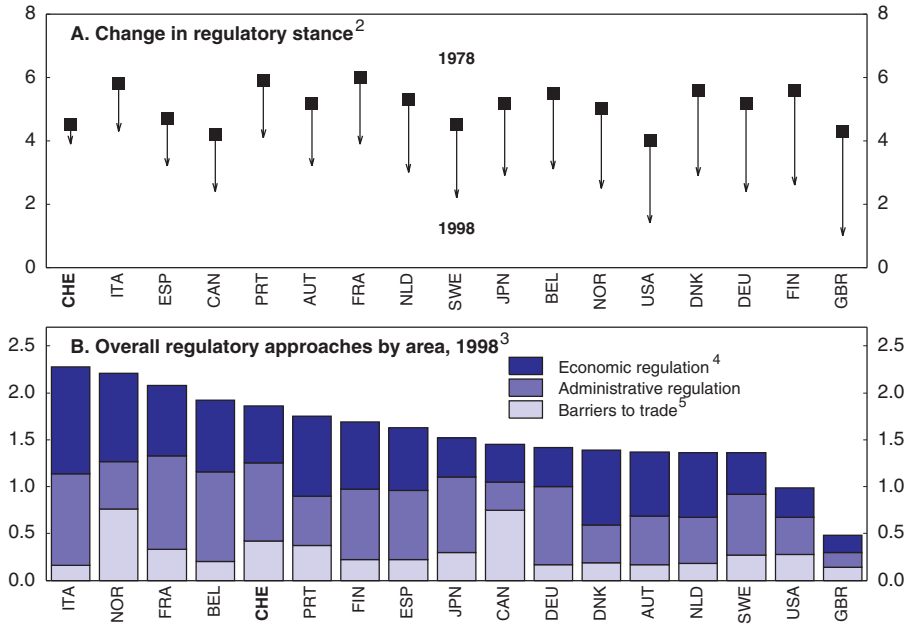
Figure 22. **Indicators of product market regulation**[1]



1. The regulatory stance is measured by a synthetic indicator ranging between 0 (least restrictive) and 6 (most restrictive).
2. Reports changes in the regulatory stance in seven non-manufacturing industries (gas, electricity, post, telecommunications, passenger air transport, railways and road freight) between 1978 and 1998.
3. Indicator of economy-wide product market regulations.
4. Includes barriers to competition and state control.
5. Includes trade and FDI restrictions.

Source: Nicoletti *et al.* (2001), "Product and Labour Market Interactions in OECD countries", *OECD Economics Department Working Paper,* No. 312, OECD, Paris.

cases have represented a substitute for exports, have increased the balance on investment income, which reached 7¼ per cent of GDP on average between 1998 and 2002, helping to raise Swiss national income. However, annual average GNP growth, despite exceeding that of GDP by about ¼ of a percentage point between 1990 and 2002, does not make up for the growth differential with the OECD country average, which reached about 1½ percentage points during this period.[14]

Legislation and implementation

The competition law has been reformed…

Legislation on competition is recent in comparison with other countries, the main texts being adopted in the mid-1990s in the context of the economic

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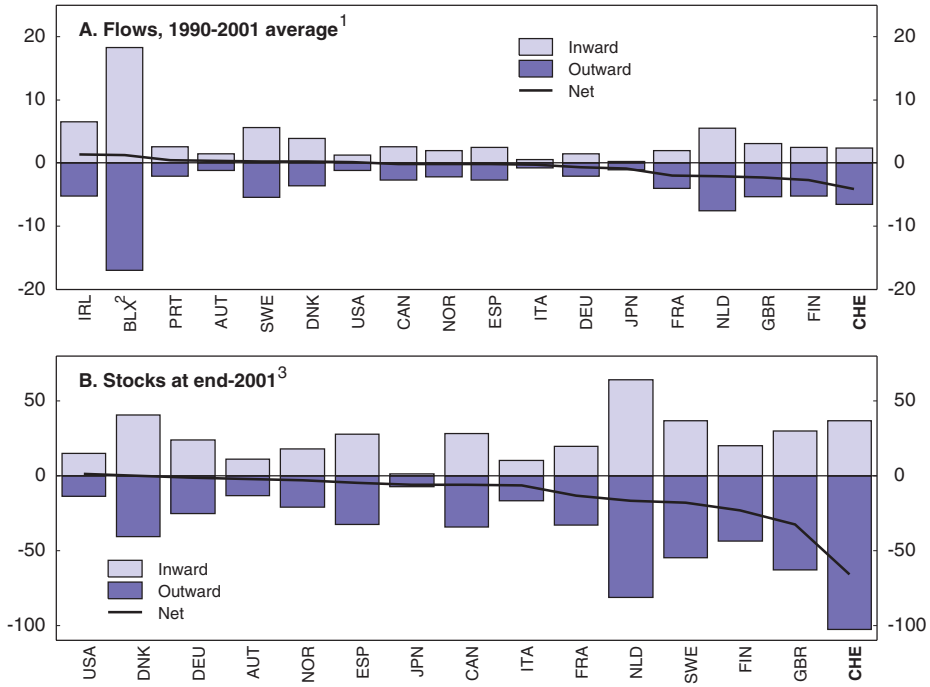
Table 17. **Specialisation patterns and indicators of innovation in selected countries**

	Switzerland	Austria	France	Germany	Italy	EU[1]	Japan	USA
Industrial structure, 2000								
(per cent of total value added)								
Agriculture	**1.2**	2.2	2.8	1.2	2.8	2.5	1.3	1.4
Mining and manufacturing	**17.9**	21.1	18.0	22.4	21.1	19.9	21.2	16.8
of which:								
Chemical, rubber, plastics and fuel products	**3.4**	3.0	3.5	3.5	2.8	3.1	3.2	2.7
Machinery and equipment	**6.2**	4.9	3.5	6.6	4.4	4.5	6.3	4.1
Electricity, gas and water	**2.4**	2.3	2.0	1.8	2.1	2.2	3.6	2.2
Construction	**5.2**	7.8	4.6	5.1	4.8	5.5	7.1	4.7
Business sector services	**46.8**	46.6	49.5	48.2	50.1	48.0	45.2	55.0
Distribution	**12.5**	12.8	10.2	11.3	13.1	11.5	13.2	17.4
Restaurants and hotels	**2.7**	4.1	2.8	1.3	3.6	2.9	n.a.[5]	0.9
Finance and insurance	**16.1**	6.8	5.0	4.5	6.2	5.2	5.9	8.7
Other business services	**15.4**	16.1	25.3	25.1	19.9	21.7	20.0	21.3
Community services	**20.7**	20.0	23.1	21.3	19.1	21.6	21.6	21.3
Indicators of innovation activity								
Investment in knowledge,[2] 1998	**4.8**	3.5	4.1	4.2	2.1	3.8	4.7	6.0
Business sector R&D:								
Expenditure, 2000[3]	**2.0**	1.1	1.4	1.8	0.5	1.2	2.1	2.0
Share of total R&D spending, 2000[4]	**69.1**	40.2	52.5	65.8	43.0	55.8	72.4	69.3
Number of patents per million population, 1998	**112.9**	32.7	36.2	68.4	12.0	35.3	79.3	52.7
Per cent of ICT patents in all patents, 1998	**7.3**	6.2	14.8	9.8	6.0	13.1	21.3	19.3

1. The industrial structure pertains to the euro area, excluding Ireland and Luxembourg; 1999 data.
2. Total investment in higher tertiary education, R&D and software. Per cent of GDP.
3. Business R&D expenditure as a percentage of GDP; 1998 data for Austria.
4. 1996 data for Italy.
5. Include in distribution.

Source: OFS; OECD, OECD *Science, Technology and Industry Outlook*, 2002 and STAN database.

Figure 23. **Foreign direct investment in OECD countries**
As a percentage of GDP



1. 1998-2001 for the outward flows of Ireland.
2. Belgium-Luxembourg.
3. Or latest available year.

Source: OECD, *International Direct Investment Statistics*.

"revitalisation programme". The texts mainly concern the Federal Act on Cartels and other Restraints of Competition (LCart), adopted in 1995; the Regulation on Business Merger Control (1996); the Domestic Market Act (1995) and the Act on Technical Hindrances to Trade (1996). LCart, which was the first legislation to give the competition authorities decision-making powers,[15] was amended in June 2003 and these should enter into force in April 2004. The main amendments concerned the introduction of direct sanctions and a leniency programme, which will considerably reinforce the measures available to combat anti-competitive behaviour. Prior to this reform, sanctions could only be imposed in the event of repeated offences, which never occurred. Competition law has thus moved closer to EU law and to that of other OECD countries.

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The competition authority, which comprises the Competition Commission (or COMCO, the decision-making body) and its Secretariat (investigation body), is responsible for applying Lcart which consists mainly of rendering decisions on antitrust and merger cases. COMCO can also make recommendations to the political authorities and formulate opinions on general questions relating to competition. COMCO is also empowered to ensure that the Confederation, cantons, communes and other bodies with public responsibilities comply with the provisions of the Domestic Market Act (DMA). It can, under this heading, make recommendations to them concerning legislation that is under consideration or already exists, and carry out enquiries.

There are bodies other than COMCO that may intervene on competition matters, notably in connection with appeals procedures.[16] Amongst these bodies, the Price Surveillance Authority has particular responsibility for monitoring price trends and intervening in sectors where there are exemptions with respect to the direct application of LCart and where there is little or no competition, because prices are controlled or set by powerful companies in the absence of effective competition (the health sector is an example). The objective is to prevent excessive prices from increasing or being maintained.

COMCO currently has fifteen members, including six representatives of interest groups appointed by the Federal Council. All can have other appointments, so that they can, for example, sit on company boards, as some of them do. LCart stipulates that a COMCO member has to decline to give an opinion if he is liable to face a conflict of interest, but despite this provision, it seems that the independence of COMCO's decision-making is insufficiently guaranteed (Neven and von Ungern-Sternberg, 1998),[17] and the problem is going to become more acute with the application of direct sanctions and the leniency programme. With regard to being on company boards, no measures were taken as part of the 2003 revision of LCart other than to introduce a provision requiring members of COMCO to declare their interests in advance.[18]

In view of the scale of the tasks outlined by the law, the resources available to the competition authorities are limited, particularly since the latter have also to oversee compliance with the DMA. The COMCO Secretariat numbers some 45 full-time staff, which is low in international comparison even when taking into account the 15 additional posts, which should be created (Figure 24). Combining the resources and functions entrusted to COMCO and the Price Surveillance Authority could make for better utilisation of available resources.

... but still differs from that of other countries

Competition law does not include any general rule on sectoral exemptions (like in the European Union) and it applies to private and public companies and events whose effects are felt in Switzerland, even if they take place abroad.

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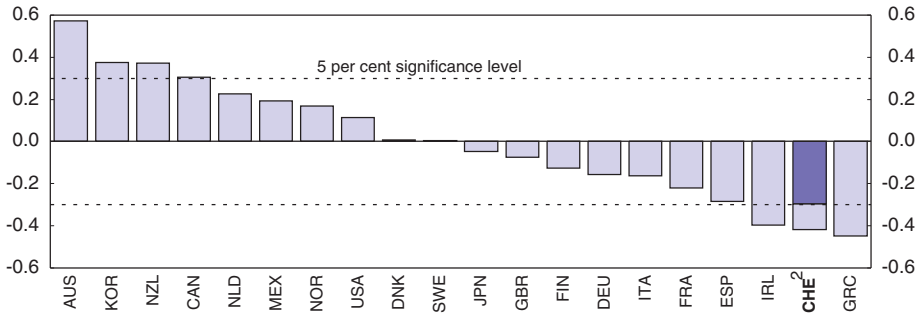
Figure 24. **Staff resources of competition authorities**[1]



1. The effect of the size of the economy on staff/GDP ratios has been estimated by the following equation:
   log (staff/GDP) = 0.513173 – 0.38325 * log (GDP)
   (1.5) (–3.2)
   (t-statistics in parentheses); S.E. = 0.30; R^2 = 0.36
   The chart shows the residuals after control for effects of country size.
2. The darker colour represents the Swiss position if the staff of the competition agency is increased to 60 instead of the current 45 persons.

Source: OECD.

However, many areas, which are subject to regulations, such as imposing a state pricing system or granting special entitlements to certain companies for carrying out public tasks, are excluded from the scope of LCart. Notable amongst the sectors at least partially exempt from competition law are agriculture, health – including pharmaceuticals – and also some of the network industries which are subject to specific regulations. In addition, the law does not apply to the effects on competition that stem solely from the legislation on intellectual property. However, the 2003 revision of LCart does limit the above exemption, noting that "import restrictions based on intellectual property rights are assessed according to the provisions of the present law".

Unlike law in most other countries, where agreements are concerned, LCart is based on the abuse rather than the prohibition principle. This principle slows COMCO's decision making process with respect to hard core cartels.[19] Agreements concerning competition are only declared unlawful when they have a marked impact on competition in the market for certain goods and services and are not justified for reasons of economic efficiency or when they rule out effective competition. LCart contains a presumption of unlawfulness for horizontal price, quantity and market-sharing agreements, which brings it closer – for this type of cartel – to legislation based on the prohibition principle (Gugler and Zurkinden, 2002). The 2003 revision of LCart introduced a new presumption of unlawfulness concerning vertical agreements which impose a minimum sale price or a fixed sale price, and also regional distribution contracts when sales by other suppliers are excluded.

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The control of business mergers rests on a system that requires the competition authority to be notified before such mergers take place. Notification is mandatory above certain thresholds[20] and, even if the thresholds are not reached, when COMCO has previously established that a participating firm had a dominant position on the market concerned or a neighbouring market. COMCO can prohibit the merger, or authorise it subject to certain obligations and conditions when the merger creates or reinforces a dominant position capable of eliminating effective competition and does not improve competition on another market, the beneficial effects of which outweigh the disadvantages of the dominant position. All told, Switzerland's merger control system is more "permissive" than European law or the law of other OECD countries as only mergers liable to eliminate – and not strongly affect – competition can be prohibited or made subject to certain obligations and conditions (Venturi, 2002). However this additional criterion of eliminating effective competition has never been applied by COMCO.

The law needs to be implemented with vigour

As a Parliamentary report observes, the criteria and methods employed by COMCO and its Secretariat to decide whether competition in a particular market is effective have been subject to criticism (OPCA, 2000a). Under particular attack were the competences of the Secretariat and the shortcomings of the indicators on which COMCO's decisions are based.[21] Where, however, they have been confirmed by independent experts, these criticisms have been taken into account by the competition authorities, which have improved their methods of analysis, as a report commissioned by the Federal Department of Economy confirms (Weizsäcker, 2000).

Since LCart came into force in 1996, COMCO has handed down relatively few decisions concerning cartels and abuses of dominant position. However these decisions may have had a preventive impact on the behaviour of some business associations. Although some instances of anti-competitive behaviour did cease during the course of COMCO's investigations and do not therefore appear in the statistics, the "ponderous pace" feared by the OECD in 1999 has persisted (OECD, 1999).[22] The wait-and-see policy regarding vertical restrictions does seem, nevertheless, to have come to an end in 2002 with the adoption of a guideline concerning the appraisal of vertical agreements, followed by a more specific one concerning these agreements in the area of motor vehicle marketing (Annex IV). The two guidelines signal an increased determination to combat harmful vertical agreements (Meinhardt and Merkt, 2002). The future will show whether COMCO has actually changed tack and is applying the law strictly with respect to both horizontal cartels and vertical agreements.

When LCart was adopted, it was not foreseen that there would be so many merger operations requiring notification.[23] And with some of the Secretariat's

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resources being earmarked for merger control operations, this may also be an explanation for the relatively limited results on the antitrust front. To date, COMCO has not formally banned any company merger. However in pre-notification contacts, some companies abandoned their merger plans after the authorities expressed serious doubts. Some were authorised under obligations and conditions (UBS, for example). Company merger authorisations in relatively concentrated sectors such as the press, food and distribution have raised doubts in some circles as to the actual role of merger control procedures in Switzerland (von Ungern-Sternberg, 1999).

So far, the deterrent effect of the competition law was limited, reflecting the limitations of the law in force until 2003. Prior to then, COMCO could only impose direct administrative sanctions in the cases of mergers (if a company failed to make known an operation or did not comply with a ban) and breaches of the obligation to inform the authorities.[24] In the other cases, the sanction was indirect. In other words, when an infringement of competition was discovered, COMCO could do no more than issue a decision stating the unlawfulness of the behaviour (which can be used for claiming civil damages). The company could be sanctioned only if it did not comply with the decision once it came into force, meaning that the first infringement was not sanctioned. The 2003 revision of LCart introduced direct administrative sanctions for breaches of the law orchestrated by horizontal price, quantity and market-sharing cartels, certain types of vertical agreements and abuses of a dominant position. The amount of the fine can be up to 10 per cent of turnover in Switzerland over the previous three financial years. Along with direct administrative sanctions, a leniency programme was also introduced. These changes are welcome and signal an important step towards increasing the credibility of Swiss competition law, even though, because of the constitutional roots of the principle of abuse, it has not been possible to impose direct sanctions in every instance of the law being breached.

Where promoting competition is concerned, COMCO has submitted around ten recommendations, most of them to the Federal Council, but the majority have not been taken up by the authorities. Some of these recommendations relate to important issues such as the liberalisation of the non-life insurance sector, reimbursement by the sickness insurance system of drugs purchased abroad or parallel imports for products protected by patent rights. However, COMCO could have made more use of its authority to make recommendations in the key areas that have marked the economic and social policy debate in Switzerland in recent years (*e.g.* health, air and rail transport, energy).

LCart does not provide for any international legal co-operation or consultation and exchange of information system with foreign competition authorities.[25] Moreover, Switzerland has, to date, signed no international agreements on judicial assistance or legal co-operation with respect to competition.

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Contacts with foreign competition authorities are purely informal in nature. Despite their efforts and their presence at international meetings such, for example, as those of the OECD's Competition Law and Policy Committee, the Swiss competition authorities are somewhat isolated (*e.g.* no participation in European Competition Network). This does not facilitate the work of the Swiss competition authority.[26]

Regulatory policy

This section looks at those sectors which are subject to widespread regulation, including network industries like electricity and gas, telecommunications, railways and postal services, where liberalisation measures are currently being implemented, albeit more slowly than in neighbouring countries. Competition in service sectors is also analysed, in particular distribution sectors including parallel imports, and professional services, where the main challenge is to develop a true internal market.

Reform in network industries should be accelerated

New plans for electricity reform must be adopted rapidly

The electricity sector is composed of four vertically integrated companies that control the long-distance transmission network, and a myriad of small ones, most of them public firms owned by cantons or municipalities. Distributors enjoy *de facto* monopoly rights for regional or local areas, and set the price for electricity transmission through their networks. Their accounts are often not separated from those of local governments, and distribution is sometimes operated together with other utilities (gas and water distribution), which results in tariff structures that are not fully transparent. Revenues from electricity distribution, together with taxes on generation and transmission, which are set by cantons, are used to cross-subsidise other regional or local activities. They constitute a sizeable source of revenues for these governments (CHF 700 million a year or 0.2 per cent of GDP). The current market structure creates inefficiencies which result in electricity prices being above the OECD average (Figure 25), despite the fact that a large part of electricity is hydro-powered (60 per cent), and in a wide variation of prices across different types of consumers and across regions. Large firms can obtain favourable conditions from electricity generators, with average discounts of close to 30 per cent with respect to SMEs, while households pay on average 55 per cent more than businesses. Price differences across cantons can also be large, with some cantons paying for electricity four times more than others (*Surveillant de prix*, 2002).

In December 2000 the Swiss Parliament approved a reform that follows the broad principles of the EU reforms: accounting and operating separation of transmission from generation and distribution, freedom of access at transparent

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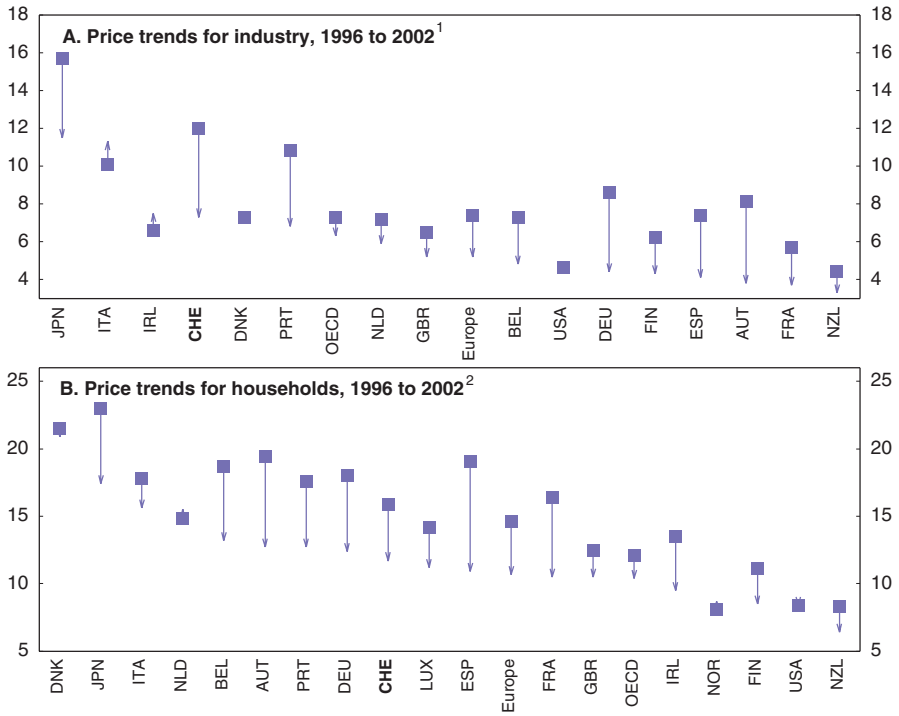
Figure 25. **Electricity prices for industry and households**
In US cents/kWh



1. 2001 for Germany, Netherlands and Spain; 2000 for Austria, Belgium, OECD Europe and OECD total.
2. 2001 for Germany and Spain; 2000 for Belgium, OECD Europe and OECD total.
Source: IEA, *Energy Prices & Taxes, 3rd quarter 2003*, IEA/OECD, Paris.

prices to the transmission network, which would be managed by an independent company, enforceable priorities for using the grid and clear rules for determining distribution prices. The government also prepared a regulation on the methodology to determine these prices. In parallel, several companies started merger negotiations. However, the reform was rejected by the people in September 2002. The main concerns of opponents to the reform were to maintain public service obligations and especially security of supply. Once the law was rejected, merger initiatives that might have helped to consolidate the sector were halted. A new commission (OSEL) has been set up to prepare a new reform law that should be discussed by Parliament by 2005 and enter into force by 2007, the date of complete liberalisation in the European Union.[27]

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The new law is bound to include the backbone of the previous reform initiative, which was not controversial, including the setting up of a wholesale electricity market, the separation of transmission from generation and distribution and access pricing to the network. This should help to reduce geographical price differences, although cantons and communes will still be able to tax the electricity sector in various ways (as foreseen in the rejected reform law). The reform is also necessary to allow an EU compatible pricing regime for electricity transmission, which would benefit companies that trade electricity with neighbouring countries. Being at the centre of Europe, the business of exporting and importing electricity is very important in Switzerland, which has a well developed transmission network (26 500 MW installed cross-border capacity) and feeds large electricity demands from other countries, especially Italy. For the time being, private arrangements are negotiated with foreign companies, but the perspective of an integrated EU electricity market requires a transparent pricing for international transmission of electricity. Some other demands of those opposing the reform during the referendum debate, which should be taken into account when drafting the new reform, should in principle not interfere with an efficient working of the market. Indeed, a strong regulator is a necessary requirement for liberalisation of a natural monopoly, while the recourse to more clean energy sources should be adopted separately with targets based on a cost-benefit analysis and could be implemented through a green certificate market applied to all renewables.[28] Finally, securing electricity supply is fully compatible with electricity reform, given the high degree of interconnection with other countries and the rejection of the nuclear moratorium in May 2003. The reform should go ahead quickly, not only because further delays prolong the period of higher prices faced by business consumers and households,[29] but also because a few cantonal governments are on the way to approve or have already adopted laws that reinforce the monopolistic power of current distributors.[30]

The gas sector should be reformed in parallel to the electricity sector

The share of natural gas as primary energy source (9.3 per cent) is well below that of other countries, gas being used for industrial and domestic needs but barely for electricity generation. As in the case of electricity, a large number of companies exist in the gas industry, many of them public with management and accounts not clearly separated from those of local governments. Natural gas is imported by Swissgas, which is owned by other gas companies. The industry is vertically integrated and works in the form of local monopolies. Third party access to the high pressure network is possible since 1963, although it has only been demanded in 2001 for the first time to transport gas from the North of Europe to Italy. Gas prices are well above the OECD average (Figure 26).

Liberalisation of the gas sector was being prepared prior to the September 2002 referendum on electricity reform, similar to the EU reforms,

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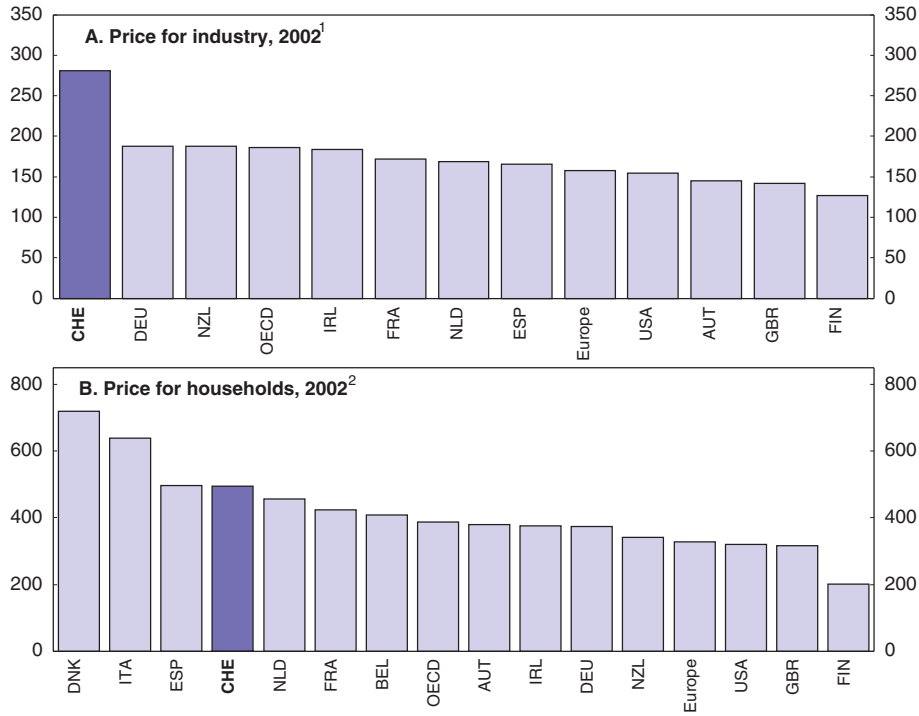
Figure 26. **Natural gas prices for industry and households**
In USD per 10 millions kcal



1. 2001 for OECD total; 2000 for Germany and OECD Europe; 1999 for Austria.
2. 2001 for OECD total; 2000 for Belgium, Germany and OECD Europe; 1999 for Italy.
Source: IEA, *Energy Prices & Taxes, 3rd quarter 2003*, IEA/OECD, Paris.

i.e. with regulated third party access of the gas network controlled by an independent regulator, and progressive opening of the market for large consumers first, and subsequently for small consumers and households. The idea was to implement it after the electricity reform was approved; however, the rejection of the electricity reform has halted liberalization plans for the gas sector. As an alternative, a partial opening of the market is being launched by gas operators with a series of agreements to set clear rules for determining prices for third party access to the network. However, though it is welcome, it is not sufficient to create a regulatory framework comparable to that of the EU countries. *First*, the agreement implies only a partial liberalisation of the market, since it would affect only large

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consumers; *second*, it does not make compulsory the separation of different activities (transport and distribution); *third*, it does not provide for an independent regulator for the sector; and fourth, it could be in conflict with anti-trust law.

Reform initiatives should be taken up again by the government in parallel to those in the electricity sector, so as to avoid delaying potential gains from liberalisation.[31] Moreover, two of the main concerns of opponents to electricity reforms – the need to ensure long-term energy supplies, and providing incentives to use renewables – would be better dealt with by matching both reform processes, which would also allow the consideration of the large potential of natural gas as a primary source for electricity production.

The unbundling of the local loop is a priority in telecommunications

Liberalisation of the telecommunications sector started in 1998 when the new telecommunications law came into effect. The old post and telecommunication operator was divided into the public post company and an enterprise holding, the telephone monopoly. This latter became Swisscom and was subsequently partially privatised.[32] As in the rest of the OECD, new competitors have appeared in fixed telephony which, together with technological advances, has resulted in a price reduction of fixed telephony services over the past five years and in prices close to the OECD average (Figure 27). The market regulator is the Communication Commission (ComCom), which is operationally independent from the government, but depends on it for its budget.

In fixed telephony and up to 2002, Swisscom's market share shrank from 100 per cent to 84 per cent for local calls, 66 per cent for national calls and 47 per cent for international calls. However, a major competition issue to be solved is the lack of unbundling of the local loop of Swisscom, as it was not explicitly included in the liberalisation law, contrary to most other OECD countries. This gives Swisscom a clear competitive advantage, since physical access of competitors to the last mile would allow them to make offers to their customers on an equal footing with the incumbent. The Federal Council, which is in favour of unbundling, has included unbundling in the interconnection regime via a decree, rather than a law. This implies that the ComCom, which has the competence to impose interconnection prices, can also impose unbundling at a given price.[33] However, Swisscom is not recognising these powers of the ComCom and has threatened to appeal to the court if the ComCom, in the context of an interconnection complaint, take a decision solely based on the decree. The Federal Tribunal might take two years to come to a decision and may conclude that a new law is needed to introduce unbundling.[34] The Federal Council is attempting a new reform of the Telecommunications Act, which might also take some time to be implemented and has still to be drafted and debated in the Parliament.

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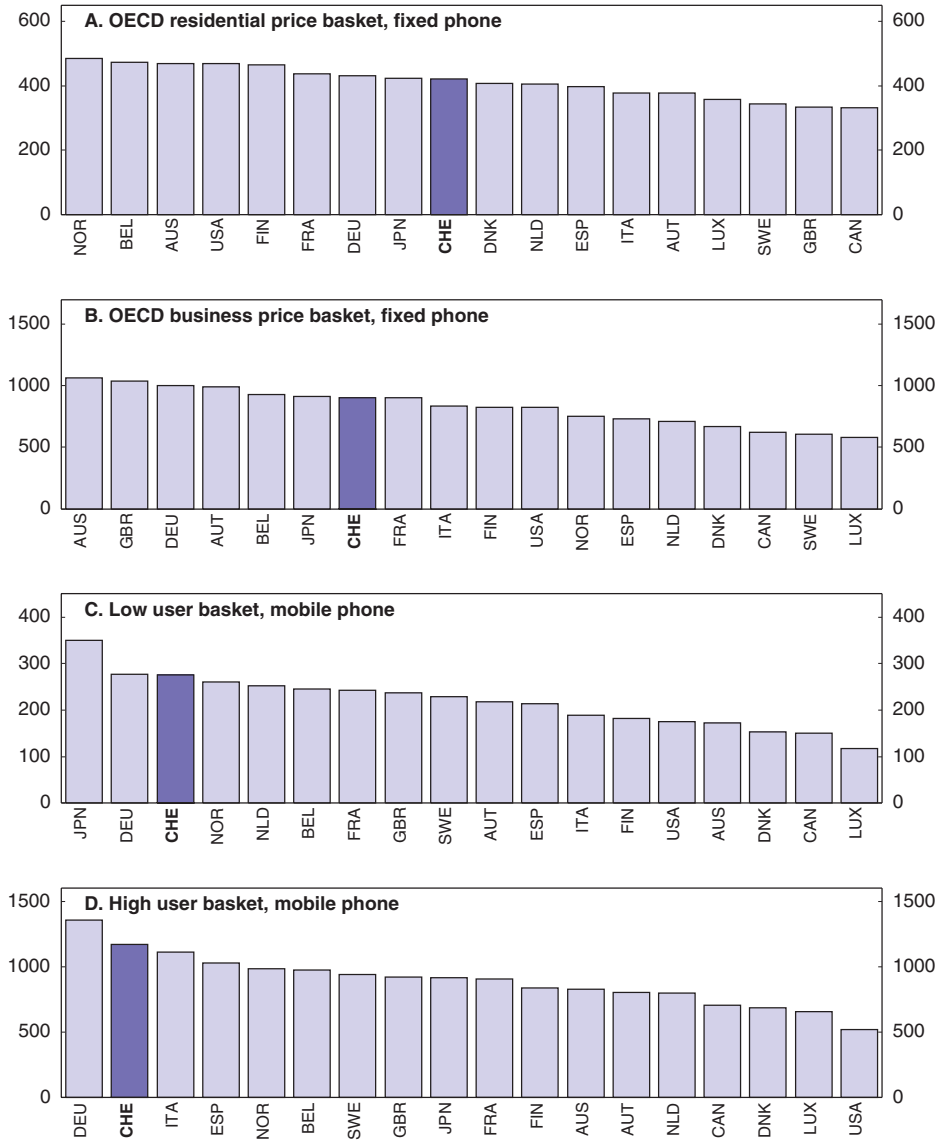
Figure 27. **Telephone prices**
Total charges in USD, August 2003



Source: OECD, Communications Outlook database.

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The unbundling of the local loop is not only key for fixed telephony, but also for competition in the business high volume data transmission market segment, which is of special relevance for the development of cheap internet services. In particular, unbundling would favour effective competition from other companies in providing DSL services to consumers at low prices. Bandwidth transmission is also provided by cable networks, which is dominated by Cablecom. Cablecom was separated from Swisscom by the government through its sale to a foreign company in order to reduce its dominant position, and offers voice and high speed transmission services at competitive prices, posing some competitive pressure on Swisscom.[35] However, even though cable infrastructure is well developed, Cablecom is unlikely to rapidly expand its telephony services to the whole population, which would require large investments.[36] In any case, even if Cablecom would provide telephony services in the whole country, a quasi-duopoly led by Swisscom and Cablecom would emerge, with other telephone companies using the Swisscom network without locally installing their own infrastructure. This would probably not ensure a rapid transmission of efficiency gains into lower final prices.

Three firms provide mobile telephony services. They include the subsidiary of Swisscom, with a higher market share (63 per cent as estimated by the Federal Communications Office) than incumbent operators in other countries due to the relatively late start of liberalisation, which gave a "first mover" advantage to Swisscom.[37] Mobile telephony prices have also declined since liberalisation started, although prices have barely changed in the last two years. Indeed, the COMCO is currently examining a case of possible horizontal price agreements between these companies.[38] By the end of 2004, three companies providing UMTS mobile telecommunications should operate, covering 50 per cent of the country, which might also raise competitive pressure in the high volume data transmission market.

Reforms in the railway sector could be more ambitious

Being at the crossroads of Europe, a large amount of land traffic passes through Switzerland. Partly for environmental reasons, and after a referendum in 1992, the Swiss Constitution incorporates a mandate to move land transport from roads to railways. The first pillar of this strategy is a large investment programme in the railway networks. The second pillar is a tax on heavy trucks, which was negotiated with the European Union in a bilateral agreement since it affects a large number of European trucks crossing the country. The third pillar aims at the liberalisation of railway transport to raise competition and thus make the switch from road transport to railways as cost efficient as possible.

A first reform entered into effect in 1999, including the incorporation of the main public company, CFF, which previously was an agency depending on

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government, and the accounting and operation separation of the network from the transport business to avoid cross subsidisation.[39] The reform also dealt with market access conditions. Merchandise transport was opened to competition, while scheduled interurban passenger transport remained a reserved domain of CFF. In regional passenger transport, the so called "principle of command" was implemented, whereby railway companies are only obliged to cover profitable lines or those lines for which they receive a "command" (and the corresponding subsidy) from a public administration (usually the canton, which is financially supported by the Confederation).[40] In principle, this should lead to more transparency, since subsidies will only be paid in advance for well defined services, instead of merely covering *ex post* deficits as in the past.

Since the reform, competition has improved, but full liberalisation is still a long way off, and there remain some obstacles to raise competition further. The major problem in *merchandise transport* concerns the slots for using the network, which are allocated by a body controlled by the largest two companies, CFF and BLS.[41] The new reform does not foresee to change the system fundamentally, but the arbitration commission will receive more competences.[42] On *international freight traffic*, a bilateral agreement with the European Union has introduced partial liberalisation of international freight transport on a reciprocal basis, allowing foreign companies to provide links with Switzerland as well as transit through Switzerland, but in some cases only in partnership with a local company. Currently, a second reform is in preparation, in parallel to that of the EU, which would withdraw this latter requirement and would allow free entry of foreign companies. In *regional passenger transport*, traditionally, more than 30 companies have provided services in the internal market, although most were tiny firms, many of them owned by cantons, while CFF had around 80 per cent of the market. The main problem in this market segment concerns the tendering of regional passenger lines. Currently cantons are not obliged to tender these lines, which are often awarded directly to specific companies, usually with public ownership (Confederation, cantons and some townships). The new reform does foresee to oblige public administrations to auction lines if the performance is notoriously inefficient. In addition, the reform also deals with the payment of subsidies whenever they go to profit making companies. Today, this is not possible since it is considered that public money paying for a universal service cannot go to "profits", benefiting private shareholders. But this is likely to prevent competition, since it eliminates the possibility of private companies to make any "normal" profit derived from their activity, and *de facto* eliminates any incentives for private providers to make offers for tendered lines, thus reserving this part of the market to publicly owned ones. The new reform is expected to change this situation and allow for subsidies being paid to private companies.

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The postal service is being opened to competition

Switzerland is among the OECD countries with a very dense postal network, with one office for every 2 128 inhabitants.[43] Some of these offices are little used, with only 25 per cent of them making a profit, resulting in an operating loss of CHF 500 million according to La Poste's calculations, the loss being covered by profitable business in the reserved area. Prices of postal services are also amongst the highest, at current exchange rates, in the OECD. After the separation of La Poste from Swiss Telecom in 1998 and the partial liberalisation of the market, new participants raised their market share to over 25 per cent. Liberalisation continues gradually, in parallel to a reform of La Poste to reduce costs. Reform plans, which are inspired by those of the European Union, have been laid out in a white paper ("Vue d'ensemble de l'évolution future du marché postal en Suisse"), which served as background to new legislation for reforming La Poste, recently adopted by Parliament. In the debate, alternative proposals have also been presented by defenders of a strong public postal service with a wide definition of universal service, making the reform of the sector a hot issue of public debate.[44]

For regulatory purposes, postal services are separated into three different market segments with different degrees of liberalisation. One area, which at present includes letters and parcels up to 2 kg, is reserved to La Poste. Under current reform plans competition will be extended in 2004 to all parcels, and by 2006 to letters of more than 100 g, although this second step of liberalisation has been made contingent on the result of analyses of reform processes in Switzerland and in the European Union, and on the condition that the universal service can be financed. The second area, which includes parcels of up to 20 kg, is open to competition, but universal service obligations apply to La Poste (but not to other competitors), implying that it is obliged to provide these services in the whole country at the same price. The third market segment is for parcels of more than 20 kg, which is completely liberalised, with no obligation for La Poste to provide it everywhere. Liberalisation for the first segment proceeds at a slower pace than in the EU, which at present reserves the monopoly to the incumbent firm in each country to letters up to 100 g and will lower this limit to 50 g by 2006. By 2009, the EU is expected to completely liberalise the market for letters (OECD, 2003b).[45]

These reforms will be complemented by a new concession regime for private postal firms. It will improve the supervision of the postal market and offer the possibility to finance, under certain provisions, the universal service. In particular, providers with more than CHF 100 000 of annual turnover should pay a charge proportional to their sales in order to compensate La Poste for providing universal service. The precise rates of this charge will be determined by the ministry, the maximum rate having been set at 3 per cent of turnover. Only if these firms extend their services to all the national territory would they be exempted. This way of financing the universal service is conditional on the accounts of La Poste,

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i.e. it would only be paid in case the public company would incur losses in the provision of the universal service in spite of operating efficiently. In order to determine this, and to avoid that La Poste cross-subsidises other segments of its activity subject to competition with the revenues received to pay universal service, a careful separation of the accounts of La Poste is needed, and must be strictly controlled by an independent regulator. Such a regulator should also ensure that automatic financing for La Poste's universal service does not result in moral hazard, eliminating the incentives for providing such service in a cost-efficient way. Before a concession charge for financing the universal service will be introduced, new business opportunities within and beyond the reserved area will have to be exploited by La Poste, and savings from restructuring the company will have to be realised. In this respect, the number of post offices has already been reduced from 3 390 to 2 951 at the end of 2002 by partly substituting unprofitable ones in remote areas by mobile offices and other reorganisations of the collection of mail. An alternative and more transparent framework for providing the universal service would be that the government calculates the net costs of universal service (including direct costs but also benefits from having nationwide delivery), and pays compensation via fiscal transfers. Such a system has been implemented, for instance, in New Zealand or Sweden. However, after an examination of different alternatives the Parliament has decided that the financing of universal service has to be funded by the sector.

Barriers to entry are restricting competition and efficiency in the food distribution business

The retail and wholesale distribution sector, which accounted for 13 per cent of value added and 17 per cent of total employment in 2000, is similar in size to the average for other countries. It is a sector that, in principle, lends itself to intense competition, especially as regards food even if economies of scale and of scope do foster concentration (Boylaud, 2000). In international comparison concentration is high in Switzerland, with two companies sharing over 75 per cent of the food distribution market. A fair amount of rivalry exists between the two firms, one of which is vertically highly integrated and mainly distributes its own products, while the other plays a key role in the distribution of branded goods. Productivity in retailing, indicators of which need to be interpreted with caution, does seem higher than the EU average (Table 18).[46] The efficiency of this sector grew more slowly than in most other countries during the 1990s (Figure 28). Although it is highly concentrated and many shops are grouped in networks, the food distribution business is still notable for the proportionately rather large number of small outlets, with slightly more employees than in other countries. Few hypermarkets have been built, and even they are relatively small.[47]

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Table 18. **Key structural features of the retail sales sector**
2001[1]

	Outlet density[2]	Employees per entreprise	Total VA per employed person[3]	Concentration in food retail[4]
Switzerland	**53.3**	**8.6**	**121.4**	**81.0**
Austria	45.6	7.6	107.9	45.0
Belgium	73.8	3.8	107.0	48.0
Denmark	46.1	8.0	89.9	52.0
Finland	44.7	5.2	123.0	69.0
France	64.3	4.3	128.3	31.0
Germany	34.2	9.2	109.3	30.0
Ireland	44.1	7.8	97.4	52.0
Italy	128.8	2.3	97.1	9.0
Netherlands	53.6	8.5	81.2	41.0
Portugal	136.6	2.6	63.2	46.0
Spain	130.0	2.9	88.7	23.0
Sweden	63.9	4.3	103.5	60.0
United Kingdom	35.9	14.9	84.8	41.0
Norway	67.7	6.0	87.7	86.0
European Union[5]	70.3	6.1	100.0	10.0
EU excl. Italy, Portugal and Spain[6]	52.2	7.1	104.6	

1. Or latest available data.
2. Number of entreprises per 10 000 inhabitants.
3. VA expressed in USD PPP terms. European Union = 100, 1999 data.
4. Market shares of the first three firms based on sales, 1996 data.
5. Non-weighted average of EU15 excluding Greece for the first three columns and including Greece but excluding Luxembourg for the fourth column.
6. Non-weighted average of EU15 excluding Greece, Italy, Portugal, and Spain.

Source: O. Boylaud and G. Nicoletti (2001), "Regulatory Reform in Retail Distribution", OECD *Economic Studies*, No. 32, 2001/1; OECD and Eurostat.

The smallness of shops means that the range of products available is more restricted than in other countries, whereas prices are higher because of agricultural protection.[48] A number of factors hinder the opening of supermarkets, which limits scale economies and creates barriers to entry for new competitors. As in other countries, environmental protection and land-use planning regulations mean that qualified local authority approval is needed to build an outlet of over 3 000 m^2. Above all, however, it is the whole range of appeal procedures that is the most serious obstacle. It can delay the building of a hypermarket by between 10 and 15 years. In fact the entry of new competitors in this market segment tends to happen as a result of businesses being taken over, but that does not appear to raise competition much.[49] Another factor limiting competition is the very high cost of setting up a shop in the centre of a town. In addition, the co-operative structure of the two main distributors results in an asymmetric setup, making it very difficult to take them over, whereas both firms have been continuously buying up other

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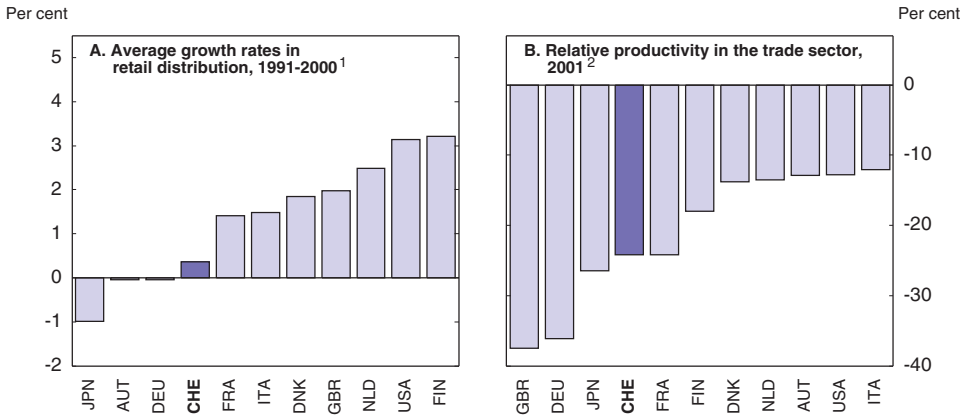
Figure 28. **Productivity in the trade sector**



1. Austria: 1996-2000; France and Japan: 1991-98; Germany: 1992-98; Switzerland: 1991-2001.
2. Productivity in the trade sector relative to total economy, in per cent. Trade sector includes wholesale, retail trade and repairs for all countries except for Japan and the United Kingdom for which restaurants and hotels are also included.

Source: OFS; Statistics Austria; OECD, Services Statistics on value added and employment and STAN database.

companies in recent years.[50] According to Kalirajan (2000), foreign firms encounter greater impediments to market entry than domestic firms. Since 1997, however, some of the obstacles to foreign firms' purchasing land have been lifted, although there are still restrictions as to the nationality and place of residence of the members of company boards. All in all, growing concentration around a food distribution duopoly and barriers to entry are hardly conducive to efficiency as it jeopardises competition.

Vertical agreements are prejudicial to competition in non-food distribution

Regulations concerning shop-opening hours also affect competition. They vary from canton to canton. In many regions regulations are quite strict compared with other countries, though in some, such as Zurich, opening hours have been liberalised or are not regulated anymore.[51] Moreover, some non-food outlets show fairly homogeneous prices, which would not seem related to strong competition but could rather result from producers making excessive use of recommended prices.[52] COMCO monitors pricing, particularly in the area of consumer electronics. The most serious distribution problem, however, relates to the high prices practised by importers and wholesalers (BAK, 2001). Vertical agreements can sometimes be used by foreign producers to segment the Swiss market from foreign markets. In these agreements, Swiss distributors and consumers are forced to purchase their goods at higher prices from approved dealers since they cannot

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contract directly with foreign distributors with lower prices (Zäch, 2002). This contributes to consumer tourism in border regions.

COMCO regulations, which since February 2002 have restricted the legality of vertical agreements in general, and those in the automobile sector more specifically, are designed to stop price discrimination. Since November 2002, for example, the competition principles applicable to automobile sales are based on the new European regulations in this area (EC 1400/2002). Whereas under the previous regime, suppliers could select dealers and allow them exclusive rights for a given region, the new regulations oblige the automobile industry to authorise parallel imports from the EEA. They also make it obligatory to choose between a selective and an exclusive distribution system, permit multiple brand selling and separate sales from after-sales service.[53] Although the regulations could have been changed more radically, given the existing limitations prevailing at the EU level, the recent reform is a step in the right direction. A transition period for the existing agreements is scheduled up to 2005. The regulation, by encouraging imports, should reduce prices of new cars, which are generally higher than in the rest of Europe even though VAT is lower (Figure 18). After publishing a communication on vertical agreements COMCO announced that it was carrying out numerous investigations in various sectors such as car distribution, farm tractors, cosmetics and electronic goods (Flückiger, 2003). In some cases, it has been possible to reach an amicable agreement with companies, eliminating the contentious clauses. However, cartel-like behaviour does seem to persist in some sectors, for instance, in the book market, which is largely supplied by imports and where two distributors hold a dominant position respectively in the French-speaking and German-speaking part of the country and can impose prices between 15 and 30 per cent higher than in Germany and France.[54]

The possibility to block parallel imports of products protected by patents is one of the reasons for price differentials.[55] A potential for arbitrage mainly exists with regard to pharmaceuticals and certain consumer durables (computers, household appliances and consumer electronics, watches and vehicles).[56] Shifting to the international exhaustion regime for patent law, such that patent law could no longer be invoked to block parallel imports has been the subject of a lot of controversy for several years (Box 4). Recently, however, the Federal Council did reject such a change. This decision was based on the results of two studies ordered by the authorities to guide their decision (Federal Council, 2002). These studies show, *first*, that liberalising parallel imports would have positive economic effects on Switzerland. However, the empirical results suggest that the gains would not exceed 0.1 per cent of GDP, despite the prices of parallel imports falling by between 14 and 32 per cent in the case of drugs and between 4 and 8 per cent for consumer goods. According to the Government's report to Parliament, this was considered a small gain. *Second*, it is also noted that introducing international exhaustion could cause difficulties in the drugs market regarding the preservation

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Box 4. Advantages and drawbacks of applying the international exhaustion principle to patent law

OECD countries have adopted different policies with respect to parallel imports: the United States applies the national exhaustion rule, while the EU, as a customs union, allows parallel imports between member countries, but not from third countries ("regional exhaustion"). A few countries such as New Zealand have adopted the international exhaustion principle while countries like Japan apply still another solution ("implied licence"), with some similarities with the international exhaustion principle.[1] As was observed in a recent report addressed by the Federal Council to Parliament (Federal Council, 2002), the question of the authorisation of parallel imports and the exhaustion of patent rights is complex since it affects competition and innovation policy, and also has to take account of the international context.

Economic theory is inconclusive concerning the welfare impact of bans on parallel imports, but it is admitted that, under certain assumptions, price discrimination may have positive economic effects. For instance, Malueg and Schwartz (1994) argue for a banning of parallel imports on the grounds that perfect price discrimination would result in net global output expansion and raise global welfare; it would notably ensure that low-price markets are provided with goods. This includes notably developing countries' markets, which in case of application of international exhaustion would possibly not be delivered anymore. Second, as long as producers maintain or are obliged to maintain price differentiation if parallel imports are authorised, parallel importers are exploiting the arbitrage opportunity and free-ride on the costs of research and development that patent holders incur. Third, high entry costs together with rapidly decreasing margins from parallel imports due to country size are likely to limit the number of parallel importers in a small country; the resulting oligopolistic market structure could leave little benefit for the customer.[2] Finally, any change in the intellectual property right system on a larger scale has to take the overall regulatory framework into consideration, in particular the balance between innovation and technology diffusion. The introduction of the international exhaustion principle carries the danger of shifting the attention of producers away from competing with innovations.

This assessment does, however, prompt a number of comments. *First*, since economic theory is inconclusive concerning the macroeconomic effects of authorising parallel imports, it is important to consider pertinent empirical work. A number of empirical studies do indicate that liberalising parallel imports has a positive impact, as suggested by the recent study ordered by the Federal Council concerning Switzerland. However, this positive macroeconomic impact seems bound to be small, as is true of any measure affecting competition only in a small part of the economy. Yet to reject this type of measure on the basis of a case-by-case evaluation runs counter to an overall strategy of stepping up competition, which is what Switzerland needs. *Second*, the main factors that make Switzerland attractive for research activity (the availability of skilled labour and quality infrastructure) would not be affected by opening up to parallel imports. Moreover, the latter would have only a very limited impact on the activity of firms carrying out their research in Switzerland, the bulk of them operating in the world market.[3]

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Box 4. **Advantages and drawbacks of applying the international exhaustion principle to patent law** (*cont.*)

T*hird*, it is the drugs market regulations that appear to be responsible for most of the big drug price differentials with other countries. Ensuring competition between these regulations could prompt reforms aimed at reducing their undesirable effects in terms of economic efficiency. F*inally*, as noted in the report ordered by the Federal Council, authorising parallel imports could be compatible with low-price selling of patented drugs to poor countries thanks to the introduction of vertical agreements including, if necessary, specific labelling of drugs to prevent them being re-exported to developed countries.

All told, if it were not possible to apply the international exhaustion principle for the reasons highlighted above, an interesting alternative that has also been suggested in the Federal Council's report would be to commence negotiations with the European Union with a view to adopting the EU principle of regional exhaustion.

1. For legal aspects, see Strauss and Katzenberger (2002).
2. This is confirmed by the case of the Swiss Trademark Law where the principle of international exhaustion applies but where the prices of many branded products are still comparatively very high.
3. The turnover in Switzerland of the big Swiss pharmaceutical companies accounts for only about 1 per cent of their world sales.

of medical safety standards. T*hird*, lifting the ban on parallel imports of patent protected products would lead to competition between different countries' regulations, thereby calling into question the choices that countries make with regard to the provision of pharmaceuticals. On the other hand, keeping a national exhaustion system makes it possible, in principle, to reduce the cost of drugs going to poor countries – as the Swiss authorities want – without harming the pharmaceuticals industry in the case of re-exports to the developed countries. L*ast*, the Government's reports to Parliament argue that authorising parallel imports would mean that less attention is paid to protecting patents, which could discourage research-intensive firms from setting up in Switzerland. That said, the Federal Council advocated using the law on cartels as a corrective if the patent holder makes improper use of the right to control imports. According to the Federal Court, this is the case if there are big price differences when a product is first put in circulation between Switzerland and other countries of similar economic and legal conditions. As COMCO observes, however, this particular legal framework only allows to enforce parallel imports on a case-by-case basis. Moreover, several years may elapse between the time when a complaint is

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received and the entry into force of a decision, with the result that the corrective instruments contained in the law on cartels are limited by comparison with the adoption of the principle of international exhaustion of patents (COMCO, 2003).

The reform of professional services requires a bolder and more comprehensive approach

Regulation of professional services is widespread in OECD countries as a means to solve consumer protection concerns or ensure a high level of professional aptitude. Such intervention is sometimes explained by the need to correct market failures, which are mostly due to information and transaction costs (OECD, 2002c). Regulatory tools vary, and include, for instance, compulsory certificates to exercise an activity, special permits to work in a given area, fixed or recommended prices etc. In many cases, however, regulations are not clearly justified by market failures and are in practice entry barriers to new professionals that limit consumer choice and reduce competition, thus pushing up prices and impeding an efficient resource allocation. In Switzerland, regulations are not only widespread but they are also in many cases the competence of cantons, resulting in segmented markets for most professions and welfare losses due to the lack of a true internal market.

The Domestic Market Act (DMA) put in force in 1996 attempted to reduce this segmentation by imposing general guidelines for cantonal regulations affecting professional activities. Specifically, the Act ruled that cantonal regulations should not discriminate against professionals of other cantons, and provided a transitional period of two years to adapt cantonal legislation that might limit free market access. However, as recognised by a Parliamentary report in 2000 (OPCA, 2000b; SECO, 2002), the Act has barely improved the situation. Although some progress has been made in certificate recognition across cantons, authorisations to exercise are still widespread, and are sometimes subject to onerous additional conditions (solvency, insurance, etc.). The DMA is a framework law, which does not foresee sector specific actions to liberalise the internal market. And it does not provide for enforcement actions against cantons which do not adapt their laws to the new principles, as it relies on individuals appealing before the tribunals. The COMCO plays only an advisory role to cantons whenever they want to consult it for writing or modifying their laws. Up to now, however, consultations have taken place in very few cases. Moreover, the COMCO is not allowed to appeal to the tribunals *ex officio* against restrictions of market access, and there are few private complaints about cantonal regulations since procedures are slow and costly, with very uncertain results. Indeed, the Federal Tribunal has given a federalist interpretation of the DMA, defending cantonal regulations when they restrict competition on the grounds that the law allows them to do so when essential public interests must be preserved. In particular, the Federal Tribunal has not recognised explicitly the freedom of establishment in the definition of free market access, and it has validated

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cantonal decisions that rejected foreign certificates for exercising a profession in a given canton even if they were recognised by other cantonal governments.

All this is reflected in effective entry barriers in several professions. Improving competition in many of these would likely have a positive impact on aggregate productivity and potential GDP growth, especially because they often concern highly-skilled workers.[57] In the case of doctors, for instance, there are strong barriers for exercising the profession and tariffs are fixed and negotiated at the cantonal level, which results in a wide price dispersion across the cantons. Paramedical professions also suffer from strong market segmentation. In the case of access to the legal profession, harmonisation across cantons has been achieved for lawyers through a federal law that took effect in June 2002, but entry to the profession continues to be restricted through a period of practice required for exercising the profession on top of holding a degree. Notaries, on the other hand, escape the application of the DMA since they are public servants, but suffer nonetheless from a segmented market since cantons may follow one of three different methods to organise the profession. In the case of architects and engineers, recommended prices are published every year by the professional association (SIA), and segmentation is still prevalent, although harmonisation at the federal level is under discussion. Other sectors, like taxi transport and restaurants and hotels, are strongly segmented with authorisations often only valid in the cantonal area. For crafts and arts, a report of the SECO found that not one of the 15 different categories in this profession is harmonised across the 26 cantons.

Although some harmonisation is possible through special federal laws for some professions, like the one for lawyers, the pace of harmonisation is slow. Inter-cantonal agreements are also slow to emerge and only apply to those cantons which voluntarily adhere to them. The alternative would be to reform the DMA. Given its lack of results due to its broad scope and lack of effective instruments, a new and more ambitious law should be approved that can be applied to all professions, based on the principle of free market access, which should be clearly defined and explicitly include freedom of establishment. In order for the law to be effective, cantons should be obliged to adapt their legislation to the new DMA, and the COMCO (or a new institution especially created for this purpose) should be given a clear role to supervise its implementation. Specifically, it should be able to intervene in the tribunals when it considers that cantonal legislation does not comply with the DMA. Additionally, initiatives should be taken to identify and harmonise those technical regulations in different sectors that vary across cantons or municipalities and are not clearly justified by specific local conditions.

The service sector should be opened to greater foreign competition

The lack of competition in the service sector, stemming in part from the segmentation of the domestic market, is heightened by the weakness of external

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competitive pressures. It is essential that the service sector, which accounts for two-thirds of GDP, becomes more open so as to stimulate productivity and economic growth. Freedom to provide services is thwarted by restrictions on labour market access and measures that are discriminatory in relation to national treatment. These include requirements involving commercial representation, nationality and authorisation to do business. The bilateral agreements concluded with the European Union in 1999 improve the situation. The agreement on the free movement of persons, which entered into force in June 2002, has made it easier to recruit European foreign employees. It also provides for the liberalisation of cross-border services provided by persons for up to 90 days per year, thereby enabling a foreign firm to work in Switzerland with its employees for that amount of time. But liberalisation is still only partial. For example, the current regulations do not allow EU life insurers to be present in the Swiss market on the same terms as domestic companies (and *vice versa*), as they are required to have an office in Switzerland and make a deposit (COMCO, 2001). In contrast, within the European Union, member country life insurance companies enjoy freedom of establishment and may freely offer their services beyond their borders without having to set up a subsidiary or a branch. Discussions have been initiated in connection with the second round of bilateral negotiations with the European Union to extend these provisions to Switzerland and to authorise reciprocal and non-discriminatory access to service markets. However, difficulties are preventing these negotiations from moving forward. For the European Union, a liberalisation agreement on services that would give competitors equal treatment must include the adoption of back-up policies. For this, a number of problems remain unresolved, including, for example, the liberalisation of the electricity market, as well as changes to competition and company law. Given these circumstances, the chances of reaching a liberalisation agreement in the service sector any time soon look rather slim, although efforts should continue to be made to push policy in this direction.

Public sector issues

As mentioned above, competition pressures are lacking in key areas of the public sector. This section analyses competition issues in public procurement and in the health sector, which together account for 20 per cent GDP. It also looks at the agricultural sector, by far the most subsidised sector of the economy.

Competition for procurement contracts must be strengthened further

The procedures for awarding procurement contracts play an important role in competition. These contracts, more than 80 per cent of which are awarded by the cantons and communes, accounted for 25 per cent of public expenditure in 2000, *i.e.* about 10 per cent of GDP, which seems low by international comparison.[58] In the last few years, the authorities have made these contracts more open to competition. The legislation governing tender procedures has been

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brought in line with international law (WTO agreement and bilateral agreement with the European Union), thereby increasing transparency requirements. Since 1996, for example, the contracting authorities have not been able to award contracts above certain thresholds without submitting them to competition from firms outside the region or foreign firms. Efforts have also been made to avoid bid-rigging between local firms, which appears widespread according to the competition authorities because the present legislation on cartels is not much of a deterrent. COMCO has stepped up its surveillance and in late 2001, for the first time, banned a bidding cartel.

A lack of reliable statistics makes it difficult, however, to quantify the positive effect of these measures. Comparing identical tenders in the Confederation between 2000 and 2002 points to appreciable reductions in prices – of close to 25 per cent on average. The number of bidders at the cantonal level has also increased under the new legislation. That said, even though the mere threat of entry, rather than actual entry in the market, from firms outside the region or the country may stimulate competition, the procurement market seems to have remained fairly closed, especially at the regional and local level. Five years after the entry into force of the new legislation, over 80 per cent of the contracts awarded in the cantons of Geneva, Vaud and Graubünden, for example, were still going to local firms, with only few contracts awarded to foreign firms. In the case of the Confederation, the foreign penetration rate of government contracts, put at between 13 and 24 per cent depending on the source, also looks low (Zogg and Duperrut, 2002).

There are a number of reasons for contracts being insufficiently open. *First*, the bulk of contracts is awarded at below the threshold values for open public tenders. *Second*, procurement contract legislation is not harmonised, either between the Confederation and the cantons[59] or between the cantons themselves, despite the existence of inter-cantonal co-ordination, and is very complex, which makes it difficult to apply for both bidders and contracting authorities.[60] The interaction between and overlapping of the norms governing procurement contracts accentuate the legal uncertainty, which hardly encourages bidders to appeal in the event of a procedure being disputed.[61] Yet proper application of the legislation rests largely on the right of appeal, the use of which is also curbed by the high costs of the legal procedures (Stöckli and Zufferey, 2002). *Third*, when international thresholds are not met, the lower thresholds above which competition applies at national level have hitherto appreciably differed across cantons. All in all, the differences in rules, thresholds and procedures have an adverse effect on the transparency required for effective competition.

The authorities have recently started to tackle the obstacles to competition stemming from the fragmented nature of public procurement legislation. The cantons and communes revised the Inter-cantonal Agreement on Public Procurement (AIMP) with the objective of harmonising thresholds and procedures with respect to contracts

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not subject to international treaties. The revised version of the AIMP was in force in only four cantons at the start of 2003, but should be introduced in the others by 2004. The Confederation, for its part, intends to clarify and simplify federal law on procurement contracts and bring it into line with cantonal legislation. Efforts are also in progress to improve the application of procurement procedures by the federal purchasing services (Vogt, 2002) and a database is being developed in order to enhance understanding of trends in procurement contracts and assess both requirements and the impact of future reforms. Clarification is also needed regarding the conditions in which state-owned companies tender for contracts without distorting competition. As in other countries, private firms are subject to taxation from which the public sector is exempt when it itself provides for its goods and services, which undermines sub-contracting and could well explain the lower share of procurement contracts than in other countries.

In late 2002, a "procurement contract" Internet portal was launched to pool the cantons' and the Confederation's calls for tender and to unify the bidding system. The site will help to increase the transparency of the market which has some 5 500 public contractors, over 98 per cent of whom operate at the cantonal and communal level. The system should encourage scale economies on standard procurement contracts and a reduction in the management costs of tendering, which should lead to lower threshold values. Rational use of public money requires that the thresholds be lowered as the transaction costs of public tendering decline. At present, many contracts fail to reach these thresholds, with the result that procurement takes place by mutual agreement or invitation. These procedures are moreover liable to be used more and more frequently by the cantons and communes because, although the AIMP has harmonised threshold values for contracts not subject to international treaties, the values have been substantially raised.[62] Moreover, for construction, the AIMP now makes the choice of procedure dependent not on the value of all the contracts needed for the work, but on the value of each contract, with a distinction being made between the shell of a building and remaining building works (Zufferey *et al.* 2002). There is thus the risk of a splitting up of contracts limiting the application of competitive procedures. Yet it is important that safeguards be put in place against discrimination in favour of local firms. The latter are traditionally preferred by the cantonal and communal authorities on grounds of proximity, but sometimes on the basis of domicile too,[63] which accentuates the fragmentation of the domestic market.

The health sector is suffering from regulatory problems

The health sector, which accounted for almost 11 per cent of GDP in 2001, is notable for very high prices in international comparison and for being heavily regulated, even if it shares this latter feature with most other OECD countries (Figure 29). Demand for medical services is channelled through an insurance

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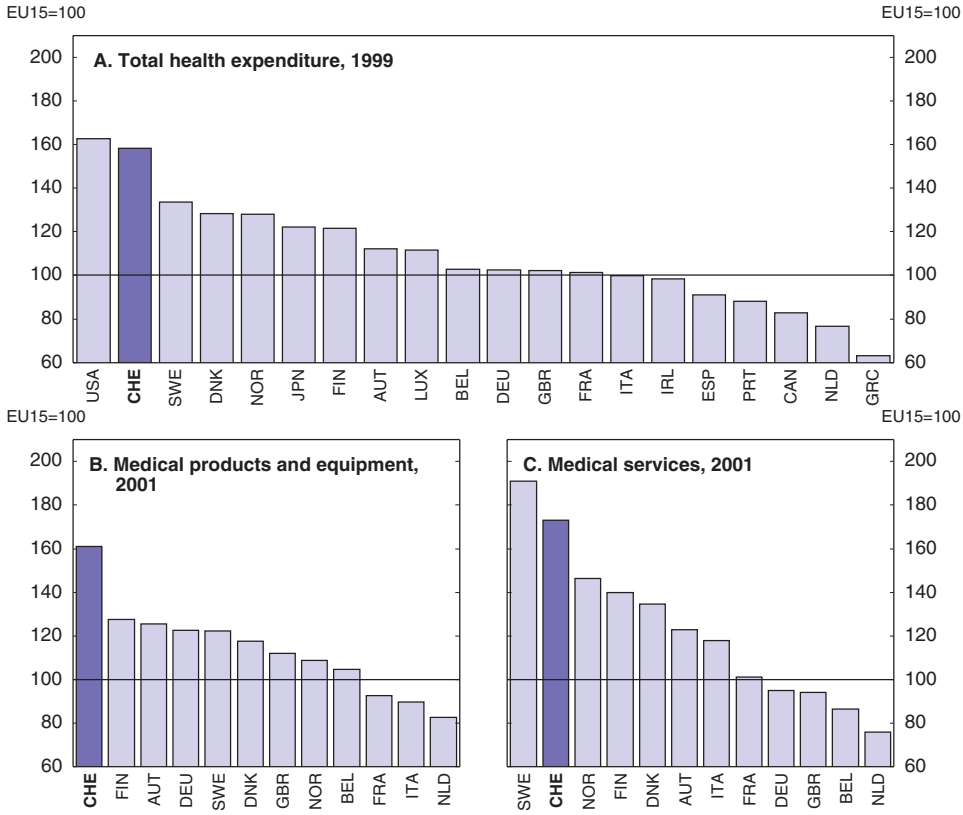
Figure 29. **Relative health price levels**



Source: Eurostat and OECD.

system that guarantees individual access to care, *i.e.* limits rationing and ensures solidarity between the insured. Restraining spending and efforts to improve efficiency therefore depend, on the one hand, on the possibilities and incentives of insurers to negotiate advantageous conditions and prices and, on the other, on the incentives that providers have to improve their efficiency. Neither prices nor supply are controlled in the current regulatory environment. In the case of drugs, for example, this is resulting in excessive growth of expenditure (Chapter II and Annex III). As economic factors (salary, interest rate levels and hours worked) and the demographic and geographical features of the market account for no more than 3 percentage points of the price differential with Germany, the United Kingdom, France and the Netherlands, which ranges from 13 to 38 per cent (Infras/Basys,

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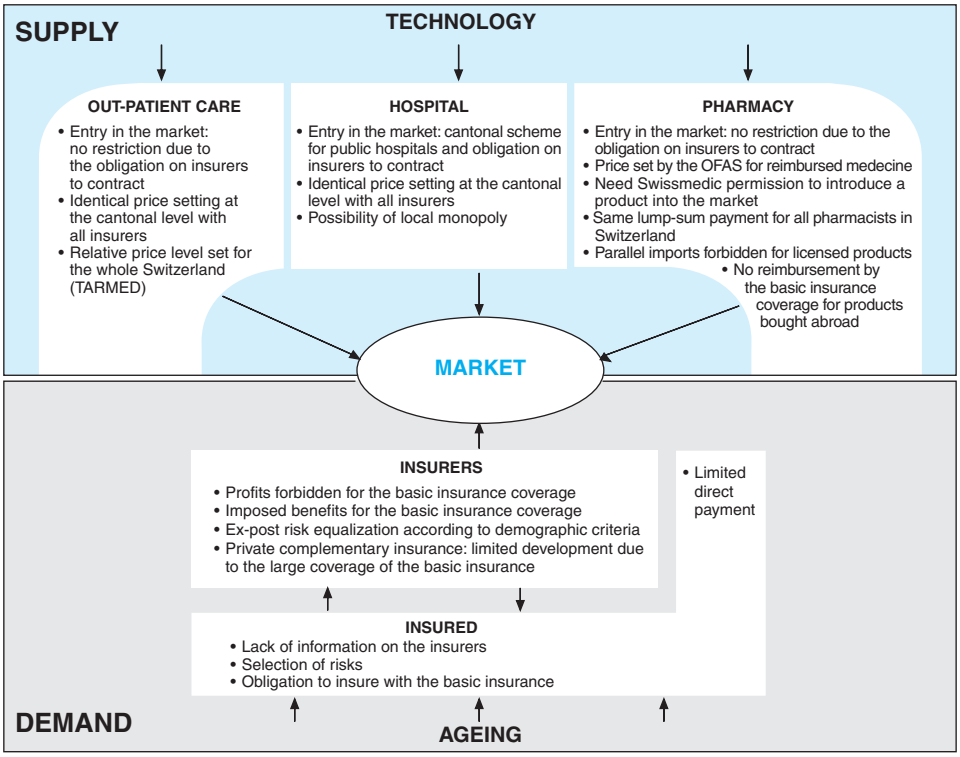
Figure 30. **Main impediments to competition in the health care and insurance markets**



Source: OECD.

2002), the remainder of the differential provides suppliers and distributors with a rent to the detriment of the insured.

There is no effective competition in basic healthcare insurance…

Effective competition between insurers is important to increase their incentives to negotiate advantageous conditions and prices with health providers. However, the compulsory basic health care insurance market is affected by regulatory problems, partly due to the need to promote solidarity (Figure 30). Despite the large differences in premiums for the basic standard contracts proposed by numerous private insurance funds, few people change their insurer, which is hardly conducive to a reduction in the differences – in contrast with trends in Germany (OECD, 2003c). The weakness of market mechanisms is partly

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due to a problem of keeping the insured informed (Colombo, 2001), but it also reflects a tendency for insurers to select risks. It would seem that bad risks have difficulty switching across funds because the latter can select them, for example by indirectly influencing their decisions to join or to leave an insurer.[64] The risk equalisation system designed to prevent such selection needs to be improved. Differences in the state of health of the insured are hard to measure on the basis of solely socio-demographic criteria currently used for equalisation. Moreover, the latter is retroactive, which hardly encourages efficiency since cost overruns in under-performing funds are partly passed on to other insurers (Spycher, 2002). The supply of products with lower cost, like healthcare networks (HMOs), is also penalised by this system, while demand is limited because the reductions in premiums awarded to households for joining HMOs are capped (at 20 per cent for the first 5 years). Lastly, the fact that profits are not allowed to be made on basic insurance curbs incentives to be efficient and innovative. Given this context, risk selection is no doubt the best strategy for insurers if they are to attract the best customers as a way of increasing their gains on the complementary insurance market.

… or between providers of medical goods and services

Competition between healthcare providers is almost non-existent because of the obligation on insurers to contract, which prevents any control over supply. All the insurers in a given canton have to accept fees from all the hospitals and private practitioners working for the basic insurance system. Moreover, the same prices, negotiated in each canton between associations of insurers and providers, apply *de facto* to all insurance funds. Since the 1996 reform, however, insurers and service providers have been legally able to sign agreements that differ from those concluded between their associations. Until now, though, very little use has been made of that possibility, with the result that price agreements are governed *de facto* by cartels (*Office fédéral des assurances sociales* or OFAS, 2001). Providers are thus not very interested in decentralised negotiations aimed, for example, at developing healthcare networks, while the cantons, which ultimately control hospital prices, seek to deal on an equal footing with the establishments that they subsidise. The end of the obligation to contract in the ambulatory sector could unblock the situation and introduce more competition by encouraging decentralised price negotiations. For that, this reform should affect in a similar way all health providers, those already in the market like those newly trained, who should be submitted to the same threat of no longer being able to practice, rather than favouring *de facto* the incumbents. In the ambulatory sector, increased price flexibility could be coupled with a reform of the pay system for doctors. They are at present being paid per item of treatment, which pushes supply. There should be more room for a system of standard payments per patient, in conjunction with the development of healthcare networks. This would help to prevent the new price system (*Tarmed*)[65] from becoming even more rigid. However, international

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experience shows that it is difficult to introduce real competition in the hospital sector (OECD, 2003c), the obstacles stemming in particular from the existence of local hospital monopolies and the difficulty for insurers to negotiate prices for medical treatment without having all the necessary information.

The drugs market is also regulated. The system for authorising the marketing of pharmaceuticals, which is state-controlled as in other countries, is complex (Annex III). The number of authorised drugs seems limited by comparison with other countries, and only 3 per cent of them are generic.[66] The financial dependence of the authority responsible for marketing approval *vis-à-vis* pharmaceutical manufacturers has been reduced recently with the creation of S*wissmedic*. Nevertheless, such dependence could still be problematic from the competition point of view (Infras/Basys, 2002). The OFAS, the authority responsible for deciding on medicines reimbursed by the compulsory sickness insurance, which account for nearly 80 per cent of sales,[67] establishes the upper price limits for all service providers in Switzerland. It fixes these maximum prices on the basis of international and internal comparisons. Since the new Law on Sickness Insurance was introduced in 1996, some convergence towards the European price level has become apparent. On the one hand, the prices of reimbursed drugs, which can be re-evaluated after expiry of patents or after fifteen years, have been lowered by OFAS through a decree.[68] On the other hand, the rapid development of the generics market favours also downward pressures on prices.[69] However, drug prices remain often higher than in other countries, the differences being generally due to the past appreciation of the exchange rate. The considerable economic weight of the pharmaceutical industry,[70] which in addition has an advantage in terms of information, seems also to weigh on its position towards OFAS, given the existing obstacles preventing effective competition from imports in this sector. The first such obstacle is the ban on parallel imports for patent-protected products, which make up 60 per cent of the market. Since 2002, the new law on therapeutic products has however authorised imports of drugs whose patents have expired. In 2000, COMCO asked the Federal Council to authorise the reimbursement of drugs and medical products purchased abroad if they are cheaper than in Switzerland and do not pose any public health risk, which is currently not possible with basic insurance. There is no decision yet. The removal of these import barriers would encourage a fall in the prices of pharmaceuticals. The introduction of the principle of prescribing active substances rather than branded products would also make doctors more independent of the pressure exerted by the pharmaceuticals industry, and it would promote the prescription of generic drugs. To that end, it would also be necessary to suppress the possibility granted to doctors by some cantons to deliver directly medical products. Lastly, there is also room for further enhancing the new system of remunerating pharmacists based since July 2001 on a lump sum payment (rather than on margins) set at the same level throughout Switzerland.

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This new system contributed to diminish the price of drugs by 10 per cent on average. The lump sum payment, the introduction of which meant that pharmacists' incomes could be separated from the products prescribed, should be negotiated on a decentralised basis.

Support to agriculture remains very high

The agricultural sector, which accounts for roughly 1½ per cent of GDP and 3 per cent of employment, is heavily subsidised, while stiff foreign trade barriers hinder competition from abroad, pushing prices far above world levels and locking resources into a sector where Switzerland does often not a have a comparative advantage. The producer support estimate (PSE), the OECD's standard measure for overall implicit and explicit subsidisation, was amongst the highest in the OECD at 73 per cent in 2000-02. It has decreased only slightly from 76 per cent since 1986-88 despite continuous reforms of agricultural policy (Figure 31). The PSE estimate includes transfers from consumers through higher prices (with an implicit tax of 62 per cent) and from the budget through direct payments and export subsidies. Instruments include target prices and quotas for milk, which is one of the most important products, deficiency payments for dairy farmers and direct payments for producers of oilseeds and sugar beet. Relatively high import tariffs affect several agro-food products while import quotas are used for products like milk, meat, potatoes, fruits, vegetables, bread grain and wine. Export subsidies support the selling of dairy products (of which more than 50 per cent is cheese), fruits and potatoes. While aid to agriculture has raised household incomes in rural areas, total income of farm households is less than 75 per cent of total income of all households, one of the lowest ratios in the OECD.

The current reform programme is laid out in the *Agricultural Policy* 2002 initiative (AP 2002), which covers 2000-03 and underpins the agricultural policy objectives included in the Constitution that stress the multifunctional nature of the sector. According to the Constitution, agriculture shall contribute substantially, through sustainable and market-oriented production, to food security, to the conservation of natural resources and to a decentralised inhabitation of the country. The AP 2002 provides for the elimination of all guaranteed prices and guaranteed processing margins (already achieved) and a re-organisation of direct payments, now more linked to an environmentally-friendly agricultural management. *General Direct Payments*, which are mainly granted in the form of area and headage payments and of historical entitlements, are now attached to the condition of compliance with environmental farm-management practice requirements, while *Ecological Direct Payments* are based on input constraints and also conditioned on environmental standards. The combined payments of all direct support amounted to CHF 2.45 billion in 2002 (0.6 per cent of GDP), 4.5 per cent more than in 2001. A new package of agricultural policy measures, *Agricultural Policy* 2007, has been

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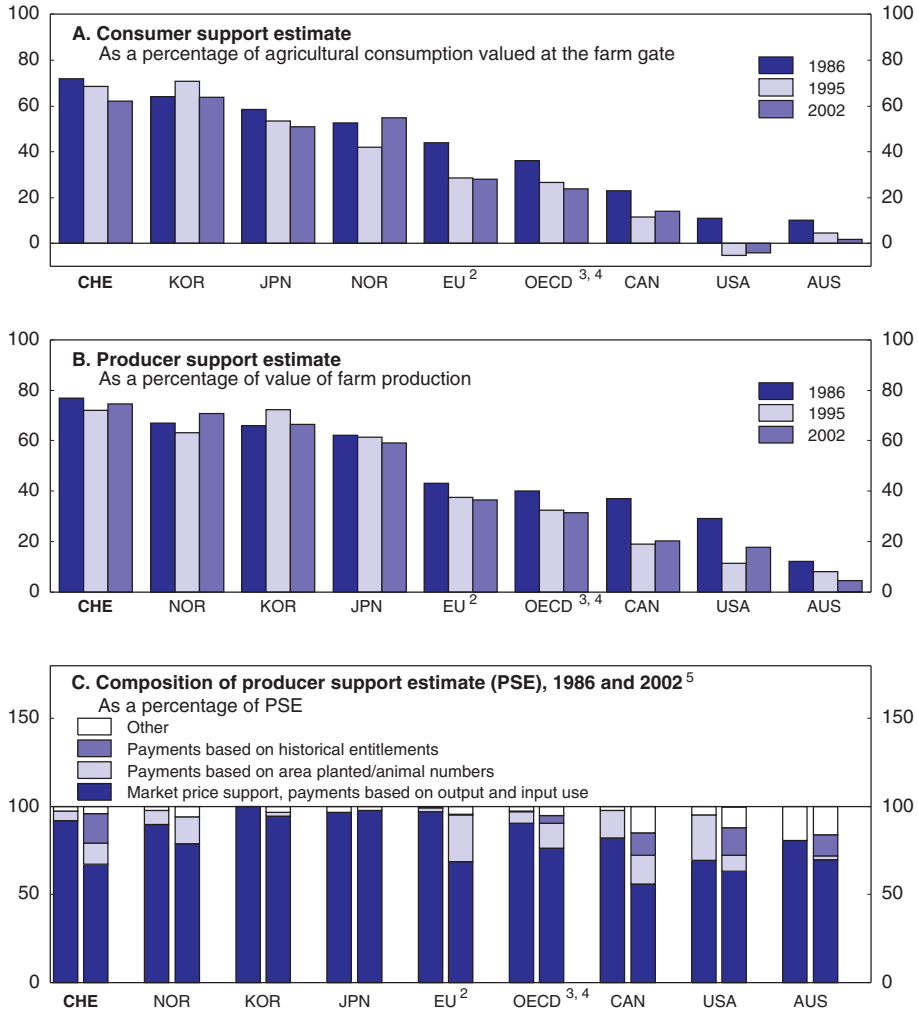
Figure 31. **Producer and consumer support estimates for agriculture**[1]

Per cent



1. For detailed explanations, see Source.
2. EU-12 for 1986, EU-15 from 1995.
3. Austria, Finland and Sweden are included in the OECD totals for all years, and in the EU from 1995.
4. OECD excludes the Czech Republic, Hungary, Poland and the Slovak Republic for 1986.
5. The first bar represents 1986 and the second 2002.

Source: OECD, *Agricultural Policies in OECD Countries – Monitoring and Evaluation*, 2003, OECD, Paris.

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approved, covering the period 2004-07 and including as key points the abolition of the milk quota system and measures to enhance rural development. Beef and pig meat import quotas should be also auctioned as from 2007.

These policy initiatives have contributed to shift support to more market friendly instruments. Since 1986-88, the combined share of market price support, output and input payments has been reduced from 91 per cent to 68 per cent of total support. This reduction in market protection has been reflected in a decrease of the price gap between world and domestic prices, although this gap stands still on average at around 200 per cent. The total level of support is still very high and has been reduced by little in recent years, while some types of direct payments have even increased in 2002. Even if the conditioning of aid to environmental goals has helped to improve environmental outcomes, continued support linked to the quantity of output or inputs encourage production and therefore delay the reallocation of resources to much more productive activities. In this respect, the process of decoupling of subsidies from production should be accelerated. The environmental benefits provided by agriculture as a public service, like a pleasing country side and the contribution to biodiversity, should be clearly identified and if possible quantified, and provided directly rather than through policies that stimulate output (OECD, 2002e). Similarly, income support in rural areas should be provided with instruments that concentrate on poorer households rather than general measures that apply to all farmers. In this sense, incentives to reallocate employment to alternative activities could help to raise incomes in rural areas and reduce future support.

General assessment and the need for further reforms

Realising the need to strengthen competition to stimulate economic growth, which has been sluggish, the authorities have launched a number of reforms. Liberalisation measures are scheduled in certain sectors, such as telecommunications and postal services. The recent revision of the Cartel Act, which provides for sanctions against anti-competitive behaviour without waiting for a second offence, crucially increases the deterrent effect of the legislation. This is an important step in the right direction. These enacted or planned reforms are, however, in many cases no more than a belated and partial catch-up on regulatory reforms in other countries, and further progress is essential. The limitation of sanctions to only certain types of illegal acts, while the most damaging ones, shows that there remains a potential to develop the Swiss competition culture further. A more broad-ranging and ambitious strategy of reform in the goods and services market is needed to give fresh impetus to the economy. It should be three-pronged and inspired by the recommendations in this report, which are spelt out in Table 19.

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Table 19. Recommendations aimed at increasing competition in the goods and services markets

A number of reforms or draft reforms aimed at strengthening competition have been launched in the last few years. In many cases, however, they were no more than a belated and partial catch-up with other countries, and further progress is essential. As the analysis contained in this report suggests, an ambitious, three-pronged reform strategy should be implemented. It is important to ensure that competition law, which could be further strengthened, is strictly applied, that competitive pressure in the sheltered sectors will increase, while certain industries should become more exposed to foreign competition.

Reform and application of competition law

- Apply the Cartel Law in a rigorous, effective and comprehensive way.
- Ensure the economic and political independence of the COMCO by eliminating members that represent economic interests.
- Increase COMCO's financial resources and personnel.
- Envisage to co-operate internationally in competition matters.
- Change the constitutional clauses that hinder the Cartel Law to apply the prohibition principle for cartels during the next revision of the Constitution.
- Consider introducing criminal sanctions as a means of punishing people responsible for the most serious anti-competitive behaviour.

Strengthening competition in the sheltered sector

Health sector

- Improve and centralise the information services about insurance funds; encourage a simpler and more homogeneous presentation of comparisons between premiums.
- Extend the risk equalisation system beyond 2005; perform the relevant calculations at least partly in a forward-looking manner and include in the system criteria linked to health risks.
- Consider allowing profits to be made in the basic insurance market.
- Abolish the obligation to contract in the ambulatory and pharmaceutical sector; increase price flexibility by means of decentralised negotiations between suppliers and insurers.
- Introduce a prescription system based on active substances rather than branded products, which should lead to greater competition on a more transparent basis.
- Remove the obstacles to effective foreign competition on the drugs market.
- Reduce the influence of the pharmaceuticals industry on authorisations for drugs to be placed on the market and on price setting with respect to reimbursed drugs. Review the way Swissmedic is financed. Include competition experts in price negotiations between OFAS and drugs manufacturers.

Procurement contracts

- Continue efforts to ensure that the existing regulations are better applied; use the potential of the new Internet site, sitemap.ch, to increase competition and lower the threshold values necessitating open public tenders.
- Limit the possibilities for splitting contracts and so avoiding competitive procedures.
- Improve the legal protection afforded by appeals against the use of non-competitive procedures to award procurement contracts.

Revision of the Domestic Market Act

- Include a specific reference to freedom of establishment in all the cantons.
- Authorise COMCO to appeal to the courts in the event of restrictive practices.

Network industries

- In telecommunications, un-bundle local loops without further delay; take care to prevent any horizontal agreements in mobile telephony.
- In the rail industry, create a network regulator independent of the two main rail companies; introduce systematic tendering for running regional passenger lines, which receive public subsidies.
- Accelerate the pace at which the postal services are being opened up to competition; create an independent Post Office regulator to ensure that the financing of the universal service does not interfere with competition in other parts of the market.

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Table 19. **Recommendations aimed at increasing competition in the goods and services markets** (*cont.*)

– In the electricity and gas sectors, liberalise the markets – including for households – in a manner compatible with the reform in the European Union; launch the reform of the gas industry now in conjunction with that of the electricity industry; reduce political interference by cantonal and local authorities in the running of the public gas and electricity companies; create strong and independent regulators guaranteeing equitable access to the market; separate environmentally friendly measures in favour of renewable energy sources from the reform of the electricity sector.

Opening up to foreign competition

– Reduce the protection and excessive assistance enjoyed by agriculture and accelerate the de-linking of subsidies from production. Clearly identify the environmental objectives, which need to be pursued directly rather than using them as an excuse for persisting with very high aid to agriculture.
– Start negotiations with the European Union quickly aimed at adopting the principle of regional exhaustion of patent rights, as is in place in the Union.
– Speed up bilateral negotiations with the European Union concerning the opening up of the service sector to foreign competition.
– Remove the barriers that still make it difficult for foreign competitors to access distribution; reduce the obstacles to building hypermarkets.

Source: OECD.

The first part of the strategy should focus on the *competition law framework*. It is necessary to ensure that competition law is rigorously applied, given the new possibilities provided by the recent reform. Since 2002, the COMCO has provided clear signals that the competition law will be applied strictly, which contrasts with the earlier more timid approach, notably when it comes to vertical restrictions. The authorities should pursue this tough approach. To succeed and to boost competition significantly, it needs to be given adequate financial resources and personnel. Their investigative capacity will have to be reinforced, the independence of its members better guaranteed by moving away from a purely militia system towards a professional system allowing the members of the COMCO, in particular the presidency, to occupy their function on a full time basis. Criminal sanctions should also be introduced to broaden the range of instruments if the existing instruments do not deter enough. Moreover, a constitutional modification which would apply the prohibition principle for cartels, would send a strong and clear signal in favour of a pro-competition climate.

As part two of the strategy, measures should be to taken to raise incentives for greater efficiency in the *sheltered sector*. Regulatory reform is especially necessary in the *health* sector, where abolition of the obligation on insurers to contract in the ambulatory and pharmaceutical sectors, plus the dismantling of healthcare price-setting cartels, would probably reduce the level of medical costs and keep a tighter rein on volume growth over the medium term. Competition in the drugs market would also be stimulated if doctors were to prescribe active substances rather than branded products and if policy with

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regard to bringing pharmaceuticals to the market were to be made less restrictive. The authorities also need to pursue their efforts to open *procurement contracts* up to competition, particularly at the cantonal and local level, by encouraging a reduction in the threshold values requiring an open public tender and by reducing the possibilities for splitting up contracts and so limiting the application of competitive procedures. Strengthening competition in the sheltered sector also means removing some of the divisions in the cantonal markets, which would be encouraged by a *reform of the* Domestic Market Act. It is important in this context to go much further than enacting a framework law. What Switzerland needs are the regulatory foundations that will provide a genuinely unified domestic market by liberalising the professional service markets, with enforcement of the freedom of establishment in all the cantons. Also the role of the COMCO needs to be strengthened so that it can appeal to the courts in the event of restrictive practices and, secondly, it should check systematically cantonal laws and insist that they are in line with the principles of the Domestic Market Act.

The reform underway in the *network industries*, which concerns only certain sectors, is heading in the right direction but does not go as far as reforms in other OECD countries. It also needs to be extended to other sectors. In telecommunications the unbundling of local loops scheduled by the federal authorities should be implemented without further delay. In the railway sector, a regulatory authority independent of the two main rail companies needs to be set up, while the obstacles to competition to run regional passenger lines – which are subsidised by the government – need to be removed. The rate at which postal services are being opened up to competition is still too slow and the creation of an independent regulator would help to ensure that universal service financing does not hamper competition. Rapid and complete liberalisation of the electricity and gas markets would also be beneficial. In the case of electricity, the reform should include the uncontroversial parts of the previous plan, such as the separation of electricity generation, transmission and distribution. Strong and independent regulators guaranteeing equitable access to the market are needed in both the electricity and the gas industries. Moreover, the management of public companies in these sectors ought not to be the subject of political interference by the cantonal and local authorities.

The third pillar of this reform strategy should focus on making certain industries more open to *foreign competition*. The first step must be to reduce the protection and excessive assistance enjoyed by *agriculture* and to speed up the de-linking of subsidies from production. Also steps to further facilitate *parallel imports* would be welcome, for instance in the area of patent-protected products, including drugs. In this connection, the new regulations authorising such imports in the automobile industry do constitute progress. Opening the *service sector* up to competition within the framework of a bilateral agreement with the European Union should also remain a priority and the negotiations now in progress should

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move on more quickly. These reforms, coupled with an effective campaign by COMCO against potentially damageable effects of vertical agreements, would strengthen the competition on the Swiss market. This is likely to result in lower prices in many areas. Such developments would also be helped by a reduction in the number of obstacles to the building of hypermarkets that constitute barriers to entry in the *distributive sector*.

Such reforms are difficult to adopt, however, because they arouse a lot of hostility in the industries that would be most affected. A further reason for hostility is the fear that the negative effects on employment due to restructuring might persist. Recent international studies (Nicoletti *et al.*, 2001) do show, however, that there is ultimately a positive link between regulatory reform and the rate of employment in the business sector. Even though the employment rate is relatively high in Switzerland, there is still room for further improvement notably for female employment (Chapter IV). Moreover, the flexibility of the Swiss labour market encourages a rapid shift of employment to the most productive industries. The strengthening of competition, with the objective of eliminating the rents enjoyed by certain sectors to the detriment of consumers, is needed if the Swiss economy is to extricate itself from the trap in which it has been caught for a number of years. The lack of competition would appear to be largely responsible for the slow growth in the last decades, with inadequate efficiency gains in the sheltered sector and the excessively high cost of living and, consequently, wage costs encouraging investment to relocate abroad.

The positive effect of the proposed reforms should be felt in a number of ways. The strengthening of competition will help to cut prices, profits and reduce wage mark-ups in the industries least exposed to competition, which should increase demand. It will encourage firms to make better use of their factors of production, which will stimulate productivity gains. These static gains, resulting from a more efficient resource allocation, will in addition be boosted by dynamic gains stemming from more intense innovative activity and better use of new technologies, which should be encouraged by a more pro-competitive framework (OECD, 2002f). Because of the complexity of the mechanisms involved and the difficulty of measuring the specific impact of the proposed measures on profits, wages and productivity, it is hard to put a precise number on the gains from reforms. It is possible, though, to calculate an order of magnitude for these effects on the basis of international comparisons and assumptions concerning the efficiency gains and also the reductions in profits and wage mark-ups that could be achieved in certain industries. The OECD Secretariat has carried out such a calculation for agriculture, health care, telecommunications, gas and electricity, and the professional and distributive services, which together account for one-third of total output.

The results of these calculations, which are set out in Table 20, suggest that the strengthening of competition in these sectors could increase the level of

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Table 20. **Assumptions and effects of sectoral deregulation**

	Total economy	Agriculture	Tele-communi-cation	Electricity	Gas	Health	Profes-sional services[1]	Trade	Other sectors
Assumptions (% change)									
Costs of intermediate inputs	–5.3	–12.5	–3.5	–6.2	–3.1	–6.5	–6.5	–8.0	–4.5
Labour costs									
Labour productivity	4.8	30.0	5.0	20.0	20.0	10.0	15.0	15.0	0.0
Wages	–2.1	–5.0	0.0	–10.0	–10.0	–15.0	–5.0	0.0	0.0
Capital costs	–0.1	0.0	0.0	–5.0	–5.0	0.0	0.0	0.0	0.0
Price mark-up[2]	–6.5	–20.0	–5.0	–20.0	–20.0	–25.0	–10.0	–20.0	0.0
Sectoral static effects (in %)									
Direct price effect	–5.9	–16.6	–3.2	–11.0	–9.4	–16.4	–10.6	–11.6	–2.3
Price-induced output effect[3]	2.9	8.3	1.6	5.5	4.7	8.2	5.3	5.8	1.2
Employment effect[4]	–1.5	–16.7	–3.2	–12.1	–12.8	–1.6	–8.4	–8.0	1.2
Estimated long-run macro-economic impact									
Output	8.0								
Domestic demand	6.7								
Employment	1.5								
Real wages	8.0								
Inflation[5]	0.0								
Memorandum item:									
Sectoral value added as a share of total GDP	100.0	3.1	2.5	1.2	0.0	10.7	3.8	12.5	66.2

1. Other business services.
2. Price mark-up is defined as gross operating balance minus depreciation divided by the sum of the cost of intermediate inputs, labour costs and depreciation.
3. Price-induced output effect is estimated on the basis of a price elasticity of demand of –0.5.
4. Sectoral employment effects include impacts of productivity growth and the price-induced output growth, but ignore the possible impact of lower wage premia.
5. Percentage points.

Source: OECD estimates.

output by some 8 per cent over 10 years, the bulk coming from an increase in potential output. That estimate, the details of which are given in Annex V, rests on the assumption of productivity gains and of reduced profits and salaries allowing a fall in prices ranging between 3 and 17 per cent in the sectors analysed, which would narrow the price differential with other countries. On the basis of these assumptions, it is estimated by means of a static analysis that the overall general price decline could reach 6 per cent.[71] The estimate takes into account the reductions in intermediate consumption brought about by the sectoral falls in prices, which are calculated by means of an input-output matrix. The increased competition, mainly expected in the sheltered sector, will indeed also reduce

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costs and strengthen the competitiveness of the exposed sector. The estimated impact of the reforms on production also takes into account the dynamic repercussions on the rest of the economy. For example, the positive effect of productivity gains on potential output will ultimately be accompanied by a rise in employment and in the stock of capital stemming from the permanent reduction in wage marks-up and corporate profits caused by increased competition. Lower prices and efficiency gains ought also to help trigger a virtuous spiral of falling costs in all sectors, higher real incomes and lower interest rates. These effects were estimated by means of a simulation carried out with the OECD's Interlink model. The impact on productivity of increased innovative activity and the use of new technologies was not taken into account, however, and nor were the more broad-ranging effects of an improved general competition policy, which should also have long-term beneficial effects on the economy.

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Notes

1. Price differentials provide only an indirect indication of the degree of competition. For example, a price difference in a given sector may not necessarily be due to a competition problem in the sector, but to the induced effect of high prices in other industries. Price comparisons are also biased by differences of indirect tax across countries. VAT rates are however lower in Switzerland than in most OECD countries.

2. Possible reasons for a high real exchange rate include the relatively strong productivity performance of the export sector and the increased demand for non tradables.

3. On the basis of the estimated relationship between the levels of living standards and the levels of prices among OECD countries, on average Swiss prices should not exceed those of the EU by more than 15 per cent because of the difference in living standards between them.

4. The lack of long-run sectoral data makes it impossible, for example, to calculate margins, wage premia or concentration indices based on output by sector in a manner comparable with other countries.

5. After rejecting EEA membership in 1992, Switzerland found itself in the company of European partners that had to a large extent eliminated technical hindrances to trade amongst themselves. In the framework of the 1990s revitalisation programme, the Federal Council put in place a strategy designed to offset these negative effects. Under the provisions of a 1996 Federal Act, technical regulations have, as far as possible, to be made compatible with those of Switzerland's main trading partners. Harmonisation has fairly well progressed, but is still insufficient *e.g.* with respect to construction products as the reference system in the EU evolves slowly.

6. One possible explanation could be the high level of compulsory spending on non-tradables, for instance health care expenditure.

7. As mentioned in Seco (2002), the degree of exposure to international trade is now higher in Austria than in Switzerland, while the reverse was true ten years ago. During the 1990s, Austria has become a member of the European Union and joined the euro area.

8. According to OECD calculations (OECD, 2002a), the average tariff on imports of agricultural products was 218 per cent in 2000, *i.e.* 3.6 times the level for the EU and 7.7 times that for the United States.

9. Although 72 per cent of drugs are imported, their prices are, on average, appreciably higher than in other countries.

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10. The legality of the vertical agreements protecting exclusive import networks was only called in question in February 2002 for the first time. Although the regulations have recently changed, numerous consumer and investment goods are still being imported via these exclusive networks because of the time it takes to put competing networks in place.

11. According to the OECD, aligning these regulations on those of the least restrictive countries would increase the stock of investment in Switzerland by nearly 35 per cent, *i.e.* three times more than the estimated rise in the OECD average (OECD, 2003a). This positive effect appears to be confirmed by the increase in direct investment in telecommunications in the late 1990s, following liberalisation in that sector.

12. According to the Bureau of Labor Statistics, in 2001 an industrial worker's average hourly wage in USD was higher in Switzerland than in any other OECD country with the exception of Norway and Germany. The difference to the European average was 20 per cent.

13. This phenomenon seems to affect both big companies and SMEs. Examples include Nestlé, a multinational which does its research in Switzerland but has relocated the bulk of its production abroad, and Logitech, initially a SME producing computer products which has successfully adopted much the same strategy of internationalisation (Seco, 2002).

14. A reservation regarding statistics is needed since Switzerland does not yet apply SNA 95. Moreover, the United States, the country with a comparable per capita income at the outset, applies also hedonic pricing in some sectors, with, possibly, a positive impact on overall growth rates. In addition, measuring growth in the service sector is increasingly difficult, and particularly so in the financial sector which in Switzerland – as a share of GDP – is considerably larger than in most other OECD countries.

15. The Acts of 1962 and 1985 only allowed the Cartel Commission to make recommendations to the Federal Department of Economy, which held the decision power.

16. In addition to the Appeals Commission for Competition Matters, there are also the Federal Court, the Federal Council, and the cantonal civil courts; the Federal Communications Commission (ComCom) also acts as a sectoral regulator.

17. Concerning the presence of interest group representatives, the report by the Parliamentary body responsible for control of the executive (OPCA) observes: "There is a risk that COMCO members appointed as representatives of specific interests may favour those interests to the detriment of the interests of maintaining competition" (OPCA, 2000a).

18. Parliament did not adopt the government's initial proposal to strengthen COMCO's independence and professionalism by reducing the number of its members and removing interest group representatives.

19. For example, a Federal Court Judgement of 14 August 2002 referring notably to the principle of abuse, annulled a COMCO decision banning a hard core cartel.

20. COMCO must be notified of business mergers when, during the financial year immediately preceding the merger, the participating firms together had a minimum turnover of CHF 2 billion or a turnover in Switzerland of at least CHF 500 million, and at least two of the participating firms had an individual turnover in Switzerland of at least CHF 100 million. Where insurance undertakings are concerned, instead of turnover it is total gross annual premiums that are taken into account, and in the case of banks and other financial intermediaries, it is gross output.

21. See, for example, Neven (1999) and Flückiger (1999).

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22. Between July 1996 and July 2003, COMCO banned 10 horizontal cartels and concluded 4 mutual agreements relating to horizontal cartels. It handed down 7 bans and concluded 2 mutual agreements relating to abuses of dominant position. Vertical restraints that have the effect of dividing the markets and preventing entry are considered to be one of the main reasons for certain price differentials between Switzerland and its neighbours and a major curb on the competitiveness of SMEs. But COMCO has banned only one vertical agreement in seven years, namely the Citroën case in 2002. See Zäch (2002), NZZ (2001) and Prümmer (2003).
23. In 1995 the Federal Council estimated the number of notifications at 10 to 15 per year, whereas since 1996 COMCO has been receiving between 30 and 35 notifications every year.
24. Since the law came into force, only 7 indirect sanctions have been imposed and there have been no indirect criminal sanctions. The latter concern natural persons responsible because of their decision-making authority in the company.
25. LCart Articles 58-59 concern the application of international agreements. Their objective is to eliminate restraints of competition that originate in Switzerland but take effect abroad.
26. As the OPCA report observes, however: "And yet, given the globalisation of the economy, exchanges of information between Swiss and foreign competition authorities are becoming a necessity" (OPCA, 2000a).
27. However, part of the liberalisation movement has already taken place in the European Union. By 2004 all businesses will be able to choose their provider. These are limit dates for EU members although several countries like the United Kingdom, Ireland, Germany and Spain have liberalised earlier on their own.
28. In order to work properly and avoid an excessive marginal cost when supply of renewables is scarce, this setup should be complemented by a cap premium over market price which, when attained, allows suppliers to use conventional sources.
29. Calculation of the COMCO point to losses for Swiss businesses of beyond CHF 800 million per year for not liberalizing the market. These calculations are based on the price differential of average electricity prices paid by Swiss firms with respect to German firms (4.65 cents per kWh) and on annual consumption (18 000 GWh) (Stoffel, 2003).
30. This is the case in the cantons of Freiburg, Jura and Obwald.
31. As in electricity, the liberalisation process will be completed in the Union by 2007, and choice of provider will be extended to all commercial consumers by mid-2004.
32. The Confederation still has a majority control of Swisscom, with 62.7 per cent of the shares.
33. In practice, the ComCom can impose interconnection prices only once negotiations between interested parties – in this case the incumbent operator and each of the competitors – do not succeed in fixing prices and conditions for interconnection, which is very likely to happen. This delays somewhat its intervention.
34. A previous decision of the Federal Tribunal on leased lines has favoured the position of Swisscom.
35. By mid-2003 there were 637 000 broad-band access lines (8.7 for every 100 inhabitants), half of them corresponding to ADSL lines and half to cable modems. However, although these figures show that internet access is evenly split between both technologies, voice access through cable modem is much less developed as it requires additional investment.

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36. In particular, to provide voice transmission cable networks have to be adapted to allow bi-directional high speed transmission.
37. The market shares of the other two operators are 17 per cent (Diax) and 14 per cent (Orange).
38. However, mobile companies have refused to reply to the questionnaires sent by the COMCO for its investigation.
39. As for the historic debt of CFF, part of it was cancelled and another part transformed into loans at zero interest rate with conditional payback clauses.
40. The principle of command is also applied to regional passenger transport by bus or ship.
41. Until now, no complaints have been brought to the attention of the arbitration commission.
42. It is thought that the new solution complies with the concept of independence for slot allocation systems which is required by EU-Directives.
43. This is equivalent to one office per 12 km^2, the highest ratio in the OECD.
44. In April 2003, the Federal Council advised the Parliament to reject the popular initiative "Post service for everybody".
45. A formal decision on this final step of liberalisation in the EU by 2009 has not been taken yet.
46. On the other hand, productivity in the distributive sector as a whole does not seem very high by international standards (Figure 28). However, the link between competition and productivity in the trade sector is somewhat ambiguous as less competition allowing higher margins could in principle lead to higher productivity.
47. There are only some twenty hypermarkets of relatively modest size in Switzerland (Le Temps, 2002). The average floor area of retail stores was only 269 m^2 in 2000.
48. The range of food products offered by the main Swiss retailers is put at 10 000, but the main distributors provide only half of that range in most of their shops. In France and Germany, the range is between 15 000 and 20 000, and in the United States between 20 000 and 25 000 (Le Temps, 2002).
49. The Carrefour group, world number two in the hypermarket sector, when it bought up Jumbo in 2001, had less than 3 per cent of the Swiss food market.
50. In early 2003, Coop, the number two food distribution company, bought up Waro and EPA.
51. Shops close at 18 h 30 or 19 h 00 in 15 out of 26 cantons, and frequently at 16 h 00 or 17 h 00 on Saturdays. In some cantons, small supermarkets that are open in the evening and on Sundays have set up in service stations. However, measures have been taken to restrict competition on opening hours in the canton of Fribourg.
52. Imposed minimum and fixed resale price maintenance are not banned *per se*, but they do presume the elimination of competition. Firms therefore publish recommended prices which are not binding.
53. In a selective distribution system, the supplier can only sell his goods and services to approved stockists or repairers. In an exclusive distribution system, the supplier gives the retailer an exclusive sales area. Multiple brand selling means that the same dealer can sell several different brands.

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54. The recent arrival of the Fnac group, which has the capacity to do without this distribution system, may however increase competitive pressure in the sector.

55. In Switzerland, patent law is subject to the national exhaustion regime, which prohibits parallel imports of patent-protected products without the approval of the holder of the patents, even if there are large price differentials with other countries. In contrast with patent rights, copyright and brand rights are subject to the international exhaustion principle which allows parallel imports. See Kraus (2003) for a discussion of the legal framework on parallel imports.

56. For patented drugs, the best prices found at international level are 40 per cent lower than in Switzerland, while for non-food consumer goods the average wholesale price differential is put at 30 per cent (Federal Council, 2002; Frontier Economics, Plaut and BAK, 2002).

57. Between 300 and 350 000 professionals work in regulated sectors. Among them, there are 43 000 engineers and architects, 21 000 doctors, 88 000 nurses and 23 000 travelling salesmen.

58. The volume of procurement contracts accounted on average for 20 per cent of GDP in the OECD countries in 1998, while the share of these markets potentially open to international competition was estimated at 7.6 per cent of GDP (OECD, 2002d).

59. The incorporation of international agreements into domestic legislation is done separately for the Confederation and the cantons, which means that the law has not evolved uniformly. Under federal law, for example, negotiations with bidders are allowed during the procedure, which is not the case for the cantons.

60. A survey carried out between January 2000 and March 2001 showed that nearly half of the Confederation's contracting bodies did not publish invitations to tender.

61. The legislation is sometimes in contradiction with the DMA (Domestic Market Act) and the law on cartels, which also apply to procurement contracts. For example, under cantonal legislation on procurement contracts, bidders have to comply with the working conditions in force at the place where the contract is carried out, but it is the DMA that determines the place of origin.

62. The maximum amount for contracts negotiable by mutual agreement was raised from CHF 50 000 to 150 000 for a service or construction supply contract and to CHF 300 000 for a contract on the shell of a building.

63. To respond to an invitation to tender in the canton of Geneva, architects and engineers have to have their head office in the canton. It is possible, however, under certain conditions, to be registered temporarily in the canton.

64. According to different surveys, only 2 per cent of insured people, mainly the young who are a good risk, changed insurers in 2000, and the level of switching decreased between 1997 and 2000 (Colombo, 2001). Insurers can encourage bad risks to leave or, alternatively, seek to attract good risks by proposing additional options at interesting prices.

65. Tarmed, which is to come into force in early 2004, will be replacing the 26 cantonal systems currently in use. It establishes prices for medical services throughout the whole of Switzerland on the basis of a point system which takes into account the time spent on each patient, the competence of the doctor and the type of treatment provided. The value of the point has to be negotiated at cantonal level.

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66. In 2000, 7 000 drugs were authorised, 2 500 of which were reimbursed under the basic sickness insurance scheme, whereas the French social security system reimbursed 20 000 products.
67. Part of the drugs reimbursed by the compulsory sickness insurance is prescribed by hospitals, which are managed by the cantons. These pharmaceutical products are also bought at cheaper prices. The share of medicines reimbursed by the compulsory sickness insurance, outside hospitals, reaches 60 per cent of sales.
68. Switzerland is the only European country which re-evaluates the prices of reimbursed drugs after expiry of patents or after fifteen years.
69. The market of generics rose by almost 40 per cent in the first half of 2003 (year-on-year) and this increase will probably be stronger in the second half of 2003.
70. 20 per cent of Swiss exports are pharmaceuticals, while Novartis, Roche and Serono have a 7 per cent share of the world market.
71. This reduction of the price level will remain limited compared with the estimated difference vis-à-vis the European average, which reaches 25 per cent if one takes into account the relatively higher Swiss standard of living. The easing of monetary policy and the endogenous exchange rate adjustment should also avoid that the downward pressure on prices will lead to deflation (Annex V).

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IV. Other structural reforms to maintain high living standards

It was argued in Chapter I that the low trend growth of GDP is to a large extent rooted in structural factors. Many of these relate to a lack of competition in product markets, which was analysed in Chapter III. This chapter looks at other areas where performance has traditionally been considered good, but where further improvements can be achieved in order to raise the speed limits of the economy. In the labour market, there is room for raising participation, which would also help in facing the challenges of an ageing population. Education policy must pursue quality improvements without increasing the already high costs. Finally, the financial system, which has been one of the engines of growth in recent years, needs to adapt swiftly to the recent financial markets slump. This chapter looks at these issues in turn, with a final section devoted to three areas of sustainable development, *i.e.* climate change, water policy and waste management. A synopsis of recommendations, including those concerning the public sector issues discussed in Chapter II, is provided in Table 21.

Labour market policy

Despite more than a decade of slow growth, *per capita* GDP in Switzerland remains comparatively high. This is explained to a large extent by the intensive use of human resources. Participation is very high and, together with unemployment rates that have rarely risen beyond 5 per cent in recent decades, results in employment rates for the 15-64 age group of 86 per cent for men and over 72 per cent for women, which are among the highest in the OECD. A well functioning labour market is crucial in this context, with no legal minimum wages and relatively flexible wage determination that allows a rapid absorption of any excess labour supply. Employment protection legislation is among the least stringent in the OECD, only behind that of the United States, the United Kingdom, Canada and Australia (Nicoletti, Scarpetta and Boylaud, 1999). However, this flexibility for adjusting employment mostly applies to the private sector. In public companies and at all levels of administration, reform is complicated by the priority given to maintaining jobs, despite the almost complete suppression of the civil servant status.

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Table 21. **Recommendations for further structural reform**
Based on previous and current Surveys and action taken since Spring 2002

	New (*italics*) and old recommendations	Actions taken
	Labour market	
2000	Lower maximum duration of unemployment benefit.	Reform of unemployment benefits reducing maximum duration for most workers, increasing the minimum contribution period for eligibility and reducing contribution rates.
2002 and 2003	Provide more work opportunities for involuntary part-timers; develop child care facilities to encourage female participation. *Provide public support to child care through targeted child care subsidies or through tax expenditures.*	None.
2003	*Ensure that the performance link due to the previous incentive system for regional placement offices is not lost.*	The *bonus-malus* incentive system for regional placement offices has been suppressed, but information on individual placement results is still provided.
2003	*Reduce contribution rates of older workers to the pension system and put them in line with those of other workers. Introduce incentives for prolonging work beyond standard age of retirement with fiscal incentives or higher accrual rates after standard retirement age.*	
	Education	
2003	*Promote earlier entrance in the education system, possibly re-assigning resources from other areas of education.*	
1999	Tackle low literacy of new immigrants.	
2003	*Put into practice reform plans for secondary education to improve quality and increase attention to worse students through re-organisation and productivity gains.*	Report jointly elaborated by the cantons with proposals for improving education quality of compulsory education.
1999	Make the financing of university education more efficient and fairer.	
2003	*Increase fees in tertiary education where private returns to education are high.*	Budgetary spending for university education rises by 4 per cent in the medium-term fiscal plan.
2003	*Increase the efficiency of the Hautes Écoles Spécialisées by streamlining the number of paths.*	
	Financial markets	
2002	Privatise cantonal banks and pursue their social policy objectives through the budget.	None.
2002	Increase further the resources of the anti-money laundering regulator and rely less on self-regulation.	Resources increased, but still limited.
2003	*Ensure that the new regulator for financial markets includes occupational pension funds and all its members are independent from the profession. Ensure that the new regulator is properly staffed.*	Planned merger of the Federal banking commission and the regulator of private insurances in a single body.

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Table 21. **Recommendations for further structural reform** (*cont.*)
Based on previous and current Surveys and action taken since Spring 2002

	New (*italics*) and old recommendations	Actions taken
	Reform the tax system	
1999	End the time-limitation of the federal government's power to raise income taxes and VAT.	Draft reform to be discussed by the Parliament.
1999	Replace the progressive tax on corporate income still prevailing in a number of cantons by a flat rate tax on corporate profits.[1]	
1999	Amend tax treatment of funds of non-residents in a way that makes them less attractive as vehicles of tax evasion.	Agreement reached with the European Union on the taxation of saving income with the planned implementation of a withholding tax on interest income accruing to EU individuals.
2003	*Consider shifting the balance of the tax structure, with a reduction of direct taxes offset by higher indirect taxes. Enhance the neutrality of* VAT.	
2003	*Lessen the chronic instability of the withholding tax on dividend (without modifying its rate or tax base).*	
	Enhance budget management and public spending efficiency	
2002 and 2003	Avoid too mechanical approach to the estimation of the output gap used for the debt containment rule. *Consider modifying the estimation method of the structural deficit to identify more clearly the non-cyclical component of the budget balance.*	Technical adjustments were brought to the output gap estimation based on an HP filter to make it more sensitive to revisions in GDP growth.
2002	Apply the debt containment rule in a process that is centred on multi-year planning.	None.
2002	Assess the fiscal costs and the main beneficiaries of tax expenditure on a yearly basis and include this information in the budget process.	None.
2002	Extend the use of outcome-oriented management methods.	
2002	Implement rapidly the proposed reform of spending assignments and fiscal equalisation schemes. Adjust conditional grants to better reflect spillover effects across jurisdictions and further consider the spending assignments (*e.g.* for early childhood and educational facilities).	Discussion of the reform proceeding. It requires a reform of the Constitution. Its implementation is not expected before 2007.
2003	*Speed up the reform to reduce fiscal administrative burdens for businesses. Consider adopting the "silence means consent" principle for delivering administrative authorisations.*	Some measures will not be implemented before 2006.
	Ageing	
2003	*Consider raising retirement age and reforming the pension indexation system to ensure the financial viability of public pension system* (AVS) *beyond* 2015.	Technical analyses have been carried out concerning the measures that might resolve the financial problem of the AVS.

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Table 21. **Recommendations for further structural reform** (*cont.*)
Based on previous and current Surveys and action taken since Spring 2002

	New (*italics*) and old recommendations	Actions taken
2003	*Curb the growth in the number of disability pensions. Consider introducing a system for awarding benefits initially on a temporary basis.*	Confederation's surveillance of the granting of invalidity pensions in the different cantons reinforced with the 4th revision of the disability insurance.
2000	Improve information and enhance transparency in pension plan accounts. *Adopt a flexible management of the minimum rate of return both upwards and downwards.*	Reports on the financial situation of both private and public pension funds are being prepared.
	Sustainable development	
2003	*Equalise carbon taxes across sectors and link them to the price of CO_2 emission permits.*	
2003	*Link support to renewable energy to either the cost of emission permits or the carbon tax.*	
2003	*Ensure that the recycling cost premia do not exceed the cost of incineration and the avoided production externalities. Examine the feasibility of introducing competition in the market for incineration.*	
2003	*Reconsider the scale of agricultural support taking into account the associated environmental externalities. Improve coordination across administrations to ensure effective application of programmes against water pollution related to agriculture.*	

1. Nine cantons still have a tax rate increasing with the rate of return on own capital.
Source: OECD.

Despite the good labour market results, some problems emerged in the 1990s related in part, but not only, to the prolonged period of economic stagnation. One such problem was the rapid rise of the unemployment rate to near 5 per cent by mid-decade, exerting pressure on the finances of the unemployment benefit (UB) system, as benefit duration was extended to provide extra revenues to those unemployed workers not finding a job during the standard period. To cope with these pressures, several reforms of the UB system were introduced, reinforcing requirements on the unemployed to participate in active labour market programmes, while job placement offices were also reformed. These reforms were analysed in detail in previous *Surveys* (OECD, 2000 and 2002a). After the extensions of the 1990s, benefit duration and contributions were cut back in 2003. Two other labour market features that have been apparent in the 1990s may have important consequences for the future if they persist. *First*, participation rates of older workers, although higher than in most other countries, have declined over the past decade, a trend that might create problems for the

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future financing of the first pillar of the pension system (Chapter II) and could reduce labour supply in a country that has often suffered from labour shortages at cyclical peaks, especially of skilled workers. *Second*, while female participation rates have increased in recent decades due to cultural reasons, the increase in the average level of education and the need of raising household incomes, notably during economic downturns, a very large share of women works part time, limiting effective female participation. The expansion of the effective female labour supply could be facilitated if the conditions to make family life compatible with full time work were improved. This section assesses these labour market issues, and is complemented by an assessment of education policy issues in the next section that may help to improve the qualifications of the labour force and thereby increase productivity growth.

Participation of older workers has declined

Employment-population ratios of those aged 50-64 are among the highest in the OECD, at 86.1 per cent for men and 62.4 per cent for women. However, while female employment for these age groups has continued to increase due to a strong cohort effect, the old-age male employment rate has decreased from 89 per cent in 1991. Declining participation and employment for male workers, combined with population ageing, could pose problems for the financing of the public pension system if it continued, and could damp potential GDP growth further, although the arrival of new immigrants following the implementation of the bilateral agreements with the European Union could limit the reduction of the labour force (Box 5).

Several interrelated factors seem to be behind the decline in participation of older workers (OECD, 2003a). *First*, adverse cyclical conditions like those experienced in the 1990s may have a greater long-term impact on older workers, since these have more difficulties in finding a new job. This is reflected in much higher long-term unemployment rates for older workers than for younger ones, despite the fact that both groups have similar overall rates of unemployment. *Second*, and also responding to cyclical factors to some extent, early retirement has increased during the 1990s, touching especially highly-skilled workers in sectors like transport and communications. Moreover, in the construction sector, with particularly hard working conditions, the rate of invalidity has increased sharply. *Third*, older workers are more costly to employers. Wages in Switzerland have a comparatively large seniority component, suggesting that wage differentials between the old and the young may not fully reflect productivity differentials (Figure 32). In addition, the second pillar of the pension system, which consists of compulsory occupational pensions, has higher contribution rates for some older workers, resulting in higher costs for enterprises (Chapter II). *Finally*, in periods of rapid technological change the human capital of older workers may deteriorate

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Box 5. The agreement with the European Union on the free movement of persons

Switzerland is one of the OECD countries with a large share of foreigners in total population (20 per cent) and in the labour force (25 per cent). In the long-term, demand for foreign workers could increase in order to face the labour force shortages linked to population ageing, although social resistance to large immigrant inflows could make this difficult. One of the seven bilateral agreements with the European Union that entered into force in June 2002 is on the free movement of persons. Under its provisions, which extend also to members of the European Free Trade Area (EFTA), the same principles on the free circulation of persons as in the EU treaties apply among signing countries on a reciprocal basis. Several transitory measures are being implemented to avoid a massive labour market entry of European citizens. During five years, the maximum number of permanent and short-term work permits will be limited, respectively, to 15 000 and 115 500 (relative to the current labour force of 4.3 million). After 2007, and for 7 years, Switzerland will keep the right to impose new limits in case of a very large inflow of workers. At the same time, Switzerland will maintain until 2004 existing controls both on the priority rules applied when assigning a job, which favour Swiss citizens and foreigners living in Switzerland over European citizens, and on wages and working conditions, in order to avoid "social dumping". No special transitory provisions apply to Swiss citizens moving to EU or EFTA countries.

In the first three months after the implementation of the agreement, applications for permanent permits have increased substantially since many workers and firms had been postponing new work applications until the entrance into force of the new agreement. Many daily commuters have also applied for permanent permits. However, during 2002 the number of permanent residents from EU-EFTA countries had only increased by 1.3 per cent, equivalent to 10 249 net entrants. The agreement seems to have allowed some firms to meet demands for certain types of workers, who are in shortage in Switzerland. With the enlargement of the Union in 2004, new transitory provisions are being negotiated regarding citizens from new member countries, although a very large inflow is not expected. For citizens coming from other countries, the number of work permits is limited to 4 000 per year for permanent permits and to 5 000 for short-term permits, also with a strict control that gives preference to residents in Switzerland (and now EU-EFTA citizens). These permits are usually provided only to qualified workers.

rapidly. All these factors suggest that there is an incentive for firms to lay off older workers or to encourage them to go on early retirement.

To avoid a further decline in the participation of older workers, the disincentives to employing older workers built into the pension system through higher contribution rates should be suppressed where they exist. Indeed, positive incentives to voluntarily extend the working life could be introduced to raise

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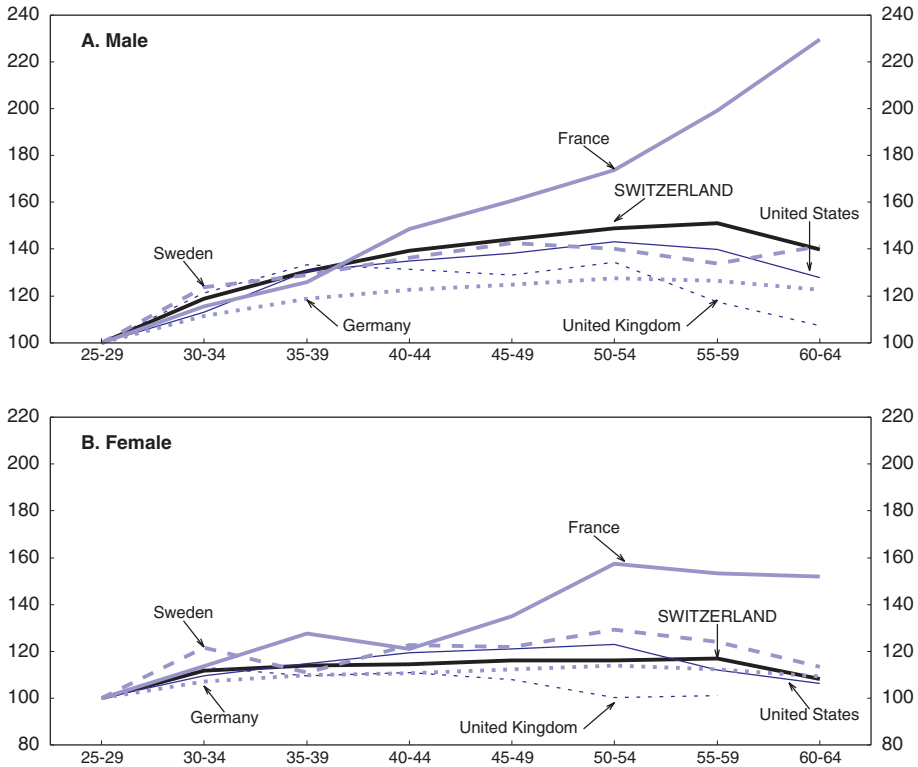
Figure 32. **Wage pattern by age in selected OECD countries**
Wage of the 25-29 year olds = 100



Source: OECD, *Ageing and Employment Policies – Switzerland*, 2003, OECD, Paris.

participation. This could be implemented either through the pension system, with for instance lower contributions for the old or higher accrual rates if they work beyond the legal retirement age, or through tax deductions for work income of those close to or beyond the retirement age. Such measures should be designed so as to be neutral or positive for the budget. This is feasible since the extra cost of fiscal incentives to change the work-leisure decision of workers in favour of additional time worked could be more than compensated by the additional tax revenues generated by the extra value added and earnings. Other complementary measures could also be envisaged, like special ALMPs designed for the old when they become unemployed, thus avoiding long unemployment spells and recourse to invalidity schemes and early retirement (OECD, 2003a).

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Table 22. **Labour market indicators**
Percentage, 2002

	Participation rates			Unemployment rates			Employment rates					
									Females			
	Total	Males	Females	Total	Males	Females	Total	Males	Unadjusted		Adjusted by hours[1]	
									Ratio	Rank	Ratio	Rank
Sweden	78.0	79.8	76.1	5.0	5.4	4.7	74.0	75.5	72.5	3	62.6	1
Portugal	72.1	79.6	64.9	4.8	4.1	5.7	68.6	76.3	61.2	8	60.8	2
Finland	77.2	79.4	74.9	10.5	10.7	10.2	69.1	70.9	67.3	5	60.7	3
Norway	80.6	83.7	77.4	4.1	4.0	4.2	77.3	80.3	74.2	1	60.0	4
Denmark	79.9	83.8	75.9	4.3	4.3	4.4	76.4	80.2	72.6	2	59.6	5
Switzerland	**81.3**	**88.7**	**73.9**	**3.0**	**2.9**	**3.1**	**78.9**	**86.2**	**71.6**	**4**	**52.4**	**6**
Austria	71.7	79.5	64.0	4.9	5.2	4.6	68.2	75.3	61.1	9	52.2	7
United Kingdom	75.3	82.3	68.3	5.1	5.6	4.4	71.5	77.7	65.3	7	51.7	8
France	69.0	75.5	62.6	8.7	7.8	9.8	62.9	69.6	56.4	11	47.2	9
Germany	71.5	78.7	64.2	8.6	8.8	8.3	65.4	71.8	58.8	10	46.6	10
Ireland	67.9	78.4	57.4	4.3	4.7	3.8	65.0	74.7	55.2	12	45.1	11
Belgium	64.1	72.6	55.4	6.9	6.3	7.8	59.7	68.1	51.1	13	41.8	12
Spain	65.8	78.8	52.6	11.1	7.7	16.3	58.4	72.8	44.0	14	40.6	13
Netherlands	76.5	84.8	67.9	2.6	2.3	2.9	74.5	82.9	65.9	6	40.5	14
Italy	61.0	74.2	48.0	9.3	7.1	12.7	55.4	68.9	41.9	15	37.3	15

1. The adjusted measure is calculated as the total weekly usual hours divided by 40 and by the working-age population.
Source: OECD based on European Labour Force Survey data.

Effective participation of women is lower than it seems

As elsewhere, female participation rates have been rising, reaching 73.9 per cent of women aged 15-64 in 2002. This trend has affected women older than 25, while younger women devote more time to study. However, this increase has been parallel to the rise in part time work, which affects 44.7 per cent of all working women against 24.1 per cent in the OECD and 8.9 per cent of Swiss men. Part-time contracts allow women who desire to work less than full time to enter the labour market. But, at the same time, such a large share of part time employment implies that a large part of human resources, sometimes of highly-skilled women, is lost. When female participation is measured in full-time equivalent, the employment rate falls to only 52.4 per cent from 71.6 per cent on a head-count basis (Table 22).

In many cases, working part time is a voluntary decision. However, in other cases, part-time work is taken when no full-time jobs are available.[1] A study by the Federal Statistical Office (OFS, 2003) indicates that the rise in labour participation of women aged 25-45 in recent years is linked to the need to increase family

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revenues, suggesting that economic incentives do play an important role in female labour supply. In this respect, the recent reform of the personal income tax that improves incentives to work for the second earner in a married couple should help to raise female participation rates (Chapter II). But the relatively low effective female participation rate may also reflect several institutional features. One is the lack of paid maternity leave, which does not yet exist in Switzerland (it was rejected in a referendum in 1999), although Parliament is discussing the creation of such a scheme (which is likely to be challenged again in a referendum). A second factor is the lack of facilities for child care (in general for children between 0 and 4), which are costly for individuals and scarcely subsidised; short hours of children in primary and secondary school in international comparison also impose a barrier to female work. Female labour supply is likely to have high wage elasticity, implying that any economic barriers to work may reduce female work substantially. International evidence shows a clear effect of early childhood care provision on female participation (OECD, 2003b). In Switzerland, up to 80 per cent of mothers of children aged 15 or less work part time, while the proportion for all women working part time is 53 per cent. All this indicates that taking care of children may be limiting female participation. Greater provision for early childhood care and schooling and of after-school care facilities would imply a large efficiency gain, since the number of adults per child involved in formal education is likely to be much lower than that for care at home.[2] In principle, this efficiency gain could be captured by the market, perhaps facilitated by firms, which supply care for their workers' children. However, there are several potential reasons for extending public support to child care and early childhood education. F*irst*, as in primary and secondary education, which is publicly financed in all OECD countries, there are externalities in formal early childhood education that may justify some public financing to encourage supply. For instance, early schooling seems to improve results later in education, and favours language skills and integration of foreign children, which seem at the root of failures in secondary education (see below). S*econd*, when wage compression is high and wages for relatively unskilled jobs (the ones required for child care) are above productivity, the emergence of market supply to cover child care needs is difficult. T*hird*, the provision of early childhood education at the municipal level suggests that there may be a spillover effect not well taken into account, since external benefits from education at these ages go usually beyond the limits of the municipality. Several options are available to provide public support to early childhood care and education. One is direct provision, which would be available by the promotion of schooling of young children. For raising female labour supply, child benefits are not optimal since they are not conditional on participation. Childcare subsidies or tax expenditures for child care expenses would be more effective and cheaper, as they could be targeted to low income families and could be made conditional on labour force participation of both parents.

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Unemployment benefits have been reformed

The unemployment rate reached a cyclical peak of 5.2 per cent in 1997, and subsequently fell until 2001, when it started to rise again. In parallel to the cyclical swings, and partly responding to them, the UB system (LACI) has been changed several times, with a general tendency to increase the duration of benefits up to the mid 1990s, but complemented by a reform that aimed at closely conditioning benefit receipt on the participation in ALMPs (the so called *activation principle*). At the same time, an extensive evaluation exercise was carried out to determine which measures are most effective; this has helped to target them better.

In 2003, a new reform of the LACI has been introduced that aims at reducing the cost of the insurance scheme for enterprises and to improve incentives for early job search of the unemployed (Annex VI). The reform reduces the basic contribution rate from 3 to 2.5 per cent and phases out the special "solidarity contribution" for highly-paid workers. This should have a positive impact on labour demand, and its introduction in a period of job losses is especially positive. To improve labour supply, the reform also extends the minimum work requirement for receiving unemployment benefits from 6 to 12 months and reduces the benefit duration for all workers below age 55.[3] This measure is likely to reinforce incentives to find a job, since the international experience shows that UB duration has a strong impact on job search efforts, with workers typically finding jobs shortly after becoming unemployed or shortly before the maximum UB duration. However, the impact of the reform on search efforts of white-collar workers, many of which have become unemployed recently due to the crisis in telecommunications and in the financial sector, is uncertain since these are highly-skilled workers with high reservation wages, and because UB replacement rates net of taxes for high income workers are unusually high in international comparison.[4] In any case, the reform should help to avoid the current increase in unemployment being transformed into long-term unemployment, thereby pushing up the NAIRU again as happened during the 1990s.

Parallel to the reform of the LACI, an incentive mechanism for placement offices (ORP) that linked their funding to placement results (*bonus-malus*), which was implemented in 2000, was suppressed in January 2003. However, information on results will continue to be published. The system assessed the performance of individual offices based on four criteria,[5] relative to the background of the labour market situation in the area. The suppression of the *bonus-malus* incentive scheme was based on the argument that it did not provide direct information to ORPs on the reasons for their relative performance, but it seems to have been effective up to now, with an increase in the number of placements relative to resources (Sheldon and de Wilde, 2002). However, these effects are difficult to disentangle from those of the activation principle. In any case, the authorities should be

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vigilant that the performance gains in job placements achieved up to now are not lost due to the lack of incentives to ORPs.

Education

School starts in Switzerland relatively late in international comparison, which is another factor impinging on the female participation rate discussed above. Compulsory school begins at age 7, and elementary school, which is not compulsory but is publicly provided, usually at age 5. However, only 5 per cent of children attend school at age 3 and less than 30 per cent at age 4, while in many European countries enrolment rates are 80 per cent or over at these ages. Primary and secondary education is mostly publicly financed and is a cantonal competence. Compulsory education extends to age 16, and afterwards 28 per cent of students prepare for access to university (one of the lowest proportions in the OECD), while 60 per cent join a dual system of professional training which combines studies with training in firms, similar to the systems in Germany and Austria. This dual system prepares students to join the labour market quickly, or to access the recently created *Hautes Écoles Spécialisées*, that currently provide applied oriented tertiary education to about 5 per cent of students, although this figure is expected to rise to over 9 per cent by 2010. Another 21 per cent of all students follow vocational training courses at the tertiary level in non-academic training institutions. The dual system is traditionally viewed as one main factor that explains the high share of skilled workers in Switzerland.

The education system is expensive

The Swiss education system is among the most expensive in the OECD. Expenditure per student in secondary education is the highest, and the second highest in primary education (after Denmark) and in tertiary education (after the United States). In current US dollars the cost differences with the OECD average is even larger, and is only partly explained by higher *per capita* GDP (Figure 33). Expenditure per student is lower in pre-primary education, and its negative gap with respect to primary education is the largest in the OECD, which together with the low enrolment rates at low ages shows that the relative investment effort in pre-primary education with respect to higher ages is very low. As a share of GDP, Switzerland is one of the OECD countries that spends most on primary and secondary schools, but is below the OECD average in tertiary education because of the relatively early labour market entry and the consequent low enrolment in universities.

High wages for teachers seem to be the main explanation for the higher cost per student than elsewhere. The number of students per teacher at school is also lower than the OECD average (except in primary education), but this is unlikely to explain a large part of the high cost. Switzerland has by far the highest wages for teachers in primary and secondary education. The minimum salary for a

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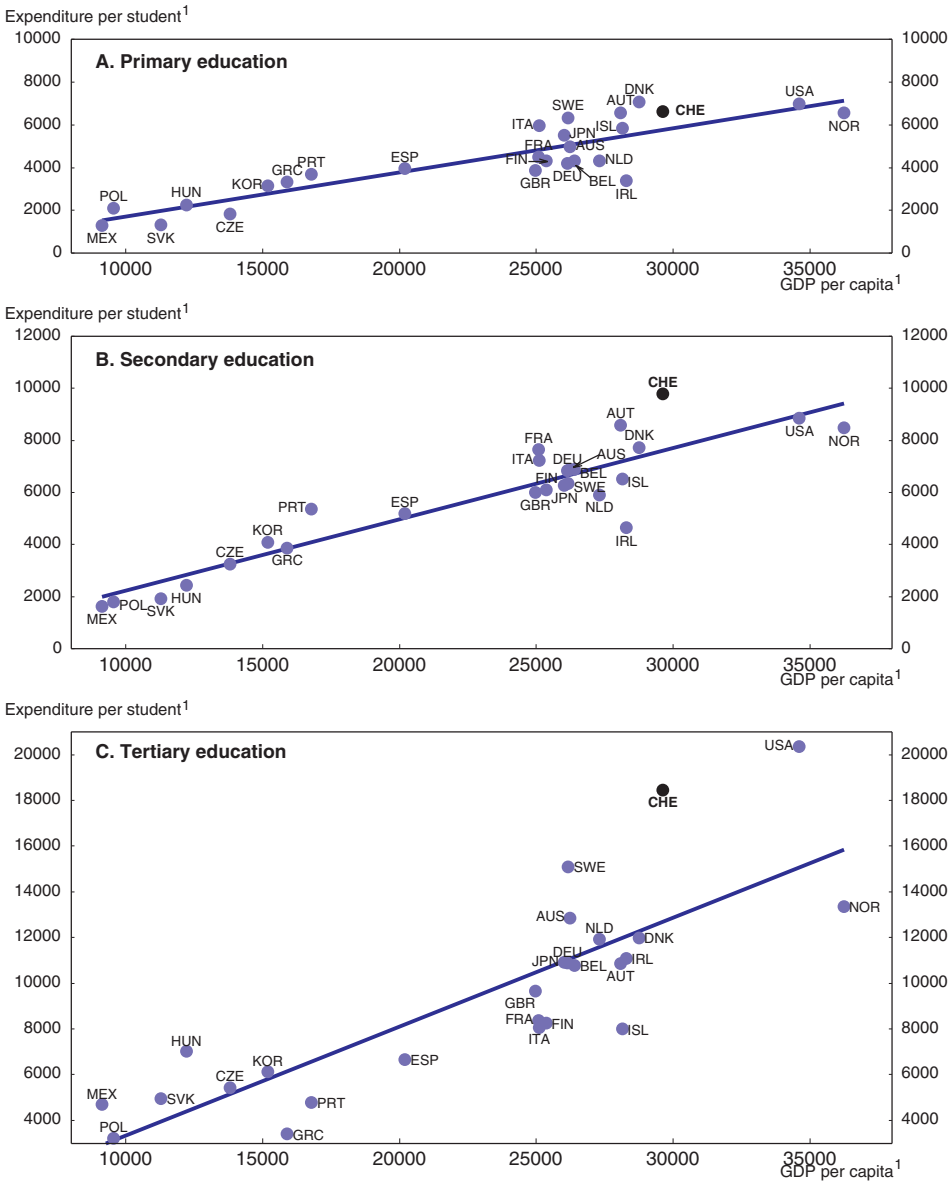
Figure 33. **Expenditure per student and GDP per capita**
2000



1. In equivalent USD converted using PPPs.
Source: OECD, *Education at a Glance – OECD Indicators 2003*, Tables B1.1 and X2.1, OECD, Paris.

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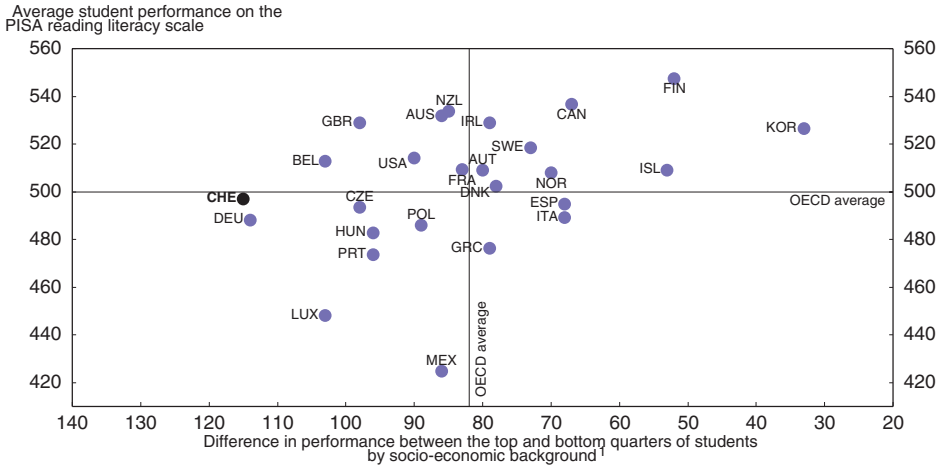
Figure 34. **Student performance and equity**
2000



1. Measured by the socio-economic index of occupational status.
Source: OECD, *Education at a Glance – OECD Indicators 2002*, Table A9.1, OECD, Paris.

teacher with 15 years experience in secondary education was USD 62 052 in 1999 in PPP terms, twice as much as the average OECD teacher's salary. Similar differences are found at other levels of compulsory education. Even after taking into account *per capita* GDP these wages are high. Part of the explanation lies in the relatively high wages of Swiss civil servants as compared to other countries, although teachers' wages also seem to be high with respect to those of other Swiss civil servants with similar qualifications.[6, 7]

The PISA *report has revealed some weaknesses in school performance*

Despite the relatively high standards traditionally attributed to the Swiss education system, the OECD PISA report has revealed some weaknesses in the results of compulsory schooling (OECD, 2002b). The study, which compares the performance of 15 year-old students across countries, found that Switzerland is 17th out of 29 countries in reading abilities and that there is a wide dispersion of results between the best and the worst students (Figure 34). A sizeable share of students (20 per cent) cannot adequately understand a written text at age 15. Many of these, even if born in Switzerland, are foreign students. Furthermore, there is a strong correlation of results with the social and economic background of parents, with children of parents from a wealthier social background and with higher levels of education attainment performing much better than those of less

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well-off families, which suggests that the school system does not properly correct for social inequalities.

These results indicate that there is ample room for improvement in primary and secondary education. They have raised concerns among the cantonal authorities, who are responsible for secondary education and they have requested a report from national experts to outline possible reforms (CDIP, 2003). The report recommends, among other things, to bring the age of scholarisation forward, increase testing of students' performance, provide enhanced autonomy for schools, implement longer hours of study (mornings and afternoons) and pay more attention to language skills of students, especially of those coming from immigrant families. These proposals broadly coincide with provisional conclusions from the PISA report at the international level, which found that countries with the best average performance are also those with more equal results across students, since they are countries where more efforts and resources are devoted to students with special difficulties. Also, the best systems seem to be those that provide more managerial independence to school heads and those which start scholarisation early. The CDIP report is a good general guide for reforms in the school system and the CDIP committee has proposed a national plan to implement it, though it has not yet resulted in specific reform initiatives, which will require co-ordination among the cantons. Most of the proposed measures would not demand additional funding, although earlier school entry and the development of special measures addressed at improving the performance of the worst students would. Given the high cost of the school system, these should come from the reallocation from other areas of education or from productivity gains in teaching. On the other hand, the extension of education at pre-primary level, apart from contributing to the improvement of quality, would pay back in the form of higher effective participation of women in the labour market (see the labour market section above).

In tertiary education there is room for more private finance

The public education system for tertiary education includes two types of public institutions, the *Hautes Écoles Spécialisées* (HES) and the *Hautes Écoles Universitaires*; there are also private education institutions providing occupational training. The HES, created in 1996 by merging a previous network of technical centres into 7 regional institutions, provide tertiary education with a strong applied bias, combined with some theoretical education. Their goal is to improve the qualification for students who have followed professional training in secondary education and to adapt workers' skills to new technologies and sectors. Although its development is relatively recent, a preliminary assessment report concludes that results have been relatively positive so far, but that the number of paths is still too large, with many of the courses attended by less than 20 students.

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In the university system (*Hautes Écoles Universitaires* and *Hautes Écoles Spécialisées*), the entry rate is, at 29 per cent, below the OECD average of 45 per cent, partly because the dual system of professional training and those institutions offering non-academic vocational training at the tertiary level of education provide alternative pathways for many young adults. University education is a competence of cantons with a financial contribution by the Confederation, which is provided through subsidies to each of the 7 universities linked to their research and teaching performance. Cantons which do not have a university contribute to their budgets through a fixed contribution per student, which varies by field of study. Given the already high cost of the system, together with the declared goal of enhancing investment in higher education, one possible way of obtaining additional resources is to increase the contribution of students to the cost of education. As argued in Blondal *et al.* (2002), the private rate of return of investment in higher education is very high in most OECD countries, with rates that surpass by far those of other types of investment, including stock market investments. Such returns would justify a higher contribution of students to the cost of education. Moreover, the public funding of higher education is regressive, since those who go to university come usually from wealthier backgrounds or, through university, will receive higher lifetime earnings. For Switzerland, private rates of return have been calculated by Weber (2003, forthcoming) using a methodology similar to that of the OECD with 2001 data. For men, the results point to high returns for the HES (around 9 per cent) and lower returns from universities (around 5 per cent), which in the latter case are below those in a number of OECD countries but still higher than the real interest rate.[8] Higher payments may reduce somewhat the incentives to participate in tertiary education, although this effect is likely to be moderate, since typically the larger costs of participating in university education are the opportunity cost of not taking up a job and the costs of living. Various mechanisms to increase private contributions to education exist: direct increase of tuition fees coupled with subsidized loans for poorer students to solve possible credit market failures, or deferred payment conditional on income after education.

Resources for education should be used better

There is a broad consensus in Switzerland on the need to improve education outcomes. Furthermore, despite the large amount already being spent on education, there appears to be widespread popular support for further increasing it. University education has been declared a priority area for the period 2004-07, with annual increases in spending by the Confederation of 4 per cent under the preliminary government plans, which are currently being discussed by Parliament in the context of the medium-term fiscal plan. This goal should be coupled with reforms that improve the efficiency of education spending. Improving the autonomy of schools may help in this respect. Likewise, more systematic evaluation of performance at all levels of education would help to channel public

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resources where they are most needed. Thought should be given to the reallocation of resources to early childhood education, which receives relatively little money, from other education levels, since it is an area where Switzerland lags other countries and formal education at early ages seems to be highly correlated with better education outcomes later in life. In tertiary education, potential efficiency gains in the HES system are large and could be achieved through the elimination of those paths for which demand is scarce, or through the merger of similar paths in different schools. Other areas where there is room for quality improvement in the education system are reviewed in a recent OECD report on tertiary education in Switzerland (OECD, 2003c).[9] Finally, higher tuition fees for tertiary education would provide additional resources, reduce drop-out rates and raise incentives to obtain a degree earlier, provided that reforms derived from the "Bologna process" make the study length more flexible.[10]

Financial sector

Financial services are one of the most dynamic sectors of the Swiss economy and account for a large share of GDP (12.7 per cent in 2000) and employment (5.5 per cent). Pushed *inter alia* by the international stock market rally, growth of value added during most of the 1990s was remarkable, reaching an average of 7.3 per cent between 1995 and 2000 and accounting for 0.8 percentage point of the average growth rate of the whole economy (1.8 per cent) (Table 23). More recently, the financial market slump since 2000 and the slowdown in activity have had a considerable adverse impact on the banking sector, while insurance companies have suffered heavily from stock market losses. This section assesses briefly the performance and medium-term prospects of the sector, including current plans to reform supervision of financial institutions, and describes the recent agreement with the European Union on taxing income from savings.

Banks have reacted to the international financial market crisis

Following the international stock market slump and the decline of corporate investment, banks' revenues shrank over the last two years, mostly because of lower commission and trading income and losses on financial assets. The banking system has reacted swiftly by cutting operating expenses by 6 per cent in 2002, mostly through reductions in personnel, with estimated job losses of 5 to 10 thousand workers (out of 125 000 workers in the sector). This reaction has not been enough to avoid a profit squeeze, with returns on assets falling from 0.9 per cent in 2000 to just below 0.5 per cent in 2002, though this lower level is not below the average ratio for the period 1987-97, before the exceptional results of 1998-2000. The decline in 2002 was more marked for the two large banks and for cantonal banks, but has affected co-operative and regional banks less, since their revenues are more dependent on the more stable lending business. According to

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Table 23. **Sectoral value added[1] over the 1990s**

Per cent

	1990-95	1995-2000	1990-2000
Average growth rate			
Non-financial firms	–0.5	1.2	0.3
Financial intermediaries	–0.4	8.3	3.8
Insurance companies (excl. social insurance)	1.6	2.3	2.0
Public administration	1.5	1.0	1.3
Social insurance	1.4	1.9	1.7
Other	2.3	1.7	2.0
Total	–0.1	1.8	0.8
Contribution to growth			
Financial sectors[2]	0.0	0.8	0.4
Non-financial sectors[3]	–0.1	1.1	0.5
	1990	1995	2000
Share of financial sectors in total value added	9.7	9.8	12.7

1. Value added at constant 1990 prices.
2. Includes financial intermediaries and insurance companies (excluding social insurance).
3. Includes non-financial firms, public administration, social insurance and other.

Source: OECD.

the SNB, credit risks have not increased in 2002, although they are expected to grow in the coming year as the recession will lead to more bankruptcies, and will probably result in an additional negative impact on profits (SNB, 2003). Capital adequacy ratios have been maintained since banks have shifted away from equity investments towards less risky assets, with weighted ratios still among the highest in the OECD. Insurance companies have also been affected by the stock market slump, with many of them suffering losses in 2001 and 2002 despite the increase in *premia* and cost cutting. However, in November 2002, the sectoral regulator rejected a proposal to loosen solvency requirements for private insurance companies. In the case of social insurance companies (those dealing with occupational pension funds and health insurance), financial losses have pushed some companies to increase *premia* and reduce conversion rates of capital into annuities (Chapter II).

Financial supervision is being reformed

Supervision of the financial system is carried out by different institutions. The Federal Banking Commission (FBC) is in charge of institutional supervision of banks, investment funds and stock markets (in the latter case complemented by partial self-regulation by the Swiss Exchange SWX). Systemic stability is assessed by the SNB. The insurance system is controlled by two bodies, the Federal Office of Private Insurance (FOPI) and the Federal Social Insurance Office dealing, respectively,

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with private insurance, and pension and health insurance companies. Issues related to money laundering in the sectors not supervised by the previously mentioned institutions are dealt with, either directly or indirectly through self-regulating bodies, in a special institution (Money Laundering Control Authority, MLCA). As to the Competition Commission (COMCO), it supervises areas covered by the Cartel Law.

In 2000, the Zufferey report recommended the merger of supervisory bodies (Zufferey, 2000). As a follow-up, the Zimmerli Commission was charged with the task of proposing concrete reforms. In a provisional report published in July 2003, the Zimmerli Commission recommended the merger of the FBC with the FOPI in a single body with several divisions (*e.g.* banking, insurance, stock exchange, etc.) and a distinct governance structure: a supervisory board, which would deal with strategic issues of supervision, and a management board that would supervise financial institutions. After the publication of this first report, the Commission is expected to examine the possibility of extending prudential surveillance to independent asset managers, introducing brokers and local currency exchange dealers. At the same time, the Commision will make a recommendation on the issue of the possible integration of the MCLA within the new supervisory body. Supervision of pension funds and of some participants in financial markets, such as independent asset managers or local currency exchange, would, under current proposals, remain outside the powers of the new supervisor. The Federal Council is expected to submit, after a consultation process, a final law proposal. Although the preliminary proposals of the Commission go in the right direction, a bolder approach to overall supervision would be desirable, in order to profit from the synergies of controlling different types of institutions that are interlinked through ownership relationships and deal in the same financial market segments. In particular, there is no reason to leave outside the new body the supervision of independent asset managers, which currently are only supervised by the MLCA for money laundering purposes.[11] It would also be desirable to include supervision of pension funds in the same institution as that of insurance companies, since both types of institutions are directly involved in managing occupational pensions. That would also facilitate the task of harmonizing legislation across them (Chapter II). Finally, the authorities should also ensure that the independence of supervisors from the market is guaranteed, with full separation of the supervisory board from professional activity in the financial sector, and that the new institution is adequately staffed to carry out its tasks.

Challenges ahead

Looking ahead, the financial sector will continue to be exposed to increasing competition from other financial centres, which will have to be met with continued productivity gains. Swiss banks will have to apply the draft agreement reached with the European Union on the taxation of savings income. The Swiss authorities

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will collect a withholding tax on non-Swiss source interest income accruing to EU resident individuals, with tax rates of 15 per cent when the agreement will enter into force (in principle in January 2005), 20 per cent after three years and 35 per cent after six years. Three quarters of the receipts from this tax will be transferred to EU countries, while the rest will be retained by the Swiss authorities to meet administrative costs. This agreement will allow Switzerland to maintain bank secrecy. However, the draft agreement provides for the exchange of information upon request in all civil or criminal cases of tax fraud or the like. It also allows the exchange of information on bank accounts for tax purposes when the taxpayer consents to the disclosure of the information. The view of the authorities is that this agreement is not likely to imply a large outflow of funds from Swiss banks to other financial centres in the next few years, given that the agreement applies to individual holders (not firms) and that the withholding tax is being phased in. However, it will have an impact on administrative costs of banks, which will collect the new tax. In addition, banks may face increased costs from the application of the new Basle Accord on capital adequacy, which is to be concluded at the end of this year and will demand additional efforts in risk evaluation methods. This may affect in particular small banks with less capacity to develop or buy sophisticated risk management models.

In the longer term Switzerland may also have to face up to the continuing pressure to provide more complete access to bank information to foreign tax authorities as illustrated by the recent statement of G7 Finance Ministers (20 September 2003). The issue of improving access to bank information for tax purposes was also recently considered by the OECD Council (September 2003). The Council noted that there was no consensus on a draft Recommendation on the matter; it further noted that 28 member countries were able to accept the common understanding of tax fraud and 26 were able to agree to take appropriate initiatives to achieve, by 31 December 2005, access to bank information for the verification of tax liabilities and other tax administration purposes. However, Switzerland's draft agreement with the EU and its opposition, together with other three OECD members, to the proposed OECD Council Recommendation show the determination of the Swiss Government to resist pressures inside and outside the OECD and to defend alternative approaches in this field.

Sustainable development

There is a growing concern that long-run sustainable development may be compromised unless measures are taken to achieve balance between economic, environmental, and social outcomes. This sections looks at three specific issues of sustainable development that are of particular importance for Switzerland. In each case, indicators are presented to measure progress and the evolution of potential problems, and an assessment is made of government policies in that area. The

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section considers Swiss policies in the realms of mitigating *climate change*, reducing *water pollution*, and improving *waste management*. It also considers whether institutional arrangements are in place to integrate policy-making across the different elements of sustainable development (Box 6).

Climate change

Main issues

In response to concern about climate change, Switzerland accepted an abatement target of reducing greenhouse gas (GHG) emissions by 2008-12 by 8 per cent compared with the 1990 level as part of the Kyoto Protocol. Policies put in place over the past decade to hold down emissions appear to have been very costly, and the main issue facing Switzerland is to reform its abatement policy to lower the costs imposed on the economic pillar of sustainable development, while at the same time ensuring that international obligations are met.

Performance

The GHG emission intensity of the Swiss economy remains exceptionally low in comparison with all other OECD countries (Table 24). The low economy-wide intensity is largely due to the large shares of hydro (55 per cent) and nuclear power (40 per cent) in electricity generation, which result in a GHG emission intensity in the energy sector being negligible and the second lowest in the OECD area. Compared with GDP, GHG emissions elsewhere in the economy are also lower than in most member countries. However, while the emission-GDP ratio has fallen in line with the EU average in transportation over the 1990s (Table 25), the decoupling of sectoral emissions from GDP growth has been comparatively weak in other non-energy sectors, especially in manufacturing where they even rose. Consequently, with no scope for reducing the GHG intensity of electricity production, the overall fall in GHG intensity was limited to only 1 per cent between 1990 and 2000, which is somewhat less than in the rest of Europe and in other countries. Projections based on unchanged policies suggest that they will fall by a further 3 per cent relative to 1990 in the coming ten years, leaving a shortfall of 4 per cent relative to the target required by the Kyoto Protocol (Figure 35).

Policies

Since 2000, the federal law on the reduction of carbon dioxide emissions has been at the heart of climate change policy. It targets a 10 per cent reduction of carbon dioxide emissions from 1990 levels by the period 2008-12 in order to attain the Kyoto target,[12] with separate sub-targets of 8 and 15 per cent reductions in emissions from the combustion of vehicle and heating fuels, respectively.[13] There are also policy targets to stimulate the use of new forms of renewable energy.

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Box 6. The integration of policies across sustainable development areas*

The new federal constitution, approved by the electorate in 1999, specifies in its preamble that the Swiss Federation must promote sustainable development. More specifically it states that the Federation and cantons shall strive to establish a durable equilibrium between nature, in particular its capacity to renew itself, and its use by man. In 2002, a second sustainable development strategy was introduced (Federal Council, 2002), establishing quantitative targets in some areas and identifying a series of actions to reach them. However, monitoring policy presents some problems because of the lack of adequate statistics in some fields, notably in the areas of air and water quality (de Montmollin and Altwegg, 2000).

The Federal Council recognises that no suitable instruments are as yet available to specifically assess the sustainable development aspects of a project or policy. In order to establish more binding criteria to judge policies, so that concerns about sustainable development can exert a greater influence on action taken at various levels and be systematically taken into account in sector policies, the Federal Council intends to examine the feasibility of creating a set of methodological instruments. These tools would be used to evaluate the effects of draft legislation, concepts and projects in terms of the three dimensions of sustainable development and to indicate potential deficiencies. The Federal Council would thus be able to assess and refine federal projects on the basis of transparent and objective criteria. It is also investigating the activities that could be subject to a sustainability assessment, and how such an assessment could be differentiated from existing or planned assessment instruments. At present, within regulatory impact analysis, policy measures are evaluated on the basis of cost-effectiveness, given a predetermined policy target, rather than using cost-benefit analysis to set jointly the target and means of implementation. Environmental impact assessments have been a requirement since 1989 for large projects, and require that the interests of various stakeholders be taken into account.

The highly federal structure demands considerable co-ordination between the federal authorities and sub-national levels of government and between sub-national levels of government. To this end, in 2001, a Sustainable Development Forum was established to bring together the various levels of government. Furthermore the population is able to express its opinion on many environmental issues through voting on referenda. For instance, an ecological tax reform was rejected in 2000, while in 2003 a proposed ban on nuclear power was rejected.

* The sections in this report dealing with climate change, water pollution and waste management are inputs into the Organisation's follow up on sustainable development as mandated by the Ministerial Council decision in May 2001.

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Table 24. **Main indicators: climate change**

Indicators of greenhouse gas (GHG) emission intensity, gram of CO_2 equivalent per USD[1] of GDP, in 1995 prices

	Level of emissions, 2000				Average annual percentage change, 1990-2000			
	Total	CO_2 from electricity	CO_2 from transport	Other	Total	CO_2 from electricity	CO_2 from transport	Other
Switzerland	**267**	**2**	**78**	**187**	**–0.9**	**–3.8**	**–0.4**	**–1.1**
Austria	403	66	96	241	–2.1	–3.2	–0.0	–2.4
Denmark	501	171	88	242	–2.4	–2.7	–0.8	–2.6
France	402	30	102	271	–2.0	–2.6	–0.0	–2.6
Germany	519	168	91	260	–3.9	–3.6	–1.1	–5.0
Italy	432	108	89	235	–1.1	–0.4	0.0	–1.7
Japan	441	132	81	229	–0.3	0.1	0.8	–1.0
Netherlands	553	138	80	335	–2.5	–1.0	–1.1	–3.3
Norway	454	3	97	354	–2.9	–1.6	–2.9	–2.9
Sweden	340	35	110	195	–1.9	–1.5	–0.8	–2.6
United Kingdom	512	137	106	268	–3.6	–4.3	–1.4	–3.9
United States	779	273	192	315	–1.9	–0.7	–1.3	–3.0
OECD total	639	201	137	307	–1.9	–0.8	–0.6	–2.9
EU	491	120	100	272	–2.4	–2.4	–0.4	–3.1

1. National currencies converted to USD using purchasing power parities.

Source: Greenhouse gas emissions from national submissions to the United Nations Framework Convention on Climate Change (UNFCCC) and national publications; carbon dioxide emissions for electricity and transport from the International Energy Agency and GDP from OECD, National Accounts database.

Until 2005 the authorities will be mainly relying on voluntary measures to obtain abatement.[14] These programmes include support for energy efficiency and renewable energy. Over the 1990s, official estimates suggest that "voluntary" measures reduced carbon dioxide emissions by up to 1½ per cent.[15] The associated costs for measures implemented in 1999 and 2000 are estimated to have differed markedly across the different programmes, but averaged at over EUR 600 per tonne of carbon abated, three times the cost of regulatory programmes in place (Federal Office of Energy, 2000). This is one of the highest abatement costs in any major GHG reduction programme in the OECD area, and it exceeds estimates of the global benefits of reducing emissions tenfold (Tol and Downing, 2000). Overall, the fiscal cost of these programmes has been small, as the bulk of the cost has been passed onto consumers through purchasing requirements. In total, the programme has cost almost 0.2 per cent of GDP, with less than one-tenth of this cost being met by fiscal outlays.

The principal reason for the high cost of measures included in the programmes is the support of (non-hydro) renewable energy. During the 1990s, the share of these energy forms in electricity generation doubled and increased

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Table 25. **GHG emissions and sectoral indicators**
Annual average percentage change, 1990-2000[1]

	Total level[2] in 2000	Total emissions	CO_2 emissions			Indicators per unit of GDP
			Per kWh electricity	Residential, per unit of private consumption	Road transport, per vehicle (km)	Electricity use
Switzerland	**53**	**–0.1**	**–4.7**	**–1.7**	**–0.3**	**1.1**
Austria	80	0.3	–3.3	–2.4	–1.8	0.1
Denmark	69	–0.1	–3.4	–4.0	–0.2	0.7
Finland	74	–0.4	–0.9	–7.0	–1.4	0.7
France	550	–0.2	–3.2	–1.7	–0.5	0.6
Germany	991	–2.1	–1.5	–4.0	0.7	–2.1
Italy	547	0.5	–1.2	–1.4	–1.1	0.8
Japan	1 386	1.1	–0.9	0.1	0.3	1.0
Netherlands	218	0.4	–2.7	–2.9	–0.1	1.8
Norway	55	0.6	0.5	–8.8	0.5	–1.9
Sweden	69	–0.2	1.0	–4.1	–1.4	–0.5
United Kingdom	649	–1.3	–3.6	–1.9	–0.5	–0.7
United States	7 001	1.3	–0.3	–1.9	–0.2	–0.4
OECD countries[3]	14 543	0.6	–0.8	–1.8	–0.3	–0.2
OECD excluding US	**7 542**	**0.1**	**–1.2**	**–1.7**	**–0.4**	**0.0**
EU countries	4 073	–0.3	–2.0	–2.4	–0.4	–0.2

1. 1991-2000 for Germany.
2. In million tonnes CO_2 equivalent.
3. All OECD countries except Korea, Mexico and Turkey.

Source: GHG from national submissions to the UNFCCC and from national sources; carbon dioxide data from the IEA; private consumption from OECD, National Accounts database.

by 50 per cent in heat generation, though they still represent a very small share of total primary energy supply (SwissEnergy, 2001). The promotion of these renewable sources of energy has been based on both investment subsidies and a price premium for producers of 0.16 cents per kWh. On average, the abatement cost for one tonne of carbon has exceeded EUR 1 600. Photovoltaic electricity, in particular, was one of the most costly programmes. Consumers are able to opt to purchase photovoltaic electricity at up to 10 times the cost of their normal electricity bill, but few consumers do this in practice (Gehrig and North, 2001). However, the authorities have recognised that funding in this sector is grossly inefficient and, with the introduction of the Energy Act in 1998, have started to reorient funding to other areas (SwissEnergy, 2002).[16] More generally, the Federal Council has proposed to Parliament that the budget used to fund all voluntary greenhouse gas reduction programmes should be reduced by two-thirds in the period from 2003 to 2006.

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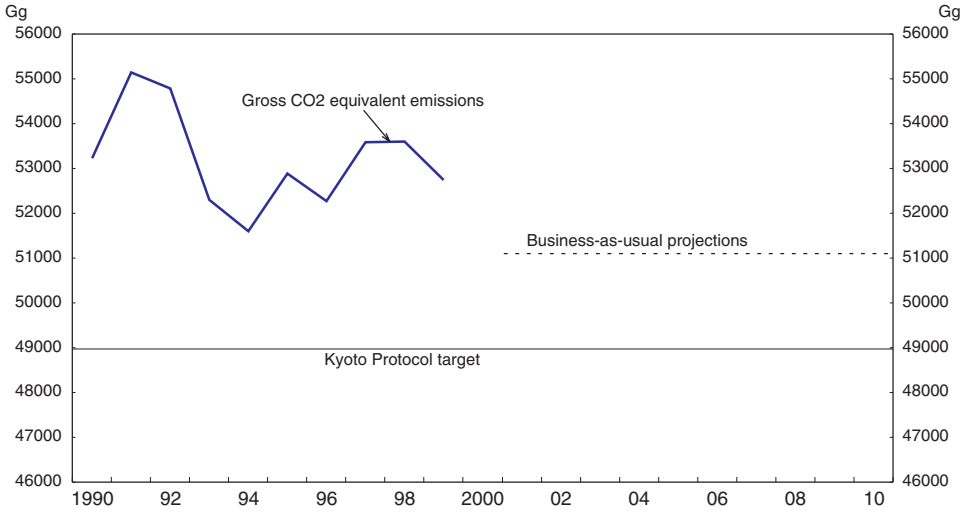
Figure 35. **GHG emissions and projections**



Source: UNFCCC.

The government has taken a number of steps that may reduce emissions from transport. The introduction of a weight-related distance fee on heavy goods vehicles in Switzerland,[17] which aims at internalising the negative impact of road traffic on health and congestion and recuperating investment costs (OECD, 2002a), may also alter the modal split in freight in favour of rail instead of road transportation so lowering GHG emissions. To further facilitate this movement, the government has undertaken major rail infrastructure projects. In addition, the authorities concluded a voluntary agreement with the association of Swiss car importers to raise the fuel efficiency standards of imported vehicles and so directly lower emissions. Such voluntary agreements have resulted in the manufacturers adopting technologies that have a high cost in terms of the carbon emissions that are avoided (Bates, 2001). In addition the government is considering a proposal to vary the vehicle import tax in line with fuel efficiency characteristics.

The law on CO_2 (carbon dioxide) provides that, in 2004, the Federal Council must review progress towards meeting the carbon dioxide target. If progress through voluntary agreements is judged insufficient to meet the reduction targets stipulated in the CO_2 law, then a tax must be introduced.[18] The original law provides that the carbon tax may be levied at a rate of up to CHF 210 (EUR 135) per tonne of carbon dioxide emitted.[19] It can be levied on either heating fuel or motor fuel or both and the tax rates need not be identical. Large emitters of

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carbon dioxide can obtain an exemption from the tax by agreeing with the government carbon dioxide emission limits. Smaller emitters can group together and sign collective agreements. Once the agreements are in place, companies will be allocated emission permits and will then be free to trade them with other companies or groups of companies. If companies do not have permits equal to their emissions in the period 2008 to 2012, they will have to pay the tax on carbon retroactively for each tonne emitted since tax exemption. The system is designed to encourage international trading both by foreseeing a link with the EU system and by using the flexibility mechanisms of the Kyoto Treaty. Permits will be allocated according to the emissions cap for the target period 2008-12 and have to be cancelled annually to offset actual emissions. Excess emission allowances cannot be carried over to subsequent years within the target period, so pushing up compliance costs. One estimate has suggested that a tax of between CHF 50 and 100 per tonne of carbon dioxide would be sufficient to meet the targets, though the precise amount will depend on the extent of the reduction agreed by large emitters (Federal Institute of Technology, 2003) and of the effectiveness of other CO_2 relevant measures implemented. The proceeds of the tax must be fully remitted: to the general population on a per capita basis and to businesses (except those exempt from the carbon tax) through reductions in social security contributions.

Conclusions

Latest assessments suggest that Switzerland will not meet its Kyoto commitment using just voluntary agreements. It seems likely that the system of carbon taxation and emission permit trading envisaged by the CO_2 law will have to be introduced. In designing this system, Parliament should favour market mechanisms (pricing) as the primary economic determinant of sustainable development. In the area of greenhouse gas emissions, an efficient system would ensure that carbon dioxide taxation is equalised across all sectors. In particular, a linkage should be established between the proposed carbon tax and the price of emission permits in the sectors that are exempted from the tax. Looking further ahead, the taxation of greenhouse gases other than carbon dioxide should be considered. It will also be important to ensure that companies that receive emission permits pay the appropriate amount of corporate tax on the value of their permits.

Waste management

Main issues

Waste management policy in Switzerland is based on the polluter pays principle and embodies a waste hierarchy approach. Since the mid-1980s, as a result of previously relatively uncontrolled land filling of waste and resulting contamination of groundwater and soil, waste management policy has put far

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Table 26. **Performance indicators: waste management**

	Municipal waste					
	Annual growth		Level	Relative to private consumption	Recycling	Composting
	Period	Per cent	Kg per capita, latest available year	Gram per USD, 1995 PPPs	Per cent of total, latest available year	
Switzerland	**1990-2000**	**1.4**	**655**	**41.8**	**32**	**14**
Austria	1990-1999	3.7	548	45.5	24	40
Denmark	1994-1999	3.5	626	57.4	25	14
France	1993-1999	1.7	524	46.4	10	8
Germany	1991-1998	–1.5	549	46.0	41	. .
Italy	1990-1999	4.0	497	40.1	7	10
Japan	1990-1999	0.2	406	34.6	13	. .
Netherlands	1991-2000	2.9	613	53.3	23	24
Norway	1992-2000	2.7	617	51.2	22	9
Sweden	1990-1998	2.8	452	47.5	25	8
United Kingdom	1990-1999	2.3	558	44.8	9	2
United States	1990-1999	1.3	764	35.7	22	6
OECD average		1.3	501	42.3	17	7
EU average		1.9	526	45.6	18	10

Source: OECD and Eurostat.

greater emphasis on promoting alternative waste treatment methods as well as reducing the generation of refuse. In the context of a continuous increase in waste, the main issues for Swiss waste management policy is to ensure that the different waste treatment options are used in a way to minimise the costs of refuse generation and its disposal both in direct economic terms and in the costs that are imposed on the natural environment.

Performance

The generation of municipal waste per unit of private consumption grew at roughly the OECD average over the 1990s, and its level is similar to that of the OECD average. However, in terms of waste generation per capita, Switzerland is second highest after the United States, reflecting its high income and consumption levels. The method of treating this refuse has changed markedly in the 1990s, the share of land filling being halved and that of recycling doubling. Indeed, at close to half of all municipal waste treatment, Switzerland has one of the highest recycling rates in the OECD area (Table 26). For some categories, such as glass and aluminium cans, recycling rates even exceed 90 per cent.

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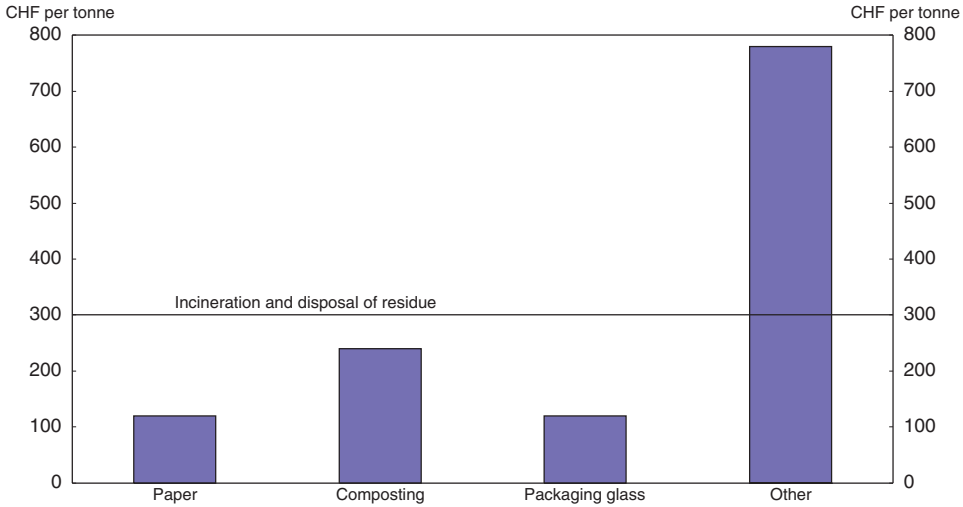
Figure 36. **Cost for recycling selected waste streams**



Source: Office fédéral de l'environnement, des forêts et du paysage.

Policies

The introduction of waste charges based on weight and volume for municipal waste collection have had measurable impacts on reducing waste generation. Around two-thirds of communes have introduced such charging and this has led to both the amount of waste generated and waste management costs falling by around 20 per cent (OFS, 2002). However, an adverse side effect is the illegal dumping of waste and the backyard incineration of around 1 to 2 per cent of municipal waste that generates twice as much dioxin as waste incinerators and so poses a small threat to health (OFEFP, 2002a). In addition illegal dumping causes environmental degradation. In recognition of these problems, the federal authorities are encouraging local authorities to cover municipal waste costs by a mix of volume or weight charges while lowering the incentive to either dump illegally or incinerate waste at home by using a degree of flat-rate charging.

In contrast to other countries with high recycling rates, national legislation does not generally set specific targets for recycling. Rather individuals are given an incentive to recycle through volume and weight-based charges for waste disposal. Indeed, the amount of material going to recycling has increased by around one-third in those communes with such charges. Furthermore, for some waste categories the costs of recycling are favourable compared with the (high) cost of incineration, the major alternative waste treatment option (Figure 36),

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mainly because of the heavy reliance on consumer delivering waste to collection points rather than the waste being collected at the consumer's residence. The extent of recycling for some waste streams is determined periodically in ordinances, but only for materials where it is deemed feasible at a reasonable cost. In effect, if costs of recycling these waste streams are excessive, incineration is the preferred waste management option. In this context, the authorities do not advocate recycling of household plastic waste (except for plastic – PET-bottles), as the costs are prohibitive and the environmental benefits limited. Recycling targets, in combination with voluntary or mandatory prepaid recycling fees,[20] have only been set for beverage containers with the agreement of most of the suppliers of these products. Actual recycling rates of over 90 per cent for glass bottles and aluminium containers considerably exceed the required rate of 75 per cent. However, a recent evaluation has shown that the recycling rate for PET-bottles was only 72 per cent, and measures will have to be taken to raise the recycling rate. However, the prepaid recycling fee set for plastic bottles, typically the most expensive material to recycle, exceeds CHF 1 000 per tonne, which is likely to far exceed the sum of private and external costs of incineration, indicating that, in this case, recycling is using more environmental, physical and human resources than it saves.[21]

Direct controls have been used extensively to favour incineration of waste. Although standards for landfill sites have been strengthened considerably since the 1980s, and international experience shows that it remains the least costly waste treatment option, the landfill of biodegradable waste has been banned since 2000. Incineration facilities, which are often coupled with energy recovery, enjoy a monopoly in the geographical area they serve that has boosted prices, which are also high because of very strict emission control standards. Indeed, the average cost of incineration is around CHF 220 (EUR 145) per tonne (excluding collection and transport) and can rise to double that sum in certain areas. Elsewhere in Europe, costs vary between EUR 40 and 100 per tonne, though prices similar to those in Switzerland are to be found in Germany and Austria. Capacity utilisation in existing incineration plants is high (97 per cent), requiring the construction of new incinerator capacity to cope with increasing waste generation and the required incineration of biodegradable waste and sewerage sludge (see below) (OFEFP, 2002b), imposing significant costs on consumers.

Conclusions

Waste management in Switzerland has made considerable progress in moving towards a system based on incineration and recycling. This has come at the cost of prohibiting the option for biodegradable waste disposal in landfills. By taking a pragmatic approach to waste management, overall costs have been contained. The federal authorities should continue to encourage local authorities to change the charging structure for municipal waste to reduce incentives for

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illegal disposal and domestic incineration, thereby reducing the externalities to the environment and health, even if it reduced the extent of recycling. Secondly, great progress in increasing recycling has been made by only selecting waste streams for which it is feasible. The authorities should consider augmenting this approach by linking the recycling *premia* for different beverage containers directly to the cost of incineration and the avoided production externalities and only recycle to the point at which it is profitable taking into account external costs and benefits. Such an approach should be generalised to other waste streams. Finally, the authorities should investigate the possibility of introducing competition in the market for incineration, which could further enhance the attractiveness of this type of waste treatment through helping to reduce costs.

Water pollution

Main issues

Water pollution is generally at low levels in Switzerland after major investment in wastewater treatment facilities and prohibiting the use of several pollutants. As point source pollution is now largely contained, the main issues facing the authorities is implementing effective measures to control diffuse agricultural sources of pollution in order to protect public health without imposing an excessive burden on society.

Performance

Water quality has improved markedly over the past few decades in certain lakes and reservoirs, notably those close to urban areas, but there has been less progress in rural areas. For example, water quality in Lake Geneva has continuously improved over the 1990s, thanks to increasingly advanced wastewater treatment facilities. At the same time, the concentrations of nitrates and, to a lesser extent, phosphates are at low levels compared with many other OECD countries but have not improved significantly over the past decade in major rivers (Figure 37 and Table 27), despite a considerable fall in the nitrogen surplus on land. Nonetheless, concentrations of nitrates in some lakes and groundwater exceed national standards in over 350 mainly agricultural communes.

Policies

Water quality objectives are set in the federal law on water protection and related ordinances, which establish some of the most stringent standards in Europe. The legislation imposes tough standards for point source emitters, who are also required to pay a discharge fee, and has required major investment in wastewater treatment facilities over the past decades. Furthermore, standards for groundwater quality are extremely stringent.[22] For example, permissible nitrate

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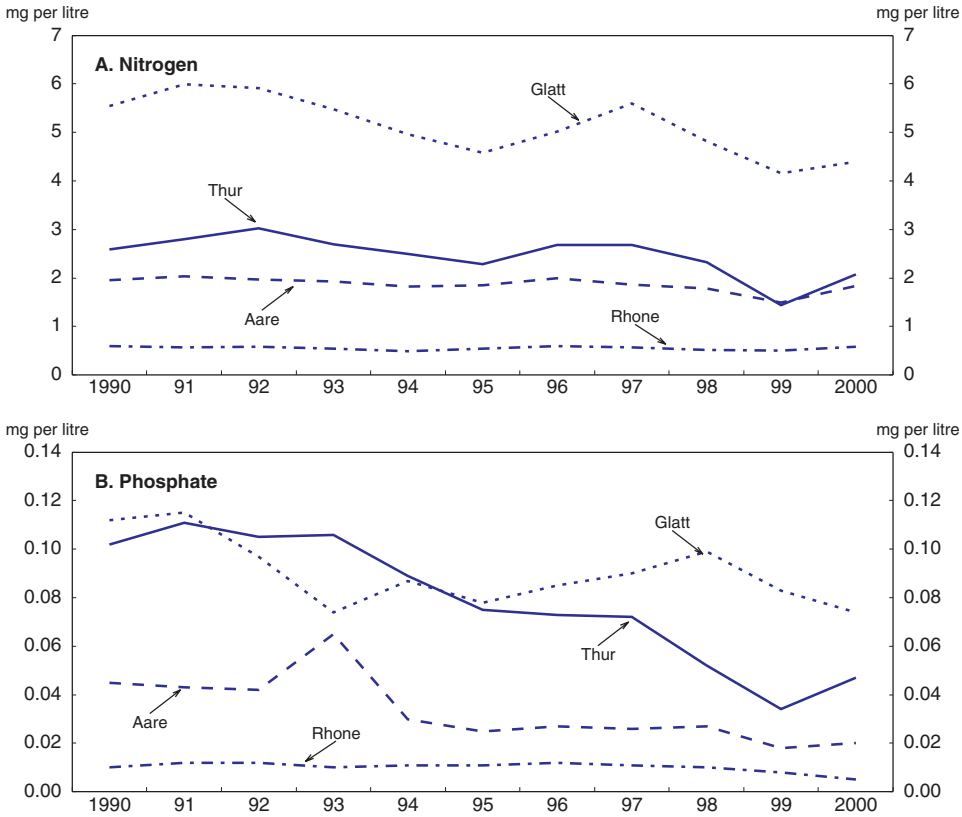
Figure 37. **Water quality in selected rivers**



Source: OFS, *Environment Switzerland 2002 – Statistics and Analysis.*

concentrations in groundwater (25 mg per litre) are half the concentrations set by many European countries and even stricter than the tolerance levels for drinking water established in Swiss foodstuff legislation (40 mg per litre). When pollutant concentrations exceed national standards, the cantonal authorities are required to identify the causes and put in place measures to bring pollutant levels in line with requirements. The approach to water quality management has also prohibited the use of certain substances, including phosphates in detergents and the use of the pesticide atrazine in some cases. More recently the application of sewerage sludge on farmland was prohibited, though in this case the main motivation was food safety rather than water pollution.[23]

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Table 27. **Indicators of water pollution**
Selected rivers

	Nitrates (mg N/litre)		Total phosphorus (mg P/litre)	
	Average 1980-85	Average last 3 years	Average 1980-85	Average last 3 years
Switzerland	**1.5**	**1.5**	**0.1**	**0.1**
Austria	1.2	1.3	0.2	0.1
Denmark	3.5	2.6	0.3	0.1
Finland	0.2	0.3	0.0	0.0
France	2.8	3.1	0.4	0.4
Germany	3.5	3.3	0.5	0.2
Italy	2.0	2.1	0.3	0.2
Netherlands	4.4	3.3	0.5	0.2
Norway	0.3	0.3	0.0	0.0
Sweden	0.4	0.5	0.0	0.1
United Kingdom	4.6	5.2	0.8	0.9
United States	1.1	. .	0.2	0.1

Source: OECD (1999), *Environmental Data Compendium*, OECD, Paris.

While the authorities have in place effective wastewater treatment facilities to adhere to the strict standards in most areas, one of the major tasks facing Switzerland is the necessary renovation of this infrastructure. The total annual costs of renovation over the next decade are estimated to be CHF 2 billion (around 0.5 per cent of GDP). Following the implementation of the polluter-pays-principle in water policy, the financing of this renovation is a matter of the local wastewater operators.[24] Most of these operators, mainly run by the communes, are small and face difficulties in raising external finances. As the financial impact on smaller and poorer local authorities can be large, these areas are allowed longer periods to implement the necessary renovation work and in some cases are granted less stringent requirements for wastewater treatment.

Diffuse pollution from agriculture in the past decades reflects the very high levels of agricultural support (OECD, 2002c), which has encouraged the use of fertilisers and pesticides. Measures of agricultural support have been changed over the past decade with beneficial effects on discharges of water pollutants. Thus, the partial switch from price support to direct payments has moderated the incentives to intensify production. Also, some support measures have been made conditional on the provision of ecological services. Cost-effectiveness analysis of possible measures to reduce nitrogen pollution suggests that reform of agricultural support is very attractive, simultaneously reducing nitrogen pollution and the cost to the economy of such support (Table 28). Overall agricultural reform would reduce nitrogen runoff from agriculture by almost a quarter and account for half of the reduction in nitrogen pollution necessary to eventually bring all

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Table 28. **Cost-effectiveness analysis of measures to reduce nitrogen run-off**

	Reduction in nitrogen run-off, thousand tonnes of nitrogen per year	Annual total cost,[1] thousand CHF per tonne of nitrogen	
		Short-tem	Long-term
Agricultural reform[2]	27-28	–12.5	–24.5
Agricultural reform with integrated production and bio farming[3]	31-32	–11	–22.5
50 per cent tax on fertilisers	7	–1.6	–1.6
Elimination of manure	7-10	16	6
Redistribution of manure	3	0	0

1. A negative total cost is a net benefit to the economy.
2. Agricultural reform includes measures, such as a reduction in price support, that would have led to a halving in the price differential for agricultural products with the European Union between 1994 and 2002.
3. Integrated production is a farming practice that attempts to minimise the application of pesticides and fertilisers in return for direct payments from the federal authorities.

Source: OFEFP (1997), "Stratégie de réductions des émissions d'azote", *Cahier de l'environnement*, No. 273.

underground water back below the very strict national standards, comply with ammonia emission targets and reduce outflows that, in most of the country, are eventually transported to the North Sea.[25]

Although the law on water protection requires cantons to set maximum amounts of fertiliser and manure that can be applied to land, the federal authorities have also launched a specific programme targeting nitrates with the participation of the cantons. Measures supported by such financing range from changing crops to ones that pose less risk of nitrogen run off to converting tilled farmland to permanent pasture in areas adjoining water bodies. Participation in this scheme has been very low with only 11 projects. As a result only a small fraction of available resources from the federal authorities has been used (3 per cent). The principal reason for the low take up of this scheme is severe co-ordination problems, as all parties concerned need to co-operate for a project to be eligible for federal financing of up to 85 per cent of costs. In the face of these co-ordination problems, the authorities are forced to take administrative measures, constraining farming practice in areas where pollution is above standards set in legislation, when the agreement of all parties to participate is lacking.

Conclusions

The management of water pollution has achieved notable results in some areas and for some pollutants following the expansion of the wastewater treatment infrastructure, tight controls on point source emissions, and prohibitions on the use of certain substances. Standards for water quality are already particularly

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rigorous in Switzerland and any new legislation in this area warrants cost-benefit analysis to ensure that additional costs of attaining new targets are not disproportionate to the benefits. Further measures are required to reduce agriculture's impact on water quality. While measures of agricultural support have been linked to ecological functions, the authorities should reconsider the scale of agricultural support taking into account the associated environmental externalities. This would require a cost-benefit analysis of agricultural subsidies. As co-ordination failures prevent the effective application of programmes supported by federal finances to reduce agricultural pollutant run-off in particularly affected areas, the authorities should reconsider this scheme. Particularly if targeted direct payments to farmers do not have the envisaged effects, a tax on nitrogen inputs, which recycles revenues to farmers on the basis of the nitrogen content of output (*i.e.* a tax on excess nitrogen input), could represent an administratively simpler means of attaining the desired improvement in water quality.

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Notes

1. According to the 2003 Swiss Labour Force Survey 30 per cent of all part-time workers (*i.e.* 9 per cent of the labour force) would have liked to work more. One third would have preferred to work on a full-time basis and two thirds would have chosen to work more hours in their part-time job.
2. In terms of national accounting, the gain would be even bigger, since care at home is not accounted as part of GDP whereas institutionalised early care is.
3. Benefit duration for unemployed workers of age 55 or older, provided that they have contributed for 18 months or more within the last two years, has not been reduced.
4. These are internal OECD calculations using the OECD Tax-Benefit Model.
5. The criteria are the following: average duration of job search of unemployed, proportion of job seekers that enter long-term unemployment, exits from long-term unemployment due to benefit exhaustion and proportion of re-registrations after a period of de-registration.
6. See OECD (2001), Table D1.2, where it is shown that in 1996 Switzerland was near the bottom of the OECD league in the wage comparison of several categories of civil servants with those of primary school teachers, implying relatively high wages for teachers.
7. The relatively high wage level might partly explain the fact that Switzerland has less recruitment problems in teaching as compared to other OECD countries.
8. Previous calculations that take into account the probability of dropouts, which are higher in universities than in other types of education, reduce the returns for university education to 3.6 per cent for men and 4.5 for women (Wolter and Weber, 1999).
9. This report includes recommendations such as strengthening nation-wide co-ordination across institutions, increasing the autonomy of senior managers in universities, improving the work conditions and remuneration of younger academic staff relative to professors and putting more emphasis on teaching capacities when recruiting staff (and not only on research potential).
10. The agreement known as the "Bologna process" harmonizes university paths across a large number of European countries.
11. The Commission is taking up this issue in 2004 in conformance with its initial mandate by the Ministry of Finance. Furthermore, an independent expert Commission for Pension Funds headed by Professor Jürg Brühwiler was charged in July 2003 with the task to analyse the supervisory system of pension funds and life insurance companies and to propose concrete reforms to optimise the current system.
12. As energy related CO_2 emissions make up for nearly 80 per cent of overall GHG emissions, a 10 per cent reduction target for carbon dioxide emissions translates into

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an 8 per cent reduction in GHG emissions. For other greenhouse gases, policies include banning the use of the synthetic gases covered by the Kyoto Protocol (HFCs, PFC, SF_6), unless there is no alternative technology available or the alternative is more environmentally harmful.

13. The percentage reduction from the business-as-usual projection for 2010 implies a 20 per cent reduction for emissions from transport fuels.
14. In addition, there are ordinances on air pollution, waste management, and environmentally hazardous substances and the national clean air strategy, which are also relevant for climate change policy.
15. Studies from other countries cast doubt on the effectiveness of voluntary agreements (OECD, 1999).
16. Until 1998, investment was limited to promoting renewable energy. The authorities are now targeting more cost efficient measures in attaining greater energy efficiency, which are largely administered by the cantonal authorities.
17. From 2001, the new heavy vehicle fee depends on the distance travelled on Swiss roads, the gross weight of the vehicle, and the pollutant emission characteristics of the vehicle.
18. A number of other factors must also be taken into account when deciding on the introduction of this tax (Epiney and Gross, 2002).
19. Such a figure is equivalent to EUR 500 per tonne of carbon.
20. Pre-paid disposal fees are also levied on waste of electronic and electrical goods (voluntary system), batteries, glass-bottles (both mandatory), tyres, and vehicles (both voluntary).
21. RDC-Environment and Pira International (2001).
22. It has, however, proved difficult to find a close connection between water quality standards and health outcomes in other countries.
23. The ban on sewerage sludge will incur an estimated cost of CHF 41 million annually arising from the costs of replacing the nutrient content in sewerage sludge and additional costs for incineration (Stadelmann *et al.*, 2002). In 2000, the cost of drying, transporting and incinerating a tonne of sewerage sludge was CHF 1 200 per tonne, while the cost of using it for agricultural purposes was CHF 360.
24. Households pay a fee covering wastewater treatment based on consumption. The fee for other point sources is based on consumption, adjusted for the supplementary costs of pollutants in their discharge.
25. In the south-west of the country watersheds feed the Mediterranean.

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Annex I

Main features of the tax package

The tax reform put before Parliament in February 2001, and voted in June 2003, contains three parts:

- A reduction and change in income taxation.
- A cut in taxation on financial transactions.
- A reform of housing taxation.

The *main component* of the tax package concerns household income taxation, which is set to be cut by CHF 1.3 billion (¼ per cent of GDP). The reform includes the introduction of a procedure for taxing married couples based on the partial splitting model,* and an increase in deductions for dependent children. The *second component* consists of a series of exemptions from stamp duty, the cost of which is put at CHF 310 million, or 10 per cent of the total amount of the tax. These exemptions, which apply to foreign institutional investors and Swiss and foreign investment funds, were largely implemented as of 2001, with the adoption of urgent measures. The latter were adopted definitively and to some degree extended to listed foreign companies. The *third component* of the reform provides for the abolition of rental values and taxation of housing, but keeps the deductions with respect to interest costs (for 10 years, with a capping system) and upkeep expenses (for an unlimited period), while home ownership savings have been made exempt from tax. The cost of these measures is CHF 480 million as far as taxes received directly by the Confederation on its own and the cantons' account are concerned.

The overall cost of the reform, put at CHF 2 billion (½ per cent of GDP), costs CHF 1.5 billion for the Confederation and CHF 500 million for the cantons. Parliament has increased the scale of the cut in taxes by comparison with the Federal Council's initial proposal, especially as regards the housing measures. The tax reform will however be the subject to a referendum, as a number of cantons consider the budgetary cost excessive. Abolishing rental values, which are also used by cantons and communes for their own taxes, will result in these administrations incurring further revenue losses put at CHF 1 billion. Moreover, the household income tax reform could lead to an additional decline by CHF 1 billion of cantonal resources, if they are *de facto* pushed to adopt a number of new deductions in order to be consistent with the federal rules. Because of this referendum, which could be organised in spring 2004, the reform will be postponed for a year or rejected, depending on the outcome of the popular vote. If adopted, the impact of the tax package on the Federal budget will probably not be felt until 2006. Also, the reduction in tax on real estate is expected to come into force in early 2008, leaving people sufficient time to adapt.

* Under this procedure, the rate of taxation is calculated by adding spouses' incomes together and dividing them by 1.9, which eases the burden on a large proportion of households without affecting the burden on taxpayers living alone.

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Annex II

The occupational benefit system: principal characteristics

Occupational benefit plans are fully funded and they occupy an important place in the Swiss pension system, of which they constitute the second pillar. Both outlays and revenues for this branch are higher than those of the AVS (the first pillar), and the assets managed by pension plans amounted to CHF 474 billion, or 117 per cent of GDP, in 2000. Institutions in this branch therefore constitute prime players in the financial market.

The Law on Occupational Benefits (LPP) instituted a minimum insurance system in 1985. Enrolment in a pension plan is compulsory for salaried workers over 24 years of age whose annual income exceeds CHF 25 320. Contributions, of which at least 50 per cent are paid by the employer, are calculated on the portion of income between this figure and three times this figure, *i.e.* CHF 75 960. The rates to calculate the contributions vary in general between 7 and 18 per cent, depending on the age of the insured person (OECD, 2000).

The contributions accumulated over the course of an individual's working life, together with the interest they generate, constitute that person's retirement capital, on which the pension is calculated. For those pension plans that are primarily financed by contributions (76 per cent of pension plans in 2000), the law requires that interest be calculated at a certain rate, the minimum level of which is set by the Federal Council. This minimum rate, which had remained stable at 4 per cent since 1985, was reduced to 3.25 per cent in 2003. Upon retirement, the accumulated capital is transformed into a lifetime annuity, using a conversion rate defined by law as a function of average life expectancy. This rate has been maintained at 7.2 per cent since 1985 (*i.e.* retirement capital of CHF 100 000 will entitle a person to an annual pension of CHF 7 200).

A large portion of the occupational benefit system in fact goes beyond these minimum standards, through optional insurance plans. Many pension plans will insure the portion of annual salaries exceeding the mandatory maximum (CHF 75 690) or, in certain cases, those below the legal minimum (CHF 25 320). Because payments to occupational benefit plans are not taxable, this "supplementary" or "beyond-requirement" regime, which is not subject to the rigid conditions of the LPP, allows income to be shielded from tax. While employers are required to pay only half of these contributions, they are voluntarily paying more than 60 per cent. Independent workers, who are not covered by the second pillar, are also able to contribute on an optional basis. Although detailed information is not available, it is estimated that "supplementary" insurance accounts for at least half of the occupational benefit system.

The second pillar is organised on a decentralised basis, involving some 9 000 foundations, of which 2 600 are registered as providers under the mandatory regime. These foundations are legally separate from their sponsoring companies. Most SMEs in fact do not have their own plan but are affiliated with collective foundations of banks or insurance companies. Individual supervision of plans, depending on the type of institution,

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is the responsibility of the cantons and the Federal Office of Social or Private Insurance. The Federal Office of Social Insurance (OFAS) maintains overall surveillance of the system.

The LPP establishes a Safety Fund, which serves as a re-insurer of the pension plans. This fund is financed on a pay-as-you-go basis: all pension plans contribute 0.03 per cent of insured income and 0.3 per cent of their retirement assets. If a pension plan is unable to meet its obligations because of insolvency, the Safety Fund will pay the mandatory portion of pensions and, within certain limits, the supplementary portion as well. The Fund's reserves stood at less than CHF 40 million at the end of 2001.

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Annex III

Main features of the Swiss health care system

The private sector accounts for a large proportion of health care funding and supply. The main features of the system are as follows:*

- The system is based on a private insurance market. There are a large number of private funds providing households with basic insurance, which is compulsory, plus optional complementary insurance. These funds are not allowed to make profits on basic insurance, which provides cover defined by the Federal Social Insurance Office (OFAS). The absence of profits in this market segment is based on the principle that health care decisions should not be guided by profitability but only by strictly medical considerations.
- For a given insurance fund, the insured pay the same premium whatever their age and sex. Premiums, which are paid individually, can however vary between funds which are in competition in the same canton. The insured can obtain reductions in their *premia* if they are ready to restrict their choice of health care provider by, for example, joining a managed care system or whatever family doctor system their insurers can propose. P*remia* can also be reduced if members are prepared to pay a bigger deductible than the scheduled minimum.
- Under the Swiss system, which operates with single *premia* per insurer, a risk equalisation mechanism has been introduced on a temporary basis (up to 2005), to take account of differences in people's medical requirements and prevent competition between funds from resulting in risk selection. In each canton, the insured are divided into 30 groups defined according to age and sex and the average costs of each group are compared with the average costs of the insured as a whole. Each insurer receives from or pays into an equalisation fund a contribution which depends on the differential between the average cost of the population he covers and the population as a whole. This risk equalisation takes place at the end of every financial year.
- The cantons and communes co-fund about 60 per cent of the hospitals, while the Confederation and the cantons partly subsidise the insurance premiums of low-income households. Hospitals are to a large extent funded on the basis of the number of days of hospitalisation, while doctors in private practice are remunerated on a payment-per-service basis.
- All insurers are bound by the obligation to contract, *i.e.* they have to accept invoices submitted by all doctors approved.

In Switzerland, two main phases need to be completed before a drug is put on the market and reimbursed by the mandatory health system: 1) a marketing approval procedure and

* See OECD (2000) for a detailed description.

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2) the inclusion in the Specialities List. Since the new Law on Therapeutic Products came into force, S*wissmedic* has been the responsible authority in the first phase. Medical products may only be distributed if a corresponding licence is granted by this authority, which examines the drugs' quality, safety and efficiency point of view. S*wissmedic* issues licences but may also withdraw them. This licence is one of the preconditions for inclusion in the Specialities List (SL), which defines the medicines reimbursed by the mandatory health insurance. In the second phase, the OFAS is the responsible authority. It decides on inclusion or exclusion from the SL and establishes the upper price limits for all service providers in Switzerland. Calculation of the prices is based on a comparison of prices abroad in Germany, the Netherlands, Denmark and the United Kingdom (and also with Italy, France and Austria), and a therapeutic cross-comparison, where comparable SL medicines are available in terms of indication or action.

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Annex IV

Comco Communications in the realm of the vertical agreements

By virtue of LCart Article 6, COMCO may stipulate in a Communication the conditions on which competition-related agreements may generally be deemed justified on grounds of economic efficiency within the meaning of LCart Article 5, par. 2. If greater judicial security is required, COMCO may also issue Communications, in analogous application of the aforementioned Article 6, to lay down other principles for interpreting the law. To date, COMCO has adopted four Communications. The first one, issued on 15 December 1997, dealt with the certification and sponsoring of sports goods.* On 4 May 1998 COMCO adopted a Communication on calculation formulas which specified the conditions justifying the use of such formulas by associations. In 2002, COMCO adopted two important Communications. The first dealt with the assessment of vertical agreements and the second with vertical agreements in the distribution of motor vehicles.

Communication of 18 February 2002 on the assessment of vertical agreements

In February 2002, COMCO adopted a Communication regarding the assessment of vertical agreements. In that Communication, the content of which is similar to Community law, COMCO deems that vertical agreements have a notable effect on competition, within the meaning of LCart Article 5, par. 1, if, *inter alias*, they:

- Impose on merchants, either directly or indirectly, fixed or minimum prices for the resale of goods or services.
- Limit a merchant's resale territory or circle of customers, either directly or indirectly.
- Limit sales to final consumers, insofar as such limitation is imposed on an approved merchant within a system of selective distribution.
- Limit cross-supply relationships between approved merchants within a selective distribution system, even if the merchants are active at different echelons of the market.
- Prevent a supplier from supplying spare parts to third parties (final consumers, repair shops, etc.) other than merchants who are parties to the agreement.
- Impose a non-compete obligation for a period exceeding five years, or exceeding one year after a vertical agreement expires.

In its Communication, COMCO stipulates that other vertical agreements are not generally deemed to have a notable effect on competition if the market shares of all participating

* This Communication dealt with the conditions justifying certification and sponsoring agreements, insofar as such agreements regulate or influence competition in the market for sports goods.

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businesses do not exceed 10 per cent in any of the relevant markets. However, the Communication also states that there can be exceptions to this principle if competition in the relevant market is limited by the cumulative effects of a number of similar vertical distribution networks operating side by side, provided that participating suppliers and merchants are effectively or potentially in competition.

Any vertical agreement that has a notable effect on competition, within the meaning of LCart Article 5, par. 1, is deemed unlawful unless justified on grounds of economic efficiency within the meaning of LCart Article 5, par. 2. The Communication states that an agreement may be justified *inter alias* if it enables a distribution network to be organised more efficiently, and if a restriction of competition is necessary to achieve that aim.

Communication on vertical agreements in motor vehicle distribution

COMCO issued a Communication on vertical agreements in motor vehicle distribution in October 2001. This Communication, which is in harmony with the new European exemption Regulation (EC) No. 1400/2002, supplements that of February 2002 on vertical agreements, stipulating certain aspects specific to the distribution of motor vehicles. The fundamental aspects of that Communication are as follows:

- It compels the motor vehicle industry to allow vehicles to be imported from the European Economic Area (EEA).
- It dissociates sales from after-sales service.
- It facilitates the sale of spare parts and parallel imports thereof.
- It allows a choice between selective and exclusive distribution systems.
- It facilitates access to spare parts, information (including training) and diagnostic equipment for independent repairers.
- It compels the motor vehicle industry to allow multi-branding, *i.e.* to allow dealers to offer more than one brand for sale in the same establishment.

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Annex V

Assessing the benefits of the regulatory reforms: methodological information

This annex provides methodological information on the quantification of the macroeconomic effects of reforms that would increase competition in the agriculture, health care, telecommunications, gas, electricity, professional services and distribution sectors, which together account for a third of total production.

The methodology employed is similar to that used in the OECD study on regulatory reform[1] (OECD, 1997). Effects were quantified in two steps. The first consisted in calibrating the possible effects of sectoral reforms, using static analysis to gauge the potential impact on prices. The second step established a correspondence between the assumptions underlying the price decline estimated for the economy as a whole and the resultant rise in production. This was done by assessing the reforms' dynamic effects via a macroeconomic model, capable of quantifying the consequences of a reduction in business margins and wage mark-ups, and of a boost to labour efficiency, that the proposed reforms are expected to produce.

First step in the quantification: calibrating the effects of sectoral reforms

Calibrating the potential effects of regulatory reform at the sectoral level entailed a set of assumptions regarding the impact on productivity, margins and wage mark-ups in the sectors concerned. In some capital-intensive sectors such as electricity or gas, assumptions of a decrease in the cost of capital were also factored in because of the economies of scale that reform-related restructuring is expected to generate. The calibration adopted in the various sectors with regard to these different assumptions was based on a variety of considerations. First, it depended on the potential effect on prices. The effect of the price cuts factored in at the sectoral level took account of price level differentials with other countries, as estimated, for example, from purchasing power parity data. This sectoral effect was quantified using a static analysis that was able to take account not only of the assumptions made with regard to margins, productivity and wages, but also of price decreases for intermediate consumption, thanks to the use of an input-output matrix.[2] Second, account was taken of the effect of regulatory reforms carried out in other countries in order to quantify their possible effects. In the case of electricity, the studies available would suggest, for example, that the reform undertaken in Germany enabled businesses to cut their prices by 15 to 20 per cent. Lastly, empirical information gathered when the various sectors were analysed – *e.g.* concerning productivity gaps with other countries – was used as well.

Overall, the assumptions made were consistent with price decreases of between 3 and 17 per cent in the sectors analysed. The sharpest decreases involved health care services and agricultural and processed food products. These sectors are the ones with the greatest downside potential, since in both cases the price level differential relative to the

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EU average is between 50 and 60 per cent. In all, the aggregate impact of the proposed sectoral reforms on the whole economy pointed to a drop in the general price level of some 6 per cent. This would be obtained through a 6½ per cent reduction in margins, productivity gains of 5 per cent and a 2 per cent compression of wage mark-ups. In addition, intermediate costs might drop by roughly 5 per cent. While significant, these cost reductions and productivity gains would still offset only a fraction of the gap in the level of these variables as compared to other countries.

Second step in the quantification: estimating the induced effect on production

The OECD's macroeconomic model Interlink was used to estimate the proposed structural reforms' dynamic effect on production.[3] A simulation of rising trend productivity and falling margins and wages, calibrated on the above estimates, was performed over a ten-year period.[4] The simulated reductions in margins and wages, combined with a widening of the output gap because of the rise in potential output, was conducive to downward price pressures. It was therefore assumed that the authorities relax monetary policy to neutralise those pressures. Interest rate cuts would also reduce the exchange rate, which would be conducive to an upturn in activity compatible with that of potential output. This easing of monetary policy was calibrated as being compliant with the National Bank's goal of price stability.

The results of this simulation show that the positive effects of regulatory reform on production, which result from productivity gains, are likely to be reinforced in the long term by rising employment and an increase in the stock of capital. The permanent reduction in wage mark-ups and business margins in fact induces a structural decrease in unemployment and stimulates potential employment, the rise being accompanied by an upturn in capital spending. This increase in the stock of capital is also underpinned by the fall in interest rates. Moreover, this dynamic favourable to the rise in potential output is stimulated by triggering a virtuous circle of lower costs and improved international competitiveness. The cumulative impact of the rise in GDP is estimated at 8 per cent after ten years as compared to the reference scenario, the bulk of these gains representing a rise in potential output. Continued price stability, combined with exchange rate depreciation exceeding 10 per cent at the end of the period, also allows an *ex post* reduction of price levels in relation to other countries. This strengthening of macroeconomic performance also widens the scope for action by general government, which benefits from a rise in tax revenue and an easing of debt service.

Clearly, an empirical exercise such as this provides only an approximation of the long-term benefits of enhanced competition as a result of the proposed reforms, and the time needed to achieve the reforms is also uncertain. Nevertheless, the estimated overall impact would appear reasonable, insofar as the positive effect of productivity gains on production should ultimately be bolstered by a rise in employment and the stock of capital. As shown by similar empirical analyses concerning European countries (Bayoumi *et al.*, 2003), increased competition should in fact reduce the market power wielded by businesses and workers that prompts them to cut supply while increasing their rents.

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Notes

1. A similar exercise assessing regulatory reforms had also been carried out in the previous survey (OECD, 2002).
2. The input-output matrix used was derived from work by the Laboratory of Applied Economics in Geneva using a database for 1990 recalibrated for 1995 by the Swiss Federal Institute of Technology in Zurich (see Lips and Nieuwkoop, 2001, for further details).
3. The structure of the Interlink model's supply block makes it possible to evaluate at an aggregate level the effect of trend output and profit and wage behaviour on activity (Turner *et al.*, 1996).
4. In order to ensure consistency with the static quantification of the reforms undertaken in the first step of this exercise, the shock was calibrated on an *ex ante* basis, *i.e.* without factoring in the impact of other (*e.g.* financial) variables describing the functioning of the rest of the economy. The interaction between the margin, wage and productivity variables was taken into consideration, however, so as to ensure adequate calibration of the shock performed with the model.

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Annex VI

The reform of the unemployment benefit system (LACI)

The measures approved in the 2003 reform of the unemployment benefit system are the following:

- The basic contribution rate was reduced from 3 to 2.5 per cent on earnings up to CHF 106 800, while the "solidarity contribution" of wages between CHF 106 800 and 267 000 was decreased from 2 per cent to 1 per cent. In January 2004 the basic contribution will be further reduced to 2 per cent and the "solidarity contribution" will disappear.*
- The minimum contribution period required to be entitled to receive the unemployment benefit (UB) was raised from 6 to 12 months. This will only apply to new unemployed, not to actual benefit recipients.
- Benefit duration will be reduced from a maximum of 520 days to 400 days, although workers above age 54 will be spared from the reduction provided they have contributed for at least 18 months. Cantons with an unemployment rate above 5 per cent may also keep the current maximum duration of 520 days. However, in that case they have to bear 20 per cent of the additional cost. The impact of this measure could be small in the short term, since most current unemployed workers have had short unemployment spells (because of the recent rise in unemployment), but it could rise quickly if the stagnation persists.

* The solidarity contribution was raised from 1 to 2 per cent at the end of the 1990s to cope with the financial problems of the LACI. At that time it was considered a temporary measure.

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Glossary of acronyms

AI	Disability insurance
AIMP	*Accord international sur les marchés publics* (Inter-cantonal Agreement on Public Procurement)
ALMPs	Active labour market programmes
AP	Agricultural Policy
AVS	First pillar of the old-age insurance
BAK	Conjuncture Institute of Basel
BLS	Bureau of Labour Statistics
CDIP	*Conférence suisse des directeurs cantonaux de l'instruction publique* (Swiss Conference of Cantonal Ministers of Education)
CFF	Federal railway company
CHF	Swiss franc
ComCom	Federal Communications Commission
COMCO	Competition Commission
CPI	Consumer Price Index
DFI	Federal Department of the Interior
DI	Disability Insurance
DMA	Domestic Market Act
EC	European Commission
ECB	European Central Bank
EEA	European Economic Area
EFTA	European Free Trade Area
EMS	*Établissements médico-sociaux* (Medico-social institutes caring for the elderly)
EU	European Union
EUR	Euro currency
FBC	Federal Banking Commission
FEER	Fundamental Equilibrium Exchange Rate
FOPI	Federal Office of Private Insurance
g	gram
GDP	Gross Domestic Product
GHG	Greenhouse gas
GNP	Gross National Product
HES	*Hautes Écoles Spécialisées*
HFCs	Hydrofluorocarbons
HMO	Health Maintenance Offices (Health-care networks)
HP	Hodrick-Prescott (filter)
INTERLINK	Macroeconomic model of the OECD Economics Department
kg	kilogram

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KOF	Federal Polytechnic School of Zurich
KWh	kilo watts per hour
LACI	Unemployment insurance law
LIBOR	London Interbank Offered Rate
LPP	Occupational benefit law
mg	milligram
MLCA	Money Laundering Control Authority
MWh	mega-watts per hour
NAIRU	Non-accelerating inflation rate of unemployment
NATREX	Natural real exchange rate
OECD	Organisation for Economic Cooperation and Development
OFAS	*Office fédéral des assurances sociales* (Federal Social Insurance Office)
OFEFP	*Office fédéral de l'environnement, des forêts et du paysage* (Swiss Agency for the Environment, Forests and Landscape)
OFS	*Office fédéral de la statistique* (Federal Statistics Office)
OPCA	*Office parlementaire de contrôle de l'administration* (Parliamentary body)
ORPs	Regional placement offices
OSEL	Organisation of the Electricity Sector
PET	Polyethylene terephthalate
PFCs	Perfluorocarbons
PISA	Programme for International Student Assessment
PPP	Purchasing Power Parity
R&D	Research and Development
SIA	Strategic Impact Assessment
SMEs	Small and medium-sized enterprises
SNB	Swiss National Bank
STEP	OECD working group on short-term projections
SWX	Swiss Exchange
UB	Unemployment benefit
UBS	Union Bank of Switzerland
UMTS	Universal Mobile Telephone System (third generation mobile telephone system)
USD	United States dollar
VAT	Value Added Tax
WTO	World Trade Organisation

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